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HY Pediatrics, by Dr Michael D Mehlman

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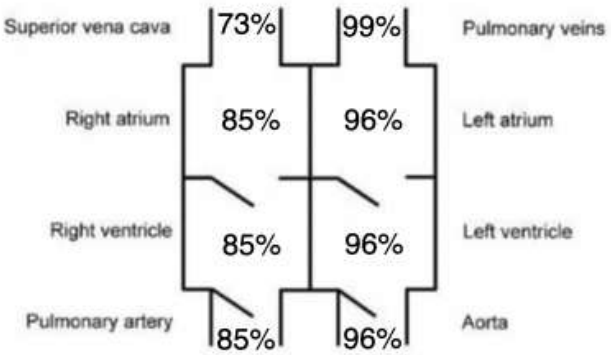
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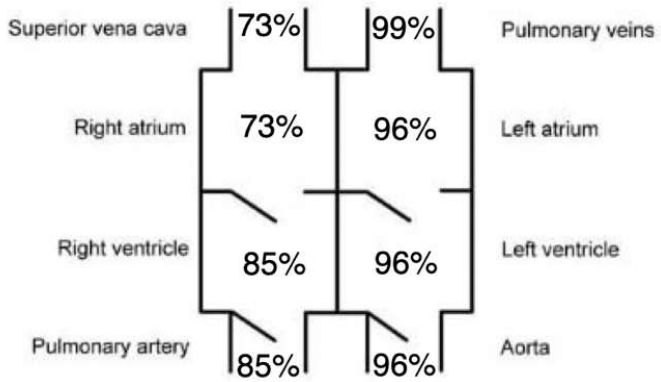
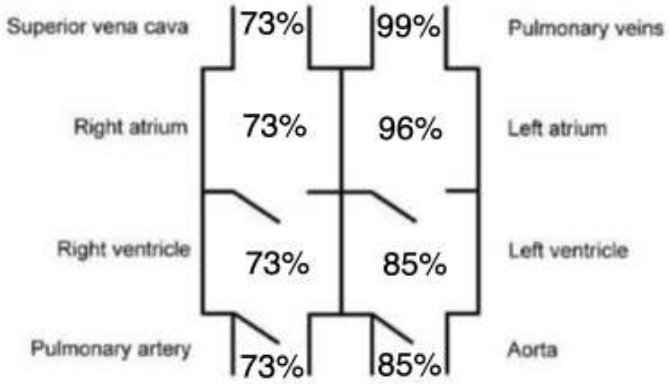
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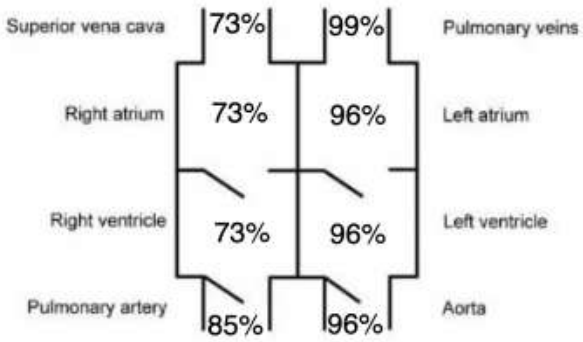
Peds Cardio

<table>
<tr><th colspan="2">HY Valvular / flow abnormalities for Peds</th></tr>
<tr><td colspan="2">- There's a lot of detail regarding the following conditions I could discuss at length, but I'm going to keep the focus on what is sufficient for the pediatrics shelf here (and Steps 2CK/3), which is the point.</td></tr>
<tr><td>Atrial septal defect</td><td>- Fixed splitting of S2
- Can sometimes be associated with a systolic flow murmur, since more blood L → R from the LA → RA means more blood flow across the pulmonic valve. So Q might say "fixed splitting of S2 and a systolic murmur."
- Sometimes can be seen in Qs as "wide, fixed splitting." I only mention this because some students get pedantic / ask about this. "Wide splitting" just means right ventricular hypertrophy. So if the Q says "wide, fixed splitting," they're saying the patient has RVH due to an ASD.
- Patent foramen ovale = ASD on USMLE. Don't confuse with patent ductus arteriosus (discussed below).
- USMLE loves asking questions showing you change in oxygen in the chambers of the heart and making you choose ASD, VSD, etc.
 

- For example, you can see above that somehow O2 increases from the SVC to the RA, which is ordinarily impossible. The only way this could occur is if an ASD is present, where oxygenated blood moved from LA → RA.</td></tr>
<tr><td>Ventricular septal defect</td><td>- Holosystolic (aka pan-systolic) murmur at lower left sternal border.
- Can be associated with a diastolic rumble or enlarged left atrium (if more blood going L → R across VSD, then more blood is returning to the LA from the lungs → LA dilatation).
- Seen as part of tetralogy of Fallot (VSD, RVH, overriding aorta, pulmonic stenosis).
- If a VSD is repaired, USMLE wants ↑ LV pressure, ↓ RV pressure, and ↓ LA pressure as the changes now seen in the heart.
- VSD does not cause cyanosis at birth. Only years later after the higher blood flow to the lungs results in pulmonary hypertension, followed by right ventricular hypertrophy and reversal R → L (Eisenmenger) does the patient become cyanotic.
- Murmur can be silent or soft at birth, followed by loud at 7 days of life. The USMLE will ask why the murmur is louder now → answer = decreased pulmonary vascular resistance – i.e., the lungs open up during the first week of life, resulting in decreased RV pressure and an increase in the L → R pressure gradient (louder murmur).
- Conversely, if they ask why the murmur was softer at birth compared to now, the answer is "increased pulmonary vascular resistance," where the lungs were still closed at the time, so there was a lesser gradient L → R (softer murmur).
- Similar to ASD Qs, USMLE loves giving you diagrams with changes in O2 between the chambers and then making you infer we have a VSD.</td></tr>
</table>

	 - NBME loves this style of Q. You can see O2 somehow increased from RA to RV. The only way this is possible is if we have a VSD where oxygenated blood moves from LV → RV.  - This one might initially appear a little more difficult. This is Eisenmenger syndrome, where we have a reversal of flow from RV → LV across the VSD. The NBME is known to show this diagram as well.
Atrioventricular septal defect	- Seen in Down syndrome. - Between the atrium and ventricle, aka "endocardial cushion defect," although this latter term can also apply to ASD and VSD in Downs.
Mitral regurgitation	- Holosystolic (pan-systolic) or just regular "systolic," 29 times out of 30. - Q on NBME 20 offline for Step 1 has MR as "mid-systolic," but I contend this is erratum. - Most USMLE questions will not mention it radiating to the axilla. - Seen **acutely** in rheumatic heart disease as a kid (valve scars over years later and becomes mitral stenosis). - Can cause **JVD** (i.e., back up all the way to the right heart); this is mentioned numerous times on the new NBMEs.
Mitral stenosis	- Described as "rumbling diastolic murmur with an opening snap"; can also be described as "decrescendo mid-late diastolic murmur" (i.e., following the opening snap). - Can cause a right-sided S4 if the pressure backs up all the way to the right heart (seen on NBMEs sometimes; this confuses students because they think S4 must be LV, but it's not the case). An S4 is a diastolic sound heard in either the LV or RV when there is diastolic stiffening due to high afterload. - 99% of mitral stenoses are due to Hx of rheumatic heart disease (i.e., the patient had rheumatic fever as a child, where at the time it was mitral regurg, but years later it has now become mitral stenosis). - One 2CK NBME Q mentions patient with history of rheumatic heart disease who, years later, now has 4/6 rumbling diastolic murmur **without**

	an opening snap; this is still mitral stenosis. Although opening snap is buzzy for MS, just be aware it's not mandatory and that this Q exists on NBME.
Mitral valve prolapse	- Most common murmur. - Described as mid-systolic click. - "Myxomatous degeneration" is buzzy term that refers to connective tissue degeneration causing MVP in Marfan and Ehlers-Danlos. - Almost always asymptomatic. On 2CK forms, they want you to know about "mitral valve prolapse syndrome," which is symptomatic MVP that presents as repeated episodes of "fleeting chest pain" on the left side in an otherwise healthy patient 20s-30s, sometimes starting in teenage years. They might say there is Hx of MI in the family, but this is MVPS, not MI. Answer on 2CK form is "no treatment necessary." - USMLE loves using MVP as a distractor in panic disorder questions, particularly on the 2CK Pysch CMS forms. They will give long paragraph about panic attack/disorder + also mention there's a mid-systolic click; they'll ask for cause of patient's presentation → answer = panic disorder, not MVP → student is confused because they say mid-systolic click, but the MVP isn't the cause of the patient's presentation; the panic disorder is; MVP's are usually incidental, benign, and asymptomatic.
Aortic regurgitation	- Decrescendo holo-diastolic (pan-diastolic) murmur; can also be described as "early diastolic murmur," or "diastolic murmur loudest after S2." - Causes wide pulse pressure (i.e., big difference between systolic and diastolic pressures, e.g., 160/50, or 120/40) → results in head-bobbing and bounding pulses (don't confuse with slow-rising pulses of aortic stenosis). - The bounding pulses can be described on NBME as "brisk upstroke with precipitous downstroke." In turn, they can just simply say, "the pulses are brisk," meaning the systolic component is strong. - I would say 4/5 times bounding pulses means AR. The other 1/5 will be PDA and AV fistulae (discussed below). Bounding pulses occur when blood quickly leaves the arterial circulation. In AR, the blood quickly collapses out of the aorta back into the LV. In PDA, it leaves the aorta and enters the ductus arteriosus; in AV fistulae, it leaves for a vein. - Can lead to volume overload on the LV and eccentric hypertrophy. - Highest yield cause on USMLE is aortic dissection → can retrograde propagate toward the aortic root causing aortic root dilatation and AR. - Even though MVP is most common in Marfan and Ehlers-Danlos, AR is second most common in these patients, since if they get aortic dissection, this can lead to AR. - Peds shelf can give you random 2-year-old with robust decrescendo diastolic murmur (answer = AR). Student says, "What's causing it though?" Great question. It's still what shelf will do. You could be aware that congenital bicuspid valve *also can cause AR in peds,* not just AS.
Aortic stenosis	- Mid-systolic murmur, or just "systolic" murmur; can also be described as "late-peaking systolic murmur with an ejection click." - Radiates to the carotids. This descriptor shows up quite frequently on NBME (way more than radiation to the axilla for MR). - Causes slow-rising pulses, aka "pulsus parvus et tardus" (don't confuse with bounding pulses of AR). - SAD → Syncope, Angina, Dyspnea; classic combination seen in AS, albeit not mandatory. If you get a question where they say systolic murmur but you're not sure of the diagnosis, if they say chest pain or fainting, you know it's AS. - Often caused by bicuspid aortic valve. **The patient need not have Turner syndrome and often won't.** Bicuspid valve is usually inherited as an autosomal dominant familial condition.

	- The bicuspid valve need not calcify in middle-age prior to the AS forming. **Bicuspid valve can present with AS murmur in child or high schooler.** - Do aortic valve replacement on 2CK if 1) cross-section of valve is <1.0 cm^2, or 2) there is SAD. They ask both of these as separate Qs where they want valve replacement. - "Supravalvular aortic stenosis" can occur in Williams syndrome (rare AD disorder with elfin-like facies).
Tricuspid regurgitation	- Will be described on USMLE as a holosystolic murmur that increases with inspiration. - Right-sided heart murmurs get worse with inspiration → diaphragm moves down → decreased intra-thoracic pressure → increased right-heart filling. - Can cause pulsatile liver. - Highest yield cause of TR on USMLE is **pulmonary hypertension / cor pulmonale.** I see this all over the NBME exams. For whatever reason, these conditions do not cause pulmonic regurg; they cause tricuspid regurg. In other words, if you see tricuspid regurg in a Q, your first thought should be pulmonary hypertension or cor pulmonale (right heart failure due to a pulmonary cause). - Bronchopulmonary dysplasia can cause cor pulmonale in infants on NBME. This is when the neonate is exposed to too much oxygen at birth, usually while in ICU and/or on home oxygen, followed by development of fibrotic lung disease → backs up to right heart.
Tricuspid stenosis	- Nonexistent murmur on USMLE. I don't think I've ever seen this assessed once on any NBME exam for Steps 1 and 2 combined. - In theory, would be a rumbling diastolic murmur similar to mitral stenosis, but would increase with inspiration since it's on the right side of the heart.
Pulmonic regurgitation	- Same as with tricuspid stenosis, this is a nonexistent murmur on USMLE. I've never seen it assessed. In theory it would be the same as aortic regurg but on the right (i.e., holo-diastolic murmur), but increases with inspiration.
Pulmonic stenosis	- Seen in tetralogy of Fallot. - Described as mid-systolic murmur, or just regular "systolic" murmur, that increases with inspiration, at the left sternal border, 2nd intercostal space. This is the theoretical location, whereas AS is the 2nd intercostal space on the right, not left. But the USMLE often isn't strict about murmur locations this way.
Patent ductus arteriosus (PDA)	- Ductus arteriosus is special vessel in fetal circulation that connects the proximal pulmonary trunk to the descending arch of the aorta. This allows for blood to bypass the high-resistance lungs *in utero*. After birth, this vessel should close, resulting in a remnant called the ligamentum arteriosum, but sometimes it does not close → PDA. - If a PDA occurs, blood moves in the neonate L→R (i.e., opposite of *in utero)* from the descending arch of the aorta to the pulmonary trunk. 

	- You can see in the above diagram, somehow the blood become more oxygenated from the RV to the pulmonary artery, which is ordinarily impossible. The only way this could have occurred is if oxygenated blood came L→R from the aorta to the pulmonary artery via a PDA. - Murmur described three ways on USMLE: 1) continuous, machinery-like murmur; 2) pan-systolic pan-diastolic murmur (meaning it's continuous throughout both systolic and diastole); and 3) to-and-fro. The latter shows up on 2CK offline NBME 6. - Classically associated with congenital rubella (HY). They'll give a kid born with a PDA and then ask what the mom experienced while pregnant; answer = arthritis and/or rash (rubella often presents as arthritis in adults). - Indomethacin (NSAID) will close the PDA. - Prostaglandin E1 is used to keep a PDA open (if a kid with congenital heart malformations is born cyanotic and we need to buy time until surgery). - An open PDA can mask cyanosis in a newborn in a variety of conditions (i.e.,., hypoplastic left heart syndrome or pre-ductal coarctation). If they tell you a kid is born with normal APGAR scores but a week later becomes cyanotic and they ask why, the answer is "closure of ductus arteriosus."
Functional (flow) murmur	- Systolic murmur seen in the setting of higher heart rate caused by infection, anemia, or pregnancy. Caused by increased flow across the pulmonic and/or aortic valves. - Known as a functional murmur because this means it goes away once the heart rate comes back down. - Seen all over 2CK Peds forms, where they try to trick you into thinking the kid has a valvular pathology of some kind, but there isn't; there will merely be an infection or simple viral infection. - Can be seen sometimes with ASD, where the patient will have fixed splitting of S2 "plus a systolic murmur" → merely higher right-sided volume, so more flow across the pulmonic valve.
Venous hum	- On 2CK Peds form; described as a murmur in the neck that abates when the kid is laid supine + the neck rotated. - Benign + don't treat.
"Ball-in-valve" murmur	- Associated with cardiac tumors (i.e., myxoma in adult, or rhabdomyoma in kids for tuberous sclerosis). - Described as a diastolic rumbling murmur that abates when the patient is re-positioned unconventionally (e.g., onto his or her right side).
Coarctation of the aorta	- Refers to narrowing of the aortic arch (this is referred to as coarctation; do not use the word stenosis to describe this). - Classically seen in Turner syndrome, but absolutely not mandatory. Shows up idiopathically in plenty of NBME Qs. I point this out because students often think the patient must have Turner syndrome. - Presents as upper extremities that have higher BP, brisk pulses, and are warmer; the lower extremities have lower BP, weak pulses, and are cooler. - Sometimes the Q can just say, "the radial pulses are brisk." → The implication is, "Well if they're saying specifically that the radial pulses are brisk, that must mean the pulses in the legs aren't." - Murmur sound not important for USMLE. Can sometimes be described as a systolic murmur heart in the infrascapular region. - Can cause LVH with left-axis deviation ECG (on NBME). - USMLE doesn't give a fuck about pre- vs post-ductal. Pre-ductal in theory will be a very sick neonate. Post-ductal will be an adult (most cases).
Tetralogy of Fallot	- 1) Pulmonic stenosis; 2) RVH; 3) overriding aorta; 4) VSD. - If you're asked which component most determines prognosis, the answer is the degree of pulmonic stenosis.

	- The child will not be cyanotic at birth, but then years later, will develop Eisenmenger syndrome (i.e., a reversal of the L→R shunt over the VSD to be R→L) and cyanosis, where the stem gives a school-age kid who squats on the playground to relieve symptoms. - Squatting ↑ afterload, which ↑ LV pressure, which ↓ the pressure gradient of the R→L shunt, thereby mitigating cyanosis. - Squatting also ↑ preload by ↑ venous return back to the right heart. But it is the effect of ↑ afterload that is most related to the ↓ in symptoms.
Persistent fetal hypertension	- Aka persistent fetal circulation. - Q will give a post-term birth at 42 or 43 weeks + meconium-stained fluid + echo of the neonate shows a R→L shunt across the foramen ovale. Student says, "Wait, but isn't the foramen ovale between the atria, and that's only open in the fetus but is supposed to close after birth?" Correct. Hence we have *persistent* fetal circulation. - Answer on USMLE will be "failure of pulmonary vasodilation." Meconium aspiration syndrome can ↓ opening of the lung vasculature, leading to ↑ right heart pressure and ↑ risk of persistent fetal circulation.
Congenital heart block	- SLE in pregnant women can cause congenital heart block in neonates. Sounds weird, but shows up on NBME somewhere.
Adams-Stokes attack	- Idiopathic arrhythmia disorder that causes transient hypoxia to the brainstem, resulting in seizure-like episodes. - EEG does not show any seizure activity.

Rheumatic heart disease (rheumatic fever) HY points

- *Strep pyogenes* (Group A Strep) oropharyngeal infection results in production of antibodies against *S. pyogenes*' M-protein that cross-react with the mitral valve (i.e., molecular mimicry; type II hypersensitivity).
- Can occur with the aortic valve in theory, but on USMLE, it is always mitral valve.
- Results in mitral regurgitation acutely and mitral stenosis late, as discussed earlier.
- Presents as **JONES** (J♥NES) → **J**oints (polyarthritis), **♥ C**arditis, subcutaneous **N**odules, **E**rythema marginatum (annular, serpent-like rash), **S**ydenham chorea (autoimmune basal ganglia dysfunction that results in dance-like movements of the limbs).
- Cutaneous Group A Strep infections don't cause rheumatic fever, but can still cause PSGN.
- Tx = penicillin.

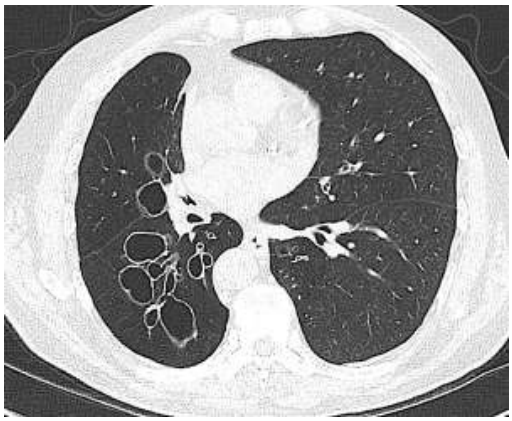


Erythema marginatum

Peds conditions confused for cardiac path	
Panic attack	- Psych forms love trying to make you think this is an MI. - They'll give you a 17-yr-old male who feels doom / like he's going to die. - Sometimes they mention in stem Hx of MI in family as distraction. - They can say patient has mid-systolic click and then they ask for cause of patient's symptoms → answer = panic disorder, not MVP. Student gets confused, but MVP is almost always asymptomatic, where panic attack is clearly cause of the patient effusively hyperventilating. - Treat with benzo.
Vasovagal syncope	- Fainting in response to stressor (e.g., emotional trigger). - Stress triggers an initial sympathetic response, which in turn triggers a compensatory parasympathetic response. This latter response is excessive in some people, where the peripheral arterioles dilate and the heart slows too much → decreased cerebral perfusion → lightheadedness/fainting. - 2CK wants you to know a tilt-table test can be used to diagnose, where a reproduction of symptoms can occur.
Costochondritis	- **Not** cardiac. This is MSK. But I'm putting this here real quick because students often confuse this for cardio. - Inflammation of cartilage at rib joints. - Will present as chest pain that worsens with palpation or when patient reaches over the head or behind the back. These two findings are clear indicators we have an MSK condition, not cardiac. - Can be idiopathic, caused by strain (e.g., at the gym), or even post-viral.
Pleurodynia	- MSK condition asked twice on 2CK material (once on FM form; also on Free 120) that has nothing to do with the lungs, despite the name. - This is viral infection (Coxsackie B) causing sharp lateral chest pain due to intercostal muscle spasm. Sometimes students choose pericarditis, etc., even though the presentations are completely disparate. - Creatine kinase can be elevated in stem due to ↑ tone of muscle.
Viral pleurisy	- Viral infection causing inflammation of the pleura (layers covering the lungs), leading to sharp chest pain. - If this is the answer, CK will be normal (unlike pleurodynia, because it's not MSK).
Breath-holding spell	- They'll say 3-year-old was having a tantrum followed by falling on the floor + appearing blue. - Child will involuntarily stop breathing following a trigger – e.g., being upset, frightened, or experiencing pain. - The child often cries or becomes upset, exhales forcefully, stops breathing, then develops cyanosis, which can sometimes be followed by brief loss of consciousness and jerking (can be mistaken for seizure). - Shows up on Peds forms so you need to know it exists.

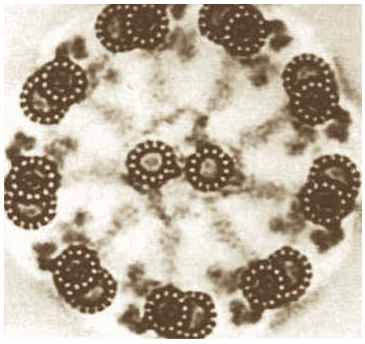
Peds Pulmonary

General HY lung conditions in Peds	
Asthma	- Bronchospasm that occurs either idiopathically/hereditarily, or in response to certain allergens or cold air. - One-third of patients with asthma only present with a dry cough and no problems breathing. This is called cough-variant asthma. It will often present as a patient with a dry cough that's worse in the winter. - Can also present as part of atopy constellation – i.e., dry cough in winter, seasonal allergies / rhinoconjunctivitis / urticaria in spring, and eczema in summer. - Aspirin-induced asthma mechanism HY for USMLE → inhibition of COX by aspirin → shunting of arachidonic acid down lipoxygenase pathway → increased leukotrienes → increased bronchoconstriction. - Samter triad = aspirin allergy, asthma (due to aspirin), nasal polyps. - "Increased expiratory phase" is a buzzy phrase that will be thrown into quite a few asthma vignettes. It is not specific for asthma and can refer to any obstructive pathology, but I just make note of it here because you'll see it quite a bit for asthma and say, "What's that mean?" → in obstructive conditions, it takes us a lot longer to exhale (FEV1 is ↓↓). - Acutely, asthma causes ↓ CO2, ↑ pH, ↔ bicarb. This is an acute respiratory alkalosis. Even though the patient is having difficulty breathing, CO2 diffuses quickly, whereas O2 diffuses slowly, so insofar as the patient's respiratory rate is ↑, CO2 will be ↓. Bicarb is unchanged because it takes the kidney about a day to alter excretion. In contrast, CO2 is ↑ in COPD because there is large amounts of hypoxic vasoconstriction due to excessive mucous (chronic bronchitis) or ↓ alveolar surface area, so CO2 can't get out, even with faster breathing. In asthma, alveolar surface area is intact, and the degree of mucous production and hypoxic vasoconstriction is not nearly as bad as chronic bronchitis. - The combination of ↓ CO2 and ↓ O2 in acute asthma attack is known as type I respiratory failure. Eventually, the patient will begin to fatigue and breathing rate will slow. This will be observed as CO2 and pH rebounding to normal levels *despite* O2 remaining low. This means the patient is in transition to a type II respiratory failure (i.e., hypoventilation), where we have ↑ CO2 and ↓ O2. USMLE wants intubation as the intervention. - 2CK Q gives vignette of asthma and then asks for best initial step in diagnosis → answer = "spirometry." The expiratory component of the curve, as discussed earlier, will appear concave. - Should be aware methacholine challenge can also be done to diagnose. This is is a muscarinic agonist that bronchoconstricts and can induce symptoms. Never give during acute episode. This can be tried between episodes. But if you're forced to choose diagnostic modality, go with simple spirometry. - USMLE cares about both outpatient management of asthma as well as acute attacks. - For outpatient management: 1) First-line Tx is β2-agonist (albuterol) for acute attacks. 2) If the Q says the patient has weekly episodes, or they ask for what could decrease risk of future episodes if patient is already on albuterol, the answer is inhaled corticosteroid (ICS; i.e., fluticasone). 3) If the combo of albuterol + ICS is insufficient, the next step is increasing the dose of the ICS. 4) If that doesn't work, adding a LABA (i.e., salmeterol) is the next step. 5) If insufficient, agents such as mast cell stabilizers, leukotriene blockers, etc., can be tried. If patient has Hx of aspirin allergy, the latter in particular can be effective.

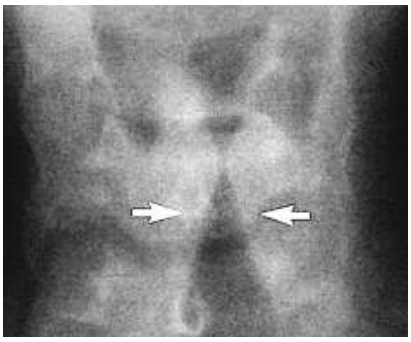
	6) Last resort for outpatient asthma is oral prednisone. It is most effective at decreasing recurrence of episodes, but we want to avoid it if at all possible because of the risk of Cushing syndrome and ↓ linear bone growth in Peds. - Acute management of severe asthma attack (i.e., not outpatient) is nebulized albuterol, oxygen, and IV methylprednisolone. Inhaled corticosteroids have no role in acute asthma management. - NBME for 2CK wants you to know that any asthma patient who requires hospital management for an acute episode must automatically be given ICS (i.e., fluticasone) on discharge. In other words, if patient isn't currently on an ICS, it must be commenced at discharge. I'm explicit here because this particular NBME Q doesn't mention albuterol anywhere in the stem, but inhaled fluticasone is the answer. - Another 2CK NBME Q gives a patient who is currently not receiving any asthma medications but who gets 2+ episodes weekly → answer = "inhaled fluticasone + albuterol." Students think this is weird because they jump straight to dual management + are even audacious enough to put the ICS before albuterol in the wording of the answer, but it's what they want. Any patient with 2+ episodes per week needs ICS in addition to the albuterol and can be commenced right away on dual therapy.
Anaphylaxis	- Acute dyspnea and bilateral wheezes in patient who was gardening (stung by a bee), who ate a particular food (e.g., peanuts), or who was commenced on a recent drug (e.g., TMP/SMX). - Vignette often gives tachycardia and low BP. - Mechanism is IgE crosslinking on surface of mast cells and basophils that leads to degranulation and histamine + prostaglandin release. Eosinophils can be recruited in response. - USMLE wants: ↓ vascular resistance, ↑ CO, ↓/↔ PCWP. - Tx = **intramuscular epinephrine** → the strong β2-agonistic effect opens the airways; the strong α1-agonistic effect constricts the arterioles and restores BP. - NBME 9 for 2CK asks, "In addition to self-injectable epinephrine therapy, what is most appropriate therapy to ↓ recurrences" → answer = **venom immunotherapy (VIT).** What this does is desensitizes the patient to the antigen by allowing him/her to develop neutralizing IgG antibodies against it.
Bronchiectasis	- Dilation of the airways (ectasia means dilation) that occurs due to "loss of musculature of the airways." - Transverse CT scan will show cystic dilation of the airways.  - Most common cause worldwide is TB. Most common cause in western countries is CF. But on USMLE, it will usually be a smoker. Asthma does not cause bronchiectasis. - Presents almost always as "cups and cups of foul-smelling sputum." - "Foul-smelling" means anaerobes, such as *Bacteroides*. - One question on 2CK Peds form gives bronchiectasis as the answer in a child who has right middle lobe syndrome, where they say there is scant white sputum (not cups and

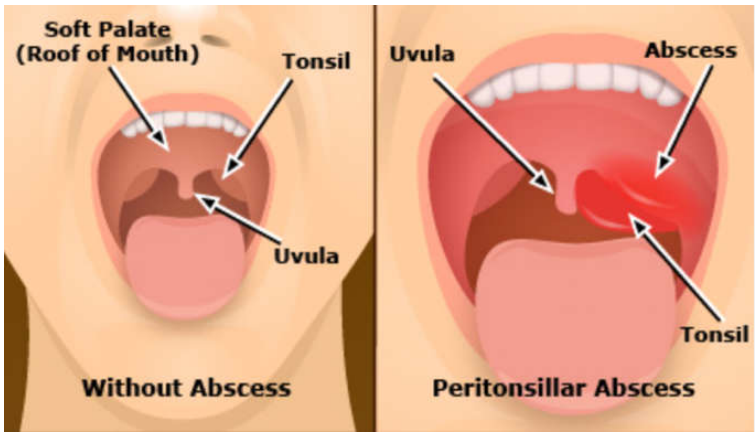
	cups) and a thin, linear opacity visualized in the right middle lobe. If you think it's weird, complain to NBME not me. - **Clubbing** tends to be seen in these patients. It's not mandatory, but it's to my observation USMLE likes it for bronchiectasis.

HY Congenital lung diseases	
Cystic fibrosis	- Autosomal recessive; chromosome 7. - Mutations in *CFTR* gene, which codes for a chloride channel. - Most common mutation is ΔF508, which is deletion of a phenylalanine. - USMLE wants "abnormal protein structure," or "abnormal protein folding," a common mechanistic answer for CF. - In the disease, the CFTR chloride channel, normally located at the cell surface, is instead sequestered at the RER. This is a HY point, where they want you to know patients with disease do *not* have the chloride channel on the cell membrane. - Disease results in impaired ability to secrete Cl^- into alveolar and pancreatic secretions. This causes greater (-) charge within the cell. Na^+ then moves through ↑ ENaC into the cell to balance charge. Water follows Na^+, resulting in dried up alveolar and pancreatic secretions. These are known as inspissated secretions. Inspissation means desiccated (dried up) within a lumen. The inspissated alveolar secretions lead to mucous plugging, airway obstruction, and recurrent pneumonias. The inspissated pancreatic secretions lead to "exocrine pancreatic insufficiency" (HY phrase) and fat-soluble vitamin malabsorption. It is to my observation that vitamin E deficiency is the highest yield vitamin deficiency due to CF across NBME questions, which will present as neuropathy in a child who has CF. - Meconium ileus HY; refers to failure to pass stool in the first 24 hours after birth. - CF can sometimes cause nasal polyps. - USMLE doesn't obsess over pathogen causes for CF pneumonia, but classically *Pseudomonas* is a culprit. Prior to age 10, *S. aureus* exceeds *Pseudomonas* as most likely organism; after age 10, *Pseudomonas* eclipses *S. aureus*. The caveat is you have to use your head though: if they say 8-year-old with pneumonia due to gram-negative rods, you know that's *Pseudomonas*. Likewise, if they say a 17-year-old with pneumonia due to gram-positive cocci in clusters, you know that's *Staph aureus*. - Sweat glands reabsorb Cl^- rather than secrete it. This is impaired in CF, so we have ↑ Cl^- in the sweat (i.e., >60 mEq/L). The sweat chloride test is more accurate for diagnosis than genotyping due to allelic heterogeneity of *CFTR* gene, where many different mutations cause the disease and can be difficult to detect with routine screening panels. - A transepithelial nasal voltage test can also be done, where there is ↑ nasal potential difference. - Male patients can have congenital bilateral absence of vas deferens (CBAVD), leading to absent sperm in a sample. - Since this is an AR condition, there is a 2/3 chance a phenotypically normal sibling of a CF patient is a carrier. This applies to any AR disorder, but USMLE likes to use CF as the archetypal example. If this 2/3-point sounds confusing, I talk about this stuff in detail in my HY Genetics PDF. - CF medications not HY for Step, but you could be aware that -caftor agents (e.g., ivacaftor, lumacaftor) facilitate with proper CFTR channel localization to the cell membrane + structural folding. Dornase-alfa

	(correct, not alpha) is a nucleotidase that softens mucous. Guaifenesin is also a mucous-softening agent.
Primary ciliary dyskinesia (Kartagener syndrome)	- Answer on USMLE for a question that sounds like CF but patient has situs inversus or dextrocardia – i.e., patient will have recurrent pneumonias and organs (or just heart) on opposite side of body. - Abnormality of cilia function due to defective dynein arm. This is a HY point, where sometimes the answer will just be "dynein." - A cilium on cross-section has a 9x2 microtubule configuration. Dynein is a molecule that is necessary for cilia function.  - Patient will sperm in sample, but motility will be decreased. This is in contrast to CF, where sperm are absent due to CBAVD. - Women can get ectopic pregnancy due to abnormal Fallopian tube cilia. - This diagnosis sounds straightforward enough, but I've seen plenty of students miss these questions. For the situs inversus, they will describe this as cardiac sounds loudest on the right, or there being a large mass palpable beneath the left costal margin (liver on the opposite side).

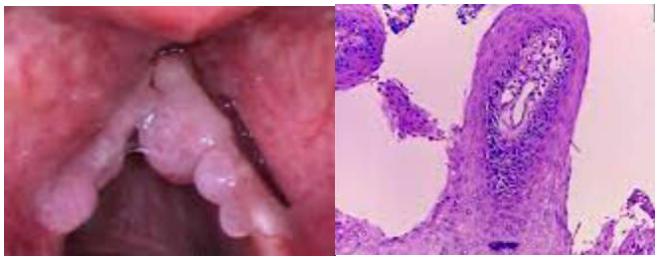
HY Infection-related stuff	
Pneumonia	- Lobar pneumonia = *Strep pneumoniae* on USMLE (right lower lobe consolidation with dullness to percussion). - Bilateral interstitial pneumonia (aka atypical pneumonia) in immunocompetent teenagers or older = *Mycoplasma* on USMLE. - Lobar pneumonia where they say "interstitial markings" and *Strep pneumo* isn't listed → answer = *Mycoplasma* (the word "interstitial" wins over location). - Bilateral interstitial / "ground-glass" pneumonia in child who's mom was IV drug user → *Pneumocystis jirovecii* pneumonia (PJP); child has HIV + contacted it via vertical transmission. - Pneumonia in 3-wk-old neonate who had conjunctivitis 1-2 weeks ago → *Chlamydia trachomatis* (the STI; drains through nasolacrimal duct to lungs). - Pneumonia in newborn first few days of life → Group B Strep (*Strep agalactiae*), which is gram (+) cocci. If they say gram (+) rods, that's instead *Listeria*. If they say gram (-) rods, that's *E. coli*. - Pneumonia in CF → *Pseudomonas* or *S. aureus*. - USMLE wants bronchoalveolar lavage for diagnosis of PJP. - USMLE wants for pneumonia: adventitious/bronchial (i.e., abnormal) breath sounds + ↑ **tactile fremitus** (air vibrates due to movement through infective consolidation within alveoli). - Community-acquired pneumonia (CAP) empiric Tx = azithromycin on 2CK (on NBME). This covers the atypicals (*Mycoplasma, Legionella, Chlamydia)* as well as *S. pneumo*.

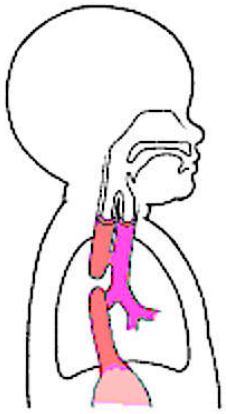
	- If patient has been on antibiotics in the past 3 months or has severe lung disease, levofloxacin (respiratory fluoroquinolone) can be given first-line. - CAP that results in sepsis or septic shock → give ceftriaxone (sometimes cefotaxime for peds). - Nosocomial pneumonia (i.e., hospital- or ventilator-acquired) requires coverage for MRSA and *Pseudomonas*. USMLE wants vancomycin PLUS either ceftazidime (a 3rd-gen cephalosporin) or cefepime (a 4th-gen). - For fungal pneumonia, Tx = fluconazole. - For fungal pneumonia + fungemia (high fever, chills) → Amphotericin B. - NBME 9 for 2CK wants you to know **sputum culture, followed by blood cultures** are done in all patients with pneumonia who are septic. What they do in this patient is give you patient who has sputum culture performed, then they ask what should be done next for diagnosis? → answer = blood culture.
Bronchiolitis	- Answer on USMLE for a kid <18 months old who has low-grade fever and bilateral wheezes. - Caused by respiratory syncytial virus (RSV). - Tx is supportive care on USMLE. Don't choose answers like ribavirin or palivizumab.
Laryngotracheal bronchitis (aka croup)	- Caused by paramyxovirus (aka parainfluenza virus). - Presents as hoarse, barking, or seal-like cough in school-age kid. The Q can say the cough gets better when his dad brings him out into the cold air. - Neck x-ray shows "steeple sign," which is sub-glottic narrowing.  - Sometimes the Q can give you easy vignette of croup, but then the answer is just "larynx" (literally inflammation of the larynx, trachea, and the bronchi). "Sub-glottic" means below the area of the vocal cords. The larynx is the area encompassing the vocal cords. - Tx is supportive. If they force you to choose an actual Tx, the answer is steroids if listed, followed by nebulized racemic epinephrine.
Epiglottitis	- Caused by *Haemophilis influenzae* ***type B***. - Seen in unvaccinated and immigrants (can be unvaccinated), as well as patients with asplenia or sickle cell (auto-splenectomy). - Can also cause meningitis. - X-ray of neck shows "thumbprint sign."

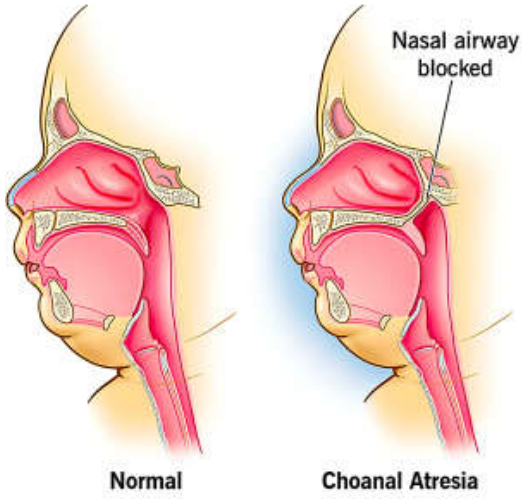
	 - Presents as child who has fever + difficulty breathing. They can say the kid is drooling and/or in tripod positioning (facilitates use of accessory muscles). - USMLE wants intubation as answer. Epiglottitis is a medical emergency that can lead to sudden occlusion of the airway. - Tx = 3rd-gen cephalosporin (cefotaxime in peds, or ceftriaxone); give rifampin to close contacts.
Peritonsillar abscess (Quinsy)	- Pus collection causing pain and swelling at posterior oropharynx. - Can be seen as complication of tonsillitis. - Patient can have **difficulty opening his/her mouth;** this contrasts with bacterial tracheitis, where the patient can fully open his/her mouth (on 2CK NBME, where this detail is important in an otherwise vague vignette). 
Bacterial tracheitis	- Diagnosis of exclusion on USMLE (i.e., you eliminate to get there). - Classically caused by *S. aureus* following a viral URTI. - Shows up on 2CK form as patient with recent viral URTI who now has stridor (suggesting upper airway narrowing) + no drooling (suggests against epiglottitis) + **can fully open the mouth without difficulty** + does not improve with racemic epinephrine (suggests against croup). - Being able to open the mouth without difficulty is important because this contrasts with peritonsillar abscess, where the patient has difficulty fully opening the mouth.
Pertussis	- Classic whooping cough presents as succession of many coughs followed by an inspiratory stridor. - What you need to know for USMLE is that this can absolutely present in an adult and that they can be vague about it, just describing it as a regular cough. The way you'll know it's pertussis, however, is they will say there's either **hypoglycemia** or **post-tussive emesis,** which means vomiting after coughing episodes.

	- Pertussis can cause super-high WBC counts in the 30-50,000-range, where there are >80% lymphocytes. *This makes it resemble ALL*. So you should know for Peds that ALL-like laboratory findings + cough = pertussis. - Q will ask number-one way to prevent → answer = vaccination (not hard, but they ask it). Pertussis is part of TDaP. The pertussis component is killed-acellular; the tetanus and diphtheria are toxoid. - Erythromycin can be given to patients with active cough; USMLE doesn't give a fuck about pertussis stages. - Close contacts should also receive erythromycin.
Tuberculosis	- *Mycobacterium tuberculosis* has unique cell wall composed of mycolic acid that is difficult to gram stain. Requires acid-fast stain. - Stem will be child living in homeless shelter or has immigrant status from endemic area. I've seen rural India and Albania as two locations in NBME Qs. - Can present similar to lung cancer, where patient can have B symptoms (i.e., fever, night sweats, weight loss) and hemoptysis. - Can cause cavitations and calcification in the lung grossly; on histo, causes caseating granulomatous inflammation. - Ghon foci/complexes are textbook descriptors for TB lesions but not assessed eponymously on USMLE. - Can cause constrictive pericarditis (can also calcify). - Disseminated TB (miliary TB) can affect multiple organ systems, leading to **psoas abscess, Pott disease** (TB infection of the vertebrae), adrenal insufficiency, meningitis, osteomyelitis, and arthritis. - First step in diagnosis is PPD test (type IV hypersensitivity). - If PPD test is (+), the next best step is CXR. - If PPD is (+) but CXR (-), next best step is "treat for latent TB, " or "give TB prophylaxis." This is isoniazid (INH) for 9 months + vitamin B6 (since INH can cause B6 deficiency). It is exceedingly HY you know that neuropathy in a patient being treated for TB has B6 deficiency. - If PPD and CXR are both (+), the next best step is "treat for active TB," which is RIPE for 2 months + RI for 4 more months (6 months total). RIPE = rifampin, isoniazid, pyrazinamide, ethambutol. - BCG vaccine is live-attenuated. USMLE wants you to know Hx of BCG vaccine does not change management based on PPD guidelines. - What is considered a positive PPD test differs depending on risk factors: **- >5mm (+):** Hx of close contact to someone with active TB; immunocompromised patient (AIDS, organ transplant recipient receiving immunosuppressants, chronic corticosteroid user); calcification on CXR. **- >10mm (+):** Health care worker or prisoner/prison worker; immigrant from endemic area; TB laboratory personnel, children <4. **- >15mm (+):** everyone else. - If a PPD test is (+), never repeat it. If it is negative, it must be repeated in 1-2 weeks (i.e., sometimes false-negatives). - If USMLE asks you how long after TB exposure will someone's sputum cultures be positive, the answer is **2-5 weeks.** - Interferon-gamma release assay (Quantiferon Gold) can be used in patients who have Hx of BCG to ↓ false (+)s, but USMLE doesn't assess it. The reason I mention it is because they want you to know *interferon-γ* is required for stimulation of alveolar macrophages to control TB. - Kids who have IFN-γ deficiency or IL-12 receptor deficiency (IL-12 normally ↑ IFN-γ) have ↑ susceptibility to TB infections. If this immuno stuff sounds confusing, I talk about this in detail in my HY Immuno PDF.

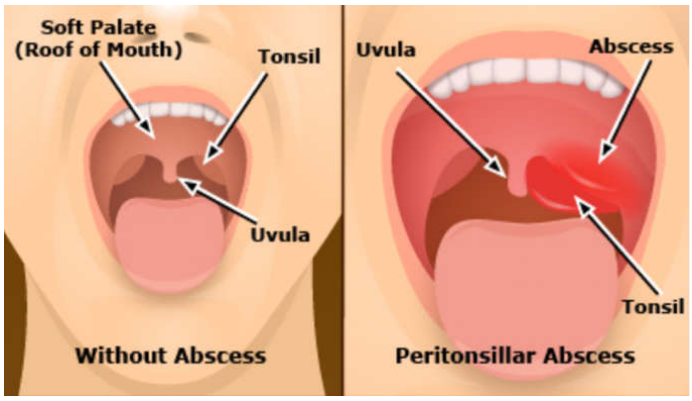
	- TNF-α is also required to suppress TB. Therefore drugs such as infliximab, adalimumab, and etanercept ↑ risk of TB, which is why they should be avoided in silicosis patients, as mentioned earlier.
Otitis media (OM)	- Infection of portion of ear just deep to tympanic membrane. - Most commonly *Strep pnuemo*. - Will present as red, **immobile** tympanic membrane. Immobility of the tympanic membrane is highly sensitive for OM, meaning that if the Q says mobility is normal, we can rule out. - "Ear tugging" can be a sign in children of either otitis media or externa. - Tx is amoxicillin or penicillin. - Augmentin (amoxicillin/clavulanate) is classically given for *recurrent* OM. So if you are forced to choose between amoxicillin/penicillin alone or Augmentin, go with the former. - For 2CK Peds, a tympanostomy tube (aka grommet) is used if the kid has >3 OM occurrences in 6 months, or >4 in a year.
Serous otitis media	- Aka otitis media with effusion. - Presents as fluid behind the tympanic membrane in a kid weeks after resolution of 1 or 2 otitis media infections. - Almost always benign and self-resolves in 4-8 weeks. Answer is observation.
Tympanic membrane perforation	- "Tympanic membrane perforation" is the answer on new 2CK NBME for 2-year-old who had 3-day Hx of viral infection followed by awakening with severe ear pain + has dried blood on ear lobe and pillow + otoscopy cannot visualize tympanic membrane because of seropurulent fluid draining from the ear canal. - Can occur due to otitis media, although vignette on NBME doesn't sound like classic OM and is as described above.
Mastoiditis	- Inflammation of mastoid bone caused by untreated otitis media. - The mastoid process is the posterior part of the temporal bone that is felt just behind the ear. - Can present as a painful ear pinna that is displaced (e.g., upward and outward). - Diagnosis is made by **CT or MRI.** X-ray is wrong answer. - 2CK IM Q gives a 2-year-old with mastoiditis where the answer is "CT of the temporal bone." Sounds wrong, since this is ↑ radiation for a kid, but it's what they want.
Myringitis	- Isolated inflammation of the tympanic membrane. - Can be bullous (i.e., bullous myringitis). - Caused by *Strep pneumo* or *Mycoplasma*.
Otitis externa (OE)	- Infection of ear superficial to tympanic membrane. - Classically caused by *Pseudomonas*. - Increased risk in swimmers and diabetics. - An NBME form has "necrotizing otitis externa" as answer for black skin within the ear canal in a patient. This is aka "malignant otitis externa." - USMLE wants "acetic acid-alcohol drops" as prophylaxis in student who does crew + continues to have water exposure. - Tx (not prophylaxis) = "topical ciprofloxacin-hydrocortisone" drops.
Sinusitis	- The answer if they tell you a school-age kid has a lingering fever after an upper respiratory tract infection (URTI) for 10-14+ days. - Whenever a URTI lingers for more than ~10ish days, you want to think about sinusitis as a differential. - A 2CK vignette gives nocturnal cough (reflects aspiration; in this case, from the sinuses) and grey membranes in the oropharynx. - The grey oropharyngeal membranes detail sounds weird, since that is normally buzzy for *Diphtheria*, but it shows up on an NBME Q where the answer is sinusitis and *Diphtheria* isn't listed.

	- IgA deficiency Qs, which presents as recurrent sinopulmonary infections, can say patient has Hx of pneumonias + presents today with sore left cheek → reflects sinusitis. - For 2CK, **CT scan** is done if chronic sinusitis >12 weeks. After CT is performed for chronic sinusitis, nasal endoscopy can be performed. - Tx is amoxicillin/clavulanate (Augmentin). This is in contrast to OM and *Strep* pharyngitis, which are treated with just amoxicillin or penicillin alone, without the clavulanate (unless recurrent).
Laryngeal papillomatosis	- Pediatric condition characterized by warts of the vocal cords. - Lesions will have papillary structures on biopsy. - Due to HPV 6/11 exposure from maternal vaginal canal. 

HY Pediatric-related respiratory stuff	
Tracheoesophageal fistula (TEF)	- HY pediatric condition in which the proximal esophagus ends in a blind pouch + the distal esophagus connects to the trachea.  - Will present as a neonate who coughs up milk with initial feeding. - Highest yield point about TEF is that diagnosis is made via **insertion of nasogastric tube** (which cannot be inserted fully because it hits the blind pouch of the esophagus). - An NBME Q wants "endoderm" as the abnormal embryology for TEF (makes sense, since esophagus is epithelial lining of the gut → endoderm).
Choanal atresia	- Weird condition in which the nasal passages don't develop patency, so the neonate is an obligate mouth-breather.

	 - Will present as a child who becomes blue/hypoxic while breast feeding (because he can't breathe through the nose), then cries/becomes pink once detaching from the breast. - Similar to TEF, diagnosis is made via insertion of nasogastric tube. - Part of **CHARGE** syndrome → **C**oloboma of the eye (hole in the eye), **H**eart defects, **A**tresia of the choanae, **R**enal defects, **G**enitourinary anomalies, **E**ar anomalies. If you're studying for Step 1, you don't need to worry about CHARGE syndrome. For 2CK Peds, you should know it.
Congenital diaphragmatic hernia	- One of the highest yield Peds conditions on USMLE. - Caused by **failure of formation of pleuroperitoneal membranes.** - Always occurs on the left, where bowel from the abdomen can herniate up into the left-chest. - Can present as ↓ bowel sounds in the abdomen + ↑ bowel sounds in the left hemithorax.  - Vignette might say there are cystic-appearing areas in the left hemithorax seen on CXR (loops of bowel).
Neonatal respiratory distress syndrome (NRDS)	- Aka hyaline membrane disease. - The answer on USMLE for respiratory distress in kid who is born <34 weeks' gestation. - Due to insufficient surfactant production by type II pneumocytes due to ↓ lamellar bodies (the specialized organelles that produce surfactant).

	- These kids have ↓ lecithin/sphingomyelin ratio (i.e., <2.0). Normally it is >2-2.4. - Another name for lecithin is dipalmitoyl phosphatidylcholine. This is asked on NBME. - USMLE can give simple vignette of NRDS and then ask for various manipulation of the ratio – i.e., "↑ sphingomyelin" might be an answer (makes sense, since this would ↓ the ratio). - ↓ surfactant production means ↓ alveolar compliance and ↑ **elastic recoil.** Surfactant is hydrophobic and normally prevents the alveoli from collapsing, so if it's deficient, the hydrophobic interactions of the alveolar walls enable ↑ collapsing/elastic recoil. - CXR shows a "reticulogranular" appearance. Very buzzy and HY. - In order to prevent NRDS, a pregnant woman giving birth <34 weeks' gestation must be ***two*** boluses of corticosteroids within 24 hours of parturition, which accelerate fetal lung maturity. For example, there is a 2CK Obgyn Q where they tell you a woman giving birth at 33 weeks' gestation was given a bolus of corticosteroids 12 hours ago. They ask for next best step → answer = "bolus of corticosteroids." Sounds weird because they said it was just done, but she needs ***two*** boluses. - For management, 2CK wants tactile stimulation first, then place under warming lights. This sequence is HY. Then give exogenous surfactant and oxygen.
Transient tachypnea of the newborn (TTN)	- The answer on USMLE when the vignette sounds like NRDS but the kid is term. For instance, they'll say neonate has difficulty breathing, but then you'll notice he's >37 weeks. - Usually seen **following C-section** or fast vaginal delivery. - Mechanism is insufficient time for the pulmonary lymphatics to clear amniotic fluid from the lungs. - CXR shows "fluid within fissure lines."
Meconium aspiration syndrome	- Newborn aspirates a mixture of meconium (the infant's first stool) and amniotic fluid into their lungs during parturition. - Can cause airway obstruction, infection, inflammation, and dyspnea. - Q will mention "dark, meconium-stained fluid." - Highest yield risk factor for USMLE is post-term birth (i.e., after 40 weeks). - Can cause persistent fetal hypertension (below).
Bronchopulmonary dysplasia	- Fibrotic lung disease in an infant caused by continued use of supplemental oxygen. - Vignette will give you a kid who was born at 26 weeks' gestation who was in ICU on oxygen + now is 4 months old and is on home oxygen. They will say ECG shows right-axis deviation +/- CXR shows increased pulmonary vascularity → this reflects RVH (ensuing cor pulmonale) from pulmonary hypertension caused by the bronchopulmonary dysplasia.
Foreign body aspiration	- Answer on 2CK Peds for kid under 2 who was crawling around on the floor who now has acute-onset respiratory distress. The vignette will then add one of two features: 1) Unilateral hyperresonance in one lung (but not pneumothorax); 2) Unilateral atelectasis + ipsilateral tracheal shift. - Exam will ask for "bronchoscopy," or "fiberoptic examination of the airways" as the answer.
Allergic rhinitis	- Not limited to kids, but can be part of atopy in patients with asthma. - "Cobblestoning" of nasal mucosa is buzzy for allergy. This same word is used to describe the tarsal conjunctiva in allergic conjunctivitis on Peds. - NBME 9 for 2CK has "use of pillow and mattress covers" as the answer. - Otherwise, USMLE wants **intra-nasal corticosteroids** as first-line med.

	- USMLE does not want anti-histamines as first-line med. - 2nd-gen H1 blockers (i.e., loratadine) are the answer if *intra-nasal* corticosteroids are not listed. Oral prednisone is a **wrong** answer. - I've seen students choose prednisone and then get irascible and say, "But you said steroids were first-line." → Yeah, bro. Intra-nasal. Not fucking oral.
Adenoid hypertrophy	- Just know that enlarged tonsils (adenoids) are a known cause of stridor in pediatrics. Answer = tonsillectomy.
Peritonsillar abscess (Quinsy)	- Pus collection causing pain and swelling at posterior oropharynx. - Can be seen as complication of tonsillitis. - Patient can have **difficulty opening his/her mouth;** this contrasts with bacterial tracheitis, where the patient can fully open his/her mouth (on 2CK NBME, where this detail is important in an otherwise vague vignette). 
Laryngomalacia	- Most common cause of stridor in infants. - Softening of airway cartilage. - Vignette will give kid under the age of 1 who has noisy breathing that is mitigated when placed prone (on stomach) or upright. - Self-resolves almost always (i.e., don't treat).
Vascular ring	- Weird cause of stridor that USMLE likes to contrast with laryngomalacia. - Vascular embryology variant can lead to airway compression. - In contrast to laryngomalacia, mitigated when neck is extended. - Tx = surgery.
Velopharyngeal insufficiency	- High-pitched, hyper-nasal, or whistling speech in patients with Hx of cleft palate due to shortened palate, with or without repair. - It's the answer on a 2CK NBME, so know that it exists. They will simply just give you vignette of cleft palate + ask what the kid is at risk for.
Cholesteatoma	- Idiopathic squamous proliferation in middle ear behind the tympanic membrane that presents as an enlarging mass. - Can cause obstructive hearing loss + grow into inner ear. - Requires surgery. - Sounds like an obscure diagnosis, but it's not. Be aware it exists for Peds.
Tracheomalacia	- Not pediatric, but I'm putting it here because I see this confused with laryngomalacia. - This is a cause of stridor in (usually) adults who have stridor developing months post-surgery. - Caused by Hx of intubation leading to softening of airway cartilage. - Just know this diagnosis / it exists + don't confuse with laryngomalacia.

Differentiating viral vs bacterial URTI

- The CENTOR criteria are exceedingly HY for Pediatrics and FM rotations (and 2CK in general).

- If 0 or 1 point, the URTI is unlikely to be bacterial (i.e., it's likely to be viral). If 2-4 points, the chance is much greater that the URTI is bacterial.

- **1) <u>Absence of</u>** cough (i.e., no cough = 1 point; if patient has cough = 0 points).
- **2)** Fever >38 C.
- **3)** Tonsillar exudates.
- **4)** Lymphadenopathy (cervical, submandibular, etc.).

- There is a version of the criteria that includes age, but on the USMLE it can cause you to get questions wrong. So just use the simplified above four points.
  - If 0-1 point, answer = "supportive care"; or "no treatment necessary"; or "warm saline gargle" (same as supportive care); or "acetaminophen." Latter is answer for 3M with viral URTI + fever on Peds NBME form 2.
  - If 2-4 points, next best step = "rapid Strep test." If rapid Strep test is negative, answer = throat culture, **NOT** sputum culture.
  - While waiting on the throat culture results, we send the patient home with amoxicillin or penicillin for presumptive Strep pharyngitis.
  - If child is, e.g., 12 years old, and develops a rash with the beta-lactam, answer = beta-lactam allergy.
  - If the vignette is of a 16-17 year-old who has been going on dates recently (there will be no confusion; the USMLE will make it clear) + gets a rash with the beta-lactam, the answer = EBV mononucleosis; therefore do a heterophile antibody test (Monospot test).
  - **EBV is the odd virus out that usually presents with all four (+) CENTOR criteria and presents like a bacterial infection.**
  - This is why it's frequently misdiagnosed as Strep pharyngitis. It is HY to know that beta-lactams given to patients with EBV may cause rash via a hypersensitivity response to the Abx in the setting of antibody production to the virus. EBV, in a patient who does not receive Abx, can cause a mild maculopapular rash. But the rash with beta-lactam + EBV causes a more intense pruritic response generally 7-10 days following Abx administration on the extensor surfaces + pressure points.

Peds Renal

Nephritic vs Nephrotic syndromes
- Nephritic conditions have blood in the urine; nephrotic syndromes don't. - You need to memorize for USMLE which HY conditions are nephritic vs nephrotic. Don't worry, I keep things real clean and concise in the tables below without the bullshit nonsense of other resources. - There are four main points you should memorize as being part of each syndrome type. **Nephritic syndromes:** 1) **Hematuria** (blood in the urine). 2) **Oliguria** (↓ urinary output; the definition is <400 mL/day, but USMLE won't assess that number). 3) **Azotemia** (↑ blood urea nitrogen; this is because urinary output is ↓). 4) **Hypertension** (↑ RAAS due to inflammation of renal microvasculature). **Nephrotic syndromes:** 1) **Proteinuria** (nephrotic level by definition is >3.5g/day, but USMLE won't assess that number). 2) **Hypoalbuminemia** (due to the proteinuria; will be <3.5 g/dL, but USMLE doesn't care about the #). 3) **Peripheral edema** (due to the hypoalbuminemia → ↓ serum oncotic pressure → transudation of fluid into interstitial spaces; severe can present with ascites; if the stem says "pre-sacral edema," this is nephrotic syndrome till proven otherwise on USMLE). 4) **Hyperlipidemia** (liver pumps out apolipoproteins in an attempt to preserve serum osmolality / oncotic pressure). - Should be noted that nephritic syndromes often have proteinuria, just usually not at nephrotic levels. Students sometimes erroneously think nephritic syndromes don't have proteinuria because it's not part of the 4-point categorization above. - You don't have to worry about the notion of which conditions are "both nephritic and nephrotic." What is most important is: simply know which conditions have hematuria and which ones don't (via tables below). - Nephrotic syndromes can ↑ risk of DVT, renal vein thrombosis, and varicocele due to loss of antithrombin III in the urine → hypercoagulable state. This is because nephrotic syndrome usually entails non-specific massive protein loss due to loss of size and charge barrier, and antithrombin is a protein. The step 1 NBME wants you to know left renal vein thrombus accretion from hypercoagulable state can cause varicocele.

HY Nephritic syndromes for Peds	
Post-streptococcal glomerulonephritis (PSGN)	- Aka "proliferative glomerulonephritis" (asked on NBME, where they give vignette of PSGN, and answer is just "proliferative glomerulonephritis"). - Answer can sometimes be written by NBME as just "acute glomerulonephritis." - The answer on USMLE for red urine 1-3 **weeks** following a sore throat caused by Group A Strep (*Strep pyogenes*). This is in contrast to IgA nephropathy (which I discuss in detail below), which is red urine **1-3 days** following a sore throat. - PSGN can be caused by skin infections. This can be impetigo (school sores), cellulitis, or erysipelas (I discuss these in detail in my HY Derm PDF). For example, the Q might say a 10-year-old has yellow crusties on his arm for the past 7 days (impetigo) + now has red urine → answer = "acute glomerulonephritis." - Type III hypersensitivity (antigen-antibody complexes that form in blood and deposit in the kidney). Don't confuse with rheumatic heart disease, which is a type II HS. - Serum complement protein C3 can be ↓ (can also be ↓ in SLE, but unrelated). - Streptolysin O or A titers will be ↑. - PANDAS is tested on Psych forms for 2CK → Pediatric Autoimmune Neuropsychiatric Disorder Associated with *Streptococci* → presents as new-onset tic/Tourette, OCD, or ADHD within weeks of Group A Strep infection.

	- Descriptors such as subepithelial deposits or "lumpy bumpy" appearance on electron microscopy (EM), etc., have been parroted across resources over the years but are essentially garbage for USMLE. - PSGN usually self-resolves in kids without sequela. But USMLE wants you to know that ↑ age is means worse prognosis, where chance of renal failure is ↑ it if occurs in adults. This is probably related to the more robust immune response resulting in more advanced renal damage.
IgA nephropathy	- Aka Berger disease; IgA deposition in renal mesangium. - Red urine 1-3 days after a sore throat. This is in contrast to PSGN, which is red urine 1-3 weeks after a sore throat. I just mentioned it above obviously. But students fuck this up despite the inculcation so I'm reiterating it like an asshole. - Caused by **viral infection,** not Group A Strep. - Can sometimes be caused by GI infections, but USMLE usually avoids this etiology unless including it in the Henoch-Schönlein purpura constellation. - Henoch-Schönlein purpura is LY for Step 1 but HY for 2CK Peds. It's a tetrad: 1) Palpable purpura (usually on buttocks/thighs). 2) IgA nephropathy (red urine). 3) Arthralgias. 4) Abdominal pain. - All 4 need not be present for HSP, but the abdo pain component here is presumably viral gastroenteritis leading to IgA nephropathy.
Alport syndrome	- X-linked disease; *mutation in* collagen IV gene. - Do not confuse with Goodpasture syndrome, which is antibodies against type IV collagen. - The answer on USMLE for a male who has red urine + an eye or ear problem (collagen IV is present in basement membranes in the kidney, ear, and eyes). - The "eye problem" can be blurry vision/cataracts; the "ear problem" will be neurosensory hearing loss (due to organ of Corti dysfunction).

HY Nephrotic syndromes for Peds	
Minimal change disease	- The answer on USMLE for otherwise unexplained edema in a kid who doesn't have blood in the urine. - Can present as peripheral edema, periorbital edema, and/or ascites. - Almost always this is pediatric. Very rarely it can be due to Hodgkin in adult. - Classically post-viral (i.e., URTI), but ~50% of vignettes won't mention that. - In other words, textbook vignette is an 8-year-old who has the sniffles for 4 days, followed by peripheral and periorbital edema a week later, without blood in the urine. Once again though, the stem need not mention the viral infection. - Called minimal change disease because light microscopy (LM) shows no abnormalities. EM, however, shows effacement of the podocytes. - Mechanism for nephrotic syndrome is "loss of size and charge barrier." - Corticosteroids are the treatment and are highly effective. - A 2CK Peds Q gives minimal change disease as etiology for spontaneous bacterial peritonitis (i.e., any cause of ascites can cause SBP; I discuss this stuff in the GIT PDF in extensive detail). - MCD is aka lipoid nephrosis (lipid droplets can be seen in urine on LM).
Focal segmental glomerulosclerosis (FSGS)	- The answer on USMLE for nephrotic syndrome in sickle cell. - Can also be caused by heroin and HIV. - Not as responsive to steroids as MCD. - FSGS literally means "some parts of some nephrons are affected."
Renal papillary necrosis (RPN)	- Renal papillae = parts of kidney where the collecting ducts meet the ureters. - The answer on USMLE for red/dark urine in a patient with sickle cell. - In contrast, if a sickle cell patient has a renal condition + no blood in the urine, the answer = FSGS, as mentioned above.

	- RPN is loss of the renal medulla from ischemia, but unlike ATN, the etiology is usually more subacute or chronic. **NSAIDs** are a notable cause in this regard, where ↓ afferent blood flow due to ↓ vasodilating prostaglandins over weeks-months causes the medullar blood flow to be slowly choked off. - RPN can occur more acutely in sickle crises, where sickling within microvasculature feeding the medulla results in focal pockets of ischemia. - Pyelonephritis can also cause RPN due to inflammatory compression of the medullar microvasculature. There's one NBME Q I can recall where renal papillary necrosis is the answer due to infection. But I'd say, overall, the highest yield point is just remembering red/dark urine in sickle cell = RPN.

Interstitial nephritis
- Exceedingly HY renal condition often confused with ATN. - Aka interstitial nephropathy, or tubulointerstitial nephritis/-nephropathy. - Think of this as "an allergic reaction of the kidney." - 4/5 Qs will be an **NSAID, β-lactam, or cephalosporin,** followed by getting a maculopapular rash and WBCs (eosinophils) in the urine. This presentation is textbook/pass-level. - Only ~50% of vignettes will mention the maculopapular rash. - The stem need not say "eosinophils" either. They can just say patient is on β-lactam + now has WBCs in the urine + no mention of rash → answer = interstitial nephritis. If they say the rash, it's even easier. - 1/5 Qs will just say a patient was on an NSAID, β-lactam, or cephalosporin and now has mild proteinuria and hematuria. They don't mention a rash or WBCs in the urine. - There is also a singular NBME Q where they just say a patient has simple peripheral edema due to an NSAID, with no mention of anything else, and the answer is interstitial nephropathy. This is more unusual, since "NSAIDs + edema" classically = pre-renal, but I've seen the NBME assess interstitial nephropathy for this as well. - NBME also can give you simple vignette of interstitial nephritis, and then ask for the location in the kidney that's affected (e.g., efferent arterioles, etc.) → answer = "renal tubule." - For example, 6F + treated with penicillin for *Strep* pharyngitis + now has maculopapular rash and eosinophils in the urine; diagnosis? → interstitial nephritis. - 16F + using naproxen (an NSAID) for 6 weeks for muscle pain from sports + has peripheral edema → answer = interstitial nephropathy. - 15F + just recently finished 10-day course of cephalexin + has mild proteinuria and hematuria → answer = interstitial nephropathy.

Other random renal conditions for Peds	
Tumor lysis syndrome	- Lysis of tumor cells (e.g., ALL) by chemoradiotherapy leads to release of uric acid and phosphate that can overwhelm the kidneys and cause crystals, namely from uric acid. - USMLE loves asking what can be given to prevent renal failure in patient undergoing chemo → answer = "allopurinol." - Or they'll ask for MOA of drug that can prevent renal failure → answer = "xanthine oxidase." - The arrows USMLE wants for what will be seen in the blood in tumor lysis syndrome is same as renal failure except we also have ↑ uric acid.
Fanconi syndrome	- Inability of the PCT to reabsorb "lots of stuff." - Shows up on NBME as an arrow Q, where they ask about the reabsorption of amino acids, glucose, bicarbonate, and phosphate, and the answer is a ↓ for all of them. - Can be idiopathic or due to consumption of expired tetracycline antibiotics.

	- Don't confuse Fanconi syndrome with Fanconi anemia. The latter is an autosomal recessive aplastic anemia with hypoplastic thumbs/radii (I've made YouTube clips discussing this).
Hartnup disease	- Inability of the PCT to reabsorb tryptophan. - Can cause niacin (B3) deficiency in theory, since tryptophan is a precursor in niacin synthesis. - This is more just textbook masturbation to be honest that's been parroted over the years. Can't say I've seen this ever assessed once. Just mentioning it here otherwise I'll get some umbrageous student popping into my DMs about it.
Cystinuria	- Cystine stones (hexagonal) can occur in young adult due to an inability to reabsorb the **COLA** (dibasic) amino acids → **C**ysteine, **O**rnithine, **L**ysine, **A**rginine. - Cystine is two cysteines put together. - Cystinuria can be diagnosed with cyanide-nitroprusside test.
Thin glomerular basement membrane disease	- Benign familial condition where patient can have mild hematuria. - Thin basement membranes allow for passage of small amounts of RBCs. - No treatment necessary.

Renal artery thrombosis
- The answer on USMLE for hypertension in a neonate following umbilical artery catheterization. USMLE simply wants "↑ renin, ↑ aldosterone" as the answer. - Sounds weird, but you need to know umbilical artery catheterization is a major risk factor for renal artery thrombosis in neonates. - 2CK NBME Q gives brief umbilical artery catheter insertion in kid born 26 weeks' gestation in order to monitor blood pressure for a pneumonia. 3 weeks later, he has ↑ BP (128/86) → answer = "↑ renin, ↑ aldosterone" as answer. - BP in term neonates should be ~60/40. In a preemie 29 weeks' gestation, it should be ~50/30 according to Google.

HY Uremia points	
- "Uremia" means "urea in the blood," but is used to refer to patients who have symptomatic renal failure with poor lab values. - On USMLE, uremic patients will have: ↑ K^+, ↓ HCO_3^-, ↓ Ca^{2+}, ↑ PO_4^{3-}; Na^+ is variable.	
Uremic pericarditis	- Friction rub in kid with ↑ BUN and Cr. - Answer = hemodialysis.
Uremic encephalopathy	- Mental status change in kid with ↑ BUN and Cr. - Answer = hemodialysis.
Uremic platelet dysfunction	- Nosebleeds and/or petechiae in patient with renal failure. - **Qualitative, not quantitative,** platelet problem. In other words, bleeding time is ↑, but platelet count is normal. - Mechanism is ↑ BUN causing impairment of platelet function. - Answer on 2CK Peds Q is written as "acquired platelet dysfunction." - Treatment = hemodialysis. - On 2CK NBME 10, Surg Q wants "initiation of dialysis" as answer in uremic patient pre-op. Improving RFTs will ↓ peri-op morbidity/mortality + ↓ risk of intra-op bleeding due to uremic platelet dysfunction.

Urolithiasis	
- Urolithiasis is broad, umbrella term that refers to both nephrolithiasis and ureterolithiasis.	
Stone type	**HY points**
Calcium	- Most common type of stone. Can be calcium oxalate or calcium phosphate, although I've never seen USMLE once assess or give a fuck about phosphate stones. - Most teenagers or young adults with idiopathic kidney stones will have "normocalcemia and hypercalciuria" – i.e., normal serum calcium but elevated urinary calcium. - **Crohn disease** and disorders causing fat malabsorption increase the risk for calcium oxalate stones (↑ fat retained in GI tract → ↑ chelation with calcium in GI tract → ↓ calcium available to bind oxalate → ↑ oxalate absorption by GI tract → ↑ urinary oxalate). - Hypercalcemia (e.g., due to primary hyperparathyroidism or malignancy) are HY etiologies. - Ethylene glycol (Anti-Freeze) and hypervitaminosis C can cause oxalate stones. - First step in prevention and treatment of stones is "adequate hydration." - **Thiazides** can be used to prevent recurrent stones by ↓ urinary calcium. NBME has as an Q that asks why → answer = ↑ reabsorption of calcium. - In theory, alkalinization of the urine can help treat oxalate stones, and acidification of the urine can help treat phosphate stones.
Struvite	- Composed of ammonium magnesium phosphate. - Formed in ↑ pH in the presence of urease (+) bacteria. - Can be seen in teenage girls with UTIs due to *Klebsiella, Serratia,* or *Proteus*. - These stones are large, ram-horn like. The image is HY.  - Apart from adequate hydration, Tx is acidification of urine with NH_4Cl + surgery.
Uric acid	- Seen in patients who have gout. Not complicated. - First-line Tx for *chronic* gout is xanthine oxidase inhibitors (i.e., allopurinol or febuxostat). They are preferred over uricosurics (meaning ↑ urinary excretion of uric acid) like probenecid and sulfinpyrazone because the latter agents ↑ risk of uric acid stones. - Probenecid and sulfinpyrazone inhibit organic anion transporter (OAT), which normally functions to ↑ absorption of uric acid. - Weird factoid about probenecid is that it can help maintain serum β-lactam levels, since the latter are excreted by OAT. There's an NBME Q around somewhere that asks about this. Sometimes old-school docs will put a patient on a β-lactam and probenecid. Don't be confused by this. The patient doesn't have gout. It's just a way of ↓ renal excretion of the β-lactam.
Cystine	- As mentioned earlier, cystine is two cysteines put together. - Cystine stones (hexagonal) can occur in Peds or young adults due to an inability to reabsorb the **COLA** (dibasic) amino acids → **C**ysteine, **O**rnithine, **L**ysine, **A**rginine. - Cystinuria can be diagnosed with cyanide-nitroprusside test.

ARPKD
- Autosomal recessive polycystic kidney disease; chromosome 6. - The answer on USMLE for cystic kidneys in pediatrics. - Can be associated with **hepatic fibrosis.** - ADPKD, in contrast, occurs in adults 30s-40s and is associated with berry aneurysms of the circle of Willis.

HY anatomic abnormalities	
Posterior urethral valves (PUV)	- Exceedingly HY for Peds for 2CK; also shows up on Step 1 NBME. - Most common genitourinary abnormality in neonatal males. - Urethra has abnormal presence of valves at the posterior (prostatic) urethra, preventing the outflow of urine. - USMLE will give Qs of varying severity. - Most severe is *in utero* oligohydramnios. - Can present as 12-hour-old neonate who hasn't yet urinated + has suprapubic mass (i.e., full bladder). - Can also present as 6-week-old boy who has full bladder (i.e., possibly slower accumulation due to only partially obstructed outflow). - Obstructed outflow ↑ risk for UTIs, cystitis, and pyelonephritis. - Apart from knowing this diagnosis, highest yield point is that we do **ultrasound followed by voiding cystourethrogram** to diagnose. - NBME Qs might already mention negative ultrasound in the vignette, or might omit it altogether, where you just select "voiding cystourethrogram."
Vesicoureteral reflux	- Congenital abnormality in which urine from the bladder can go back up into the ureter toward the kidney. - This ↑↑ risk for recurrent acute pyelonephritis → chronic pyelonephritis (as I discuss below). - Can also occur in pregnancy due to 2 reasons: 1) larger uterus in 3rd trimester can compress the ureters; 2) progesterone ↓ ureteral peristalsis. - This is why pregnant women can get pyelonephritis, and also why we always treat asymptomatic bacteriuria in pregnancy, whereas we don't treat it if woman is not pregnant.
Ureteral atresia	- Bit of a weird one, but not me being fancy. It's on NBME. - "Failure of canalization of proximal ureter" will be the answer if they tell you the renal collecting duct system is dilated, but the ureters are not dilated. - Sounds obvious, but I see students get this wrong a lot. - You need to simply know: kidneys → ureters → bladder → urethra, and if we have a congenital obstruction at any point, that could be referred to as "failure of canalization." - "Congenital ureteral obstruction" is an answer on one of the NBMEs for chronic pyelonephritis causing tubular atrophy (once again, discussed below).

Genitourinary infection-related stuff	
- For all of the STD/STI-related stuff, I cover that in the Repro/Obyn PDF. I'm keeping this small table more limited to kidney/bladder stuff.	
Acute pyelonephritis	- Infection of the kidney. "Pyelo" means kidney. - 9/10 questions will mention fever + **costovertebral angle (CVA) tenderness,** which is pain with percussion of the flank. - "Granular casts" can be seen. Do not confuse this with acute tubular necrosis. "General granular casts" can sometimes be seen in pyelonephritis and dehydration. It's on one of the 2CK IM forms, where students say, "Wait, but they say granular casts; how is this not ATN?"

	- Major risk factors are vesicoureteral reflux (especially in pregnancy) and posterior urethral valves in Peds. - Most common organism is *E. coli*. Other organisms, i.e., *Klebsiella, Serratia,* and *Proteus* cause struvite stones. - It's to my observation **bacteria can be few in the urine** in acute pyelo. This confuses some students. But it's typically what I see on NBME forms. If the infection is further down, i.e., UTI in the urethra, then bacteria are more copious. - Treatment for pyelo is ciprofloxacin or ceftriaxone (cefotaxime is correct in Peds if listed alongside ceftriaxone). USMLE is known to ask these.
Chronic pyelonephritis	- Due to recurrent acute pyelonephritis. - Almost always pediatric, where they give a 4-year-old who has a small, shrunken kidney with **tubular atrophy** and **blunting and scarring of the renal calyces.**  Kidney appears to show loss of architecture + scarring. - USMLE will show you kidney that looks similar to the above in a child with recurrent UTIs, and then the answer is just "tubular atrophy" for what is most likely to be seen on microscopic examination. - Another Q shows the same image and the answer is "congenital ureteral obstruction" as the cause. - "Thyroidization of the kidney" is a buzzy phrase that has been applied to chronic pyelo over the years, but USMLE doesn't so much care about this. What they really like is the tubular atrophy and scarring / loss of architecture of the calyces. - Chronic pyelo doesn't present with cellular infiltration the way acute pyelo does. It refers to the kidney being destroyed/shrunken from repeated prior infections.
Cystitis	- Suprapubic tenderness in female. Patient need not have fever. - *E. coli* most common cause. HY for USMLE you know **fimbriae** and **pilus proteins** facilitate *E. coli*'s attachment to urothelium. - Will have bacteria and WBCs in the urine (i.e., bacteriuria + pyuria). - Urinary nitrites and leukocyte esterase can be positive. These just reflect bacterial infections. Some students get pedantic about which organisms result in which combo of (+) or (-) findings here, but that's low-yield for USMLE. - Can be caused by **suprapubic catheters.** - **Nitrofurantoin** is HY drug to treat cystitis on USMLE.
Chronic interstitial cystitis	- Not an infection. - This is >6 weeks of suprapubic tenderness + dysuria (pain with urination) that is unexplained, where laboratory and urinary findings are negative. - They can mention **anterior vaginal wall pain** (bladder is anterior to vagina). - USMLE wants you to know you **don't treat.** Steroids are wrong answer. "Treatment" is standard placating placebo nonsense such as "education," "self-care" and "physiotherapy."
General UTI	- Classically *E. coli* infection of the urethra. - Common in females due to shorter urethras.

	- As mentioned above for cystitis, **fimbriae** and **pilus proteins** facilitate *E. coli*'s attachment to urothelium. - Inoculation of the urethra following sexual activity is most common mechanism. - Can be caused by catheters. You need to know **dysuria + Hx of catheter = UTI.** - Trimethoprim, trimethoprim/sulfamethoxazole (TMP/SMX), or nitrofurantoin are standard treatments for UTI. - For prevention, post-coital voiding confers ↓ risk of recurrence. - If post-coital voiding doesn't work, "post-coital nitrofurantoin therapy" is answer on 2CK NBME form. - If post-coital nitrofurantoin therapy doesn't work, "daily TMP/SMX prophylaxis" is the answer. Sounds absurdly wrong / like a bad idea, but it's an answer a couple times on 2CK assessments. - There's also a Q where they say a girl was treated successfully in the past with TMP/SMX for a UTI + ask how to prevent recurrent UTIs now, and they just jump straight to "daily TMP/SMX prophylaxis." - If patient is treated for a UTI and dysuria persists, the next best step is testing for *Chlamydia* and *Gonorrhea*.

HY Genitourinary injuries	
Urethral injury	- Always presents with blood at the urethral meatus. - 3 ways this presents on USMLE: 1) Blood at urethral meatus + saddle injury (dude falls on balance beam). 2) Blood at urethral meatus + pelvic fracture/instability. 3) Blood at urethral meatus + "prostate not palpable." - Next best step is **retrograde urethrogram.**
Bladder injury	- Answer on USMLE for blunt force trauma to abdomen + exquisite suprapubic tenderness + catheter insertion yields 10-30 mL of dark bloody fluid. - Next best step is **retrograde cystourethrogram.** Do not confuse with voiding cystourethrogram, which is for posterior urethral valves. - Make note that this contrasts with the voiding cystourethrogram done for posterior urethral valves. - In other words, retrograde studies are for suspected injury. Voiding study is for PUV.
Renal injury	- Will present as bruising or pain over a flank following accident. - Exceedingly HY on 2CK. Can show up in teenager hit by hockey stick or in MVA. - Blood in the urine is highly sensitive, meaning if it's not there, we can rule-out. - First step in diagnosis: "Is there gross blood in the urine?" If yes → do **CT of the abdomen with contrast** to diagnose (sounds wrong, since we're worried about kidney injury, so student says, "Why would we use contrast?" But it's what USMLE wants. **Ultrasound is wrong answer**). - If patient does not have gross blood in the urine, next best step is **urinalysis looking for microscopic blood.** If present → answer = CT of the abdomen with contrast. If negative → "no further diagnostic studies indicated."

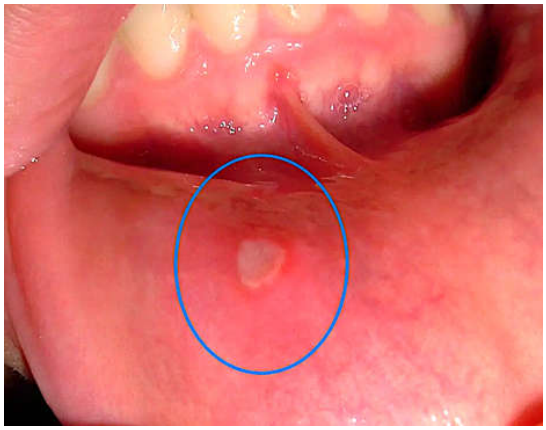
Wilms tumor (Nephroblastoma)
- The answer for painless flank mass in a kid age 2-4 years. - Does **not** present in newborns. If they give you a 12-hour-old male, for instance, with a midline mass, that is posterior urethral valves causing a full bladder (let alone the fact they say "midline," not flank). I've seen tons of students select Wilms for this, and I'm like, is the kidney in your midline? - If they give you a kid 2-4 years with a midline mass, that is neuroblastoma. - For whatever magical reason, you need to know Wilms tumor is sometimes caused by mutations on **chromosome 11.** They ask this factoid on 2CK as well.

There are various Wilms tumor syndromes you need to be aware of:

- **Beckwith-Wiedemann syndrome**
  - Wilms tumor + macrosomia (big baby >4,000g) + hemi-hypertrophy (half of the body is bigger than the other) + macroglossia + omphalocele + hypoglycemia.
  - Hard 2CK Peds Q gives newborn with macrosomia + hemi-hypertrophy + macroglossia + omphalocele; they don't mention a Wilms tumor (makes sense, since we said we won't see it in newborns); they ask for what else could be seen in this patient → answer = "hypoglycemia." I say hard Q because Wilms plays no role in the vignette.
- **WAGR syndrome**
  - **W**ilms tumor, **A**niridia (iris abnormalities), Genitourinary anomalies, Retardation.
  - Usually an easy Q, where they mention "aniridia" straight up in the vignette and then the answer is just "Wilms tumor" for what the kid can go on to develop.
- **Denys-Drash syndrome**
  - Ambiguous genitalia + Wilms tumor.

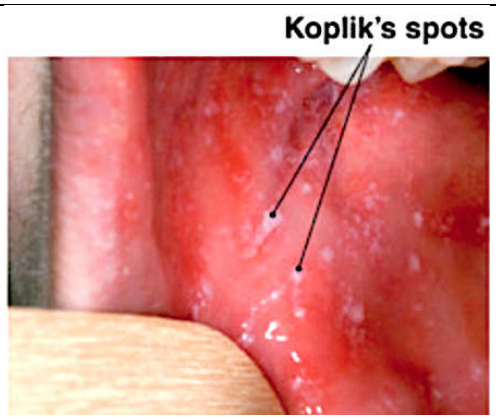
Peds Gastro

Disorders affecting lips/oral cavity for Peds	
Peutz-Jeghers syndrome	- Combination of perioral melanosis (fancy word for hyperpigmentation around the lips/mouth) + **hamartomatous** colonic polyps.  - USMLE will show image of lips pretty much always, and then they'll ask for what kind of polyps are seen in the colon (i.e., hyperplastic, tubulovillous, etc.), and the answer is just "hamartomatous."
Osler-Weber-Rendu	- Aka hereditary hemorrhagic telangiectasia; autosomal dominant. - NBME loves showing a mouth or fingernail picture of telangiectasias.  - Q will give nosebleeds in a 12-yr-old + show you the above pic. - GI bleeding can occur leading to anemia. - In adults, there can be high-output cardiac failure due to pulmonary AV fistulae.
Plummer-Vinson syndrome	- Triad of iron deficiency anemia + esophageal webs (dysphagia) + angular cheilitis (cracked corners of mouth). 

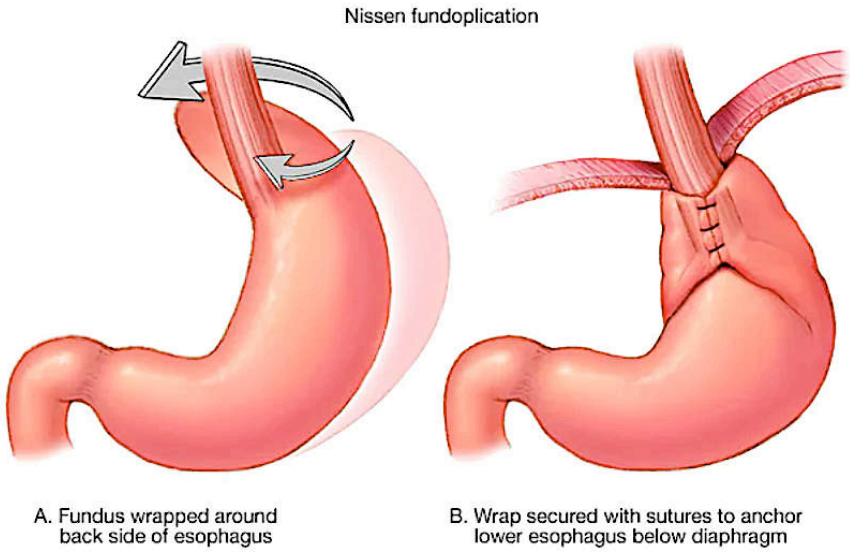
	- NBME Q can also mention pica (iron deficiency sign where patient eats clay, starch, or ice), or they can show spoon-shaped nails (koilonychia), which are a sign of severe iron deficiency.
Lip psoriasis	- Just know it's possible. USMLE can mention it on upper lip or forehead, and this somehow confuses students, where they think it has to be on extensors only. 
Aphthous ulcer	- Aka canker sore. - Usually appears as very painful lesion, self-resolving lesion on labial mucosa. - Not infective; idiopathic; thought to have very mild autoimmune association; sometimes associated with certain triggers like spicy food. 
Kawasaki disease	- Vasculitis that causes **5+ days of fever** + injected (red) eyes and/or lips/tongue + cervical lymphadenopathy + edema of dorsa of the hands + **desquamation of palms/soles** (often mentioned as palms/soles "rash," but not true rash). 

	- Students obsess over coronary artery aneurysms as though they're so HY. They're not. USMLE basically never mentions them. - Tx = aspirin + IVIG. - Only condition in Pediatrics where aspirin is OK to give. This is followed by intravenous immunoglobulin. - We normally AVOID aspirin in children and teens because of risk of Reye syndrome.
Perioral impetigo	- Often confused with herpes.  - If they show you image of young kid in particular with lip lesions, it's usually impetigo caused by *S. aureus* or Group A Strep (*S. pyogenes*). - Can lead to PSGN (if caused by Strep), as discussed in the HY Renal PDF. - *Staph aureus* exceeds Group A Strep for both bullous *and* non-bullous types (bullous generally implies *Staph*). - Tx for impetigo = **topical mupirocin** (exceedingly HY). - If orals given, use dicloxacillin, cephalexin, or Augmentin, but USMLE loves topical mupirocin for impetigo.
Herpes labialis	- Caused by HSV1 or 2. USMLE doesn't give a fuck about HSV1 being the lips and HSV2 being the genitals. Bunch of nonsense perpetuated by other resources.  - Primary infection presents with fever and regional lymphadenopathy. Recurrences aren't as severe. - HSV goes latent in sensory nerves. - Treat with acyclovir → DNA polymerase inhibitor causing chain termination. HSV resistance to it will be due to altered thymidine kinase.
Scarlet fever	- Caused by Group A Strep. - Causes strawberry tongue / red lips + salmon-pink maculopapular body rash.

	 - Treatment is penicillin to prevent rheumatic fever (type II HS); can also lead to PSGN (type III HS). I discussed this stuff in the HY Cardio and Renal PDFs.
Hand-foot-mouth	- Caused by Coxsackie A (an RNA virus under picornaviridae). - Causes benign, but contagious, lesions on, you guessed it LOL! – the hands, feet, and mouth. - Usually pediatric, but can present in adults (i.e., daycare workers, parents).  - Don't confuse with coxsackie B, which can cause dilated cardiomyopathy, diabetes type I, and pleurodynia.
Herpangina	- Also caused by coxsackie A. - Presents as oropharyngeal vesicles or sores.  - Can occur with or without hand-foot-mouth.
Koplik spots	- Caused by measles. - White/blue-ish spots on buccal mucosa.

	 - One of the ways to distinguish measles (rubeola) from rubella (German measles). - The same way you can memorize Koplik spots as unique to measles, you should memorize sub-occipital and post-auricular lymphadenophaty as unique to rubella. - Both measles and rubella have a head-to-toe masculopapular body rash. - Both can present with fever, cough, coryza, conjunctivitis. These findings are non-specific for viral infection; it's been erroneously attributed to only measles over the years, but I've it show up in numerous NBME Qs for a variety of viruses).
Oropharyngeal candidiasis	- Painless white plaque on palate or tongue that bleeds when scraped.  - Seen in immunocompromised children with AIDS or Hx of organ transplant (i.e., on immunosuppressants). - Can be seen in children with asthma who use inhaled corticosteroids. Patients need to rinse their mouths with water after use to ↓ candida risk. - Tx with nystatin mouthwash. - Odynophagia (painful swallowing) in an immunocompromised patient is Candidal esophagitis until proven otherwise.

HY Esophageal conditions for USMLE	
Pediatric GERD	- Pediatric GERD will present as coughing up milk 2-3x daily. It will not be described as high-energy or forceful (i.e., pyloric stenosis); it will not be bilious (i.e., duodenal atresia; annular pancreas); it will not occur as choking or spitting up during the first feed (i.e., tracheoesophageal fistula). - Answer to pediatric GERD Q is "immature lower esophageal sphincter". It's not complicated, but I've seen students say, "Wait, you can get GERD in kids?" - Tx for Peds GERD is positional change and thickened feeds (addition of dry rice to formula) prior to any use of PPIs. - Nissen fundoplication is used last resort, but is asked on NBMEs.

	 - Peds form gives infant with severe GERD and fundoplication is answer.
Eosinophilic esophagitis	- Idiopathic allergic condition of the esophagus that presents with difficulty swallowing. - Can be associated with asthma (i.e., dysphagia in patient with Hx of asthma). - Can occur in infants up through adults. - Biopsy shows eosinophils in epithelium (it's like no shit).  - Endoscopy shows "trachealization" of the esophagus with concentric rings.

Bilirubin uptake/secretion diseases	
Gilbert	- Pronounced "Jeel-BEAR, not "GILL-burt." Gastroenterologist I met once tripped out over med students pronouncing this wrong. - Partial deficiency of bilirubin uptake enzyme at the liver (UDP glucuronosyltransferase). - Presents as isolated ↑ indirect bilirubin with yellow eyes in teenager or young adult with stress factor, such as studying for exams, or recent surgery/trauma. The patient will otherwise be completely healthy. - No treatment necessary.
Crigler-Najjar	- Near-absence of UDP-glucuronosyltransferase causing ↑↑ indirect bilirubin in neonate. - Nonexistent yieldness but students ask about this.
Biliary atresia	- One of the highest yield Peds diagnoses. Answer on USMLE for ↑↑ direct bilirubin in a kid under the age of 6 weeks. - Caused by lack of development of the intrahepatic bile ductules and biliary tree. - Ultrasound will be done first, but USMLE wants **liver biopsy** to confirm the diagnosis. - Kasai procedure is done to treat, followed by liver transplant if unsuccessful.

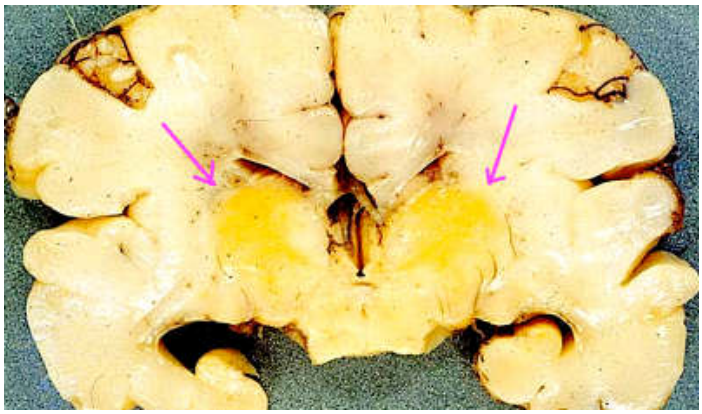
Pathologic vs physiologic jaundice

- In regard to neonatal labs (i.e., first month of life), I should note that hematocrit % can be in the 50s (NR 45-61; in adults, NR is 40-50). 2CK, for instance, will often show Hct as 56% in a newborn and the student is like "Wow that's high!" where they think there's neonatal polycythemia or some other pathology, but it's actually normal. Unconjugated jaundice can occur in neonates because of ↑ RBC turnover where HbF is replaced with HbA.

- Normal bilirubin should be ~1.0 mg/dL.
- Direct bilirubin should be ~0.1 mg/dL.
- **Pathologic jaundice** in peds = if *any* of the following is/are present:

1) Any jaundice on the first day of life (first 24 hours of life), period = pathologic.
2) Jaundice present **after** one week if term, or after two weeks if preterm = pathologic.
3) Total bilirubin >15 mg/dL.
4) Direct bilirubin >10% of total bilirubin, *even if total bilirubin is <15 mg/dL.*
5) Rate of change of increase in bilirubin >0.5 mg/dL/hour.

- If none of the above 5 is met but the neonate has jaundice, the diagnosis is **physiologic jaundice.**
- If the kid has pathologic jaundice, we then want to explore the causes (all discussed below).
- Tx for pathologic jaundice = **phototherapy first, followed by exchange transfusion;** some literature makes a case for IVIG after phototherapy, but USMLE has exchange transfusion as correct, without listing IVIG.
- If ↑ bilirubin in the neonate accumulates in the CNS grey matter, this is called **kernicterus** and can cause irreversible neurologic damage. In the gross specimen below, bilirubin has deposited in the basal ganglia (pink arrows pointing to yellow areas). USMLE will show similar image, where the answer is just "putamen."



Breastmilk jaundice	- Due to β-glucuronidase in breast milk, which leads to de-conjugation of intestinal bilirubin + increased enterohepatic circulation. - Jaundice starts on day 3-5 and peaks at 2-3 weeks. - Tx = stop breastfeeding for ~48 hours (and do bottle feeding), which leads to a rapid ↓ in bilirubin; once breastfeeding is resumed, bilirubin might rise, but not back to pathologic levels.
Breastfeeding jaundice	- Insufficient feeding (e.g., failure of suckling, etc.) + decreased milk intake leads to reduced intestinal clearance of bilirubin + increased enterohepatic circulation. - Jaundice peaks at 3-5 days. - Tx = formula feeding (fluid + caloric supplementation). - What USMLE can do is tell you neonate has difficulty attaching to the breast during feeds + has jaundice, and you're like "Omg breastfeeding jaundice!" But then you'll see that 0/5 criteria are met for pathologic jaundice, and the answer to the Q is just "physiologic jaundice."

Spontaneous bacterial peritonitis (SBP)
- One of the highest yield diagnoses on 2CK. - Bacterial infection of peritoneal fluid by mixed enteric flora. - The answer on USMLE when they give you diffuse abdominal pain and fever in one of the three following scenarios: 1) cirrhosis; 2) recent peritoneal dialysis; 3) nephrotic syndrome. - There is a 2CK Peds Q where they say kid with minimal change disease (nephrotic syndrome) has abdo pain + fever → answer = spontaneous bacterial peritonitis. - Next best step in diagnosis is abdominal **paracentesis**. This refers to aspiration of fluid from the peritoneal cavity. Do not confuse this with pericardiocentesis. "Paracentesis" as an answer shows up everywhere, and I've seen students avoid it because they're like, "What? I thought that meant pericardiocentesis." - After the paracentesis is done, USMLE wants a very specific order for what to do next. **Choose "white cell count and differential" first if it's listed, followed by "gram stain and culture of the fluid."** - This order is important because "gram stain and culture of the fluid" is a correct answer in one Q but wrong answer in another Q, where "white cell count and differential" is correct to do first. - The reason white cell count and differential is done first is because SBP is diagnosed when paracentesis shows >250 WBCs/μL. - There is a new 2CK Q where they give ascites in patient with cirrhosis, but do not mention fever or abdominal pain. However, they say paracentesis shows 900 WBC//μL → answer = "antibiotic therapy." It should be made clear though that ~6/7 SBP Qs will give abdo pain and fever. - Treatment is ceftriaxone (if cefotaxime is also listed, choose this for Peds over ceftriaxone).

Splenectomy
- Done in setting of trauma (splenic laceration), hereditary spherocytosis, or ITP. - "Autosplenectomy" refers to loss of spleen due to repeated microinfarcts in sickle cell. - "Asplenia" can refer to anyone who's had splenectomy, autosplenectomy, or who didn't develop a spleen from birth (Ivemark syndrome on 2CK NBME). - The white pulp of the spleen ordinarily contains 50% of the immune system's reservoir of macrophages. - Patients are susceptive to encapsulated organisms because the spleen has a high phagocytic capacity, where opsonization with C3b and IgG, followed by phagocytosis in the white pulp, is how they are cleared. - Patients need vaccines against *Strep pneumo, Neisseria meningitidis,* and *Haemophilus influenzae* type B. - Penicillin prophylaxis is given to patients with asplenia until age 5, or until 1 year post-surgical splenectomy. If the USMLE forces you to choose which organism this protects against the most, the answer is *Strep pneumo.* - However, if USMLE vignette tells you a patient with asplenia misses a penicillin prophylaxis dose + now has sepsis, the Tx is a 3rd generation cephalosporin (i.e., cefotaxime or ceftriaxone), not penicillin. The latter is only for prophylaxis. - Howell-Jolly bodies (nuclear remnants) are seen within RBCs on a smear in patients with asplenia. This is because senescent and abnormal RBCs are ordinarily cleared out by the red pulp of the spleen. - USMLE wants you to know the lifespan of a RBC is 4 months (118 days). They ask this on NBME as how long it will take for all carbon monoxide to be cleared out of the blood in the setting of exposure (i.e., not until all RBCs exposed are removed).

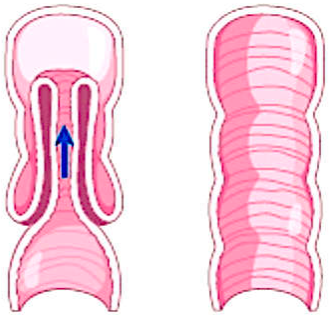
HY GI Diagnostic modalities for Peds	
Nasogastric tube	- "Insertion of NG tube" is the answer for tracheoesophageal fistula and choanal atresia.
Upper GI series	- Congenital midgut volvulus in pediatrics; will show a corkscrew. - This is a contrast swallow followed by X-rays to visualize the upper GI tract.
Abdominal ultrasound	- Pyloric stenosis. - Intussusception Dx, then do enema (definitively diagnostic and therapeutic).
CT of abdomen	- Almost always wrong answer in Peds because of ↑↑ radiation. However, I've seen this in two scenarios in Peds: - Renal injury, as discussed earlier.

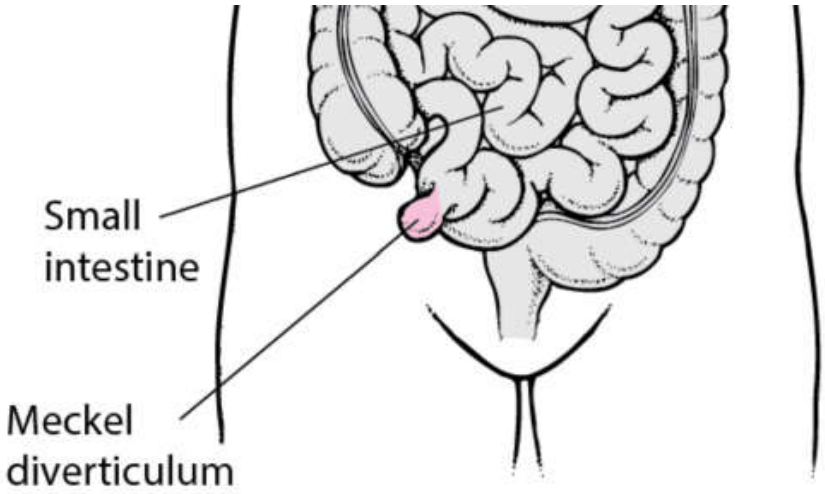
	- Blunt force trauma to abdomen accompanied by severe pain or visible hematoma. There is 2CK Peds Q where they say kid bumps his stomach on corner of table + 2 days later has growing hematoma + pancreatic enzymes → answer = CT, possibly to look for traumatic pancreatitis. - Never do abdo CT in unstable (i.e., ↓ BP) patient. Always go straight to laparotomy. This includes patients with low BP who achieve temporarily normal BP with fluids (i.e., fluids are only a temporizing measure [i.e., buying time]).
Abdominal x-ray	- Duodenal atresia (double-bubble sign). - Congenital diaphragmatic hernia (bowel gas in left hemithorax). - Necrotizing enterocolitis (pneumatosis intestinalis). - Toxic megacolon if patient is stable.
Paracentesis	- Used for spontaneous bacterial peritonitis. - As mentioned earlier, choose "white cell count and differential" before "gram stain and culture of the fluid." Do not confuse with pericardiocentesis.
Meckel scan	- Radiocontrast uptake scan that localizes to the diverticulum at terminal ileum.
Colonoscopy	- If kid has IBD (UC or Crohn), commence 8 years after the diagnosis was made and then do every 2-3 years. - For Peutz-Jeghers, start at age 8, then every 1-2 years. - For FAP, start at age 10-12 and do every 1-2 years. Then do prophylactic proctocolectomy at age 18 (on NBME). - If patient has HNPCC (Lynch syndrome), start at 20-25, then do every 1-2 years.

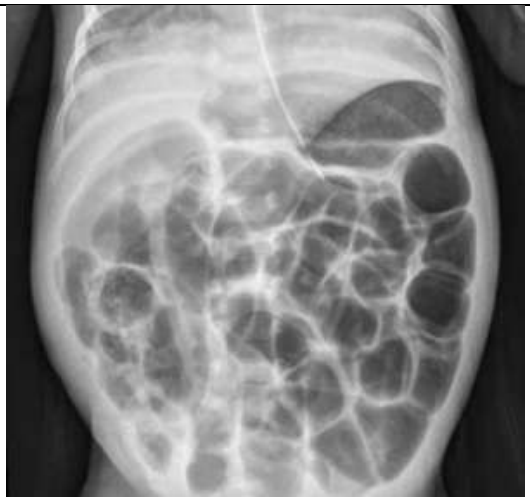
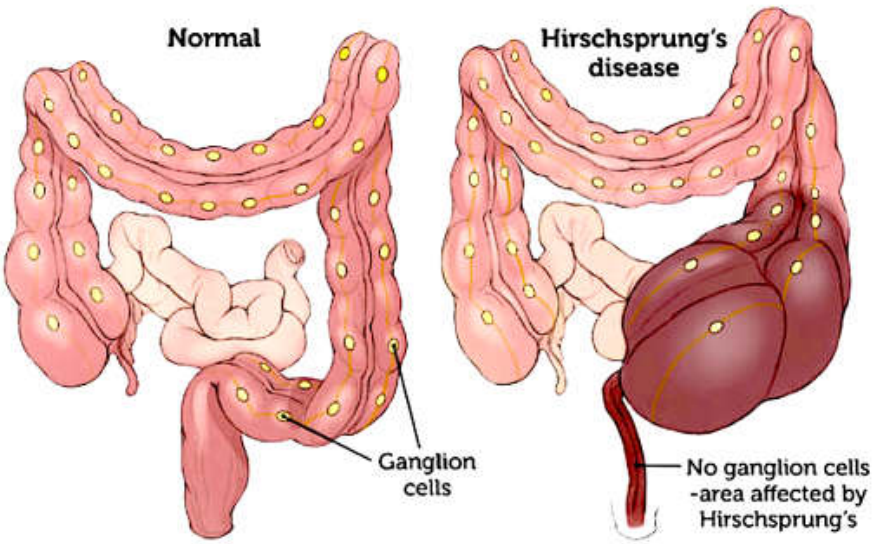
HY Polyp conditions for Peds	
Lynch syndrome	- Hereditary non-polyposis colorectal cancer (HNPCC). - Mismatch repair genes *MSH2/6, MLH1, PMS2.* - Mutations cause "microsatellite instability." - Colonic polyps/cancer; also associated with gynecologic cancer. - Start colonoscopy at age 20-25, then do every 1-2 years.
Familial adenomatous polyposis	- Familial Adenomatous Polyposis (FAP); chromosome 5; AD. - Hundreds to thousands of polyps on colonoscopy; 100% cancer risk. - Start colonoscopy at age 10-12 and do every 1-2 years. Then do prophylactic proctocolectomy at age 18 (on NBME). - FAP + soft tissue (e.g., lipoma) or bone tumors (e.g., of the skull) = Gardner syndrome. - FAP + CNS tumors = Turcot syndrome.
Peutz-Jeghers	- As talked about at the start of this PDF, this is combo of perioral melanosis and **hamartomatous** colonic polyps. - Start colonoscopy at age 8, then do every 1-2 years.
Juvenile polyposis	- Shows up on a 2CK NBME. - Q will tell you there's a teenager (i.e., juvenile, LOL!) with intermittent bleeding per rectum + colonoscopy shows scattered polyps + biopsy shows **"dilated, cystic, mucus-filled glands** with abundant lamina propria and inflammatory infiltrates" → answer = juvenile polyposis. - Only question I've seen on it, but it's on new 2CK NBME so I have to mention it.
Hyperplastic polyps	- All you need to know is these are **not pre-cancerous /** have **no dysplasia.**

HY Peds GI diagnoses	
Pyloric stenosis	- Forceful/projectile **non**-bilious vomiting in neonate days to weeks old. Obstruction is above level of duodenum, so we won't see bile. - Students fixate on exact age of the kid for pyloric stenosis versus duodenal atresia. USMLE doesn't give a fuck and will give variable ages. - Hypertrophic pylorus; "olive-shape mass" in abdomen is buzzy but rarely seen in Qs.

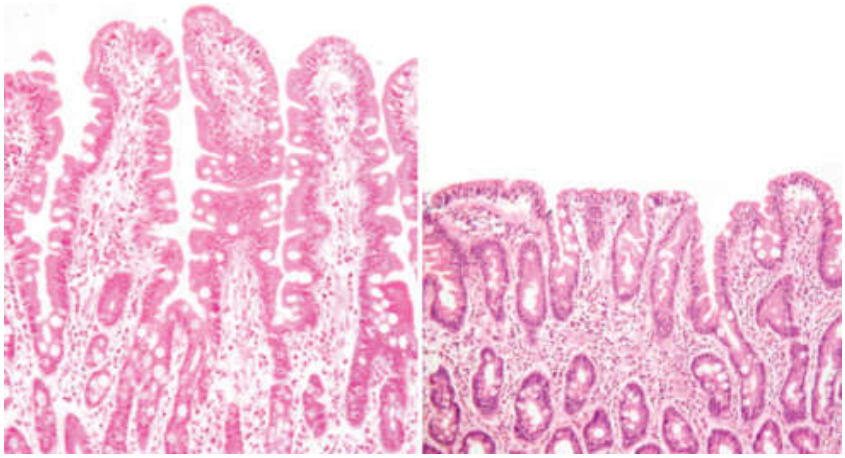
	- **Ultrasound** done to diagnose; myotomy to treat. - USMLE wants you to know this is almost always a one-off / sporadic developmental defect, where the neonate will not have a broader syndrome. In contrast, duodenal atresia is usually Down syndrome.
Duodenal atresia	- Bilious vomiting in neonate. Bilious = obstruction at level of duodenum or lower. - Associated with **Down syndrome,** albeit not mandatory. - Will cause **double-bubble sign** on **abdominal x-ray.** 
Annular pancreas	- Another cause of duodenal obstruction with double-bubble sign, but much more rare than duodenal atresia. - Step 1 is pass/fail now, but you could be aware that this is caused by "abnormal migration of ventral pancreatic bud." This was the type of garbage we memorized back in the numerical Step 1 days. Mentioning it here because I think it still floats around on the Step.
Intussusception vs Midgut volvulus	- Distinguishing between intussusception and congenital midgut volvulus is annoying as fuck, so I'll do my best to elucidate below based on what I've seen across NBMEs. Abdominal exam: - Intussusception can present with an **abdominal mass,** which can be described as "sausage-shape." This will usually occur in the RLQ. - Midgut volvulus is usually first month of life + has **no overt mass.** - "Dance sign" for retraction of RLQ in intussusception is floated around in the literature quite a bit, but I haven't seen USMLE ever give a fuck or make this a conspicuous point. It's the **mass versus no-mass** that I see as important. Imaging: - Midgut volvulus shows corkscrew appearance of small bowel on upper-GI series. This is ultra-HY. - Intussusception shows a target sign or sausage-mass on ultrasound. However, USMLE doesn't tend to mention imaging often for intussusception. - If abdominal x-ray is performed, dilated loops of small bowel with air-fluid levels may be seen **in both.** This is a non-specific finding and is buzzy for intestinal obstruction. Do not confuse with air-fluid levels on CXR, which mean pulmonary abscess. Age: - Intussusception is usually 3 months – 2 years. - Midgut volvulus is usually first month of life, with 90% under age 1. - However, the fact that there's overlap causes annoyance answering USMLE Qs and I see this as a less reliable variable.

	- There is a volvulus Q on Peds CMS form 6 where they give a 5-month-old (age range more for intussusception), but they tell you overtly an upper-GI series is performed and show you the image of the corkscrew. They also explicitly say there are no masses. So I see this as: abdominal exam and imaging are buzzy + age not as reliable. My view on how to distinguish for NBME/USMLE: - Both intussusception and midgut volvulus can present with bilious vomiting, air-fluid levels on abdominal x-ray, bloody stools, and intermittent crying/vomiting/squatting. Once again, they sound very similar. So I've more or less been able to converge on the combo of **abdominal mass as most important, followed by buzzy imaging as second-most important, followed by age.** - Now I will talk about each condition separately below.
Intussusception	- Intussusception is telescoping of the bowel into itself; 90% occurs in **jejunum or ileum** of small bowel. There is an NBME Q that basically gives a vague one-liner for intussusception where they ask the location, and the answer is just "jejunum," since it's the only small bowel option listed.  **Intussusception** **Normal** - Almost always 3 months – 2 years of age. - Idiopathic, but can be triggered by viral infection or rotavirus vaccination, where mesenteric lymphadenopathy can act as a "lead point" for the intestinal telescoping. Underlying Meckel diverticulum is also a common lead point. - In ultra-rare scenarios, intussusception in elderly can occur due to colon cancer. - Classic presentation is intermittent/colicky abdominal pain, where the kid will have episodes of drawing the legs to the chest or squatting, with blood in the stool +/- bilious vomiting. - **An abdominal mass** (i.e., sausage-shaped mass) is a key feature, but not in all Qs. - Diagnosis is first done with ultrasound, which may show a sausage mass or target sign (I've never seen NBME show the image, so they don't care). After the ultrasound, **enema is both diagnostic and therapeutic.** USMLE doesn't care what kind of enema is used; air-contrast enema is classic, but you will see all different types of enemas as correct answers. - "Air enema with ultrasound" is the answer for intussusception on 2CK Surg form 7.
Midgut volvulus	- Failure of rotation of proximal bowel. - Midgut volvulus is usually first month of life, with 90% under age 1. - There is **no abdominal mass.** - Upper-GI series shows corkscrew appearance. - Treatment is surgical.

	 - As example, USMLE will show you above image in 5-month old where you say, "Ok, that's age more for intussusception." But then they'll tell you upper-GI series is performed + they show you the corkscrew → clearly midgut volvulus, not intussusception. Answer will be "failure of rotation of proximal bowel," or "gut malrotation."
Meckel diverticulum	- Outpouching near the terminal ileum. - True diverticulum. Contains all layers of bowel (mucosa, submucosa, muscule). - Caused by incomplete obliteration of vitelline (omphalomesenteric) duct.  - Highest yield point for USMLE is that it can contain heterotopic gastric or pancreatic mucosa, resulting in bleeding and **blood in the stool.** - Another important point is that Meckel **need not be pediatrics.** This notion of rule of 2s (i.e., 2 feet from terminal ileum, 2 types of heterotopic tissue, age 2) is garbage with respect to age for USMLE. There are Qs where it shows up in adults and students get them wrong saying, "Wait, I thought it was supposed to be kids?!" → No. USMLE will happily give you a 19-year-old with Meckel. - Diagnosed with Meckel (Tc99) scan, which localizes to terminal ileum. - Only treated surgically if symptomatic with bleeding. Otherwise can be left alone.
Necrotizing enterocolitis	- Necrotic bowel occurring in premature neonates born <32 weeks' gestation. - Presents with pneumatosis intestinalis (air in bowel wall); resembles bubbles.

	 - USMLE will show you above image in kid born, e.g., at 26 weeks' gestation, and then answer is just necrotizing enterocolitis. Not hard. - It's to my observation necrotizing enterocolitis usually shows up as a *wrong* answer choice on NBMEs, where I see students pick it in, e.g., an adult, and I'm like, "Dude you see that in fucking hyper-preemies." And they're like Oh. So you need to remember this as specifically a premature neonate condition.
Hirschsprung	- Failure of migration of neural crest cells distally within the colon. - Causes an aganglionic distal segment that results in ↑ anal sphincter tone.  - Often associated with Down syndrome. - Presents usually as failure to pass stool at birth, as chronic constipation in an infant, depending on the severity. But do not confuse with "meconium ileus," which is a term that refers to failure to pass stool at birth specifically due to cystic fibrosis. - Diagnose with anal manometry showing ↑ anal sphincter tone, followed by biopsy.
Cystic fibrosis	- Discussed in extensive detail in the HY Pulm PDF. - Just a reiteration that meconium ileus + **exocrine pancreatic insufficiency** are exceedingly HY. - Secretions within the pancreatic ducts are inspissated (meaning desiccated / dried up within a lumen), making them sticky. This means the enzymes can't make their way to the duodenum → fat-soluble vitamin malabsorption → NBME exams love vitamin E deficiency in CF in particular (presents as neuropathy).

	- D-xylose test is normal, since the intestinal lining/architecture is intact. Any malabsorption that occurs is due to mere paucity of enzymes in the small bowel. Recall that abnormal D-xylose test would be Celiac and Crohn on USMLE.
Biliary atresia	- As discussed earlier, ↑↑ direct bilirubin in a kid under the age of 6 weeks. - Caused by lack of development of the intrahepatic bile ductules and biliary tree. - Ultrasound will be done first, but USMLE wants **liver biopsy** to confirm the diagnosis. - Kasai procedure is done to treat, followed by liver transplant if unsuccessful.

Autoimmune intestinal absorptive disorders	
Celiac disease	- Intolerance to gluten (i.e., to gliadin proteins found in wheat, oats, rye, and barley; but not rice). - Causes type-IV hypersensitivity response where T cells attack the small intestinal villi, resulting in **flattening of the villi** and malabsorption. The biopsy/image is pass-level. Below, we have normal villi on the left, and flattening of villi in Celiac on the right.  - Even though the disorder is T-cell-mediated destruction of the villi, patients still develop antibodies which are HY: anti-endomysial (aka anti-gliadin) and anti-tissue transglutaminase IgA. - Patients with concurrent IgA deficiency will have false (-) results for tissue transglutaminase IgA antibody. Since “autoimmune diseases go together,” and “autoimmune and immunodeficiency syndromes go together” patients with IgA deficiency have 15x greater likelihood of developing Celiac. - Celiac presents as vague bloating/diarrhea in patients who might not be able to pinpoint what they’re eating to cause the symptoms. - I’d say one of the highest yield points on USMLE regarding Celiac is that it can cause **iron deficiency anemia.** This is because iron absorption in the duodenum is impaired from the flattened villi. - For example, you might get a difficult/vague vignette where you’re not sure if the diagnosis is Celiac or lactase deficiency, but you see that hemoglobin is low, and you’re like “Boom. Celiac.” - Associated with dermatitis herpetiformis, which presents as itchy, vesicular lesions on extensor areas. This causes “IgA deposition at dermal papillae.”

	 - D-xylose test is abnormal because the intestinal lining/architecture is abnormal (i.e., we have flattening of villi). - Increased risk of gastrointestinal T-cell lymphoma in Celiac. This is called enteropathy-associated T-Cell lymphoma (EATL). - Diagnosis is made via antibody screening, followed by duodenal biopsy as confirmatory. - Treatment of Celiac is simply with gluten-free diet.
Lactose intolerance	- Deficiency of brush border disaccharidase (i.e., aka lactase deficiency). - Brush border is located at "tips of the villi." This is in contrast to stem cells of the GI tract, which are located at the "base of the crypts." USMLE asks the locations of these things. - Diarrhea/bloating in response to dairy. - In contrast to Celiac disease, there is **no iron deficiency anemia.** - ↑ incidence in Asians, but can also develop idiopathically starting in teenage years-onward. When a young adult has a new-onset diarrhea-related condition, it's usually lactose intolerance, as lactase production can ↓ with age. But once again, USMLE can trick you, so I'll be an asshole and reiterate that if you see ↓ Hb, choose Celiac instead. - Secondary lactose intolerance can occur following viral gastroenteritis (e.g., rotavirus or Norwalk virus) → sloughing of the tips of villi → vignette will say new-onset diarrhea/bloating following gastro illness 1-2 weeks ago. This is only transient and patient will recover. This is HY cause of lactose intolerance. - **Biopsy is normal.** This is in contrast to the flattened villi in Celiac. - Diagnosis is made using hydrogen breath test or detecting ↓ stool pH. - D-xylose test is normal, since the intestinal lining/architecture is intact. - Treatment is with avoidance of lactose-containing products, or with lactase pills.
Milk protein allergy	- HY for 2CK Peds but nonexistent on Step 1. - Biggest risk factor is not being exclusively breastfed for the first 6 months of life. - Will present as **blood in the stool** in a child who they say is, e.g., 4 months old, who was started on formula 3 weeks ago. - Treatment is switching to **hydrolyzed casein formula.** Switching to soy-based formula is wrong fucking answer. There is up to 50% crossover of allergy cases with kids who have milk- and soy-protein allergy. - Vignette can say kid was started on either a cow-milk or soy formula when symptoms started. It doesn't matter. Just choose hydrolyzed casein as answer.

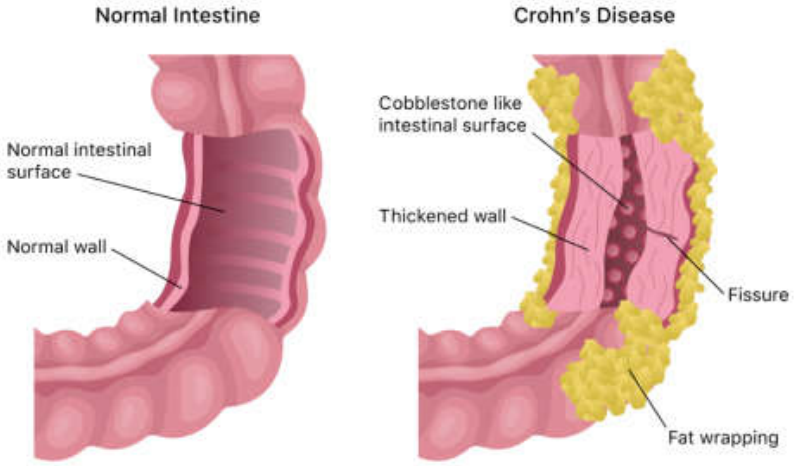
<table>
<tr><th colspan="2">Inflammatory bowel disease (IBD)</th></tr>
<tr><td colspan="2">- Refers to both ulcerative colitis (UC) and Crohn.
- Both can present with bloody, mucoid stools.
- Both are associated with HLA-B27 → PAIR → Psoriasis, Ankylosing spondylitis, IBD, Reactive arthritis.
- For example, if a patient has psoriasis + bloody stools, you say, “The bloody stools are probably IBD.” Or likewise, if patient with known IBD has lower back pain that’s worse in the morning, you say, “that’s probably ankylosing spondylitis.”
- Both UC and Crohn can be associated with other autoimmune diseases unrelated to HLA-B27, like vitiligo.
- Both can cause anterior uveitis (non-specific finding seen in many autoimmune diseases).
- Both have ↑ risk of colon cancer if the colon is involved. But if you’re forced to choose, UC has > risk than Crohn because the colon is always involved in UC but not always in Crohn.
- Both are treated with 5-ASA NSAID compounds (mesalamine / sulfasalazine) or steroids. If they ask first Tx, go with mesalamine or sulfasalazine, whichever they list (they won’t list both) over steroids. Q on 2CK IM form 8 has “prednisone therapy” as answer for Crohn, but a 5-ASA isn’t listed.
- USMLE wants you to know anti-TNF-α agents (i.e., infliximab, adalimumab, etanercept) are used in IBD in patients who fail initial Tx with 5-ASAs and steroids.
- If they ask you for which cytokine can be anti-inflammatory for IBD, the answer is IL-10. Sounds weird, but just know IL-10 and TGF-β are mostly anti-inflammatory mediators. Don’t worry about Th1 vs Th2 nonsense.</td></tr>
<tr><td>Ulcerative colitis</td><td>- Rectum-ascending.
- Not transmural – i.e., only affects mucosa and submucosa. This means fistulae are not seen. If they say fistulae to the skin overlying the anus or to any organ, this means transmural involvement (Crohn).
- Colonoscopy will show pseudopolyps and crypt abscesses. You don’t need to know what these look like. You just need to know they = UC.
- Barium enema shows “lead-pipe appearance” due to loss of haustra. This detail is very important for UC.
 
Descending colon resembles a “lead pipe”

- Can cause primary sclerosing cholangitis, as discussed earlier. This can be associated with pANCA antibodies.
- Pyoderma gangrenosum is a necrotic skin lesion seen rarely in patients with UC. There is an NBME Q that describes this as an “ulcer with necrotic debris.”
- Can cause toxic megacolon, which presents as SIRS and sometimes low BP in a patient with UC. They might say abdominal x-ray shows a 12-cm cecum (NR 3-8).
- If the patient has normal BP, NBME for 2CK wants steroids first for toxic megacolon. If the patient is unstable, go straight to laparotomy.
- Colectomy is sometimes performed in patients with severe UC, but very rarely in Crohn.</td></tr>
<tr><td>Crohn disease</td><td>- “Mouth to anus” – i.e., can occur anywhere in GI tract. USMLE loves giving mouth ulcers in Crohn. But terminal ileum is highest yield location.
- Transmural – i.e., can cause anal fistulae + to other organs.
- Colonoscopy shows skip lesions, where there is alternating diseased vs normal bowel segments, with cobblestone ulcers.</td></tr>
</table>

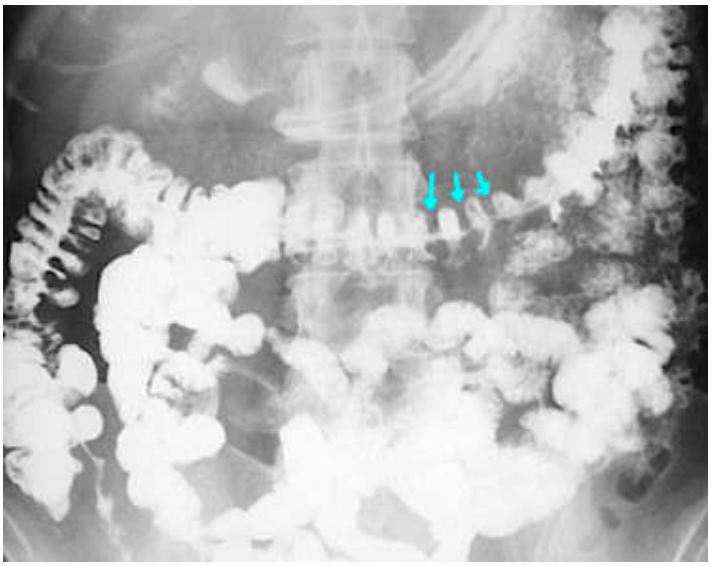


Skip lesion (left part normal + right part inflamed and cobblestoned)

- "Creeping fat" is buzzy term that can be seen on NBME, which refers to intestinal fat migration that wraps around the bowel.



- Barium enema shows "string sign," where inflamed segments are narrowed in comparison to normal bowel.



- Biopsy shows **non-caseating granulomas.** Very HY for USMLE you know that Crohn + sarcoidosis both have non-caseating granulomas.

	- Sometimes associated with erythema nodosum. Not specific for Crohn in any regard, but tends to have ↑ association, whereas UC is pyoderma gangrenosum. - Can cause anti-saccharomyces cerevisiae (yeast) antibodies. This is on a 2CK NBME, where they say (-) for these Abs, but (+) for pANCA, where answer is UC. - Intestinal malabsorption can occur, resulting in B12 deficiency most commonly due to terminal ileum being classic inflammatory location. - Impaired fat absorption can result in ↑ calcium oxalate urolithiasis, as discussed in the HY Renal PDF.

Irritable bowel syndrome (IBS)
- Don't confuse with IBD. IBS is psychosomatic (i.e., psych-related) condition. - Classic vignette will be a woman 20s-40s with stress factors who usually has alternating diarrhea / constipation; can also present as bloating or cramping. - Key detail is that symptoms are relieved with bowel motions. - NBME assesses "smooth muscle hypersensitivity" as the answer for the mechanism for IBS. - Treatment for diarrhea-predominant IBS is loperamide, which is an opioid that causes constipation. USMLE will give vignette of IBS with diarrhea, and then the answer is just "mu-opioid receptor agonist."

Appendicitis
- Triad of RLQ pain, fever, and vomiting (Murphy's triad). - Pain starts at epigastrium (visceral pain) and migrates toward RLQ (parietal pain). USMLE will ask why there is movement of the pain → answer = "inflammation of the parietal peritoneum." - Pain at McBurney's point (1/3 of the way from the anterior superior iliac spine to the umbilicus). - (+) Rovsing sign (pain felt in RLQ upon palpation of LLQ). - "Inflammatory mass and fecalith in the cecum" is phrase used on 2CK NBME form to describe appendicitis. - Diagnosis is done via ultrasound first, and sometimes CT. However, I have not seen USMLE assess either of these, likely because their utility is debated. What I have seen is **laparoscopic removal in the stable patient.** - USMLE will give ethics/consent Q, where they say patient has operation performed for ovarian procedure but it is seen that she has an acutely inflamed appendix → answer = "remove due to necessity of medical emergency." - However, if the scenario is reversed and, while performing an appendectomy, the surgeon notices a suspicious lesion on, e.g., one of the ovaries, the answer is "do not biopsy." Consent must first be obtained for all non-emergencies.

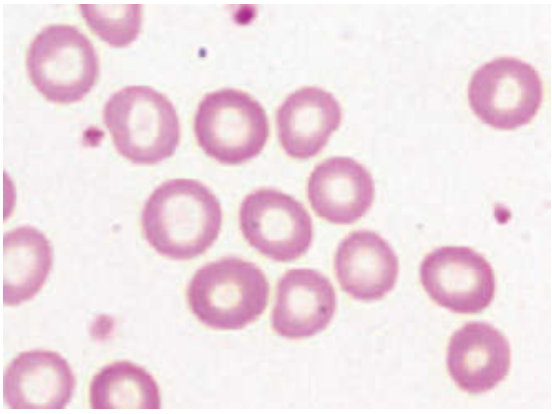
HY colorectal / proctocolonic diagnoses	
Anal fissure	- Young adult who has painful bowel movements +/- blood in the stool. - The key detail is **they refuse the rectal exam because the pain is so bad.** - For whatever reason, they can also say there's an associated skin tag. I've seen this more than once, where the student says, "What's with the skin tag?" No fucking idea. - If Step 1 forces you to choose a location for anal fissure, the answer is "posterior in the midline." Stupid, but it's asked. - Mechanism can be "increased anal sphincter tone." - Tx on 2CK Surg is **sitz bath.** - If the USMLE forces you to choose a med, topical nitrates or diltiazem is used.
Pilonidal cyst/abscess	- Answer on USMLE for cystic mass at the superior aspect of the gluteal cleft. - Contains hair; often caused by ingrown hairs. - Tx on 2CK Surg = incision and drainage.
Perianal abscess	- Answer on USMLE for painful, erythematous mass near anal verge.

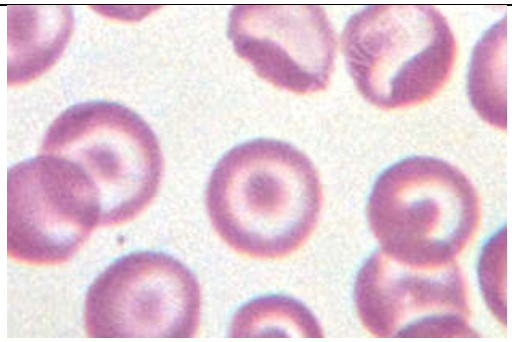
	- Increased risk in Crohn and diabetes. - Tx on 2CK Surg = incision and drainage.

Acetaminophen hepatotoxicity	
Acetaminophen	**- Acetaminophen overdose most important for USMLE. *N*-acetylcysteine is antidote.** - Breakdown of acetaminophen produces metabolite called NAPQI that depletes reduced glutathione. The latter normally mops up free radicals / prevents oxidation. - Patient will OD on acetaminophen and be asymptomatic for 24-28 hours, then will proceed to get ALT and AST in the tens of thousands + require liver transplant or death. - "*N*-acetylcysteine regenerates reduced glutathione." Memorize that. - Oxidized glutathione had disulfide bond (-S—S-); reduced glutathione has thiol (-SH). - Activated charcoal should apparently be administered to patients who present very acutely after massive ingestion, as this can ↓ acetaminophen absorption. But I've never seen this on NBME. It tends to be more Qbank that's gotchya-style this way. I've only ever seen *N*-acetylcysteine assessed across NBMEs.

HY Peds GI infections	
- Point isn't to be an endless list. I'm staying HY here for Peds. - If you want broader list of GI infections, go to my HY Gastro PDF.	
EHEC/Shigella	- Hemolytic uremic syndrome. - Toxin causes endothelial damage in renal afferent arterioles that leads to consumption of platelets + shearing of RBCs flying past the platelet plugs. - Presents as triad of 1) thrombocytopenia, 2) ↓ Hb with schistocytes, 3) renal issues (i.e., ↑ RFTs and/or hematuria). - 2CK Peds Q gives Q where "Shigellosis" is the answer and HUS is wrong in kid with bloody diarrhea, but it's easy if you simply see that the classic HUS triad isn't there. In other words, GI infection with blood in the stool isn't automatically HUS.
Yersinia enterocolitica	- Causes bloody diarrhea + **appendicitis-like (i.e., RLQ) pain** in Peds. - Arthritis can also occur, but this finding is more common in adults. - The RLQ pain is from mesenteric adenitis or terminal ileitis.
Staph aureus	- *S. aureus* pre-formed heat-stable toxin is acquired from two main sources on NBME: 1) various meats sitting under a heat lamp / out for long periods of time, e.g., at a buffet; 2) dairy products like creams, custards, potato salad. - Notably causes **vomiting 1-6 hours** after consumption. This is notable, as the symptoms occur rather quickly. Diarrhea, both watery and bloody, can occur, but is not mandatory. The main crux is the vomiting.
Rotavirus	- Most common cause of watery diarrhea in unvaccinated children < 5 years. - Vaccine normally given orally at 2, 4, and 6 months of age. - Double-stranded, segmented RNA (NBME asks it). - Wheel-shaped (also on NBME).
Norwalk virus	- Most common cause of watery diarrhea in adults and rotavirus-vaccinated children. - Cruise ships and business conferences are buzzy places to acquire (fecal-oral); basically any place with high density of people.
Enterobius vermicularis	- Ingested. Causes perianal itching / (+) tape test in children. (Enterobiasis)
Ascaris lumbricoides	- Ingested. Giant roundworm; causes intestinal obstruction. (Ascariasis)
Hookworms	- Acquired through soil/one's feet. "Hookworms" refers to *Ancylostoma duodenale* and *Necator americanus*. - Cause **microcytic anemia / ↓ hemoglobin** due to sucking blood in GI tract. - Peds form will say a kid was running around barefoot in rural Louisiana + has high eosinophils + microcytic anemia → Q asks how the causal organism was contracted → answer = "through the skin."

Peds Heme/Onc

<table>
<tr><th colspan="4">Iron deficiency anemia vs thalassemia</th></tr>
<tr><th>Variable</th><th>IDA</th><th>Thalassemia</th><th>HY points</th></tr>
<tr><td>Hb</td><td>↓</td><td>↓</td><td rowspan="6">- Thalassemia means abnormal synthesis of either the α or β globin chain. It presents two ways on USMLE:
1) As a microcytic anemia despite normal iron + ferritin; or
2) As a patient who has a microcytic anemia despite iron supplementation.
- It is extremely HY for Peds you know that a big risk factor for IDA in children age 2 or younger is consumption of >24 ounces of milk daily.
- The detail sounds strange / nitpicky but shows up a lot in Qs. For example, they'll say an 18-month old is eating regular table food + drinks whole milk daily + they ask what deficiency should be checked for → answer = iron; wrong answer is vitamin D; the latter is ↑ risk in children exclusively breastfed, usually <6 months of age.
- Celiac disease is also exceedingly HY cause of IDA in Peds, as mentioned earlier.</td></tr>
<tr><td>MCV</td><td>↓</td><td>↓</td></tr>
<tr><td>Serum iron</td><td>↓</td><td>Normal</td></tr>
<tr><td>Ferritin</td><td>↓</td><td>Normal</td></tr>
<tr><td>Hb electrophoresis</td><td>Normal</td><td>α: normal;
β: ↑ HbA2 + HbF
(See discussion at bottom of table)</td></tr>
<tr><td>What happens if we give iron?</td><td>Improvement</td><td>No improvement (HY)</td></tr>
<tr><td>RDW</td><td>↑ (HY)</td><td>↓/Normal (HY)</td><td>- Red cell distribution width.
- All you need to know is, basically always, ↑ RDW means IDA straight-up 9/10 Qs.
- The only Q where ↑ RDW is seen in thalassemia is on CMS FM form 4, where IDA is wrong, but they give ferritin within the normal range, so IDA isn't possible. But the ↑ RDW is essentially an erratum, as this is rare as fuck.
- Therefore we can essentially say that ↑ RDW is sensitive but not specific for IDA, where if RDW isn't ↑, we can rule-out IDA, but be careful about automatically saying "must be IDA" if RDW is ↑.</td></tr>
<tr><td>Blood smear</td><td>Pale RBCs</td><td>Target cells</td><td>- IDA causes pale RBCs (i.e., ↑ central pallor). Not dramatic.
 
- Target cells are highly buzzy and pass-level for thalassemia. Probably 14/15 Qs on USMLE mentioning target cells are thalassemia (α or β).</td></tr>
</table>

			 - There is one NBME Q with target cells in a splenectomy patient, but they're a background finding and don't facilitate answering the Q in any way.
Tx	Iron	Transfusions if severe	- Oral iron is ideal over parenteral (IV) or intramuscular iron always. I mention this because answers such as "parenteral iron" are wrong/distractors basically always. I'll say to a student, "You want to give this patient IV iron?" - Transfusional siderosis (secondary hemochromatosis) is sophisticated way of saying iron overload due to repeated blood transfusions, e.g., in β-thalassemia major. Each RBC transfusion contains iron. Use chelators (e.g., deferoxamine) to treat.

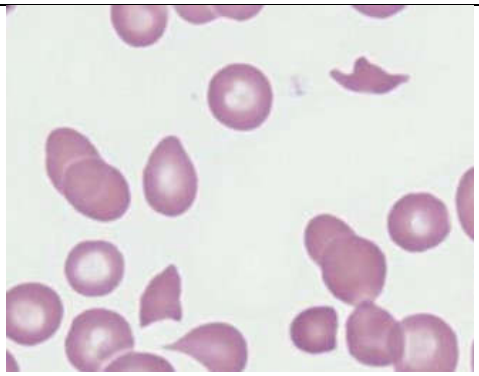
- Thalassemia is impaired synthesis of either the α- or β-globin chain in hemoglobin.
- α-thalassemia can have 1-4 alleles mutated; β has 2. Don't worry about obscure intermediate types, etc.
- 1 α-thalassemia mutation will be asymptomatic adult, where they have an incidental microcytic anemia picked up on antenatal screening (Obgyn forms love thalassemia). 2 α mutations will be adults with fatigue akin to IDA. 3 α mutations (HbH disease; β4 tetramer) will present as a sick kid; 4 α mutations (Hb Barts; γ4 tetramer) will be fatal *in utero*.
- 1 β-thalassemia mutation (β-thalassemia minor) will be an adult who is asymptomatic or has mild fatigue (similar to 1 or 2 α mutations).
- β-thalassemia major is 2 mutations and will present as a sick kid (similar to 3 α-thalassemia mutations).
- It is exceedingly HY you know that β-thalassemia has ↑ HbA2 (α2/δ2) and HbF (α2/γ2); α-thalassemia (1 or 2 mutations) has a normal hemoglobin electrophoresis.
- Sick children with thalassemia are known to get "chipmunk facies/skull" and hepatosplenomegaly occur due to ↑↑ extramedullary hematopoiesis.



Anemia of chronic disease (AoCD)
- The answer on Peds for anemia (↓ Hb) in child who has **renal failure,** an autoimmune disease (i.e., JRA, IBD, SLE), or malignancy. - Child will have low iron but **normal ferritin** (or sometimes slightly high). Ferritin is the most sensitive marker of iron stores, which means these patients are not iron deficient. - There's two main reasons iron is low in AoCD: 1) RBC production is decreased because cytokines (namely IL-6 and TNF-α) block iron release from storage sites, and 2) hepcidin, a molecule produced by the liver, decreases gut absorption of iron and promotes its storage in cells instead. - It's to my observation that **renal failure** is the highest yield cause of AoCD on USMLE. This is due to "cytokine-mediated EPO deficiency." This is an answer on an NBME form. EPO is the treatment for AoCD only if renal failure is the etiology. Otherwise the Tx is merely addressing the underlying condition (e.g., the RA), and EPO is a wrong answer. - For conditions like JRA, the Tx is "no specific measures indicated." - There is an AoCD NBME Q for RA where they ask how to treat and answer is "no specific measures indicated." Student says, "Wait, I thought we were supposed to address the underlying condition, so I was confused by this answer choice." I agree. So don't take it up with me. Take it up with NBME. - One of the highest yield points I can communicate is that even though classic MCV is normal in AoCD, **it will be ↓ in ~50% of NBME Qs.** I've seen students say, "Oh it can't be AoCD cuz MCV is low." No. USMLE loves giving low MCV in AoCD, especially on 2CK forms. I've seen MCVs of 72 and 75 (NR 80-100) in JRA Qs.

HY Core Heme DDx for Peds	
ITP	- Idiopathic/immune thrombocytopenic purpura. **- Antibodies against glycoproteins IIb/IIIa on platelets.** **- Causes ↓ low platelets; ↑ bleeding time; no change PT and PTT.** - Bleeding time is high because platelets are fucked up. PT and PTT and normal because these refer to clotting factors, which have nothing to do with ITP. - Normal platelet count is 150-450,000/μL. Platelets in ITP Qs are usually <100,000. - Precipitated by viral infection, where antibodies against viral proteins cross-react with platelets (disrupts platelet aggregation); type II hypersensitivity. - Viral infection **will be asymptomatic in ~50% of vignettes.** In other words, the Q need not mention the viral infection, so don't get confused if you don't see it. - Tx = steroids, then IVIG, then splenectomy, in that order. - Qbank has suggested that no Tx is necessary in some mild ITP Peds cases, or that in severe cases, IVIG is given before steroids. Absolute nonsense. On USMLE, they want steroids first for any case out there. - If splenectomy is performed and platelets return to normal range, but then 6 months later they fall again, the answer is "accessory spleen." Sounds weird, but a small % of people have a second spleen the size of a pea that can grow in size if the main spleen is removed.
vWD	- von Willebrand disease; **autosomal dominant.** - von Willebrand factor (vWF) normally bridges glycoprotein Ib on platelets to vascular endothelium / underlying collagen (platelet adhesion). **- ↑ bleeding time; ↑ PTT (only half the time); no change PT; normal platelets.** - ↓ vWF means ↓ platelet function (and hence ↑ BT), but platelet count is unchanged. - vWF also has secondary role of stabilizing factor VIII in plasma; since we have ↓ vWF, the intrinsic pathway gets fucked up, so we have ↑ PTT. But bear in mind, this effect on factor VIII *is only a mere secondary* role, so PTT isn't always ↑. It's to my observation that only ~50% of vWD vignettes give ↑ PTT. PT is normal because the extrinsic pathway is fine. - 8/10 Qs presents as a teenager or young adult with a mix of one platelet problem and one clotting factor problem. - Platelet problem = mild, cutaneous findings (i.e., petechiae, bruising) + epistaxis.

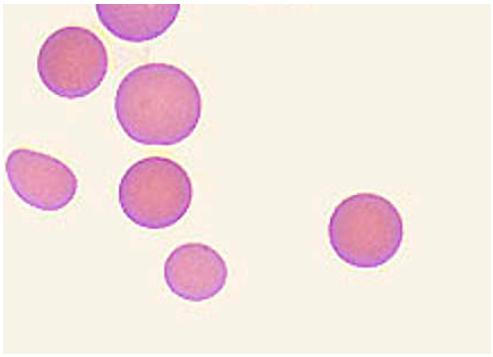
	- Clotting factor problem = excessive bleeding with tooth extraction + heavy menses. (Hemarthroses are also clotting factor but are seen in hemophilia, not vWD, as I explain below). - For example, the Q can give 17-year-old girl who has nosebleeds + heavy periods; or 17-year-old girl who has bruising + Hx of excessive bleeding with tooth extraction (i.e., we have one platelet problem + one clotting factor problem). This is 8/10 Qs I'd say. - 1/10 Qs might give only isolated platelet or clotting factor problem (i.e., only say the patient has Hx of excessive bleeding with tooth extraction), but they'll also say *one of the parents* has Hx of nosebleeds → so you say, "Oh ok. vWD is AD, so that's your combo of platelet problem + clotting factor problem linked via autosomal dominant inheritance." - 1/10 Qs will say the patient has a cut on the finger that takes longer to heal than normal + has heavy menses + normal platelets + has normal PT and PTT + they don't mention BT. The cut on the finger is their way of saying BT is elevated. Despite only an isolated ↑ in BT here, since platelets are normal, we know it's not ITP and is instead vWD. - Ristocetin cofactor assay is abnormal. All you need to know is that this is some test that is abnormal if platelet adhesion is fucked up. Tx = desmopressin (DDAVP), which can in theory help boost functional vWD levels.
Hemophilia A+B	- Deficiency of factors VIII and IX, respectively. - **Both X-linked recessive.** - Isolated ↑ PTT; BT and PT are normal. - Factors VIII and IX are in the intrinsic pathway, so PTT is ↑. Extrinsic pathway is unaffected so PT is normal. Platelets have nothing to do with this condition, which is why bleeding time is normal. - Will present two ways on USMLE: 1) School-age boy with hemarthrosis who has isolated ↑ PTT. They can also say he has a maternal uncle who died years ago from minor head trauma. 2) Neonate who has excessive bleeding with circumcision. - Hemophilia A is treated with DDAVP and factor VIII replacement. - Hemophilia B is treated with factor IX replacement. DDAVP doesn't work apparently. - If the Q tells you a patient with hemophilia A has persistently elevated PTT despite factor VIII supplementation, the answer is "check for antibodies against factor VIII." Sometimes patients can develop resistance against exogenous factor replacements.
Vitamin K deficiency	- **Causes ↑ PT, ↑ PTT, no change bleeding time.** - Vitamin K is cofactor for gamma-glutamyl carboxylase, which is an enzyme that activates clotting factors II, VII, IX, and X, as well as anti-clotting proteins C and S. - Bleeding time is normal because platelets aren't affected. - Classic Peds vignette is a neonate (often born at home, but not mandatory) who has bleeding from umbilical stump +/- retinal hemorrhages. The latter can sound like shaken-baby syndrome, but bleeding from the umbilical stump is buzzy for vitamin K deficiency.
DIC	- Disseminated intravascular coagulation. - Runaway effect of clotting factor/platelet consumption, precipitated by multifarious etiologies such as trauma, sepsis, amniotic fluid embolism, and treatment of AML. **- ↑ BT, ↑ PT, ↑ PTT, ↑ D-dimer, ↑ plasmin activity.** **- ↓ fibrinogen, ↓ platelets, ↓ clotting factors.** - Fibrinogen decreased because it is converted to fibrin. - Plasmin breaks down fibrin, so more fibrin means more plasmin is upregulated in an attempt to dissolve it. - D-dimer = fibrin degradation products; since more fibrin is being broken down, fibrin degradation products (D-dimer) ↑. - "Bleeding from catheter/IV sites" is 9 times out of 10 synonymous with DIC. However on one 2CK Surg Q, dilutional thrombocytopenia secondary to ↑ blood transfusions presents the same. - Blood smear shows **schistocytes.**

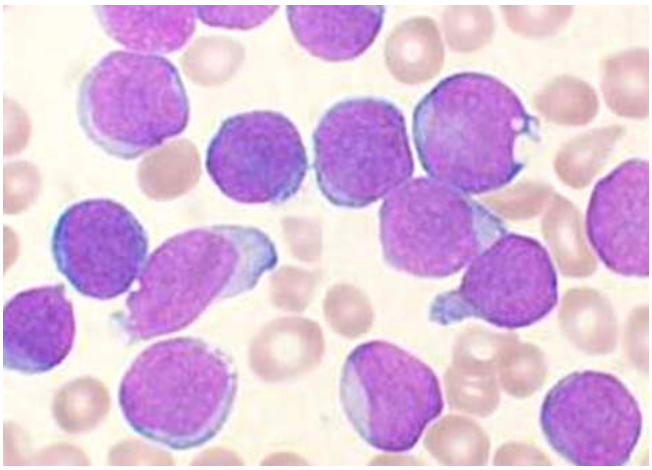
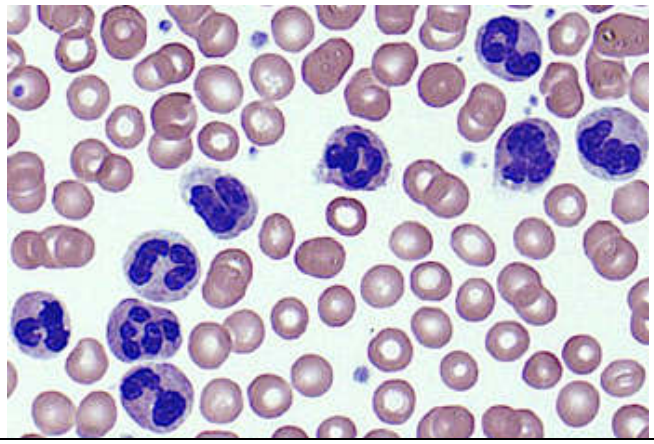
	
HUS	- Hemolytic uremic syndrome. - Presents as triad of: 1) thrombocytopenia; 2) hemolytic anemia with **schistocytes;** and 3) renal insufficiency with or without hematuria. - The combination of thrombocytopenia + schistocytosis = microangiopathic hemolytic anemia (MAHA). - Mechanism is: E. coli (EHEC O157:H7) and Shigella both secrete toxins (Shiga-like toxin and Shiga toxin, respectively) that cause inflammation and damage of renal microvasculature → platelets adhere to plug damaged endothelium → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells).
TTP	- Thrombotic thrombocytopenic purpura. - Presents as HUS triad + fever and neurologic signs. In other words: - Presents as pentad of 1) thrombocytopenia; 2) **schistocytes;** 3) renal insufficiency; 4) fever; 5) neurologic signs. - TTP is caused by antibodies against, or a mutation in, a protein called ADAMTS13, which is a metalloproteinase that breaks down vWF multimers → unable to break down platelet aggregates that plug renal microvasculature → platelet aggregations protrude into vascular lumen causing shearing of RBCs flying past → fragmentation of RBCs (schistocytes, aka helmet cells). - Can present as a stroke-like presentation in young woman, where they just want "schistocytes" as the answer (Step 1 NBME). - 2CK NBME has "plasmapheresis" as Tx.
Sickle cell	- **Autosomal recessive;** glutamic acid → valine missense mutation on the beta-chain. - Carrier status (one mutation) is referred to as sickle cell trait, which is less severe than sickle cell anemia (two mutations). - Sickle cell crises can present as **abdominal or chest pain,** as well as dactylitis (inflammation of the fingers). - The abdominal pain, in particular, as the initial presentation for sickle crisis has become HY on NBME exams. A new 2CK form gives RLQ pain + low Hb in child from Libya, where "blood smear" is the answer for what will confirm the diagnosis. The Q is tricky because it almost sounds like appendicitis. - Sickling notably occurs with dehydration + increased acidity. 

	- Diagnosed with hemoglobin electrophoresis to detect HbS. - Offline NBME Q asks which of the following best describes the molecular basis for sickling in a patient → answer = "gain of stabilizing hydrophobic interactions in the deoxygenated form of hemoglobin S." This is because valine is more hydrophobic than glutamic acid, so we have hydrophobic interactions enabling sickling. - Another NBME has "beta chain slips into a complimentary hydrophobic pocket on the alpha chain" as answer for sickling. - **Hydroxyurea ↑ HbF** and can ↓ recurrence of sickle crises. Sounds weird, but the mere presence of more HbF simply means there's fractionally less HbS floating around.

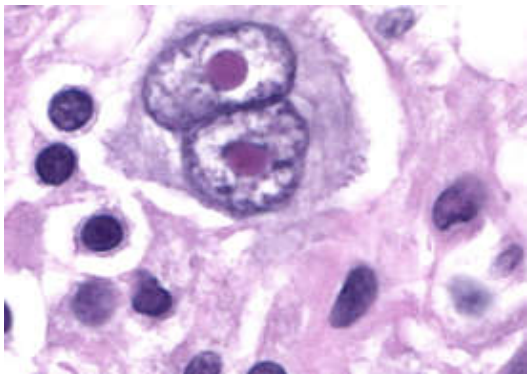
Other anemias	
Aplastic anemia	- Means all cell lines (i.e., RBCs, WBCs, platelets) are ↓. - Can be due to Parvo B19, where it infects not just myeloid progenitors, causing neutropenia, but also erythroid progenitors and megakaryocytes, causing ↓ RBCs and platelets, respectively. - Can also be due to other viruses, such as HepA (weird, but on CMS Peds form 5). - Can be chemo- and/or radiotherapy-induced. - Fanconi anemia is rare, AR aplastic anemia that, for whatever reason, also presents with hypo-/aplastic thumbs/radii.
Pure RBC aplasia	- Means only RBCs are ↓. - Can be caused by Parvo B19. - Thymoma can cause pure-RBC aplasia (as well as myasthenia gravis). - Diamond-Blackfan anemia is rare pure-RBC aplasia that presents with triphalangeal thumbs.

HY Neutropenia conditions	
- A HY general point for USMLE is that neutropenia can present as **mouth ulcers, fever, and sore throat.** - Questions will not usually come out and say, "Yes, the kid has neutropenia." The situation might be, e.g., a kid who's undergoing chemotherapy for AML who now has mouth ulcers and/or sore throat.	
Neutropenic fever (Febrile neutropenia)	- Neutropenia + fever, LOL! - Can occur in the setting of child who has chemo- or radiotherapy-induced pancytopenia (i.e., RBCs, WBCs, and platelets are all ↓), or viral-induced neutropenia (i.e., Parvovirus B19), where you see that the patient has fever and neutrophils are low (+/- RBCs and platelets). - Medical emergency (i.e., the patient has infection but no ability to fight it). USMLE wants "IV broad-spectrum antibiotics" as the next best step. - After Abx, the USMLE wants granulocyte colony-stimulating factor (i.e., G-CSF, or filgrastim) as the answer. - I've never seen granulocyte transfusion as a correct answer on USMLE.
Viral-induced neutropenia	- Parvovirus B19 can infect myeloid precursors and cause neutropenia – i.e., viral-induced neutropenia. - Other viruses can cause formation of antibodies against neutrophils, similar to how they can cause antibodies against platelets in ITP.
Cyclic neutropenia	- Gene mutation causing episodes of fever + mouth ulcers in a kid age 1-3. These episodes typically occur every 21 days, but the Q might not specify this number and will just say, "There have been several episodes like this before."

<table>
<tr><th colspan="2">Spherocytosis</th></tr>
<tr><td>Hereditary</td><td>- Autosomal dominant (one of the highest yield inheritance patterns on USMLE).
- Heterozygous mutation in ankyrin, spectrin, or band proteins, causing an RBC cell membrane/cytoskeletal defect. This causes the RBC to deform away from the normal biconcave disc and into a spherical shape.
- The USMLE might give you hereditary spherocytosis and then the answer is just “cell membrane,” or “cytoskeleton,” or “cytoskeletal defect.”
- Can be described as RBCs lacking central pallor. If they show you a smear, they look full and red.
 
- Causes a normochromic, normocytic anemia (asked on an NBME directly).
- Can cause pigment stone cholelithiasis due to the spleen identifying the spherocytes as abnormal and increasing RBC turnover.
- RBC mean corpuscular hemoglobin concentration (MCHC) can be ↑. This is not specific to spherocytosis but is nevertheless a rare lab parameter to see in Qs. It’s to my observation that if they mention it, 4/5 times the Dx is spherocytosis.
- Diagnose via osmotic fragility test and eosin-5-maleimide.
- Coombs test is negative (meaning we don’t have antibodies against RBCs). This is important in terms of contrasting with spherocytosis due to hemolysis (as I discuss below).
- Treatment is splenectomy (to ↓ RBC turnover). The answer can appear on USMLE as “splenectomy and cholecystectomy” together.
- USMLE can mention a child who has a low Hb + one of the parents had splenectomy +/- cholecystectomy performed in the past for “an anemia” → answer = hereditary spherocytosis. I also point out to the student the AD inheritance pattern based on one of the parents having the condition as well.
- Q can also just ask straight-up which heme condition is often treated with splenectomy, and the answer is hereditary spherocytosis; students sometimes erroneously choose sickle cell. Remember, sickle cells causes autosplenectomy due to repeated microinfarcts within splenic microvasculature; it is not treated with splenectomy.</td></tr>
<tr><td>Hemolytic anemia</td><td>- Spherocytosis is not always due to hereditary spherocytosis and can sometimes be seen with drug- or infection-induced hemolysis.
- In this case, the Coombs test is (+), meaning we have antibodies against RBCs. In contrast, with hereditary spherocytosis, the mechanism has nothing to do with antibodies, so Coombs test is (-).
- Antibodies binding to RBCs can trigger complement-mediated damage, which can lead to deformation of the RBC membrane and cytoskeleton.
- USMLE will give kid with recent viral infection who has low Hb and spherocytes on a smear + positive Coombs; answer = hemolytic anemia, not hereditary spherocytosis (i.e., answer = “no specific mutation”).</td></tr>
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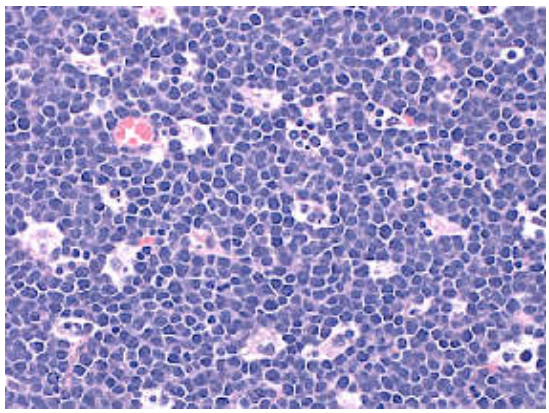
ALL vs pertussis vs leukemoid reaction	
ALL	- Acute lymphoblastic leukemia. - The answer on USMLE for a kid with leukemia (i.e., pre-adolescent). - Q will give you kid who's, e.g., 6-years-old, and has leukocyte count of 60,000/μL, where it's 90% lymphocytes. - **An important DDx of ALL is pertussis,** that for whatever reason, can cause an ALL-like picture where WBCs can be 30-40,000+, where it's 90% lymphocytes. This is called reactive lymphocytosis. The distinction with ALL is that, in pertussis, they'll say there's a cough and/or post-tussive emesis (vomiting after cough) and/or hypoglycemia (HY finding in pertussis). - ALL can be CD10 and TdT positive. Dumb, but I've received feedback of them showing up. - ↑ Risk with Down syndrome. The Q can give you standard vignette of Downs and then ask for what child is at greatest risk of developing → answer = "excess lymphoblasts."  ALL; lymphoblasts appear smooth and relatively uniform.
Leukemoid reaction	- Exaggerated immune response to infection in which WBCs go ↑↑ and can hence resemble leukemia. - As I mentioned before, "-oid" means "looks like but ain't," so the patient's lab can look like leukemia (because the WBCs are so high), but it's not leukemia. - Normal leukocyte count is 4-11,000/μL. Typical infections might cause WBCs to go ↑ to the teens or 20,000s/μL. But when WBC levels creep into the 30s, this becomes increasingly rare, even for patients who are septic, so we want to start thinking about leukemia. This is why if the patient doesn't turn out to have leukemia we call it leukemoid reaction. - This will be the answer on USMLE for patient who has **infection + WBCs >30,000/μL, PLUS** they give you a blood smear that shows neutrophils or they say leukocyte ALP is elevated (in CML it's low). 

	- This is the smear the USMLE will show you if they want leukemoid reaction. This is showing you **neutrophilia** (i.e., just lots of neutrophils). They can also write the answer as **reactive granulocytosis,** or as "increased release of leukocytes from bone marrow post-mitotic reserve pool." - A typical leukemoid reaction Q might give you a UTI + WBCs 32,000/μL + the above smear. Not hard.

Lymphomas for Peds	
- A lymphoma is a cancer of the lymphatic system. - Usually starts in lymph nodes, but can also be present in other areas of the lymphatic system, such as the spleen and bone marrow. - Almost always B-cell in origin. - Lymphomas are categorized as Hodgkin or non-Hodgkin (NHL). - Epstein-Barr virus, HIV, general immunosuppression (i.e., chronic corticosteroids or immunosuppressant drugs), and autoimmune disease (e.g., SLE, Hashimoto) increase the risk for lymphomas.	
Hodgkin	- Characterized by Reed-Sternberg cells, which are CD15/30 (+) B cells with a characteristic "owl eye" appearance.  - 100% of questions give a painless lateral neck mass or facial swelling (lymphadenopathy). - The USMLE Q will then give at least one additional finding of: 1) hepatomegaly; 2) mediastinal mass (not thymoma; this is mediastinal lymphadenopathy); 3) Virchow node (Troisier sign of malignancy; palpable left supraclavicular lymph node); this is not limited to gastric cancers; I've seen an NBME Q where they give this latter finding for Hodgkin. - B symptoms (i.e., fever, night sweats, weight loss) are common but not mandatory in vignettes. - Question can give normal leukocyte count and differential. An NBME Q gives WBCs of 10,000/μL (NR 4-11,000) and lymphocytes of 33% (NR 25-33). This is because lymphomas are cancers of lymphatic tissue. They're not leukemias, which are cancerous WBCs floating around the blood. - Nodular-sclerosing type tends to favor women. - ↑ Normal B cells relative to abnormal RS cells = better prognosis. For example, leukocyte-rich Hodgkin has ↑ normal B cells in comparison to RS cells and has better prognosis; in contrast, leukocyte-deplete Hodgkin has comparatively more RS cells and worse prognosis.
Burkitt	- t(8;14) translocation of c-myc gene; important Peds lymphoma type. - Presents classically as jaw lesion in African boy, or as abdominal lesion.



- Histo is buzzy “starry sky” appearance, which is a background of basophilic (purple) B cells with translucent macrophages.



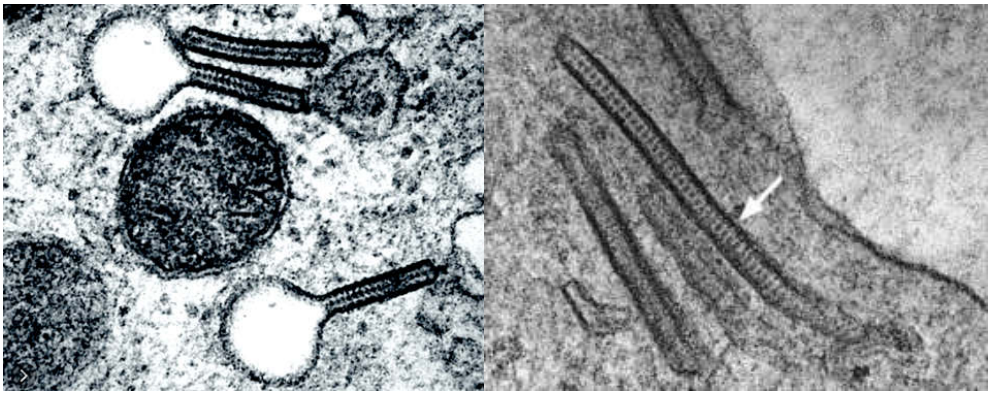
Infectious mononucleosis

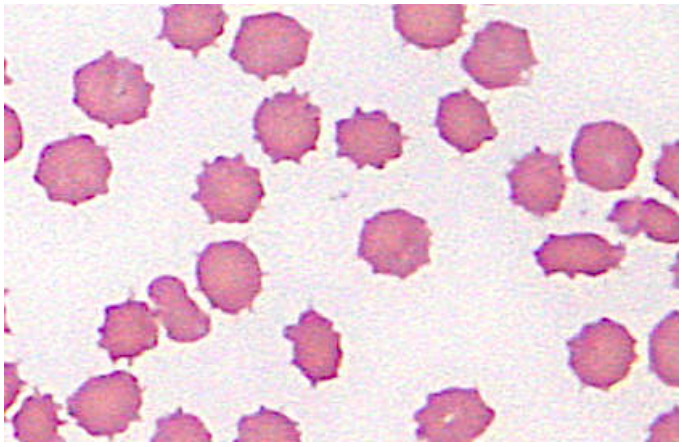
- Usually caused by EBV, but can also be CMV.
- The virus invades B cells, which then stimulates the immune system to produce CD8+ T cells attacking the viral-infected B cells. In mono, these CD8+ T cells are called “atypical lymphocytes,” or reactive T cells.
- Primary infection presents usually in teenager (or young adult) with fever >38 C, lymphadenopathy, tonsillar exudates, and lack of cough, making the presentation appear bacterial (I talked about CENTOR earlier). As a result, it is often misdiagnosed as *Strep* pharyngitis.
- If amoxicillin or penicillin is given to treat EBV, this can cause a rash. This is not to be confused with a rash caused by allergy to beta-lactams. If pre-adolescent receives beta-lactam and gets a rash, that is likely beta-lactam allergy. If a patient adolescent or older gets a rash, we do a heterophile antibody (Monospot) test as next best step in diagnosis, as EBV mono is more likely.
- Following the primary infection, mono can present as recurrent episodes of extreme fatigue that arise at interval-separations of months to years.

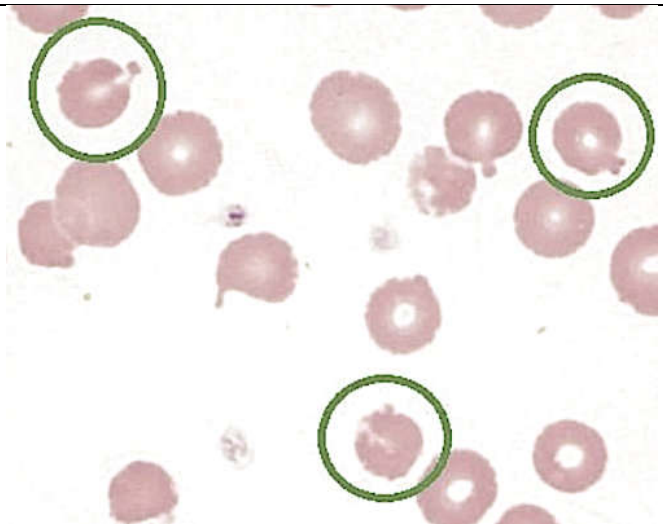
Langerhans cell histiocytosis (LCH)

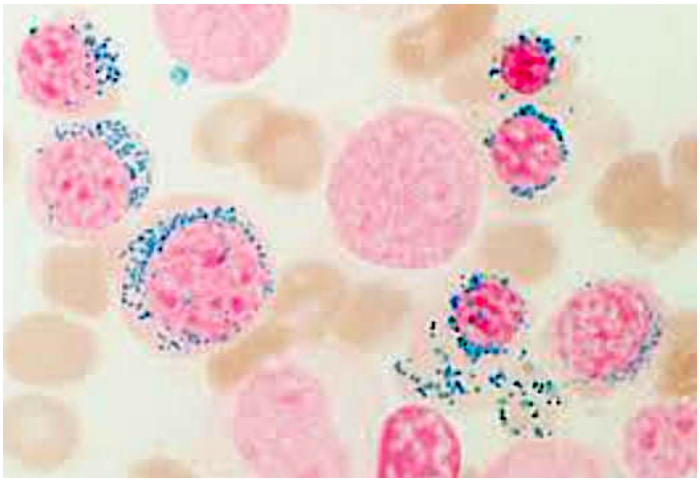
- Overproduction of Langerhans cells, which are the dendritic cell of the skin. A dendritic cell is a strict antigen-presenting cell (APC), which phagocytosis antigen, migrates to lymph nodes and the spleen, and presents it to CD4+ T cells. Macrophages and B-cells are also APCs, but they are not dendritic cells because they have additional functions – i.e., macrophages destroy antigens; B cells mature into plasma cells and produce antibodies.
- The Langerhans cells that over-proliferate do so notably in bone, causing **lytic lesions** (e.g., of the skull).

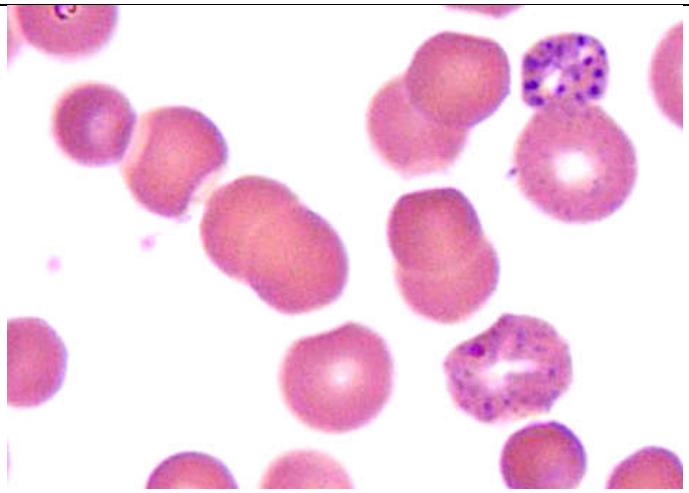
- Birbeck granules are tennis racquet- (left image below) or rod-shaped (right image below) inclusions within the Langerhans cells, visualized on electron microscopy. Students remember the granules are tennis racquet-shaped, but don't know the rod-shape. There is an LCH Q on NBME where they give the rod-shaped Birbeck granules, and students are confused because they say, "But wait, they're not tennis racquet-shaped though."



Glucose-6-phosphate dehydrogenase (G6PD) deficiency	
G6PD deficiency	- Glucose-6-phosphate dehydrogenase deficiency; **X-linked recessive.** - Most common cause of hemolysis due to an enzyme deficiency. - Presents as hemolysis and jaundice in boy who recently received a drug or had a viral infection. - HY agents that can precipitate oxidative damage of RBCs in G6PD are primaquine, dapsone, and sulfa drugs, and fava beans. - Vignette might be 12-year-old boy with scleral icterus following a viral infection. Labs might show increased unconjugated hyperbilirubinemia and LDH (due to hemolysis). RBCs are packed with LDH. High LDH can be the USMLE's way of saying there is hemolysis. - G6PD is an enzyme necessary to produce NADPH, which is a reducing agent that protects RBC membranes from oxidative damage. - Bite cells and Heinz bodies are buzzy and HY. Bite cells are RBCs with bite-like notches caused by oxidative damage. Heinz bodies are clumps of oxidized/denatured hemoglobin.  Bite cells

	 Heinz bodies
Pyruvate kinase	- Second most common cause of hemolysis due to an enzyme deficiency. - Leads to ↓ ATP production → ↓ activity of RBC Na/K-ATPase pumps → ↓ Na^+ pumped out of RBC → water stays with Na^+ → cellular swelling + lysis.

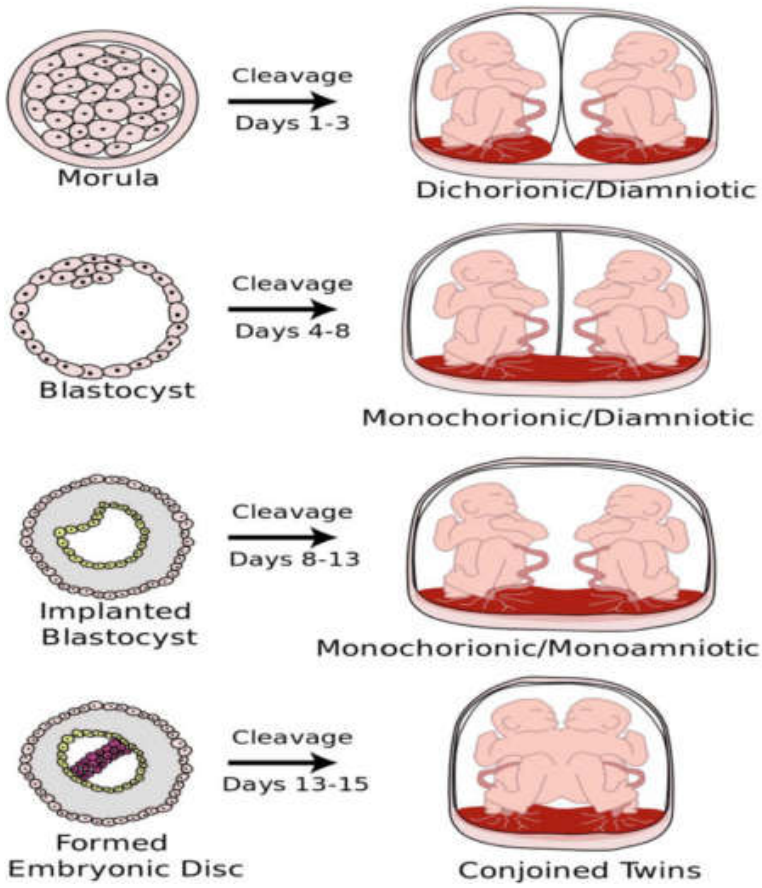
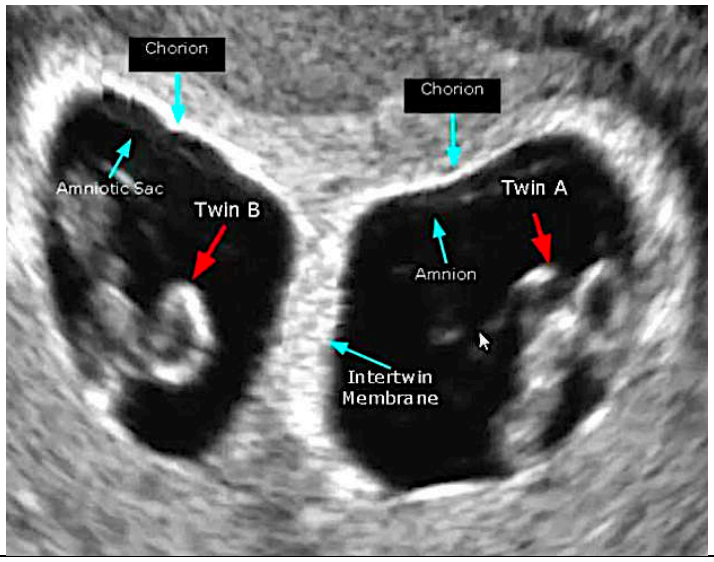
Heme synthesis disorders	
Sideroblastic anemia	- Impaired first step of heme synthesis due to deficiency or impairment of δ-ALA synthase. - Can be X-linked recessive and present in Peds (in adults, can be alcoholism). - Failure to incorporate iron into heme. This leads to "ringed sideroblasts," which are erythroblasts (RBC precursors) where iron accumulates inside the mitochondria, where the latter form a ring around the nucleus. - The iron-laden macrophages stain blue with Prussian blue stain, making this a very buzzy and pass-level image for USMLE. 
Lead poisoning	- Results in impairment of heme synthesis by inhibiting both δ-ALA dehydratase and ferrochelatase; leads to buildup of δ-ALA. - The stereotypical vignette of a 2-year-old eating paint chips in an old house leading to mental deterioration is too easy and maybe only ~1/3 of lead questions. - You need to know lead poisoning can also cause a **microcytic anemia.**

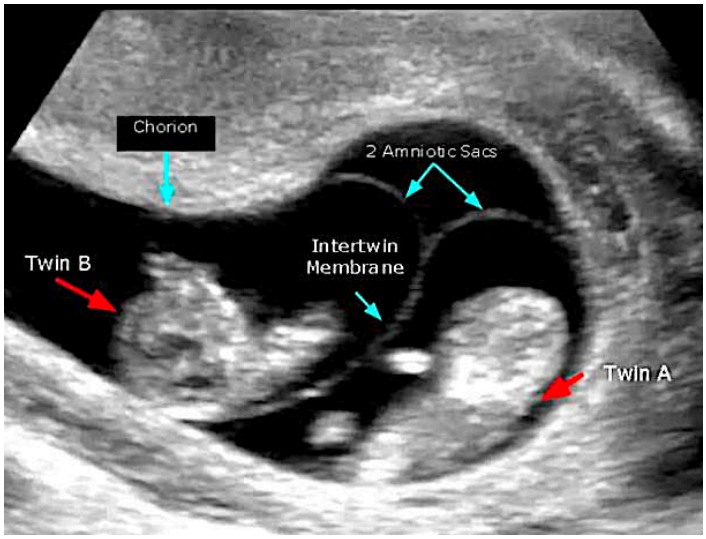
	 - Can cause basophilic stippling of RBCs (rRNA precipitates). - Tx is succimer, calcium EDTA, or dimercaprol.
AIP	- Acute intermittent porphyria. - Caused by deficiency of porphobilinogen deaminase; causes buildup of porphobilinogen and δ-ALA. - Answer on USMLE for the heme synthesis disorder where patient has red/pink/port wine colored urine and abdominal pain. - Patients can sometimes have miscellaneous neurologic findings, but not mandatory. - Treatment is hematin and glucose.
PCT	- Porphyria cutanea tarda. - Caused by deficiency of uroporphyrinogen decarboxylase, leading to increased urinary uroporphyrins. - Answer on USMLE for the heme synthesis disorder where patient has photosensitivity / blistering. - Urine can appear “tea-colored” / dark. Don’t confuse this with the red/pink/port wine colored urine in AIP.

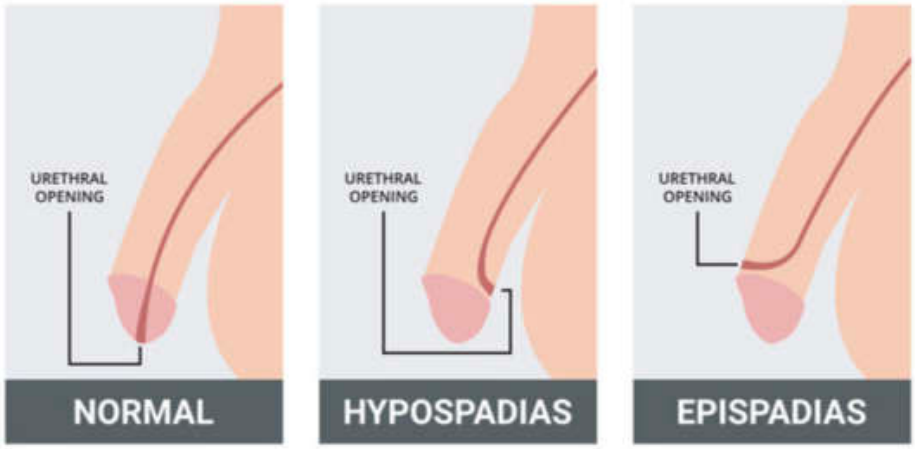
Hemolytic disease of newborn types	
Rhesus (Rh) type	- ~85% of people have Rh antigen on their RBCs – i.e., Rh(+). Women who are in the ~15% that are Rh(-) are at risk of developing IgG antibodies against Rh antigen if the fetus is Rh(+) and the circulations mix inadvertently. - When the mother develops antibodies against Rh (i.e., has become sensitized to it), we call this “Rh isoimmunization.” NBME will throw this word around a bit, so you need to know it means “she’s developed IgG antibodies against Rh.” - Mechanism: Rh(-) mom in first pregnancy is exposed to fetal blood that is Rh(+); mom makes Abs against Rh; subsequent pregnancy results in IgG against Rh crossing placenta and targeting fetal Rh(+) RBCs, leading to hemolysis in the fetus. - Rh status is checked at the first antenatal screening at 8-10 weeks. If the mother is Rh(-), then give RhoGAM at 28 weeks as well as at parturition. It must also be given if there are any interventions (e.g., amniocentesis), or if there’s complications like spontaneous abortion or abruptio placentae. - If 2nd pregnancy onward, if Rh (-) woman is found to have titers against Rh, do not give RhoGAM during the pregnancy, since it’s too late. - Kleihauer-Betke (KB) test is done in setting of hemorrhage or suspected mixing of maternal and fetal circulations in Rh(-) women. The mother’s blood is drawn, and the test works by exploiting the resistance of HbF to acid, allowing fetal RBCs to be distinguished from maternal cells on a blood smear.

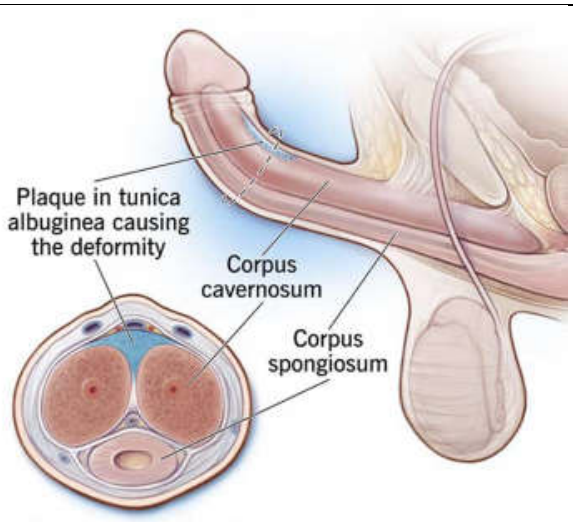
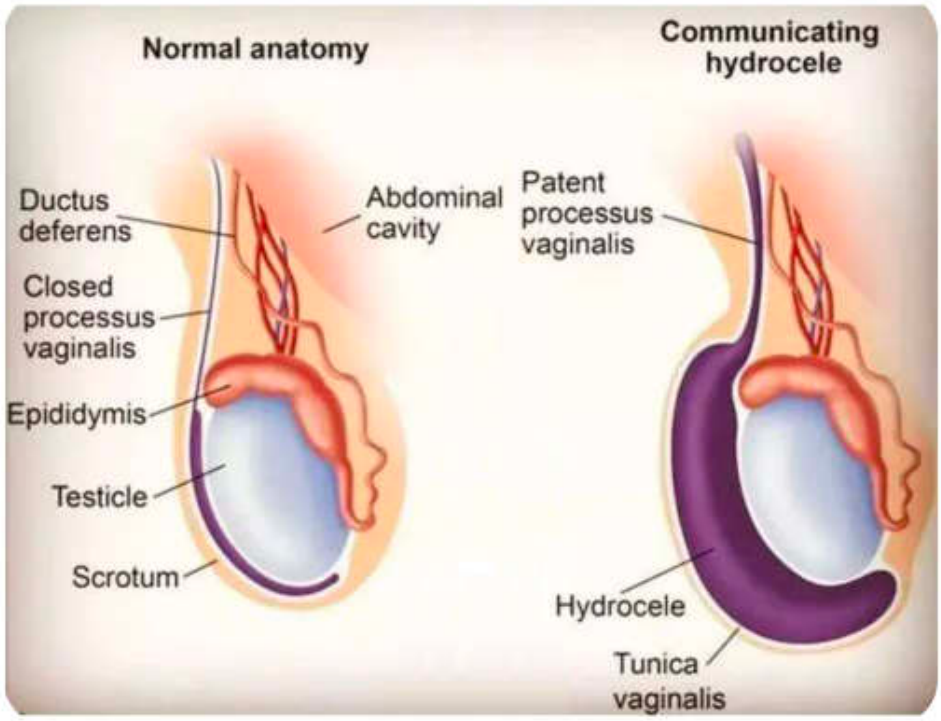
	Depending on the fraction of fetal RBCs present, this determines the dose of RhoGAM necessary. - Another 2CK Q gives ↑ **fetal HR** in 3rd trimester in woman who's in 2nd pregnancy who has (+) "anti-D titers." They ask reason why fetal HR is ↑ answer = "Rh isoimmuniziation" → destruction of fetal RBCs → ↓ oxygen delivery within fetal circulation → ↑ HR to compensate.
ABO type	- Most anti-A and -B Abs people have to opposing blood types are IgM, but people with O blood can have fractionally greater IgG; if mother with O blood has higher % of anti-A and -B Abs that are IgG, fetus can be symptomatic. - USMLE will give you O+ mom usually in first pregnancy and A or B fetus (can occur in Rh- women in second pregnancies, but the USMLE wants to assess you specifically know the ABO type of HDN, so they'll give O+ mom in first pregnancy).
Kell, Duffy, Kidd type	- Minor antigens beyond Rh (i.e., Kell, Duffy, Kidd) can sometimes cause HDN. - The process for isoimmunization is same as Rh. - What will happen is, an A+/B+/AB+ mother will have a second pregnancy where fetal hemolysis is occurring, but based on her blood type, you know it's not ABO or Rh type HDN. - The Q will say her blood titer for, e.g., Kell is high. The next best step is "check Kell status of the father." - In other words, since the mother has titers against Kell, you know she's Kell negative. So if the father's RBCs are Kell(+), the fetus is likely Kell(+), and that would explain the HDN.

Peds Repro

<table>
<tr><th colspan="2">Twinning</th></tr>
<tr><td colspan="2"> 
</td></tr>
<tr><td>Dichorionic / diamniotic</td><td>- Refers to dizygotic (fraternal) twins, where there are two zygotes to start.
- Refers to monozygotic (identical) twins, where the conceptus splits ~1-3 days post-fertilization.
- # of chorions = # of placentas, so we have two placentas in Di-Di twins.
- The NBME Q can say there is a “thick dividing membrane” between the fetuses.
 
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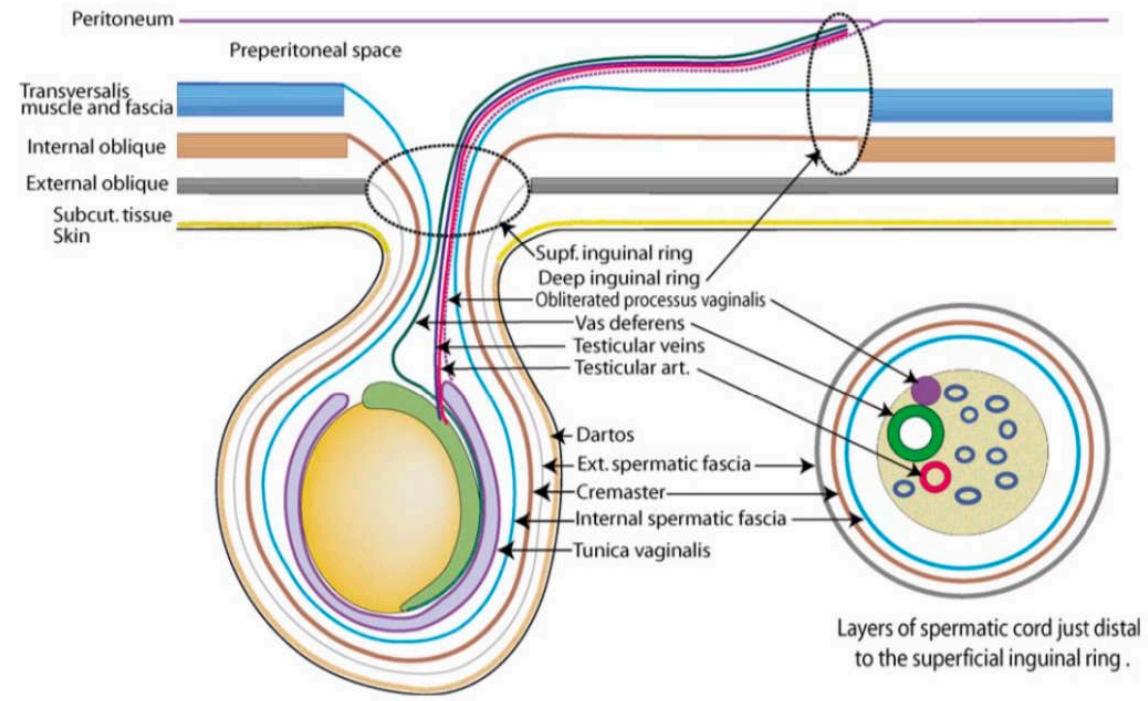
Monochorionic / diamniotic	- Refers to monozygotic twins, where the conceptus splits ~4-8 days post-fertilization. - There is no thick dividing membrane between the fetuses since they lie within the same chorion. 
Monochorionic / monoamniotic	- Refers to monozygotic twins, where the conceptus splits ~8-13 days post-fertilization. - High risk of twin-twin transfusion syndrome due to entanglement of fetal vessels. 
Conjoined twins	- Refers to monozygotic twins, where the conceptus splits ~13 days. 

Penile/testicular disorders	
Hypospadias	- Urethral meatus opens on underside of penis (ventral aspect). 
Epispadias	- Urethral meatus opens on top of penis (dorsal aspect).
Phimosis	- Cannot retract the prepuce (foreskin). - In other words, cannot pull it back to expose the glans.
Paraphimosis	- Cannot reduce the prepuce. - In other words, once it is retracted, cannot replace it back over the glans. - Just remember that paraphimosis is the bad one / emergency.
Cryptorchidism	- Undescended testis. - Increased risk of infertility (↑ temperature within abdomen → ↓ spermatogenesis). - Increased risk of testicular carcinoma. - USMLE wants no treatment prior to age 6 months; after 6 months, perform orchidopexy.
Testicular torsion	- Acutely painful testis; medical emergency; can lead to ischemia and loss of testis. Can be caused by abnormality of the gubernaculum, which is a ligamentous cord that anchors the testis within the scrotum. This leads to "bell-clapper deformity," where improper anchoring = ↑ risk of rotation. - Negative Prehn sign, which is failure of relief of pain upon lifting of the scrotum. Apparently this is an unreliable test in real life, but NBME vignettes can still mention it. In contrast, a (+) Prehn (relief of pain upon lifting the scrotum) is more indicative of epididymitis. - Negative cremasteric reflex, which is failure of the scrotal skin to retract upon palpation of the medial thigh. In contrast, a (+) cremasteric reflex (scrotal skin retracts) is more indicative of epididymitis. Normally, when the inner thigh is stroked, the cremaster muscle contracts and pulls up on the ipsilateral testis. In torsion, the twisted spermatic cord can impair the nerve fibers responsible for triggering the reflex. - USMLE wants Doppler ultrasonography as next best step, followed by surgery.
Epididymitis	- Inflammation/infection of epididymis. - (+) Prehn sign and (+) cremasteric reflex. - Chlamydia and gonorrhea are most common causes in males <35. - Over 35, *E. coli* is most likely.
Torsion of appendix testis	- Not the same as torsion of testis. - Asked on a 2CK Peds form. - Q will tell you there's a kid with a painful testis + the superior pole is blue + there's an intact cremasteric reflex ("regular torsion" it's not intact). - "Blue dot" sign = blue superior pole of testis. - Not my opinion if you think it's nitpicky or weird. Take it up with the NBME exam.
Peyronie disease	- Curvature deformity of penis caused by fibrous plaque within tunica albuginea.

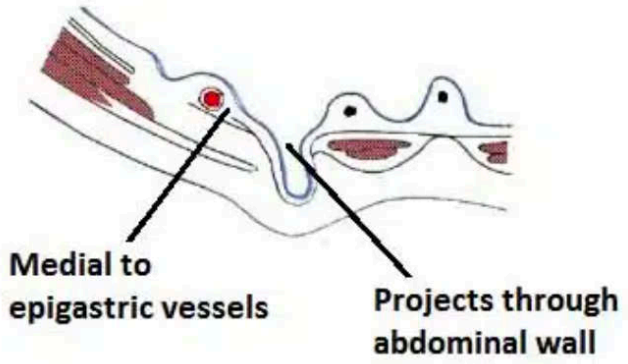
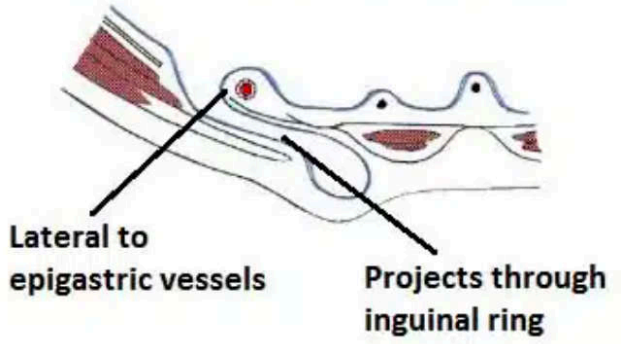
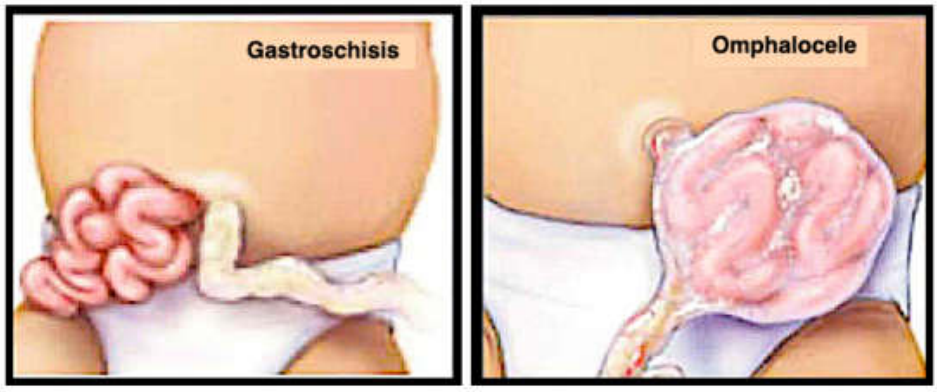
	 - Doesn't have to be treated unless causing functional impairment.
Orchitis	- Inflammation/infection of the testis. - Mumps is HY cause in unvaccinated / immigrants (sometimes implies unvaccinated). - If not caused by mumps virus, then the bacterial causes are same as epididymitis and prostatitis for young vs old (i.e., STIs vs *E. coli*, respectively).
Priapism	- Erection lasting longer than 4 hours. - Can be caused by sickle cell or by PDE-5 inhibitors (i.e., tadalafil/sildenafil). - Injection of alpha-1 agonist (i.e., phenylephrine) into the erectile tissue is treatment → decreases blood flow due to blood vessel constriction.
Hydrocele	- Serous fluid collection within the scrotum between the layers of the tunica vaginalis surrounding the testis. - Mechanism is patent processes vaginalis. - The processus vaginalis is an embryonic outpouching of peritoneum that descends into the scrotum. It normally obliterates shortly after birth, but if it remains open, can cause hydrocele or indirect inguinal hernia.  - Transilluminates when a light is shone to it. - USMLE wants no treatment before the age of 1; most spontaneously close.
Spermatocele	- Obscure serous fluid collection within the epididymis containing sperm.

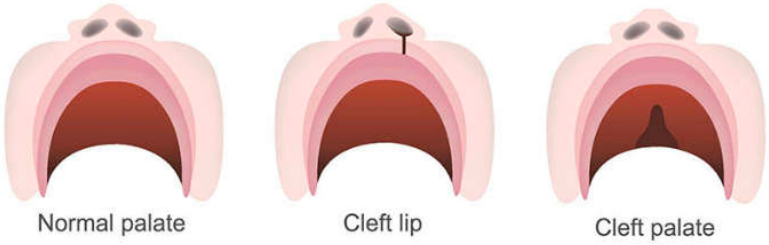
	- Occurs usually in adult men (rather than neonates as with hydrocele). - Transilluminates similar to hydrocele. But once again, it's an older adult, not neonate.
Varicocele	- Dilation/congestion of pampiniform venous plexus draining the scrotum. - Does not transilluminate (in contrast to hydro- and spermatoceles). - Can be described as buzzy "bag of worms." Sounds too easy, but an NBME Q literally uses this colloquialism. - Can also be described as a heavy, dragging scrotum lower on the left. - Occurs almost always on the left side due to the anatomy of venous drainage of the testis. - On the left, we have left testicular vein → left renal vein → IVC. - On the right, we have right testicular vein → IVC ("right to IVC"). - The left testicular vein drains at a 90-degree angle into the left renal vein, creating a proclivity for hydrostatic pressure backup ipsilaterally. - There is a 2CK NBME Q where bilateral varicocele is the answer. Highly obscure, but just be aware it's somehow possible/exists. - Can occur due to left renal vein thrombosis in setting of nephrotic syndrome (loss of antithrombin III in the urine → hypercoagulable state).

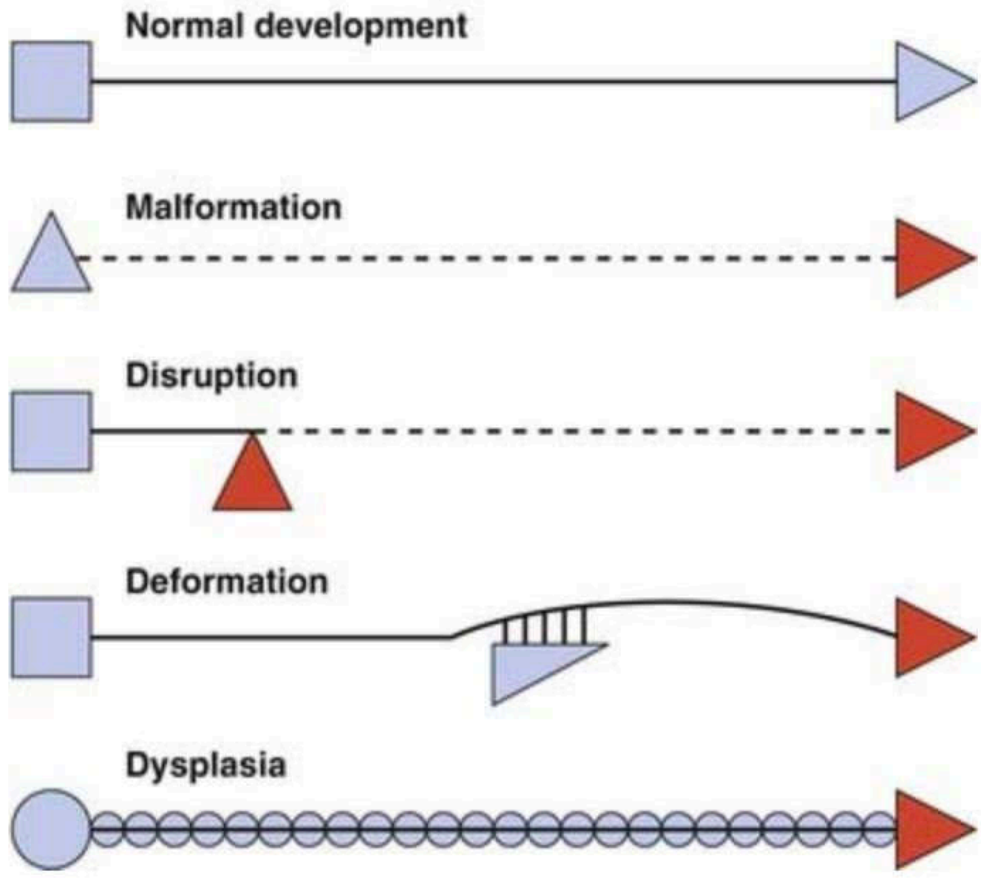
Spermatic cord + HY abdominal hernias



Direct inguinal hernia	- Abdominal contents herniate medial to inferior epigastric vessels within an area known as Hesselbach triangle, which is bordered by the inguinal ligament (inferior border), rectus abdominis muscle (medial border), and the inferior epigastric vessels (lateral border). - The weakening of the abdominal wall can be idiopathic (aging), chronic straining, or previous surgical incisions.

	
Indirect inguinal hernia	- Abdominal contents herniate lateral to inferior epigastric vessels due to a patent processes vaginalis. - Follows the path of the inguinal canal into the scrotum. 
Umbilical hernia	- Small bowel herniation through the umbilicus. - Can occur at any age, but I've seen this show up on NBME in **congenital hypothyroidism** (i.e., cretinism). - Not to be confused with omphalocele in neonates. An umbilical hernia is completely covered by skin. In contrast, an omphalocele is a herniation merely covered by a thin, translucent layer of peritoneal membrane. It can be idiopathic, but is often associated with the Trisomy 13 and 18, and Beckwith-Wiedemann syndrome.  - I discuss more hernia types in detail in the HY Anatomy/MSK PDF. The above ones are most relevant to Peds.

Cleft lip/palate	
	
Cleft lip	- Failure of fusion of the maxillary and nasal prominences. - Polygenic; occurs 1 in 1,000-5,000 depending on population. - Cause on USMLE is “one-off developmental anomaly” (i.e., it is usually not part of a greater syndromic condition). This is same for pyloric stenosis and ASD/VSD. - If a couple has child with cleft lip or palate, the chance of having 2nd child with it is 3-5%. - If one of the parents has cleft lip or palate, chance of any child having it is 10-15%.
Cleft palate	- Failure of fusion of the lateral palatine processes (aka palatal shelves).

Birth defects	
	
Malformation	- Intrinsic/innate abnormality in the structure of an organ or body part due to genetic factors or early environmental influences like teratogens. - Spina bifida, cleft lip/palate, congenital heart defects, phocomelia (severe flipper-like malformation of limbs; caused by thalidomide).
Deformation	- An initially normal organ or body part becomes abnormally shaped later in development, usually due to mechanical forces, such as uterine compression or oligohydramnios. - Club foot (talipes equinovarus), congenital hip dysplasia, plagiocephaly (abnormal head shape).

Disruption	- An initially normal organ or body part is interrupted by a destructive event, leading to the breakdown of a body structure. - Amniotic band syndrome → fibrous bands from the amniotic membrane classically entangle a fetal limb or fingers, leading to amputation.

Testicular/ovarian neoplasms for Peds	
Yolk sac tumor	- Aka endodermal sinus tumor. - The answer on USMLE for ovarian tumor in pediatrics (usually up to age 3). - Secretes AFP as tumor marker. - Schiller-Duval bodies on histo, which resemble glomeruli.
Embryonal carcinoma	- Aggressive and rapidly growing. - Undifferentiated/anaplastic cells on histo. - Secretes both AFP and hCG together as tumor markers. - Usually occurs in adult men, but can be seen as ovarian tumor also. The key for USMLE is that it's the answer if AFP and hCG are both high, whereas if only AFP is high, then the answer is yolk sac tumor. You can also remember yolk sac tumor is kids, whereas embryonal cancer is usually adults.
Seminoma	- Most common testicular cancer; occurs teenage years and older. - Large, uniform cells with watery cytoplasm. - Chemo- and radiosensitive. - Placental ALP can be a tumor marker.
Dysgerminoma	- Seminoma equivalent in females. - Large uniform cells with clear cytoplasm (same as seminoma). - Highly sensitive to chemo- and radiotherapy (same as seminoma). - Can occur idiopathically or in Turner syndrome. - LDH is a tumor marker.
Dermoid cyst	- Aka mature cystic teratoma. - Contains tissue from multiple germ layers – i.e., hair, skin, teeth, etc. - Can calcify.

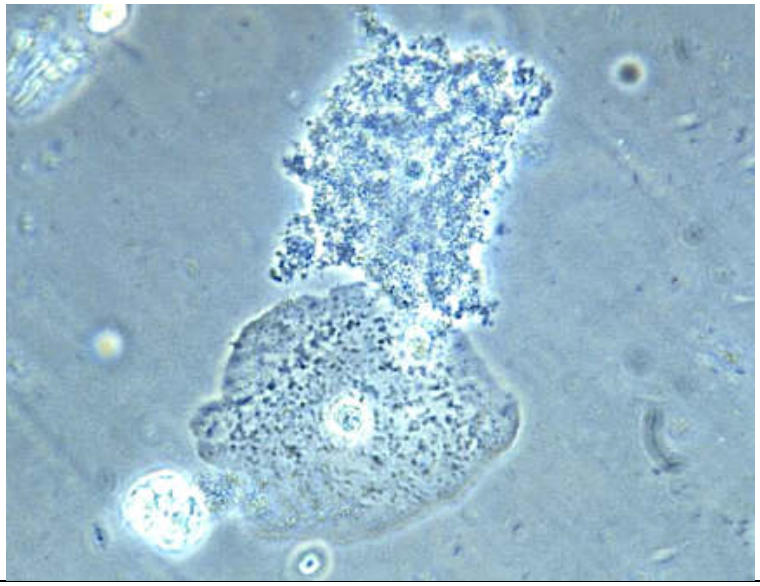
Vulvovaginal lesions	
Condylomata acuminata	- Aka warts; caused by HPV 6/11. - Present as painless, skin-colored or slightly hyperpigmented cauliflower-like popular lesions. - I discuss all of the STDs in more detail later.
Sarcoma botryoides	- Rare as fuck. - Rhabdomyosarcoma of the vagina. - Seen in pediatrics as bunches of grapes protruding from the vagina.
Bartholin gland cyst	- Shows up as a tender/painful bump at a 4- or 8-o'clock position on the vulva. - If the cyst becomes infected (i.e., warm and red), we call it Bartholin gland abscess. - USMLE wants "polymicrobial" as the most likely organism. - For cysts, sitz bath + warm compresses are first Tx. - For overt abscesses, drainage is the answer.
Vaginal foreign body	- Vignette will be a child with foul-smelling discharge from the vagina. - There will be no signs of physical trauma or lacerations (means not abuse).
Sexual abuse	- Can occur in children or elderly (I've seen both on NBME). - Vignette likes to mention lacerations. - There may or may not be discharge.

Breast conditions for Peds	
Fibrocystic change	- Benign cystic alteration in breast tissue in women 20s-50s, but also occurs in teenagers. - Shows up three ways on NBMEs: 1) Highly buzzy/classic presentation is a woman 20s-30s who has bilateral, waxing/waning breast tenderness over the course of her menstrual cycle; 2) Unilateral tenderness or sharp pain in one breast (students are often surprised that it's not always bilateral). - You do not need to biopsy or do ultrasound. It is a clinical diagnosis and Tx is symptomatic only (i.e., warm showers, evening primrose oil). - In theory, biopsy of fibrocystic change will show some buzzy findings: "sclerosing adenosis," apocrine metaplasia," and/or "blue dome cysts."
Virginal breast hypertrophy	- Virginal breast hypertrophy is benign asymmetric growth of the breasts that can occur in young women. USMLE will try to trick you into thinking it's sinister by saying a first-degree relative had breast cancer. - They will say 16-year-old girl has large breast lump under the nipple on one side. Answer is just follow-up / observe.
Gynecomastia	- Physiologic gynecomastia can occur in teenage boys on USMLE, where there can be ↑ breast tissue, either tender or not, occurring uni- or bilaterally. Just observe. - Can occur as a result of ↑ estrogen in liver failure, or due to certain drugs like cimetidine, spironolactone, ketoconazole, or even marijuana. - Caused by direct estrogen effect on breast. It is not caused by prolactin.
Necrosis	- Non-enzymatic fat necrosis of breast due to trauma; can calcify. - Don't confuse with enzymatic fat necrosis, which is acute pancreatitis. - Can occur due to overt trauma (e.g., getting hit by softball) or sometimes by tight-fitting sports bras.

Other ovarian conditions for Peds	
Ruptured ovarian cyst	- Presents as sudden, sharp pain in an adnexa (adnexa = combo of Fallopian tube + ovary). - The Q will often say there is 10-15 mL of serosanguineous fluid in pouch of Douglas on culdocentesis (means aspiration of pouch of Douglas). - The Q will say there is no mass / will not mention mass. - Treated symptomatically (i.e., NSAID). Do not do laparoscopy.
Adnexal/ovarian torsion	- 3/4 Qs will give intermittent adnexal pain that becomes constant over hours to weeks (I've seen NBME give both time frames). - 1/4 Qs will just say few-hour Hx of constant pain without the intermittent detail first. - Presents as "large" adnexal mass (i.e., 8 x 12-cm). - There will be no mention of fluid in pouch of Douglas. - Treatment is laparoscopic detorsion. - Tricky detail is that the biggest risk factor is a pre-existing anatomic abnormality (e.g., cyst). So if they say Hx of cyst + now has large adnexal mass and pain, that is torsion, not ruptured cyst.

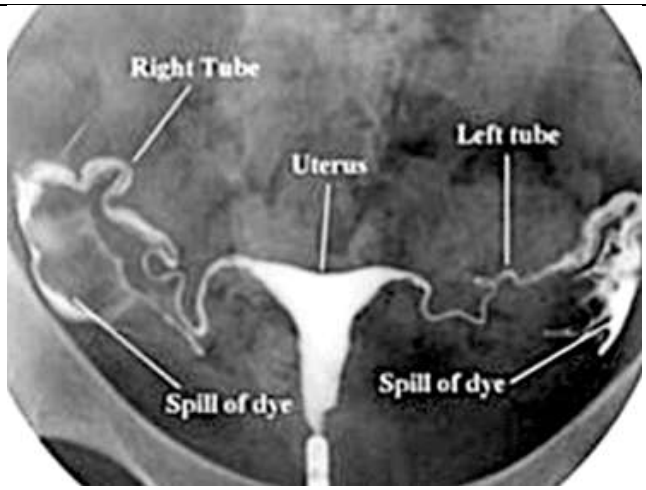
HY STDs	
- All STDs are fair game for teenagers.	
Human papilloma virus	- HPV 6/11 cause condylomata acuminata (warts). This is not limited to the genitalia and can cause laryngeal papillomatosis in neonates (warts of the vocal cords), which is asked on NBME. - HPV 16/18 cause squamous cell carcinoma of genitalia/anus; risk of overt SCC is ↑ in immunocompromised (i.e., HIV in MSM) and heavy smoking.

Chlamydia trachomatis	- The "regular" STD are *Chlamydia* D-K strains. - Causes mucopurulent discharge. - Can advance to pelvic inflammatory disease (PID) in females, which is when infection causes inflammation and scarring of Fallopian tubes, leading to ↑ risk of ectopic pregnancy. - Obligate intracellular, so cannot be grown. Discharge will show WBCs under light microscopy with **no organisms.** If the vignette tells you no organisms grow, this is pass-level for *Chlamydia*. - Treat with stat/one-off oral dose of **azithromycin;** can cause GI disturbance. - Can also be treated with one-week of BID (i.e., twice/day) **doxycycline;** cannot be taken with dairy or divalent cations (impaired absorption); can also cause photosensitivity; considered to be slightly more efficacious than azithromycin but much more annoying and arduous to take. - USMLE won't play trivia as to which drug is first-line; there will only be one correct answer. For example, doxy is not given in pregnancy, so if you're forced to choose between the two, in this case you know it's azithromycin (or even sometimes erythromycin in pregnancy). - Doxy isn't given in pregnancy because it can cause teeth discoloration in the eventual neonate. - *Chlamydia* can cause reactive arthritis (triad of urethritis, arthritis, and eye-itis – i.e., any inflammation of the eye, e.g., conjunctivitis, anterior uveitis, etc.). - To prevent ophthalmia neonatorum (neonatal conjunctivitis), treat the female while she is pregnant; to Tx the actual condition in neonates, give oral erythromycin. Conjunctivitis in a neonate can lead to *Chlamydia* pneumonia. - *Chlamydia* A-C are not STDs and cause trachoma (cause of blindness in Africa). - *Chlamydia* L1-3 cause lymphogranuloma venereum (anal strictures).
Neisseria gonorrhea	- Causes mucopurulent discharge, same as *Chlamydia.* - Gram-negative diplococci on light microscopy. - Same as with *Chlamydia*, can advance to PID, with ↑ risk of ectopic pregnancy. - Doesn't cause reactive arthritis; it causes gonococcal arthritis, which will present one of two ways on NBME: 1) monoarthritis of the knee; or 2) triad of mono- or polyarthritis + **cutaneous papules/pustules** + tenosynovitis (inflammation of tendon sheaths; stems like to give deQuervain tenosynovitis). - Treatment is IM ceftriaxone. - Always cotreat for *Chlamydia.* In other words, if the gram-(-) diplococci are seen under LM, there's no way to know if *Chlamydia* is also there or not since the latter shows no organisms, so if a patient has *Gonorrhea*, we the proper Tx is IM ceftriaxone, PLUS either oral azithromycin or doxycycline. - If patient develops PID despite having been treated early with ceftriaxone for *Gonorrhea*, the answer for why this happened can be "Hx of improper antibiotic treatment," where the patient was supposed to be cotreated for *Chlamydia* with azithromycin but was only given the ceftriaxone for *Gonorrhea.* - If patient presents with PID who's septic (i.e., high fever, tachy, high WBCs), USMLE wants "admit to hospital + IV antibiotics," not the outpatient combo of IM + oral. - 2CK Obgyn form assesses that if an asymptomatic patient comes in after a partner tested positive for *Gonorrhea* or *Chlamydia*, the answer is give treatment without waiting for test results. - Erythromycin ointment is used on neonates as prophylaxis for conjunctivitis; if neonate already has it, give IM cefotaxime (preferred 3rd-gen cephalosporin in peds if listed).
Trichomonas vaginalis	- Causes trichomoniasis. - Flagellated protozoan seen on wet mount. - Presents as yellow-green discharge.

	- Can cause “strawberry cervix,” or punctate hemorrhages on the cervix. If they don’t say this, they can sometimes say yellow-green discharge + a vaginal canal that is erythematous. - Treat with metronidazole for patient **and partner** (high rate of reinfection).
Gardnerella vaginalis	- Causes bacterial vaginosis. - Gram-negative rod that causes a thin grey/watery discharge. - Positive whiff test (KOH prep causes fish-like odor). - Clue cells exceedingly HY (squamous epithelial cells studded with bacteria). They want you to know this image for USMLE: 
Candida spp.	- Causes candidiasis. - Buzzy thick, white, cottage cheese-like discharge in ~2/3 of questions. - The other ~1/3 of Qs will mention an itchy/erythematous vaginal canal without any overt discharge (in contrast to trichomoniasis which can present with erythema of the vagina but has characteristic yellow-green discharge). - Treat with topical nystatin or oral fluconazole. - Step 1 NBME says oral Tx is given + wants MOA for drug inhibition → answer = “P-450-mediated demethylation reaction,” where fluconazole inhibits 14α-demethylase in the conversion of lanosterol to ergosterol.
Treponema pallidum	- Causes Syphilis. - Spirochete (spiral-shaped bacterium) visible under dark-field microscopy. - Primary syphilis = painless chancre (painless ulcer) on genitalia. - Secondary syphilis = 6 weeks to 6 months after appearance and disappearance of the initial chancre, patient can get body rash that **includes palms + soles,** and condylomata lata (painless genital plaques). - Tertiary syphilis = years later, patient can get gummas (appear as painless chancres but are on other areas of the body such as the face/nose), arthritis, and ascending aortitis (tree-barking of vasa vasorum). - Neurosyphilis **can occur at any stage;** it is not sequential where we have 1 → 2 → 3 → neurosyphilis. There is a 2CK Neuro Q that gives neurosyphilis in an 18-year-old. - Neurosyphilis presents as tabes dorsalis (obliteration of dorsal columns, with loss of vibration/proprioception + a positive Romberg sign, where patient falls over when standing with eyes closed), Argyll-Robertson pupil (i.e., “prostitute pupil”; accommodates but doesn’t react), and “stroke without hypertension” (i.e., sometimes findings akin to stroke but in a younger patient). - Diagnosis of primary syphilis is made via visualizing the spirochetes from a chancre scraping under dark-field microscopy.

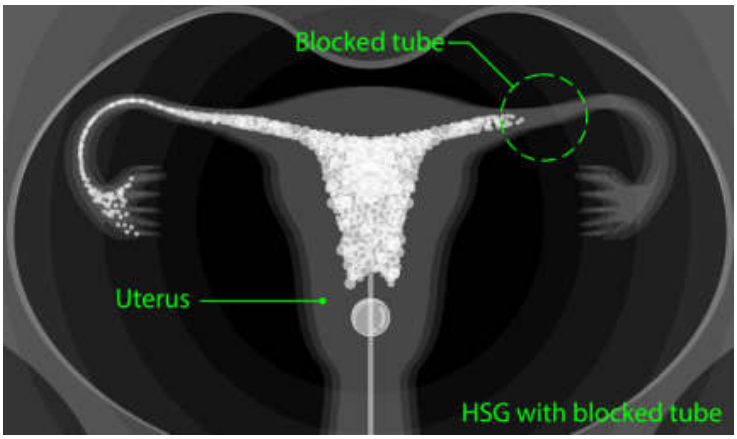
	- Diagnosis of secondary, tertiary, and neurosyphilis can be done with serology, where a VDRL is done first (sensitive but not specific); a FTA/RPR is done as confirmatory. - Patients with SLE who have anti-phospholipid syndrome can get false-positive VDRL tests. - USMLE will show you 24-year-old male with rash on his back + KOH prep is negative + ask what's most likely to diagnose → answer = FTA. - Treatment for all syphilis types is penicillin. - If patient has Hx of anaphylaxis to beta-lactams but is pregnant or has tertiary or neurosyphilis, the answer is desensitize + give penicillin. This is because penicillin is the most efficacious and needs to be given in severe cases. - If patient has Hx of mere rash to beta-lactams, but not anaphylaxis, then the beta-lactam can be given anyway.
HSV 1/2	- Herpes simplex virus 1/2. - Causes painful vesicular lesions that recur at varying intervals (usually months). - Primary infection is most severe, often with fever, regional lymphadenopathy, burning/stinging/itching pain (herpetic neuralgia), and many vesicles. - Recurrences are often less severe and preceded by herpetic neuralgia. - HSV1/2 can also cause encephalitis (confusion + blood in CSF due to temporal lobe hemorrhage) and herpetic whitlow (vesicle[s] on the finger). - Viral culture can be negative in stem (not 100% sensitive). - Treat with acyclovir (or valacyclovir).
Haemophilus ducreyi	- Causes chancroid, which is a painful ulcer. - Gram-negative rod. - "-Oid" means looks like but ain't. So it looks like a syphilitic chancre, but it's not. - The syphilitic chancre is painless; the *H. ducreyi* lesion is painful. - Often a wrong/distractor answer for HSV Qs, where students get trigger-happy and erroneously choose the weird answer (*H. ducreyi*). - Chancroid will be the answer if they tell you there's a single painful genital lesion in someone who went abroad, classically backpacking in Africa or South America. - If they tell you there's a single, small painful lesion, *but that it's a recurrence*, this is HSV, not *H. ducreyi*. The latter is bacterial and doesn't cause recurrences the way HSV does; HSV can rarely appear as a single vesicle. - There is a 2CK NBME Q where answer is actually *H. ducreyi*, but I once again caution that this is usually a wrong answer, so be careful. But I have seen it correct as a one-off. - USMLE won't assess treatment, but either azithromycin or ceftriaxone is considered first-line.

Pelvic inflammatory disease (PID)
- As discussed earlier, is caused by either *Chlamydia* or *Gonorrhea*. - When the infection ascends the uterus and Fallopian tubes, it can lead to inflammation and scarring, thereby increasing the risk for ectopic pregnancy (i.e., loss of Fallopian tube cilia leads to premature implantation usually in the ampulla of the Fallopian tube). - Adnexal and **cervical motion tenderness** are buzzy. But particularly the latter. If they say cervical motion tenderness, you know right away they're talking about PID. - USMLE wants you to know that the Fallopian tubes are normally open on both ends, where the release of the ovum from the ruptured Graafian follicle is "caught" by the fimbriae of the Fallopian tube, prior to its transit to the uterus. Because the Fallopian tubes are open on both ends, when a hysterosalpingogram is performed, spillage of dye bilaterally into the peritoneal cavity *is normal.*



Normal hysterosalpingogram

- If there is Hx of PID and the Fallopian tube is scarred, it may be sealed, where the hysterosalpingogram shows failure of spillage.



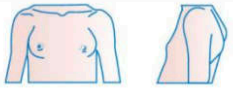
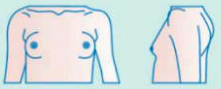
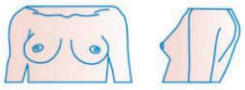
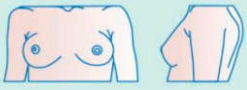
- If the Q tells you a girl has PID and is treated with antibiotics but has persistent fever and adnexal pain, next best step = ultrasound to look for **tubo-ovarian abscess,** which is a potential sequela of PID.
- Extension of PID to the liver capsule is called Fitz-Hugh-Curtis syndrome.

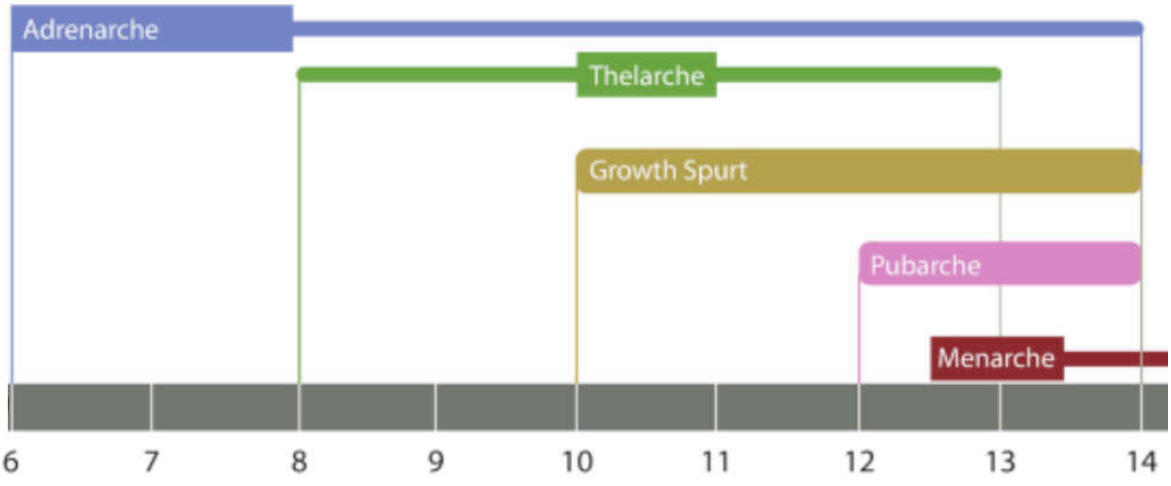
TORCHeS infections	
- Refers to congenital infections in the neonate due to infection in the mother while pregnant.	
Toxoplasmosis	- Presents as HY triad in neonate of 1) hydrocephalus, 2) chorioretinitis, and 3) intracranial calcifications. - They don't have to say the mom sat in a litter box with her cat during pregnancy. - Can be acquired from pork consumption.
Other (Parvovirus B19, VZV)	- Parvo B19 can cause aplastic anemia *in utero* or in the neonate, with increased risk in sickle cell. - Aplastic anemia = all 3 hematologic cell lines are down (i.e., RBCs, WBCs, and platelets). - Can sometimes cause a pure RBC aplasia (i.e., only RBCs down). - The term "fetal hydrops" refers to heart failure *in utero,* and can suggest severe anemia, such as with maternal Parvo infection. In other words, the stem can say the mom had a flu-like illness or cold, followed by hydrops in the fetus → answer = Parvo. - Q can ask how to confirm Parvo infection in neonate if suspected → answer = check Parvo IgM titers in neonate. - Varicella (VZV) during pregnancy can cause microcephaly and limb hypoplasia.

	- If mother contracts VZV for the first time between 5 days prior to parturition until 2 days after, Varicella immunoglobulin must be given to the neonate. - Kids with congenital VZV syndrome (as well as those who are immunocompromised), are at increased risk of pediatric shingles (just know this Dx exists / "is a thing"). - If neonate is exposed to child with active varicella, if the mother of the neonate previously had chickenpox or was vaccinated, you do **not** need to give VZV IVIG to the neonate. This is because the neonate has passive immunity from the transplacental IgG. - Should be noted that pregnant women who contract VZV can get **pneumonia** from it (sounds weird, but it's a factoid you could be aware of).
Rubella	- Presents as patent ductus arteriosus (PDA) in a neonate. Exceedingly HY / pass-level. - Cataracts and deafness also possible. - Q doesn't need to say mom had rash while pregnant. Sometimes pregnant women get only arthritis. This is not unique to rubella, but I'm just mentioning it because the stem can say she had arthritis but no rash, and student is like "Well it couldn't have been rubella in the mom then." No. Adults sometimes get only arthritis if they contract rubella or measles. - MMR vaccine is live-attenuated and is contraindicated during pregnancy due to theoretical risk to the fetus. If a woman inadvertently receives the vaccine while pregnant or within the month prior to pregnancy, it is not an indication for abortion, but risks to the fetus are increased and proper monitoring is important.
CMV	- Congenital cytomegalovirus is a diagnosis of exclusion on NBME Qs, as per my observation (i.e., we eliminate to get there). - Congenital deafness, hepatomegaly, "blueberry muffin rash," and intracranial calcifications are HY, but these are non-specific and seen in other conditions too. - The USMLE Q might say, "Kid is born with intracranial calcifications, hepatomegaly, and deafness." And the student says, "Aren't calcifications toxo?" Yes, but toxo is the strict triad as mentioned above. The student might also say, "Well can't deafness be rubella or syphilis." Yes, but for rubella they always mention PDA, and for syphilis they always mention unique findings like tooth or shin abnormalities (discussed below). So we eliminate to get there and are left with CMV. It's a bit circuitous, but it's what I've observed they do with CMV Qs. - CMV causes owl-eye appearance of cells due to intranuclear inclusions. - Tx = ganciclovir. MOA is DNA polymerase inhibitor. Mechanism of resistance is alterations to viral thymidine kinase.
Herpes/HIV	- Vaginal HSV1/2 infection in mother can lead to vertical transmission, increasing risk for encephalitis in neonate. - If a pregnant woman experiences prodromal symptoms (i.e., tingling, burning, etc.), even if no visible lesions are present, C-section is still recommended. - If a pregnant woman has predictable intervals of vesicular episodes, acyclovir is often given within 4-6 weeks of parturition to decrease risk of peripartum episode. - HIV in pregnancy is HY. Most important point is that highly active anti-retroviral therapy (HAART) is started immediately in any HIV patient regardless of CD4 count and that it is three-drug therapy. - Efavirenz (an NNRTI) is avoided in pregnancy. - TMP/SMX is avoided in first-trimester, or if woman is immediately trying to conceive, even if CD4 count is under 200. It is okay to use in 2nd and 3rd trimesters. - Intrapartum zidovudine is given to HIV (+) mothers + C-section is performed. - Zidovudine is given to the neonate within 6-12 hours of birth + given for 6 weeks. - HIV is present in breastmilk. It is generally recommended to avoid breastfeeding if mother is HIV (+). If the mother's viral load is undetectable and she is on continued HAART, transmission to neonate might not occur, but it is still advised against.
Syphilis	- Highest yield point is that it can cause tooth abnormalities (mulberry molars/incisors). - Can cause "saber shins" (bone abnormalities), saddle nose, deafness, and cataracts.

Pubertal development

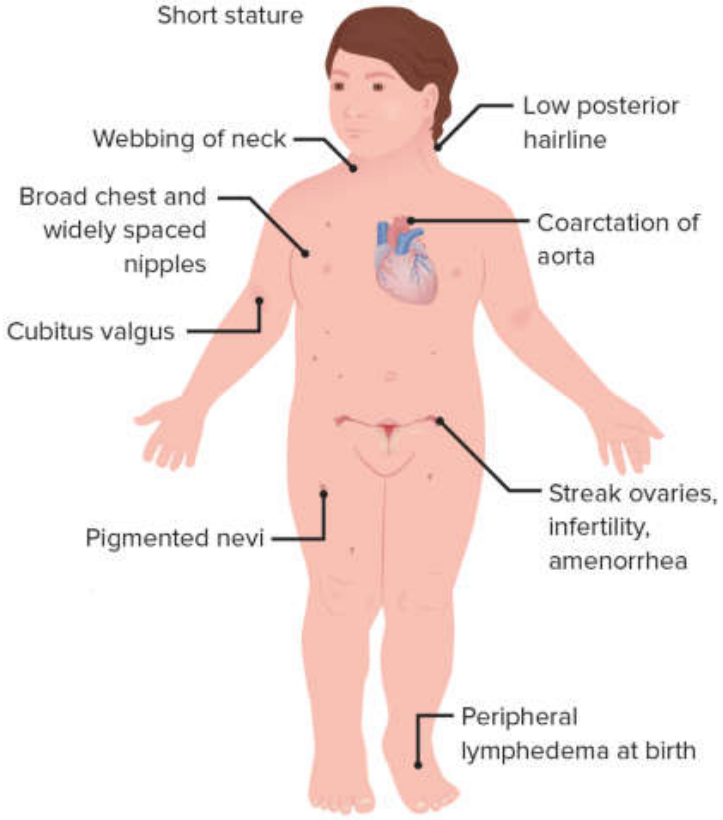
- USMLE wants you to know that at Tanner stage 3, “menarche is imminent.” That is an answer straight-up on one of the 2CK Obgyn forms, but there are also various Repro Qs where mom will bring in her daughter who’s 13/14, who’s never had a menstrual period, and they’ll say she’s either Tanner stage 2 or 3, and the answer is just “schedule follow-up in 6 months.”
- Turner syndrome can present with Tanner stage 1-2 breasts and 4-5 pubic/axillary hair. The latter are due to the effects of adrenal, rather than ovarian, androgens. In other words, don’t be confused if you get a Turner syndrome Q and they say Tanner stage 4 pubic hair.
- In androgen insensitivity syndrome (46XY karyotype, but female phenotype), breast development is normal (i.e., Tanner 4-5), but pubic/axillary hair are Tanner stage 1 (i.e., scant/absent).
- Bone age refers to degree of maturation of a child’s bones via radiographic examination.
- USMLE wants you to know that bone age < chronologic age means the short child will catch up (i.e., the growth chart is merely right-shfited). This is called constitutional short stature. The vignette will usually say both parents are average height. For example, a 14-yr-old boy is shortest in his class but has a bone age of 12, meaning is skeleton is aged 12, so he’ll catch up later.
- For constitutional short stature, sometimes instead of mentioning bone age, they’ll mention Tanner stage. For example, a 14-yr-old boy is shortest in his class + pubic hair is Tanner stage 2. The implication is he’s still very early in development and will probably catch up.
- Bone age = chronologic age means genuine short stature. I’ve seen this in Turner syndrome Qs, where the female is usually < 5 feet.

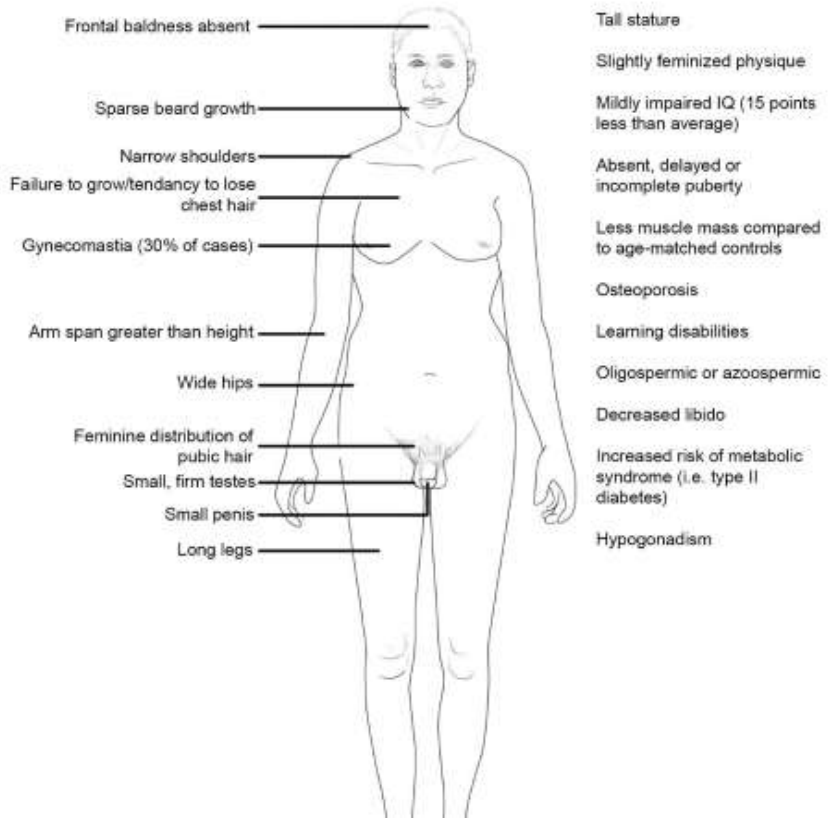
Tanner stage	Male genital appearance	Male genital description	Female pubic hair appearance	Pubic hair description	Breast appearance	Breast description
1		Testicular volume <3ml		No pubic hair		Elevation of papilla only
2		Testicular volume <3ml, change in texture to scrotal skin		Sparse growth chiefly along the labia/base of penis		Breast bud stage
3		Increase in size of penis with further testicular enlargement		Darker, coarser, more curled hair		Enlargement of breast and areola
4		Further enlargment of penis and testicles with development of glans penis		Adult type hair over a smaller area		Projection of the areola and papilla
5		Adult size and shape		Spread to the medial surface of the thighs		Recession of the areola to the contour of the breast, projection of papilla only



- A mnemonic to remember the above sequence is “Tap em” (i.e., TAPM).
- Thelarche: Onset of breast development.
- Adrenarche: ↑ production of adrenal androgens, leading to the development of pubic and axillary hair.

- Pubarche: Pubic hair growth
- Menarche: Onset of menstruation.
- USMLE asks straight-up what is most likely to occur first in a pre-pubescent female → answer = breast bud development.

Sex disorders	
Turner syndrome	 - 45XO karyotype; female phenotype. - Since there is only one X chromosome, the Q can say there is no Barr body visible on electron microscopy. - Infertile due to streak ovaries. In Turner, this is colloquially referred to as "menopause before menarche." - Can still have children with IVF using donor egg + exogenous hormones (asked sometimes on behavioral/psych Qs). - LH and FSH both ↑ due to primary hypogonadism (i.e., ↓ negative-feedback at hypothalamus and anterior pituitary due to ↓ ovarian hormones and inhibin). - Short stature (usually < 5 feet), Tanner stage 1 breast development ("shield chest"), cystic hygroma (webbed neck due to lymphatic insufficiency; asked on NBME); scattered nevi (confuses students for things like NF1, but I don't know what to say; you need to know scattered nevi are seen in Turner); normal pubic and axillary hair (Tanner 4-5). - ↑ risk of dysgerminoma developing from the streak ovaries; prophylactic oophorectomy is recommended, especially if there is Y-chromosome material present in rarer mosaic forms of Turner. - If Q tells you there's a fraction of a female's cells that are 46XX and another fraction that's 45XO (i.e., somatic mosaic Turner), the answer for the mechanism is "post-fertilization mitotic error." - Coarctation of the aorta + bicuspid aortic valve (aortic stenosis) HY.

Klinefelter syndrome	 - 47XXY karyotype; male phenotype. - Tall, eunuchoid body shape; small testes; gynecomastia; reduced IQ. - Q might say there is presence of electron-dense mass in the cell on electron microscopy (i.e., a Barr body). - Testes are present but are small; there is dysgenesis of the seminiferous tubules, leading to ↑ LH and FSH due to ↓ negative-feedback from ↓ androgens and inhibin. - Estrogen/androgen ratio is ↑ due to dysgenesis of Leydig cells (which make androgen). Absolute estrogen is also ↑ due to ↑ adipose tissue (therefore ↑ aromatase). - ↑ risk of breast cancer; risk of testicular cancer is only very subtly ↑, so prophylactic orchiectomy is not indicated.
Androgen insensitivity syndrome (AIS)	- 46XY karyotype; female phenotype. - Defect in androgen receptor. Since DHT is needed for external male development, the patient appears female. - Testosterone and DHT are normal or elevated due to ↓ negative-feedback at the hypothalamus and anterior pituitary → ↑ LH production → ↑ testicular androgen production. - The testes in AIS are usually located within the abdomen or inguinal canal. - Even though there is resistance to androgens, testes still develop because the the *SRY* gene on the Y chromosome induces differentiation of the primitive gonads into testes. This differentiation isn't dependent on androgens. - The testes produce Müllerian-inhibitor factor, so there is no uterus or Fallopian tubes in AIS. - Even though the gonads also produce androgen, since we have resistance to it, the karyotypic male appears female (i.e., phenotypically female). - Breast development is Tanner 4-5 but pubic/axillary hair is Tanner 1. - USMLE vignette will say: 15-year-old girl is brought to the physician by her mother for not yet having a menstrual period + vagina ends in blind pouch + pubic/axillary hair is scant/absent → answer = AIS. - Next best step = ultrasound first if listed, then karyotype.

	- ↑ risk of testicular malignancy (dysgerminoma), so prophylactic orchiectomy is indicated after puberty (maximizes estrogen effects for phenotypic female development), unless evidence of malignancy is seen prior.
Müllerian agenesis	- Aka Mayer-Rokitansky-Küster-Hauser syndrome. USMLE will never assess this name, but students have asked about this before. - 46XX karyotype. - Presents identical to AIS on USMLE except they'll say pubic/axillary hair is normal/coarse. - In other words, 15-year-old girl + never had a menstrual period + vagina ends in blind pouch + pubic/axillary hair is coarse → answer = Müllerian agenesis. - Ovaries are present, since no Y chromosome is present (i.e., no *SRY* gene to convert primitive gonads into testes, so ovaries as the default ensue). - But since the Müllerian (paramesonephric) ducts don't develop, the Fallopian tubes, uterus, cervix, and upper vagina don't develop. - Next best step = ultrasound first if listed, then karyotype.
MIF deficiency	- Müllerian-inhibitory factor deficiency. - 46XY karyotype; phenotypically male with testes (i.e., internal and external structures are male), but failure of regression of the Müllerian ducts means a uterus, Fallopian tubes, cervix, and upper vagina are present. - The Müllerian structures are usually located in the pelvis between the bladder and rectum (similar to location for uterus in a female), or in the inguinal canal. They are often discovered incidentally during orchidopexy for cryptorchidism. - In other words, the Q can say there are undescended testes, where upon surgical reduction into the scrotum, a primitive uterus and Fallopian tubes are discovered.
5α-reductase deficiency	- 46XY karyotype; phenotypically female, or neonate with ambiguous genitalia. - 5α-reductase is enzyme that converts testosterone → DHT. If there is a deficiency and DHT is ↓, male external structures don't develop, so phenotype is female. - Presents as "phallus (penis) at age 12." This is buzzy. They will say a 12-year-old girl has grown 4 inches in the past 4 months + has acne + hair on upper lip + clitoral hood is 3-4 cm → answer = 5α-reductase deficiency. - At puberty, the ↑↑ in testosterone can partially override the enzyme deficiency, leading to threshold production of DHT where male secondary sex characteristics can develop.
SRY gene translocation	- 46XX karyotype; phenotypically male. - The *SRY* gene from the Y chromosome is translocated usually onto one of the X chromosomes, resulting in formation of testes instead of ovaries. - The presence of testes leads to testosterone and DHT → develops male external structures; MIF production → no Müllerian structures develop. - Testes are usually small, and there can be penile abnormalities like hypo- or epispadias. - The USMLE will tell you a neonate appears male but is confirmed 46XX karyotype → answer = *SRY* gene translocation. - Rarely, this condition can be seen in 46XY karyotype, where the *SRY* gene is lost, rather than gained. Testes are absent, and instead there are streak ovaries + Müllerian structures.
Swyer syndrome	- 46XY; phenotypically female. - Caused by failure of the primitive gonads to develop into testes, but they don't develop into ovaries either. They develop into streak gonads. - ↓ androgen production causes external female phenotype. Also leads to scanty/absent pubic/axillary hair. - Absence of MIF (since no testes) means Müllerian structures develop. - Presents similar to AIS – i.e., phenotypic female with scanty/absent pubic/axillary hair, but there are Müllerian structures (i.e., uterus, Fallopian tubes, cervix, and upper vagina) and ↓ androgens, whereas in AIS, Müllerian

	structures are absent and androgens are normal or ↑ (tissues are just insensitive to them).

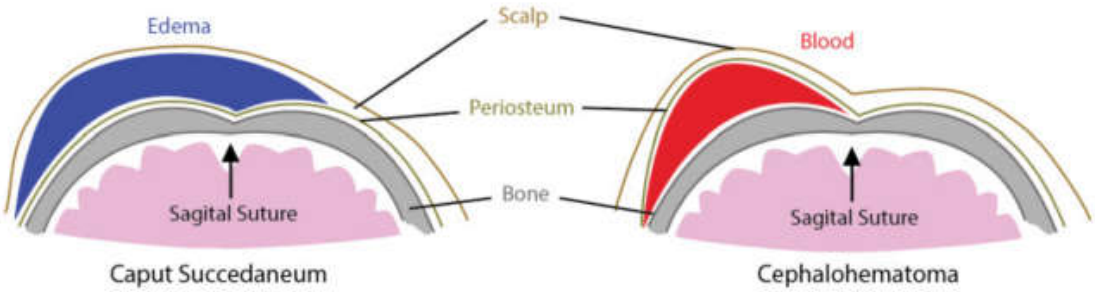
Imperforate hymen	
- The hymen is a thin membrane that partially covers the vaginal opening in females. It normally has a small hole that allows the passage of menstrual blood out of the vagina. The size and shape of the hymenal opening can vary between females. - Imperforate hymen is when the hymen completely covers the vaginal opening, which can lead to hematocolpos and hematometra.	
Hematocolpos	- Blood accumulation within the vaginal canal due to imperforate hymen. - USMLE Q will give a teenage girl who's never had a menstrual period. Physical exam shows a "blueish bulge" behind the hymen. The Q might say the girl has had monthly pain (i.e., she has in fact had menses, but blood is blocked from leaving the vagina due to the imperforate hymen). - Answer is either hematocolpos or "cruciate incision of the hymen."
Hematometra	- Blood backing up all the way to the uterus causing uterine distention. - Q will give what sounds like imperforate hymen + hematocolpos but you'll notice the blood pressure is low → distention of the uterus can trigger a vagal response leading to low BP.

HY Trisomy disorders	
Trisomy 21	- Down syndrome; caused by having 3 chromosome 21s, instead of the normal 2. - 95% due to extra chromosome 21 as a result of meiotic nondisjunction in advanced maternal age (1 in 350 at age 35; 1 in 100 at age 40; 1 in 10 at age 50). - 4% due to Robertsonian translocation – i.e., obscure process where long arm of chromosome 21 attaches to another chromosome (usually 14) and is passed to conceptus along with two normal chromosome 21s; the result is the conceptus inherits 3 chromosome 21s. - 1% is mosaic Down, which is due to post-fertilization mitotic (not meiotic) error. In this case, not all cells of the conceptus contain the trisomy. - Most common genetic cause of mental retardation; second most common genetic cause is Fragile X. Most common cause of mental retardation overall is fetal alcohol syndrome. - Associated with flattened facies; slanted palpebral fissures; epicanthal folds; single palmar crease (this finding non-specific but still often mentioned in Down); large gap between the first and second toes; Hirschsprung disease; duodenal atresia (double bubble sign on AXR); endocardial cushion defects (AVSD, VSD, or ASD); pulmonary artery malformations; acute lymphoblastic leukemia; congenital hypothyroidism; Eustachian tube atresia; Brushfield spots (iris lesions). - First trimester tri-screen (8-10 weeks): ↑ nuchal translucency, ↑ hCG, ↓ **PAPP-A**). Students often remember something is abnormal about PAPP-A, but they forget the direction, so remember that it's ↓, not ↑. - Although not technically part of the tri-screen, a hypoplastic nasal bone is also an important finding in Downs (reflects flattened facies). - Second-trimester quad screen (↓ AFP, ↑ hCG, ↓ estriol, ↑ inhibin A). As discussed earlier, just remember that the ones that have "Hs" are ↑ (i.e., hCG and inhibin A both have Hs, so those are the ones that are ↑).
Trisomy 18	- Edward syndrome. - Low-yield for USMLE. But you have to know it exists. - Low-set ears; prominent occiput; rocker-bottom feet; clenched hands / overlapping fingers; omphalocele. - Some students get emotional about the Edward and Patau syndrome quad screens, but USMLE doesn't give a fuck. But in theory the arrows are all ↓.

	- Usually fatal *in utero* or shortly after birth.
Trisomy 13	- Patau syndrome. Rare as fuck. - Holoprosencephaly (cylopia) / failure of halves of brain to separate; polysyndactyly. - Quad screen arrows all ↓, except inhibin normal. USMLE doesn't care though. - Usually fatal *in utero* or shortly after birth.

Oligo- vs polyhydramnios	
- Certain topics have "grey-area" overlap with Obgyn shelf, such as this one.	
Polyhydramnios	- Excessive amniotic fluid. - Important maternal causes: - **GDM.** ↑ glucose in the fetus pulls water with it through the fetal kidneys, causing ↑ urinary volume. - **Fetal hypoxia.** This could be due to maternal anemia, HDN, or infections like Parvo. If there is ↓ oxygen delivery from the mom, fetal RBC hemolysis, or ↓ fetal RBC production, the fetal hypoxia results in a compensatory ↑ in fetal cardiac output in an attempt to oxygenate tissues. One way this is accomplished is by dilating the renal vessels, which lowers resistance on the peripheral vasculature, but it also simultaneously increases renal blood flow and urine production. - **Multiple gestation pregnancies.** This is due to combined ↑ urinary output in the setting of multiple fetuses. - Important fetal causes: - **Fetal hydrops** (severe edema in the fetus usually as a result of heart failure). In the setting of Parvo B19 infection, fetal anemia, or HDN, cardiac output ↑ to compensate, leading to ↑ renal perfusion and ↑ urine production. In the setting of fetal heart failure where cardiac output is ↓, ↑ venous pressure will cause edema, where the kidneys attempt to rid the body of excess fluid by ↑ urinary output.
Oligohydramnios	- Insufficient amniotic fluid. - Important maternal causes: - **Uteroplacental insufficiency.** Reduced blood flow to the fetus as a result of smoking, preeclampsia, and SLE (discussed in below tables) can result in compromised renal function and urinary output. This is likely related to impaired nutrient delivery in the setting of minimally sufficient oxygenation, where renal cardiac output does not increase (since oxygenation is sufficient), but nutrient delivery is low (so renal function is reduced). - Important fetal causes: - **Posterior urethral valves.** Most common genitourinary abnormality in neonatal males. "Posterior" refers to the prostatic urethra developing thin membranes, or "valves," that prevent urinary outflow. This can lead to obstructive uropathy with vesicoureteral reflux and hydronephrosis. Severity can vary; some Qs give severe oligohydramnios; other Qs are male neonate hours old who hasn't urinated, or male neonate weeks old with urinary retention. - **Renal agenesis (Potter sequence).** Failure of kidney development → ↓ urine output → ↓ amniotic fluid → ↓ swallowing of fluid by fetus → impaired lung development → pulmonary hypoplasia. The ↓ amniotic fluid also leads to flattened facies and rocker-bottom feet.

Pregnancy in Peds
- USMLE wants you to know pregnancy can occur in Peds *before the first menses*. They can give classic Sx of pregnancy (i.e., morning nausea, bloating, breast tenderness, etc.) in 13-year-old who's never had a period → next best step = check serum β-hCG. - Reproductive / Obgyn care should be granted to minors without the need for parental consent, including use of OCPs. In the USA, the exact laws differ by state, but when in doubt, parental consent is not required. - Abortion is generally not performed with parental consent.

Caput succedaneum vs cephalohematoma	
	
Caput succedaneum	- Swelling of an infant's scalp appearing shortly after delivery. - Soft, puffy, and sometimes bruised area that can cross suture lines. - Relatively common; can occur with vaginal deliveries due to pressure exerted on the fetal head by the vaginal canal. - Not threatening; resolves on its own within a few days post-birth without Tx.
Cephalohematoma	- Collection of blood between a baby's skull and the periosteum (membrane covering the skull). - Presents as a raised, fluctuant, and well-defined mass that is confined to suture lines (in contrast to caput succedaneum, which can cross them). - Typically caused by vacuum or forceps delivery. - Usually do not pose a risk to the infant and self-resolve within weeks to months. - ↑ risk of jaundice as the blood is reabsorbed, due to ↑ bilirubin levels.

Polycystic ovarian syndrome (PCOS)
- High BMI female → insulin resistance. - Insulin resistance causes abnormal GnRH pulsation → leads to ↑ LH and ↓ FSH. This is often truncated as just saying there's an ↑ LH/FSH ratio. Some students think FSH is also ↑, but it's just the ratio that's ↑. That's wrong. FSH is low. - LH normally acts on the theca lutein cells of the ovaries to make androgens. Since LH is ↑, we get hirsutism. - FSH normally stimulates follicular development. Since FSH is ↓, we have poor follicular development, leading to failure of a Graafian follicle to rupture during ovulation. The unruptured follicle is retained as a follicular cyst. - Failure of ovulation is called anovulation. This term is exceedingly HY on USMLE. It presents as a female with irregular periods. - Anovulation is a broader term than PCOS, as it can be due to other conditions as well, such as hypothyroidism and Cushing syndrome. But when you hear the word "anovulation" alone, it is usually used as synonymous for the same mechanism as PCOS – i.e., high-BMI female who has irregular periods due to abnormal GnRH pulsation causing an ↑ LH/FSH ratio, with the ↓ FSH causing failure of follicular rupture. - So high-BMI female + irregular periods = anovulation till proven otherwise. - Anovulation + hirsutism = PCOS.

- In clinical practice, part of the PCOS diagnosis requires ultrasounds showing 11+ cysts bilaterally (Amsterdam criteria). But USMLE doesn't assess this. You just need to know anovulation + hirsutism = PCOS.
- Normally when a follicle ruptures, the remnant is called the corpus luteum, which secretes progesterone.
- Progesterone inhibits growth of endometrium; estrogen stimulates growth of endometrium.
- Women who have anovulation have ↓ progesterone production because they don't form a corpus luteum.
- This means they have ↑ estrogen in comparison to progesterone. We call this **unopposed estrogen.** This is one of the highest yield phrases for USMLE.
- Unopposed estrogen means ↑ risk of endometrial hyperplasia and, in turn, ↑ risk of endometrial carcinoma.
- The Q can give you a high-BMI female who's post-menopausal + has vaginal bleeding. Answer is just straight-up endometrial biopsy. Student asks how we know it's endometrial cancer. My response is, if she's overweight, this implies she was probably overweight in the past, which implies she's had history of anovulatory cycles and endometrial hyperplasia, leading to ↑ endometrial cancer risk.
- Because insulin resistance is the basis for PCOS, patients are at ↑ risk of developing type II diabetes.

- As mentioned above, hypothyroidism and Cushing syndrome are also HY causes of anovulation on USMLE.
- ↑ Glucocorticoids in Cushing syndrome can cause insulin resistance and anovulation, where the diagnosis can appear like PCOS. The difference is PCOS is idiopathic in response to high BMI – i.e., it is not caused by a known secondary etiology like Cushing syndrome, even though the presentations can be similar.
- Hypothyroidism leads to ↑ thyroid-releasing hormone (TRH), which stimulates prolactin, which causes abnormal GnRH pulsation. Even though this is the mechanism, the caveat I issue is that USMLE does not directly assess this, and it is infinitely more important you know that the mechanism for prolactin secretion is ↓ dopamine, or overt dopamine-2 receptor antagonism (i.e., for severance of pituitary stalk Qs).

Progesterone withdrawal test

- Progesterone withdrawal test is the process of giving progesterone followed by seeing if bleeding occurs following its withdrawal. It can be used to help diagnose the cause of oligo- or amenorrhea.
- As discussed earlier, menses are caused by the presence of progesterone followed by its withdrawal.
- If bleeding occurs (i.e., positive progesterone withdrawal test), it means estrogen levels are normal and the endometrial lining builds up just fine, and that anovulation is the cause of the irregular menses.
- In PCOS, for instance, since we don't have a corpus luteum and progesterone secretion, we don't have the sequence of "progesterone present followed by withdrawal," so we can't get regular menses. However, since estrogen levels are normal in PCOS, the endometrial lining can grow without a problem, so if progesterone is given exogenously, followed by its withdrawal, we will see bleeding.
- In conditions like Turner (where estrogen is low) or Asherman (scarred uterus that can't grow), giving progesterone won't induce bleeding since the endometrial lining hasn't developed. The bleeding will only occur if estrogen is present in normal amounts to allow for endometrial growth, where the administration, followed by withdrawal, of exogenous progesterone essentially induces forced menses.

Primary dysmenorrhea vs endometriosis

Primary dysmenorrhea	- Dysmenorrhea = period pain. - Primary dysmenorrhea = "normal period pain"; benign. - Mechanism is prostaglandin PGF2α hypersecretion. - During menstruation, the endometrial cells release high levels of prostaglandins, which cause uterine contractions and pain. The prostaglandins can also make their way into the systemic circulation, leading to headache and nausea. - This is the answer on USMLE for period pain under the age of 20. The Q can say, girl who is 17 has menstrual pain so bad she has to miss class + physical exam shows no abnormalities → answer = primary dysmenorrhea. - The answer can also be written simply as "prostaglandin secretion." - The key detail regarding primary dysmenorrhea is that **the physical exam is normal.** - In endometriosis, the physical exam is abnormal + is always over age 20. - Treatment is NSAID. OCP is wrong answer if listed alongside NSAID (on NBME).

Endometriosis	- Growth of endometrial tissue outside the uterus. Most common location is the ovary, but can also grow in other locations like the pouch of Douglas. - Mechanism is thought to be retrograde menstrual flow through the Fallopian tubes + seeding onto the ovaries or peritoneum. - Causes severe menstrual pain over age 20 + **an abnormal physical exam.** - The abnormal physical exam can be any miscellaneous finding – i.e., there is no specific finding you have to memorize. They can say findings such as nodularity of the uterosacral ligaments, a fixed retroverted uterus (due to adhesions from lesions), or pelvic tenderness. - In contrast, severe period pain + normal physical exam = primary dysmenorrhea. - In other words: - 23F + period pain so bad she has to miss work + normal physical exam → answer = prostaglandin secretion / primary dysmenorrhea. - 23F + period pain so bad she has to miss work + physical exam shows nodularity of the uterosacral ligaments → answer = endometriosis. - Descriptors such as pain with defecation (due to pouch of Douglas lesions) or dyspareunia (pain during sex) are highly buzzy and pass-level but too easy, so are often omitted from NBME questions, as per my observation. - USMLE wants diagnostic laparoscopy as next best step. - NSAIDs and OCPs are short-term measures, but definitive Tx is laparoscopic removal of lesions.

APGAR scoring

SCORE	0 points	1 point	2 points
Appearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
Activity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apneic	Slow, irregular breathing	Strong cry

- The 1-minute APGAR score is used to evaluate conditions at birth, evaluating heart rate, respiratory effort, muscle tone, reflexes, and color, with scores ranging from 0 to 10, indicating the need for immediate medical attention if low.
- The 5-minute APGAR score reflects response to resuscitation; more accurate indication of the baby's overall health and adjustment to life outside the womb.
- Scores >7 are typically considered normal.
- **Acrocyanosis** = bluish discoloration of neonatal extremities at birth (i.e., 1 point) for appearance. USMLE wants you to know this term. Next best step is place under warming lights, followed by tactile stimulation.

Notable Teratogens	
- Agent that causes malformations in the fetus. - Not an exhaustive list. Point is to be HY, not superfluous. - **Fetus is most susceptible at 3-8 weeks' gestation.**	
Agent	**Defect(s) in the fetus**
ACE inhibitors	- Renal issues.
Aminoglycosides	- Hearing issues.
Anti-epileptics	- Neural tube defects (disrupt folate metabolism). - Fetal hydrantoin syndrome (fingernail/digital hypoplasia, facial anomalies).
Carbimazole/methimazole	- Aplasia cutis congenita, choanal atresia, esophageal atresia.
Diethylstilbestrol (DES)	- Vaginal clear cell carcinoma 30-40 years later.
Isotretinoin	- Craniofacial and cardiac abnormalities; cleft lip/palate.
Lithium	- Ebstein anomaly (atrialization of right ventricle).
Methotrexate	- Neural tube defects.
NSAIDs	- Premature closure of ductus arteriosus; oligohydramnios; inhibited labor.
Ribavirin	- Craniofacial anomalies; growth deficits.
Tetracyclines	- Fetal teeth discoloration; inhibition of bone growth.
Thalidomide	- Phocomelia (severe flipper-like malformation of limbs). - Was used frequently in the 1950s-60s to treat nausea in pregnancy.
Warfarin	- Bone and facial anomalies. - CNS defects like Dandy-Walker malformation (HY Neuroanatomy PDF). - Bleeding diathesis in fetus.

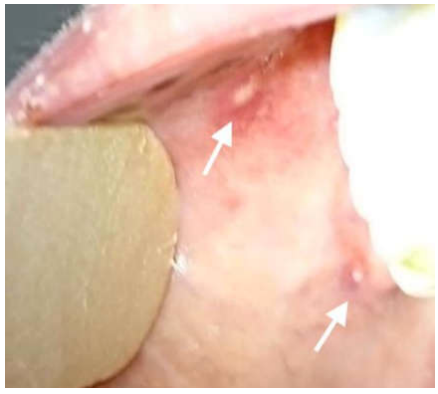
Substance abuse in pregnancy	
Agent	**Defect(s) in the fetus**
Alcohol	- Fetal alcohol syndrome; most common cause of mental retardation. - Heart/lung defects (defective neural crest migration). - **Smooth, flat, and/or elongated philtrum.** - Thin vermillion border (thin upper lip). - Widely spaced eyes (hypertelorism), midface hypoplasia, short nose.
Amphetamines + cocaine	- IUGR; cognitive deficits. - NBME can give long vignette of drug use, followed by ask what is most likely to occur in the pregnancy → answer = "preterm delivery and birth." Sounds non-specific and vague, but it's still *most likely* in comparison to most other things. - Cocaine in particular also increases risk of abruptio placentae.
Cocaine	- Neural tube defects (disrupt folate metabolism). - Fetal hydrantoin syndrome (fingernail/digital hypoplasia, facial anomalies).
Heroin/opioids	- Neonatal heroin/opioid withdrawal. - Piloerection (goosebumps), yawning, rhinorrhea.
MDMA (ecstasy)	- IUGR; cognitive deficits.
Smoking	- IUGR and reduced IQ.

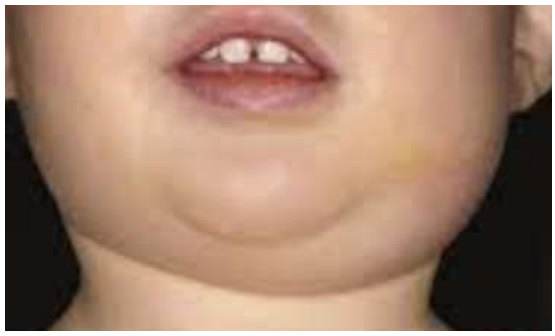
Peds Derm

HY Derm conditions for Peds	
Tinea capitis	 - Cause is dermatophytes (i.e., *Microsporum*; *Trichophyton*). - Classically circular, scaly area of scaly alopecia. - Tx = oral griseofulvin for patient only; "patient + classmates" is wrong answer on NBME.
Tinea corporis	 - Aka ringworm. - Q will often mention a dog at home. - Tx = topical miconazole or clotrimazole. - Tx for tinea pedis (athlete's foot) is topical terbinafine or topical -azole.
Cellulitis	

	- *Staph aureus* exceeds *Strep pyogenes* (Group A Strep) as causal organism; must give beta-lactamase-resistant beta-lactam in the methicillin class (i.e., dicloxacillin, flucloxacillin) or first-generation cephalosporin (i.e., cephalexin, cephazolin), or Augmentin (amoxicillin-**clavulanate**). - Amoxicillin and penicillin alone are wrong answers. - 90% of community *Staph* (i.e., MSSA) produces beta-lactamase, so amoxicillin and penicillin alone will not work if *Staph* is the cause.
Erysipelas	 - Infection of upper dermis and superficial lymphatics. - Caused by *Strep pyogenes* (Group A Strep), which eclipses *Staph aureus* for erysipelas. - Looks worse than cellulitis but is more superficial / "not as bad"; has characteristic "fiery red" appearance and may appear well-demarcated with raised edges. - Although Group A Strep > *Staph* for erysipelas, Tx is same as cellulitis (oral dicloxacillin, cephalexin, or Augmentin) because Staph can still cause it. Penicillin alone can be used for *Strep* pharyngitis.
Acne	 - Caused by *Propionibacterium acnes*; not difficult, but I've seen enough students select tinea faciei (fungal infection of face) for simple acne. - First-line Tx for acne on USMLE is topical retinoids (i.e., topical tretinoin; **not** oral isotretinoin; latter is only for severe acne). Topical retinoids (vitamin A) inhibit sebum production; they cause photosensitivity and desquamation (peeling). - Topical benzoyl peroxide is second-line for acne (although often co-administered with topical tretinoin). It clears pores and kills bacteria. - Topical clindamycin can be used if topical retinoids and benzoyl peroxide are insufficient; if topical antibiotic is insufficient, oral tetracycline is used; the latter causes blistering photosensitivity. - Last resort is oral isotretinoin; must do beta-hCG (pregnancy test) before commencement due to teratogenicity; oral isotretinoin does **not** cause problems

	with sperm in men; topical retinoids in both men and women do not cause teratogenicity.
Meningococcus	 - *Neisseria meningitidis;* gram-negative diplococcus. - Causes meningitis + characteristic non-blanching rash. - Low BP can be endotoxic shock, but student should bear in mind Waterhouse-Friderichsen syndrome is often asked; give hydrocortisone to increase BP after normal saline is administered.
Herpetic whitlow	 - One of the highest-yield spot-diagnoses on USMLE. - HSV1/2 infection of the finger. - In children, can be acquired via touching mother's cold sore (e.g., during breastfeeding). - In adults, classically dental workers. - Tx = topical and oral acyclovir.
Lyme disease	

	- *Borellia burgdorferi;* a spirochete. - Rash is erythema chronicum migrans (classic target rash); the rash need not be a target on USMLE; it can merely be circular with no clearing; but the target is classic. - USMLE can give two side by side images: 1) circular rash on limb that is not a target; 2) Bells palsy → student needs to infer this is Lyme disease even though rash isn't a target. - Tx age <8 (and pregnant women) = amoxicillin or 3rd-gen ceph, not doxycycline.
Scabies	 - *Sarcoptes scabiei.* - Lesions classically described as "linear burrows". - Can become superinfected with *Staph aureus.* - Tx = topical permethrin. - Disseminated scabies can occur in HIV patients; Tx is oral ivermectin. - Topical permethrin is also treatment for pediculosis (lice).
Bed bugs	 - Caused by an insect called *Cimex*. - Grows as clusters of itchy lesions on the limbs or torso.
Measles	

	- Aka rubeola; causes a head-to-toe macular popular rash. - Image shows Koplik spots (pathognomonic whiteish lesions on buccal mucosa). - Immigrant Hx on USMLE sometimes implies unvaccinated status. - Can rarely cause subacute sclerosing panencephalitis (reactivation of latent infection in the CNS in teenagers). - Rubella also causes a head-to-toe macular popular rash, but rather than Koplik spots, suboccipital and post-auricular lymphadenopathy (tenderness on back of head and behind ears) is characteristic.
Mumps	 - Causes POM → Parotitis, Orchitis, Meningitis. - Doesn't typically cause rash.
Fifth disease	 - The name of the "slapped cheek" facial erythema caused by Parvo B19. - Next best step = check serum IgM titers; if IgM titers are not listed, choose bone marrow biopsy. - Increased risk of aplastic anemia in sickle cell. - Once child has developed the red cheeks, he/she has immunologically cleared the illness (i.e., if they turn it into a behavioral science Q, tell parents to chill the fuck out / relax because the child has cleared the virus).
Roseola	

	- Caused by HHV6; described as “spiking fever followed by a rash” – i.e., child will have high fever for 2-3 days, followed by a rash.
Eczema herpeticum	 - HSV1/2 infection superimposed on eczema. - Often self-inoculated from touching cold sore then cracked skin of eczema. - Herpes infection can present with fever, lymphadenopathy, and tingling, burning, and painful rash (herpetic neuralgia). - Tx = acyclovir.
Dermatitis herpetiformis	 - Extra-intestinal cutaneous eruption that can occur in patients with Celiac disease. - Despite the name, this is not actual herpes. It is IgA deposition at dermal papillae. - Don’t confuse with eczema herpeticum.
Pediatric shingles	 - You need to know that pediatric shingles “is a thing.” - Classically seen in immunocompromised children (e.g., due to chemotherapy).

	- Shingles is aka “herpes zoster” and is caused by varicella zoster virus (VZV); herpes zoster is **not** a virus name; herpes zoster literally is just another name for shingles, which is caused by VZV (human herpes virus 3; HHV3), not HSV1/2 (HHV1/2).
Pityriasis rosea	 - Caused by HHV6 or 7; self-limiting. - Starts as Herald patch (larger pink ellipse above), usually on the back or trunk, then spreads upward onto the shoulder blades (“Christmas tree distribution”); USMLE will show you image and expect you can make spot-diagnosis. - Can occur in teenagers, although more common in the 20s.
Tinea versicolor	 - Caused by *Malassezia furfur*; fungus causes spotty hypopigmentation; patient usually lives in sub-tropical areas like Florida. - Can occur in teenagers who surf / go to the beach. - Tx = topical selenium.
Molloscum contagiosum	

	- Very HY spot-diagnosis for Peds; caused by Poxvirus. - Classically skin-colored or reddish papules with central umbilication. - USMLE likes giving vignette where kid went to a recent pool party.
Mycobacterium marinum	 - Unusual diagnosis but asked. - Vignette will say kid recently went to aquarium or water park + now has red lesions on the hand/arm. - Can occur in adults as well who work at aquariums or water parks.
MAI	 - *Mycobacterium avium intracellulare* (MAI); can cause lymphadenitis with classic reddish/violaceous lesion on the neck or pre-auricularly.
Eczema	 - Type I (immediate); can occur as part of atopy (i.e., asthma in winter, hay fever / rhinoconjunctivitis in spring; eczema in summer). - Remember that 1/3 of asthma patients have cough-variant asthma (i.e., only have dry cough, usually worse in the winter or with exercise).

	- Treat with oil-based emollient and topical corticosteroids; if steroids used >5-7 days continuously, thinning of the dermis may occur.
Urticaria	 - Aka hives; type I hypersensitivity; can be precipitated by various allergens, including pollen, pet dander, and drugs.
Erythema multiforme	 - Immune complex-mediated rash (type III hypersensitivity) with many etiologies; HSV1/2 infection is classic cause.
Aphthous ulcer	 - Aka aphthous stomatitis; no Tx necessary; **not** related to HSV1/2; etiology is multifactorial; can be precipitated by allergens such as spice; T cell-mediated.
Aplasia cutis congenita	

	 - Absence of skin on an area of scalp; can be caused by teratogens such as methimazole; dumb/seemingly pedantic detail, I know, but it shows up on real deal.
Hereditary angioedema	 - Deficiency of C1 esterase *inhibitor* (not C1 esterase alone). - Characterized by recurrent swelling of various bodily regions. - Tx = Danazol (androgen receptor partial agonist causes liver to produce more C1 esterase inhibitor).
NF1	 - Neurofibromatosis type I; autosomal dominant; one of the phakomatoses (neurocutaneous disorders → NF1/2, VHL, TSC, Sturge-Weber). - Image shows café au lait spot. - NF1 → café au lait spots, neurofibromas, axillary/groin freckling, optic glioma, pheochromocytoma.
Tuberous sclerosis	- Autosomal dominant; one of the phakomatoses. - Image shows adenoma sebaceum (angiofibromas). - Intracranial/periventricular nodules (“tubers” – i.e,. hamartomas), adenoma sebaceum (angiofibromas), cardiac rhabdomyoma, lymphangeoleiomyomatosis, subungual fibromas, renal angiomyolipoma.

	 Image shows adenoma sebaceum
Sturge-Weber	 - Classically Port Wine-stain birthmark (nevus flammeus), but can be described as "violaceous papules in a temporal distribution." - Associated with leptomeningeal angioma (causing seizure) and glaucoma. - Condition is not inherited and is due to somatic mosaicism.
McCune-Albright syndrome	 - Triad of "coast of Maine" café au lait spots (above image), polyostotic fibrous dysplasia (bone is replaced by fibrous tissue), and endocrine hypersecretion (classically precocious puberty).
Blue nevus	- Aka Mongolian spot. - Type of benign birthmark where dermal melanocytes fail to migrate superficially to stratum basale. - More common in Asians and blacks.

	- Often mistaken for child abuse. - Schedule routine follow-up; wrong answer is contacting child protective services. 
Acanthosis nigricans	 - Brown/black velvety skin, usually along nape of neck. - Associated with hyperinsulinemia and type II diabetes almost always. - Can also very rarely be seen in patients with underlying, visceral malignancy.
Strawberry hemangioma	 - Benign capillary tumor that will grow slightly then regress spontaneously within a few years. - No Tx necessary unless causing functional impairment. - Don’t confuse with cherry hemangioma, which is due to sun exposure in elderly and presents as a tiny cherry-colored papule.
Preauricular pit	

	- Preauricular pits are benign finding, usually not associated with any congenital disorder. - No Tx necessary. Can become infected; do surgical closure if frequent infections.
Kasabach-Merritt syndrome	 - Aka infantile hemangioma with thrombocytopenia; asked several times on 2CK Peds assessment. - Child will have thrombocytopenia due to platelet sequestration within the lesion. - Large and deforming; don't confuse with strawberry hemangioma. - Tx is surgical.
Cigarette burn	 - Q will show you image similar to above and then the answer will be "contact child protect services"; can also be on face / resemble impetigo if at different stages of healing.
Milia	 - Aka "milk spots"; clogged eccrine ducts; common, benign finding in babies. - No Tx necessary; - "Exfoliative cleanser" and "topical low-dose corticosteroid" = wrong answers.

Peds Neuro

Pediatric seizures	
Febrile seizure	- Fever can precipitate idiopathic seizure in 2-4% of children ages 6 months - 5 years. - About a two-fold risk progression to epilepsy compared to general population. - Tx is with benzodiazepine. - Febrile seizures lasting longer than 10 minutes, seizures that are recurrent within 24 hours, or focal neurologic signs more significantly ↑ risk of progression to epilepsy.
Absence seizure	- Vignette will be a kid staring off into space in class for 30 seconds spacing out, sometimes with rapid blinking. - This is considered loss of consciousness. - EEG shows symmetric 3-Hz spike-and-wave discharges. - Tx is ethosuximide (thalamic calcium channel blocker).
West syndrome	- Aka infantile spasms; X-linked recessive. - Epilepsy syndrome in infants characterized by – you'd never guess it – spasms. - Causes an abnormal EEG pattern called hypsarrhythmia, which is chaotic pattern. - Leads to intellectual disability. - Treatment is with ACTH (obscure, but apparently ↑ endogenous cortisol, which can mitigate the progression).
Lennox-Gastaut	- Severe childhood-onset epilepsy characterized by near-daily seizures and cognitive decline (hyperoralism is a sign of cognitive regression [babies put things in their mouths]). - Poor prognosis, with 5% mortality rate in childhood; 80-90% persistence of seizures into adulthood.
JME	- Juvenile myoclonic epilepsy. - Characterized by myoclonic jerks (usually hypnagogic and/or hypnopompic) that progress to tonic-clonic seizures after several months. - Age of onset is usually 10-16, but can also start in adulthood. - Tx is valproic acid.

Peds conditions misconstrued as seizures	
Adams-Stokes attack	- Idiopathic arrhythmia disorder in Peds that causes transient hypoxia to the brainstem, resulting in seizure-like episodes. - EEG does not show any seizure activity. - Q will give you kid with miscellaneous arrhythmia description + describe twitching that sounds like seizure → answer = Adams-Stokes attack.
Breath-holding spell	- They'll say 3-year-old was having a tantrum followed by falling on the floor + appearing blue. - Child will involuntarily stop breathing following a trigger – e.g., being upset, frightened, or experiencing pain. - The child often cries or becomes upset, exhales forcefully, stops breathing, then develops cyanosis, which can sometimes be followed by brief loss of consciousness and jerking (can be mistaken for seizure). - Shows up on Peds forms so you need to know it exists.

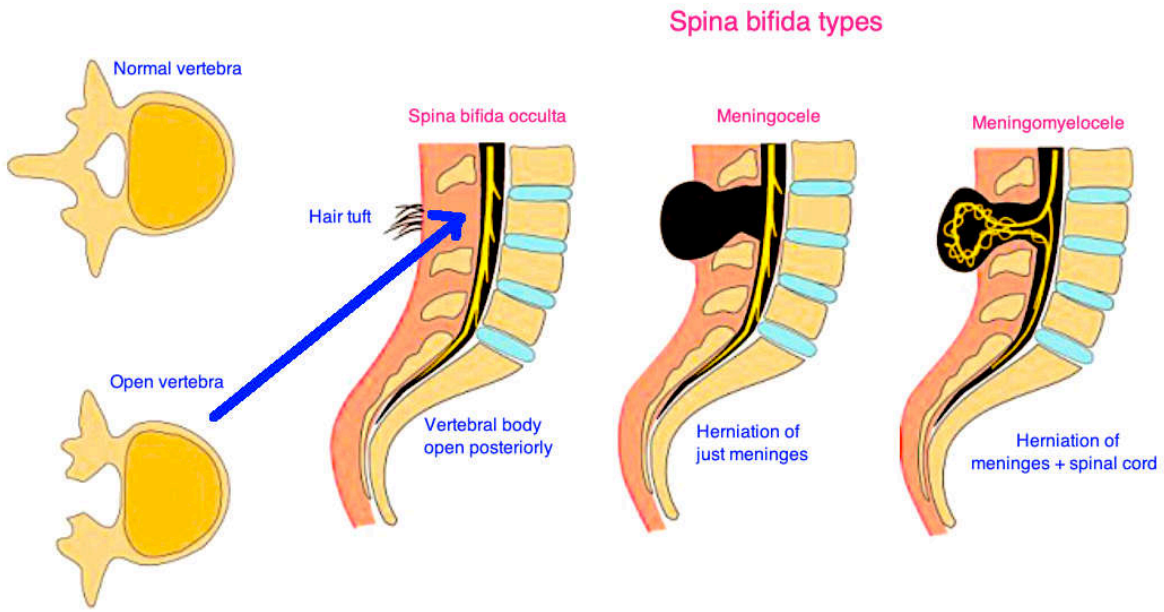
Cerebral palsy
- Motor dysfunction and intellectual disability of varying severity, posited to be due to intrapartum conditions such as hypoxia. - Will show up on USMLE as spastic paraparesis in a child who has "scissoring" of the legs.

Hydrocephalus	
Communicating	- CSF can flow freely between the ventricles and subarachnoid space around the brain and spinal cord. - The answer for meningitis causing hydrocephalus, where the mechanism is "failure of CSF reabsorption by the arachnoid granulations."
Non-communicating	- Aka obstructive hydrocephalus. - When there is a blockage preventing flow of CSF between the ventricles or subarachnoid space. - There is an NBME Q where they give a tumor of the spinal cord in the setting of hydrocephalus, and the answer is obstructive hydrocephalus. - Can be caused by aqueductal stenosis (narrowing of the cerebral aqueduct of Sylvius between the 3rd and 4th ventricles), which is the most common cause of congenital hydrocephalus. - Can be caused by colloid cyst, which is a benign fluid-filled growth at the anterior roof of the third ventricle, just posterior to the foramen of Monro.
NPH	- Normal pressure hydrocephalus. - "Wet, wobbly, wacky" +/- Parkinsonism. - In other words: urinary incontinence, ataxia, and cognitive dysfunction. - Due to impingement on the corona radiata (asked on an NBME), which is a white-matter bundle that connects the motor cortex superiorly to the internal capsule inferiorly. - Mechanism for urinary incontinence = "failure to inhibit the voiding reflex."
Congenital	- Aqueductal stenosis is most common cause. - Most important cause on NBME, in my opinion however, is congenital toxoplasmosis, which presents as a triad of 1) hydrocephalus, 2) chorioretinitis, and 3) intracranial calcifications.
Ex-vacuo	- Appearance of enlarged lateral ventricles on imaging without true hydrocephalus. - Can be caused by loss of surrounding brain matter. - Alzheimer and schizophrenia are important causes on USMLE, where enlargement of the lateral ventricles is a finding that can sometimes show up in question stems.

Autoimmune peds neuropathies	
Bell's palsy	- LMN lesion of CN VII (facial nerve). - Paralysis of ipsilateral facial muscles. - HY for Lyme disease for Peds. Q will show a target rash + give Bell's palsy. - If the Q asks for next best step, the answer is do serology for Lyme disease. If this isn't listed (or the patient doesn't live in endemic area, such as Africa), choose "no further diagnostic studies indicated." Nerve conduction studies are wrong (this is on NBME). Bell's palsy is a clinical diagnosis. - IV steroids, followed by a 10-21-day taper of oral steroids is indicated to minimize the immune response.
PANDAS	- Pediatric Autoimmune Neuropsychiatric disorder Associated with Streptococci. - In rare cases, Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask will most likely diagnose etiology for the disorder → answer = "anti-streptolysin O titers."
Sydenham chorea	- Autoimmune movement disorder seen as part of rheumatic fever. - Chorea = fast, purposeless, jerky movements. - Vignette will give child who has rheumatic fever and then ask about the cause of the movement disorder, and the answer is just "autoimmune." - I discuss rheumatic heart disease in detail in the HY Cardio PDF.

Pediatric brain tumors	
Pilocytic astrocytoma	- Most common brain tumor in kids. - Has solid (white) and cystic (black) components on imaging. - Not aggressive.
Medulloblastoma	- Aggressive brain tumor of cerebellum. - Can cause truncal ataxia if located at vermis, and ipsilateral limb ataxia if lateral.
Craniopharyngioma	- Most common pituitary tumor in children and teenagers. - Bitemporal hemianopsia (can occur with any pituitary lesion). - Embryo is that it originates from the "roof of the primitive oropharynx," or "roof of the primitive oral cavity." - Can calcify and contain cholesterol crystals; high rate of recurrence. - Don't confuse that embryo with "endoderm of foramen cecum," which refers to thyroglossal duct cyst.
Pinealoma	- Can cause vertical gaze palsy in kids (Parinaud syndrome).
Hamartomas	- The "tubers" in tuberous sclerosis are hamartomas, which are described as periventricular or cortical nodules on imaging.

Spina bifida



- HY point is that neural tube defects (spina bifida) occur **between 3-4 weeks gestation.**
- The pregnant woman needs adequate folic acid (vitamin B9) prior to and during early pregnancy for prevention.
- Antiepileptics (i.e., particularly valproic acid, but also phenytoin and carbamazepine) are also associated with neural tube defects.
- Neural tube defects lead to increased AFP and acetylcholinesterase in CSF.

HY Peds neuro conditions	
Spinal muscular atrophy	- Aka Werdnig-Hoffman syndrome. - Genetic disorder affecting anterior horns. - Presents with profound hypotonia, absent reflexes, and **tongue fasciculations.**
Cerebral palsy	- Motor dysfunction and intellectual disability of varying severity, posited to be due to intrapartum conditions such as hypoxia. - Will show up on USMLE as spastic paraparesis in a child who has “scissoring” of the legs.
Chiari 1	- Downward displacement of the cerebellar tonsils into the foramen magnum. - Usually teenager or young adult. - Presents with headaches and neck pain, but can also be asymptomatic.
Chiari 2	- Herniation of both the cerebellum and brainstem through the foramen magnum and is commonly associated with meningomyelocele. - Presents as a child with dysphagia and motor dysfunction.
Dandy-walker	- Absent cerebellar vermis + cystic dilation of 4th ventricle.
Fetal alcohol syndrome	- Most common cause of mental retardation. - Heart/lung defects (defective neural crest migration). - **Smooth, flat, and/or elongated philtrum.** - Thin vermillion border (thin upper lip). - Widely spaced eyes (hypertelorism), midface hypoplasia, short nose.
Phenylketonuria	- Autosomal recessive; deficiency of phenylalanine hydroxylase (or rarely THB, which is cofactor for the enzyme), causing inability to convert phenylalanine into tyrosine. - Screened for on heel-prick test at birth, since failure to detect PKU will result in mental retardation. - Phenylalanine-deficient diet must be given. - Presents as child with “mousy” or “musty” body odor + appears lighter skin than siblings (can cause partial albinism, since tyrosine is normally converted into melanin, and tyrosine can’t be synthesized).
Maple syrup urine disease	- Autosomal recessive; deficiency of branched-chain ketoacid dehydrogenase. - Inability to breakdown leucine, isoleucine, and valine. - Causes a maple syrup-scent in the diapers from the urine. - Mental retardation and neurologic abnormalities if untreated.
Cretinism	- Aka congenital hypothyroidism. - Most common causes are iodine deficiency in the mother during pregnancy (worldwide) and fetal thyroid dysgenesis (western countries). - Leads to mental retardation (poor myelin sheath development), impaired bone growth, hypotonia, macroglossia, and protuberant abdomen (due to umbilical hernia; can be confused with kwashiorkor, which is protein-calorie malnutrition causing ascites). - Screened for at birth using the heel-prick test to prevent exacerbation.
Waardenburg syndrome	- Congenital sensorineural hearing loss + pigmentary changes of hair and skin. - Shows up on NBME as young child with “white forelock” of hair. - Answer = failure of neural crest migration.
Double cortex syndrome	- Obscure condition asked on USMLE students are hysterical about. - Genetic condition where neurons are unable to migrate outwardly to reach their destination in the cortex, resulting in two grey matter layers (cortex) instead of one. - Mechanism USMLE wants is “failure of radial migration.”
Rett syndrome	- Neurologic degeneration that occurs exclusively in female infants. - “Hand-wringing” or “hand-flapping” is buzzy finding. - Mental regression can present as putting objects in one’s mouth (something babies normally do).
Cri-du-Chat	- Deletion on short arm of chromosome 5. - Results in distinctive cat-like cry during infancy, which is the source of the syndrome's name (French for "cry of the cat"). - Mental retardation, distinctive facial features, and microcephaly.
Friedreich ataxia	- Autosomal recessive GAA TNR disorder.

	- Causes ataxia, kyphoscoliosis, cardiomyopathy, pes cavus (high-arched feet), hammer toes, and early-onset type II diabetes.
Angelman	- "Happy puppet" – i.e., child presents with laughter. - Paternal imprinting → gene coming from dad is normally silenced; mom's gene is supposed to be expressed but is deleted or mutated. **- "Mom is not an angel"** because mom's gene is deleted. - Even if dad's gene is healthy, since it is normally silenced, it will not be expressed. - Can also be caused by paternal uniparental disomy. Since the gene is normally paternally imprinted, if both alleles are inherited from dad, both will be silenced; disease occurs because there's no maternal allele available to be expressed.
Prader-Willi	- Mental retardation + hyperphagia. - Maternal imprinting → gene coming from mom is normally silenced; dad's gene is supposed to be expressed but is deleted or mutated (**"Willi hates his dad"** because dad's gene is deleted). - Even if mom's gene is healthy, since it is normally silenced, it will not be expressed. - Can also be caused by maternal uniparental disomy. Since the gene is normally maternally imprinted, if both alleles are inherited from mom, both will be silenced; disease occurs because there's no paternal allele available to be expressed.

Meningitis	
- Presents as high fever + **stiff neck** and photophobia.	
Bacterial	- Usually Group B Strep in neonates (gram-positive cocci). - CSF shows ↑ Protein, ↓ glucose, ↑ neutrophils (polymorphonuclear cells). - NBME form has
Fungal	- *Cryptococcus neoformans* in immunocompromised patients (i.e., pediatric HIV). - CSF shows ↑ Protein, ↓ glucose, ↑ lymphocytes.
Viral	- Aka aseptic meningitis. - Commonly echovirus (if vaccinated); can also be mumps (if unvaccinated). - CSF shows normal protein (or a tad elevated), normal glucose, ↑ lymphocytes.
Encephalitis	
- Presents as high fever + **confusion.**	
HSV 1/2	- Herpes encephalitis can occur following vertical transmission. - **RBCs in the CSF** due to temporal lobe hemorrhage. - Non-contrast CT of the head is often negative, so don't be confused by this. However they can say EEG shows abnormal spikes over the temporal region. - Other CSF findings same as viral meningitis. - I've only ever seen one NBME Q on 2CK Neuro CMS form where RBCs were absent in CSF in herpes encephalitis, but you could still get the answer because other CSF findings were viral + no other virus listed.
Step 2CK/3 only	
- Do lumbar puncture before antibiotics, as the latter first can interfere with culture results. - Four indications for doing CT before LP in suspected meningitis / encephalitis: 1) Focal neurologic signs. 2) Seizure. 3) Papilledema, or the optic fundi cannot be visualized. 4) Confusion that interferes with neurologic exam / reduced Glasgow score. - The reason for CT before LP in the above scenarios is because of the risk of intracranial mass lesion, where if LP is performed, which acutely ↓ intracranial pressure, herniation and death might occur. - 2CK NBME Q gives point #3 above, where answer is CT, not LP, and many students get it wrong. - Empiric Tx for Peds meningitis is cefotaxime + vancomycin. Steroids can be added if confirmed *S. pneumo*. - For Group B Strep only, choose ampicillin + gentamicin, or ampicillin + cefotaxime; the combo of vancomycin + ceftriaxone is wrong on the NBME for GBS. - 2CK NBME Q has amp + gent as correct over vanc + ceftriaxone for GBS.	

- We normally avoid ceftriaxone in the first month of life and prefer cefotaxime as the 3rd-gen cephalosporin. The exact age of the transition is not so important. What you need to know is, if you get a Peds Q and both cefotaxime and ceftriaxone are listed, cefotaxime is what we generally prefer. - Ceftriaxone, compared to cefotaxime, displaces bilirubin from albumin, thereby ↑ risk of neonatal hyperbilirubinemia. It can also cause calcium deposits in the lungs and kidney.

Toxin-induced neuropathies for Peds	
Botulism	- Causes flaccid paralysis. - Botulin toxin of *Clostridium botulinum* inhibits release of presynaptic acetylcholine, which is normally stimulatory at muscles. - Can present as floppy baby syndrome; can also cause cranial nerve palsies. - Acquired as spores in honey in infants under 1; acquired as pre-formed toxin from canned goods in anyone older.
Tetanus	- Causes spastic paralysis. - Tetanus toxin of *Clostridium tetani* inhibits release of presynaptic neurotransmitters GABA and glycine, which are normally inhibitory. - Can present as opisthotonos (arched back) and trismus (lock-jaw). - Presents in two patients on USMLE: 1) neonate born at home whose umbilical cord was cut with a kitchen knife + tied with twine; 2) random dude who cut himself in back yard. - DtaP given at 2, 4, 6, months, then again at 15-18 months, then again at 4-6 years. - School-age kids require Tdap booster at 11-12 years, followed by Td booster every 10 years thereafter.

Rinne + Weber tests	
Rinne test	- A vibrating tuning fork is placed against the mastoid bone (bone conduction; BC). Once the patient can no longer hear it, it is removed from the bone and placed in front of the ear canal (air conduction; AC). - For both normal hearing and sensorineural hearing loss, AC > BC. - For conductive hearing loss, BC > AC.
Weber test	- A vibrating tuning fork is placed on the center of the forehead or top of the head, and the patient is asked if they hear the sound equally in both ears or if it's louder in one ear. - If it is louder in one ear, this is called “lateralization” to that ear.
Normal hearing	- Rinne shows air > bone conduction; Weber does not lateralize.
Conductive on left	- Rinne shows bone > air conduction; Weber lateralizes (is louder) to left.
Conductive on right	- Rinne shows bone > air conduction; Weber lateralizes to right.
Sensorineural on left	- Rinne shows air > bone conduction; Weber lateralizes to right.
Sensorineural on right	- Rinne shows air > bone conduction; Weber lateralizes to left.

Peds Psych

Depressive disorders	
Major depressive disorder (MDD)	- At least 5 out of 9 SIGECAPS must be present for at least a 2-week period. S – Sleep disturbance (insomnia or hypersomnia). I – Interest loss. G – Guilt or feelings of worthlessness. E – Energy loss. C – Concentration problems. A – Appetite changes (usually causes weight loss, but sometimes weight gain). P – Psychomotor agitation (restlessness; slowed speech or movements). S – Suicidal ideation. - The symptoms must cause socio-occupational impairment and must not be attributed to a substance or another medication condition. - Terminal insomnia (patient wakes up too early in the morning) can be seen. - Tx is with drugs such as SSRIs, CBT, and occasionally ECT.
Depression due to a medical condition	- Hypothyroidism is important cause of low mood and apathy. There is 2CK Peds Q where they give Hashimoto in a 14-year-old. - Can occur after severe diagnoses, e.g., ALL.
Atypical depression	- Hypersomnolence + hyperphagia in the setting of depression that is reactive / ameliorates when positive circumstances present themselves. - This latter aspect is called “mood reactivity,” where positive events of experiences can significantly improve mood; this distinguishes atypical depression from other forms of depression, where mood usually remains persistently low regardless of external circumstances. - Tx = CBT, SSRIs, or monoamine oxidase inhibitors (MAOIs). - MAOIs are considered to be highly efficacious but are not usually used first-line because of serotonin syndrome risk (discussed later).

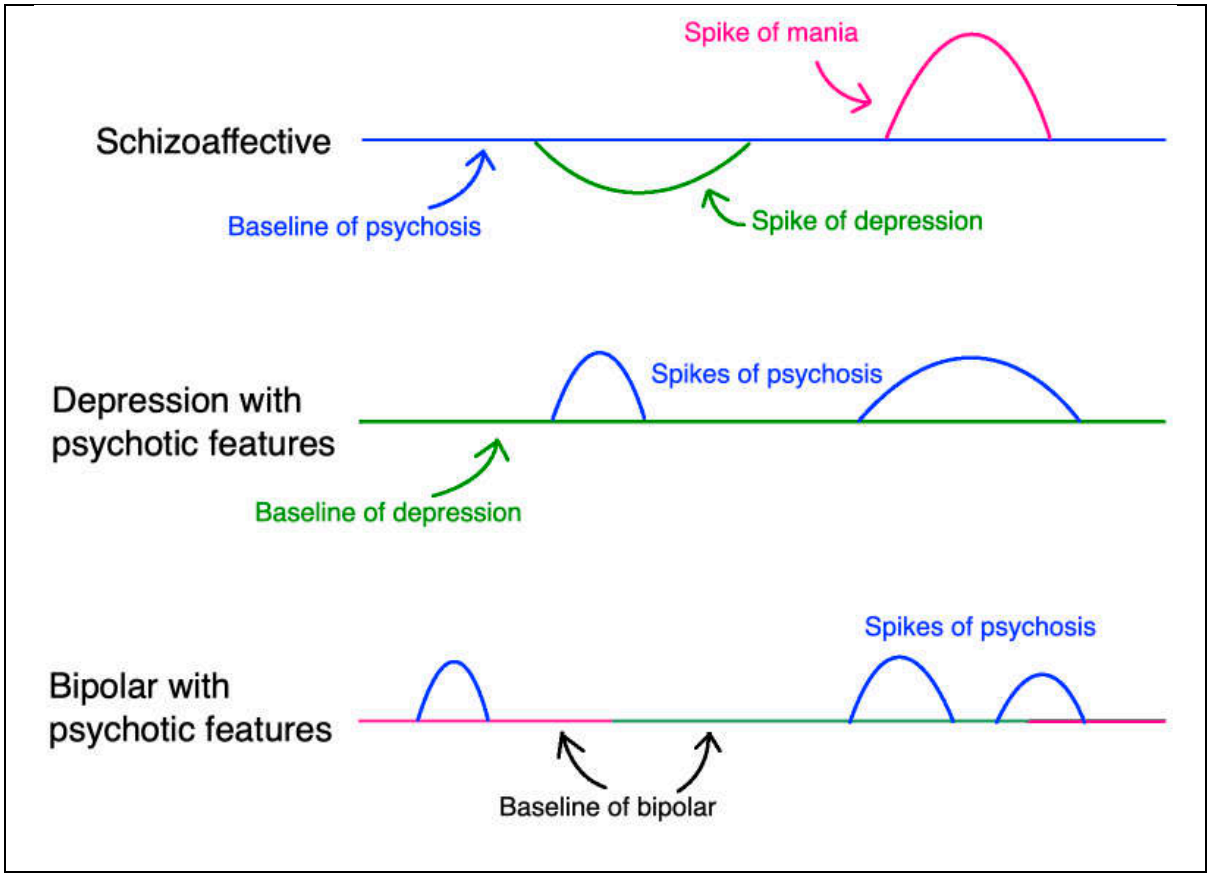
Dysthymia and cyclothymia	
Dysthymia	- At least 2 years of depressed mood without fulfilling MDD criteria.
Cyclothymia	- At least 2 years of oscillation between depressed mood and hypomania (discussed below).

Bipolar disorder I and II	
- **DIGFAST** is the mnemonic for remembering how mania and hypomania present: - **D** – Distractibility (easily distracted / not able to focus). - **I** – Impulsivity (e.g., shopping sprees, reckless driving). - **G** – Grandiosity (inflated self-esteem / having sense of special powers or gifts). - **F** – Flight of ideas (rapidly shifting thoughts/ideas, sometimes with lacking coherence). - **A** – Activity increase (goal-directed activity; sexual activity; psychomotor agitation). - **S** – Sleep deficit (decreased need for sleep). - **T** – Talkativeness (pressured speech; term means haphazard loquaciousness, sometimes without pausing).	
Mania	- State of intensely elevated, expansive, or irritable mood, often with marked impairment in judgment. - Usually lasts longer than a week and causes socio-occupational dysfunction (i.e., the patient works as dentist and can’t go to work / has problems at work). - Can sometimes cause psychotic features and require hospitalization. - ↑↑ risk of suicide.
Hypomania	- Usually lasts longer than 4 days and does not cause socio-occupational dysfunction (i.e., the Q says patient can go to school no issues). - Does not cause psychotic features and does not require hospitalization.

Bipolar I	- Diagnosed if patient has at least one manic episode, with or without a depressive episode.
Bipolar II	- Diagnosed if patient has at least one hypomanic episode + at least one major depressive episode. The patient must have never had a full-blown manic episode. - The reason at least one major depressive episode is required for diagnosis is because a hypomanic episode might be part of cyclothymia, rather than bipolar II, if the patient has merely experienced depressed mood in the past without a full-blown depressive episode.
- Treatment on USMLE is lithium or valproic acid. - Lithium can cause nephrogenic diabetes insipidus, hypothyroidism, and tremor. It is also a teratogen (causes Ebstein anomaly). - Valproic acid causes neural tube defects (interference with folate metabolism), tremor, and hepatoxicity. If the Q gives you increased LFTs + tremor, choose valproic acid over lithium.	

Psychotic disorders	
Schizophrenia	- Four points for diagnosis. 1) At least 6 months in duration. 2) Two or more of the following: - Hallucinations (always auditory on USMLE; hearing voices that aren't there). - Delusions (false beliefs; can be "bizarre," which refers to stuff like demons, aliens, and "the lord"). - Disorganized speech (often nonsensical or muffled speech). - Disorganized or catatonic behavior. - Negative symptoms (flattened/blunted affect, anhedonia [inability to feel pleasure], diminished speech). 3) Must have socio-occupational dysfunction (i.e., significant impairment in school, interpersonal relationships, or self-care). 4) Not attributed to a mood disorder (i.e., depression or bipolar), substance use, or another medical condition.
Schizophreniform	- Same as schizophrenia but 1-6 months in duration.
Brief psychotic disorder	- Same as schizophrenia but <1 month in duration.

Commonly confused conditions	
Schizoaffective	- At least two weeks of psychosis without a mood disorder (i.e., neither depression nor mania/hypomania), followed by periods of time with simultaneous psychotic and mood disorder. - For example, the Q will say a student has been hearing voices for the past 6 weeks + has been staying up all night for the past month (i.e., 4 weeks of mania or hypomania + at least 2 weeks of psychosis alone). - The name of the condition starts with "schizo," so you can remember that psychosis is the baseline of the disorder (i.e., what the patient will have as his/her underlying baseline). Then the mood disorder spikes on top of it.
Depression with psychotic features	- Depression, rather than psychosis, is the baseline. The patient will have periods of time in which he/she only experiences major depression without any type of psychosis. Then the psychosis periodically spikes on top of the depression.
Bipolar with psychotic features	- Bipolar is the baseline, where the patient will have periods of either depression *or* mania alone, followed by psychosis spiking on top of it. - If the patient has periods of mania alone followed by a simultaneous spike of psychosis, this makes the diagnosis easier. - If the patient has a period of depression alone followed by spiking psychosis, it can appear like depression with psychotic features, except the patient will *also* have a history of at least one manic episode in the past.



Delusion disorder
- One or more non-bizarre delusion, without other signs of psychosis. - By non-bizarre, as discussed earlier, this refers to potentially plausible scenarios (e.g., people at school are stealing from or trying to undermine you) that are not true, whereas bizarre on USMLE is anything that involves aliens, demons, the heavens, the lord, etc. - The person's behavior and thinking, apart from the delusion itself, tend to be normal. - Tx = CBT, anti-psychotics (e.g., risperidone), mood-stabilizers (lithium), and/or anti-depressants (SSRIs).

Anxiety disorders	
Generalized anxiety disorder (GAD)	- 6+ months of excessive and persistent worry about various aspects of life, often accompanied by physical symptoms like restlessness, fatigue, and muscle tension. - USMLE might give teenage girl who is worried about many things for 6+ months. - Patient can also have concurrent mood or psychotic disorder. In this case, the diagnosis becomes, e.g., "GAD with comorbid MDD." - Tx for GAD is CBT and SSRIs. - Buspirone (serotonin receptor agonist) can also be used.
Panic disorder	- Panic disorder is diagnosed if patient has 2 or more panic attacks + at least one month of worry about having more attacks. - A panic attack is an episode of intense fear that triggers severe physical reactions, even though there is no real danger or apparent cause. - USMLE will usually give patient with hyperventilation, where he/she feels like he/she is going to die or is having an MI. The USMLE is obsessed with making you think panic attack is cardiac. Don't get fooled. - Vignette can mention a mid-systolic click (mitral valve prolapse) + ask you the cause of the patient's symptoms → answer = panic disorder, not mitral valve prolapse.

	Student is confused because they say, "But wait, the patient has a mid-systolic click though." You're right. But the MVP itself isn't the cause of the patient's presentation. MVPs are common in the population and almost always asymptomatic. - Tx = breathing exercises to encourage the patient to stop hyperventilating. Breathing into a paper bag is wrong answer. Often times they won't have breathing exercises as an answer, where benzo is correct.
Social phobia	- Fear of being judged, negatively evaluated, or rejected in a social or performance scenario. Presents on USMLE as fear of public speaking. - USMLE wants beta-blocker (propranolol or atenolol) as 1st-line Tx. - If Q gives you an asthma patient, choose benzo. They make this distinction on NBME. - CBT for longer-term management.
Specific phobia	- Fear of a specific object or situation leading to significant distress or functional impairment. High-yield example on USMLE is fear of flying. - USMLE wants benzo for acute relief; CBT for longer-term.
Agoraphobia	- Fear of being in places or situations from which escape might be difficult or embarrassing, or where help might not be available if one were to have a panic attack. - This usually refers to crowded, open spaces.
Separation anxiety disorder	- Excessive distress with separation from home or major attachment figure. - Peaks at 12-18 months of age and usually subsides by 2-3 years, but children can still get this up into their teenage years. - USMLE can give a vignette of a child going to summer camp or school who gets stomach aches on arrival.
Selective mutism	- Disorder where patient fails to speak in certain situations, such as when confronted with new people, despite being able to speak in other situations.

PTSD vs Acute stress disorder	
PTSD	- Post-traumatic stress disorder; presents as at least 1 month of reliving a traumatic event, usually with flashbacks or nightmares about the event. - Treatment includes CBT and group therapy. - SSRIs if medication used.
Acute stress disorder	- Same as PTSD, but presenting for 3 days to <1 month since the traumatic event.

Adjustment disorder
- Socio-occupational dysfunction starting within 3 months of one specific stressor (i.e., a breakup, loss of job, death, illness). - Usually lasts <6 months. If longer than 6 months, it is called chronic adjustment disorder. - The key is the single stressor, whereas GAD is many, or no one specific, stressor. - The patient cannot have any psychotic symptoms but can have a mood disorder, where we call it "adjustment disorder with depressed mood," or "adjustment disorder with anxious mood," etc. I've seen NBMEs, particularly on 2CK psych forms, write answers like this. - First-line Tx is CBT and SSRIs.

Obsessive-compulsive disorder (OCD)
- Obsessions = Unwanted and intrusive thoughts, images, or urges that cause significant anxiety or distress. - Example is constantly feeling the need to touch a doorknob correctly before opening the door. - Compulsions = Repetitive behaviors a person performs in response to an obsession. - Example is touching the doorknob repeatedly in order to make sure it is touched "correctly."

- OCD is persistent obsessions and/or compulsions causing the patient significant distress or interfering with the patient's socio-occupational functioning. The patient need not have both in order to be diagnosed – i.e., a patient can just have either obsessions or compulsions alone.
- A key point that distinguishes OCD from OCDP (obsessive-compulsive personality disorder) is that OCD is ego-dystonic, whereas OCDP is ego-syntonic – meaning, in OCD, the patient doesn't like it / feels distressed; in OCDP, the patient is content / doesn't view anything as wrong.
- Tx = CBT. If drugs, use SSRI or clomipramine (a TCA). The latter shows up on 2CK Psych form and is a mini-factoid about OCD (i.e., you can use the TCA clomipramine to treat it).

Attention-deficit/hyperactivity disorder (ADHD)
- Inattentiveness (i.e., distractibility, not listening when spoken to) and/or hyperactivity (i.e., fidgeting, inability to remain seated or engage in activities quietly) *at home and at school.* The pattern of behavior must be consistent across venues and daily situations the child is immersed in. - Tx = 1st-line is stimulant meds, such as methylphenidate (Ritalin). Other non-stimulant meds, such as atomoxetine can be attempted 2nd-line. - External to meds, family therapy, behavioral skills training, and parenting skills training can be pursued. - Teenagers or adults can undergo CBT.

Conduct vs oppositional defiant disorder	
Oppositional defiant	- Argumentative and defiant behavior, often leading to problems at school with teachers and/or problems with grades. - The vignette can sound like normal teenage behavior, but the emphasis will be that there is impediment to social and/or scholastic progression. - Another key point is that there is no law-breaking. If the vignette mentions anything about crimes, then the USMLE wants conduct disorder instead.
Conduct disorder	- >6-month pattern of law-breaking starting age 17 or younger. - Vignette might give a teenager who engages in criminal behavior, such as killing a cat, destroying property, or engaging in theft. - USMLE won't necessarily present to you a pattern of ongoing behavior, but rather just a snapshot of a child + ask for the diagnosis – i.e., 14-year-old killed a cat; what's the most likely diagnosis → answer = conduct disorder (because what he did is a crime); oppositional defiant disorder is wrong answer.

Autism spectrum disorder (ASD)
- Neurodevelopmental condition characterized by repetitive patterns of behavior, interests, or activities, and persistent challenges in social communication and interaction. - Usually starts before age 3; more common in boys. - Child demonstrates difficulty understanding social cues, displays atypical speech patterns or body language, and/or intensely focus on specific activities. - Interest in linear patterns – i.e., trains, train tracks, and organizing toys in straight lines, can be seen. - Repetitive motions (stimming), such as rocking back and forth, are characteristic. - Patients with ASD might have reduced IQ at the expense of enhanced abilities elsewhere (e.g., painting). - Asperger disorder is a variant of ASD where IQ is normal or increased.

Eating disorders	
Anorexia	- Intense preoccupation with maintaining low body weight and an intense fear of gaining weight. BMI must be low (i.e., <18.5) for diagnosis. - Purging can be seen (not limited to bulimia). - Vignette can give girl who runs 12 miles per day and barely eats. - USMLE likes metatarsal stress fractures due to decrease bone density. - The Q can ask what the patient is at risk for later in life → answer = osteoporosis (reduced adipose → reduced estrogen). - Amenorrhea in females can be seen but is no longer mandatory for diagnosis. USMLE wants "abnormal GnRH pulsation" as the mechanism for amenorrhea in anorexia, where LH and FSH are both decreased. This is called hypogonadotropic, or central, amenorrhea. - BMI <15 or overt malnutrition is indication for hospital admission. - Tx is CBT and/or SSRI. Mirtazapine is an α2-antagonist that stimulates appetite and is used for patients with depression and anorexia. - Avoid bupropion in patients with eating disorders (lowers seizure threshold). - Most common cause of death is ventricular fibrillation due to hypokalemia. - Olanzapine can be used in theory (anti-psychotic that ↑ appetite).
Bulimia	- Binge eating followed by compensatory behaviors such as vomiting, excessive exercise, or the use of laxatives. - BMI is normal or increased (unlike anorexia, where it must be decreased). - Tx is CBT and/or SSRI.
Binge-eating disorder	- Frequent episodes of consuming large amounts of food in a short time period, often in secret; usually accompanied by feelings of guilt or shame. - Unlike bulimia, these episodes are not regularly followed by purging, excessive exercise, or other compensatory behaviors. - Tx is CBT and/or SSRI.

Dissociative disorders	
Dissociative identity disorder (DID)	- Presence of two or more distinct identities that take control of the patient's behavior, with memory gaps occurring during these transitions. - Individuals with DID often report experiences of "time loss" or being told of actions they don't recall performing.
Dissociative amnesia	- Inability to recall important personal information, typically related to a traumatic or stressful event, that goes beyond repression. The amnesia can result in loss of identity and/or life history. - Dissociative fugue is a subtype of dissociative amnesia. - Fugue = memory loss + travel, where the patient will have traveled to a destination (e.g., parking lot 50 miles from his/her home + doesn't know who he/she is when the police arrive).
Depersonalization / Derealization Disorder	- Patients feel that they are detached from themselves (depersonalization) or that the world around them is unreal or dreamlike (derealization). - Experiences can be distressing, but the patient is aware that these perceptions are not reflective of reality.

Somatic symptom and related disorders	
- Symptom generation and motive are both unconscious.	
Somatic symptom disorder	- One or more chronic somatic symptoms (e.g., GI disturbance/pain, adnexal pain, shortness of breath, neuropathic pain) that are distressing or result in socio-occupational dysfunction, but in which medical examination and diagnostic interventions cannot find any abnormalities.

Conversion disorder	- Neurologic deficits (i.e., paralysis, loss of function, numbness) usually following a stressor, e.g., loss of job, argument with spouse, etc. - Patient has La belle indifference, which refers to apparent lack of care or concern about the deficit (i.e., the patient supposedly can't move his arm but doesn't seem fazed by it). - Example is a child being dropped off at summer camp + says he can't move his finger → child is likely stressed by being separated from parents. - Hoarseness of voice shows up on NBME form where answer is conversion disorder. - NBME also gives guy with blindness in one eye, but there's no relative afferent pupillary defect (indicating CN II isn't fucked up), so Dx = conversion disorder. If you're confused about RAPD, go to HY Neuro PDF.
Illness anxiety disorder	- Formerly known as hypochondriasis. - Preoccupation with fear of having or acquiring a serious illness; worries persist even after medical reassurances of no significant health concern/risk.

Factitious disorders	
- Symptom generation is conscious/intentional; motive is unconscious. - Production of symptoms in factitious disorders is driven by a need to be seen as ill or injured, rather than for a tangible benefit such as drugs, money, or time out of prison. - Seeking medical attention/care is known as attempting to seek "primary gain," whereas something concrete like cash or time out of prison is "secondary gain."	
Factitious disorder imposed on self	- Formerly known as Munchausen syndrome. - Intentional production / feigning of physical symptoms (e.g., GI pain). - The primary motivation is to assume the sick role and receive medical attention, care, or sympathy. - Willing to undergo invasive procedures. The Q might say patient has history of appendectomy.
Factitious disorder imposed on another	- Formerly known as Munchausen syndrome by proxy. - Feigning symptoms in another person in order to achieve medical attention. - Considered a form of abuse. Can occur with parents or caretakers toward their children, elderly, or those in their care. - Needs to be reported.
Factitious thyrotoxicosis	- Injection of thyroid hormone. Aka surreptitious thyrotoxicosis. - USMLE vignette will usually make this an endocrine-type Q (see HY Endocrine PDF), where it need not be attached to the patient's desire to seek medical care (i.e., the patient could just be attempting to lose weight), however some patients will inject as a means to acquire medical attention.
Factitious use of glycemic meds	- Use of insulin or hypoglycemic meds (e.g., sulfonylureas) to achieve hypoglycemia or weight loss. - USMLE will usually ask this in the setting of C-peptide levels (i.e., low C-peptide indicates exogenous insulin; high C-peptide means either secretagogue abuse like sulfonylureas, or insulinoma). - Some patients use these meds to achieve medical attention.

Malingering
- Symptom generation and motive both intentional. - Aim is secondary gain – i.e., money, drugs, time out of school/detention. - Patient is not willing to undergo invasive procedures. For example, a child complains of severe RLQ pain and feigns a rigid abdomen to get out of detention, but once at hospital, refuses to undergo appendectomy.

Sleep disorders	
Narcolepsy	- Chronic sleep disorder affecting the brain's ability to regulate sleep-wake cycles normally. - Characterized by excessive daytime sleepiness, sudden episodes of muscle weakness (cataplexy), sleep paralysis, and hallucinations. The latter tend to occur prior to sleep (hypnagogic) or upon waking up (hypnopompic). - Thought to be due to deficiency of a neurotransmitter called orexin (aka hypocretin), which normally promotes wakefulness. - Treatment is modafinil (dopamine reuptake inhibitor + promotes release of orexin from hypothalamus).
Sleep apnea	- Can be obstructive (i.e., usually from obesity) or central (i.e., brain-related). - Chronic fatigue and poor oxygenation can lead to dysthymia / depression. The answer on NBME is "mood disorder due to a medical condition." - Polysomnography (sleep study) is what USMLE wants to diagnose. - When obstructive sleep apnea progresses to the point that the patient is a chronic CO2 retainer with pulmonary hypertension and/or cor pulmonale, we call it obesity hypoventilation syndrome (Pickwickian syndrome). If you're confused about the cardio, go to the HY Cardio PDF.
Circadian rhythm sleep-wake disorder	- Misalignment between the individual's sleep pattern and the societal norm or natural environment. - Common types include delayed sleep phase disorder (going to bed / waking up later than desired), advanced sleep phase disorder (going to bed / waking up earlier than desired), and shift work disorder (struggle with sleep due to abnormal work hours).
Somnambulism	- Sleep walking. - Occurs during Stage 3 (deep) sleep.
Nightmare disorder	- Occurs during REM sleep. - Dreams often involve threats to survival or self-esteem, leading to awakenings and distress. - Episodes can be remembered by the patient after awakening since they occur during REM sleep.
Sleep terror disorder	- Often mistaken for nightmares; more common in male children and abates by pre-adolescence. - Occur in Stage 3 (delta wave) sleep. - Episodes where the patient somnambulates and can wake up at different location, often in a panic; can demonstrate violent behavior, such as pulling on the mother's arm to go outside, followed by awakening from the episode with no memory of what happened. - This is in contrast to nightmare disorder, where the patient remembers a dream + cannot somnambulate due to muscle paralysis during REM.
Restless leg syndrome	- Idiopathic, irresistible urge to move legs while in bed/sleeping. - Most common cause is iron deficiency anemia. First step is checking the patient's serum iron and ferritin. - If iron studies are normal, gabapentin and D2 agonists (ropinirole, pramipexole) can be used. - USMLE wants you to know that patients with RLS have increased risk of developing Parkinson disease, which makes sense since D2 agonists help, indicating a potential problem with dopamine signaling or production in some patients.
Insomnia	- Inability to obtain adequate sleep, either due to difficulty falling asleep, staying asleep, or waking up too early, leading to impaired daytime functioning. - USMLE wants benzo (usually clonazepam) for acute insomnia. - Terminal insomnia can occur in depression (waking up too early).

Tic disorders	
Tourette	- Neurologic disorder characterized by repetitive, involuntary movements and vocalizations (i.e., tics). - Motor tics range from mere eye blinking, head jerking, or shoulder shrugging to more complex movements such as hopping, twirling, or even mimicking others' actions. - Vocal (phonic) tics include throat clearing, sniffing, or humming. More complex vocal tics can involve repeating words or phrases, or shouting inappropriate words (coprolalia). - Haloperidol (typical anti-psychotic; D2 antagonist) and clonidine (α2 agonist) are two agents used to Tx that show up on NBME.
Provisional tic disorder	- In pediatrics it is normal for the kid to sometimes have a singular tic that lasts up to a year (e.g., unusual tongue or lip gesture). - There is very low chance of it progressing to chronic tic disorder / Tourette. - USMLE wants follow-up in 3-6 months as the next best step (i.e., observe).
PANDAS	- Pediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococci, where Group A Strep pharyngitis can cause a tic disorder, OCD, or ADHD in the weeks following infection. - USMLE will give you kid who had sore throat two weeks ago + now has a new-onset tic, ADHD, or OCD, and they will ask what will most likely diagnose etiology for the disorder → answer = "anti-streptolysin O titers."
Drug-induced	- Stimulants such as methylphenidate, amphetamines, and caffeine can cause tics. - Tardive dyskinesia due to anti-psychotics is considered a type of tic disorder.

Miscellaneous body disorders	
Trichotillomania	- Pulling out one's hair; can be anywhere (i.e., eyebrows, scalp, etc.). - Patient may sometimes also eat his/her hair, leading to gastric bezoar (asked on NBME exam). This is an undigestible mass that can lead to bowel obstruction.
Skin-picking disorder	- Aka excoriation disorder. - Patient picks at his/her skin, leading to skin lesions.
Body dysmorphic disorder	- Excessive preoccupation with one or more aspects of one's appearance, leading to socio-occupational dysfunction.
- CBT can be attempted for all of these conditions, otherwise anti-depressants like SSRIs can be tried.	

Important child abuse findings
- Spiral fractures (rotational/twisting force applied to bone). - Posterior rib fractures (squeezing). - Circular burns (cigarettes). - Burns sparing flexor regions (from being dipped in hot water [child flexes limbs to ↓ exposed surface area). - Retinal detachment / hemorrhages. - Subdural hematoma (shaken baby syndrome). - **Avoidance of eye contact** (if they say Hx of meningitis, think instead neurosensory hearing loss).

Important developmental milestones
- What USMLE will do is ask "normal" or "delayed" for Gross motor, Fine motor, Verbal, and Social. - Highest yield points regarding milestones: - Gross motor: ○ Head up at 3 months (can support his/her own head unassisted).   ○ Turn over front to back at 4-5 months.   ○ Sits up at 6 months.   ○ Crawls by 9 months.

- Stands/walks alongside sofa at 10-11 months.
- Walks at 12-15 months.

- Fine Motor:
  - Pincer grasp at 9 months.
  - Stacks 3, 6, 9 blocks at age 1, 2, 3, respectively.
  - Copies circle at 3-4 years.
  - Copies square at 4-5 years.
  - Copies triangle at 5-6 years.
- Language development:
  - "Mama" and "dada" by 1 year.
  - Recognizes own name by 1 year.
  - 2-word sentences at age 2.
  - 900 words at age 3.
- Social:
  - Social smile at 2-3 months.
  - Stranger anxiety at 6-9 months.
  - Separation anxiety at 9-14 months.
  - Imaginary friends are normal at age 4-6 (Peds shelf likes this; no this is not psychosis).
  - Repeated questioning about death is normal age 8-10.
  - Parallel play age 2-3 (kids play side by side); cooperative play age 3-4.

Peds bladder/bowel control	
Enuresis	- Nocturnal enuresis (i.e., wetting the bed) is normal on occasion up through age 5. - First step in ↓ recurrence is behavioral science-type answers (i.e., ↓ stressors + ↑ time spent with child). Q might say child wets the bed after a sibling is diagnosed with ALL (i.e., stressor). - After behavioral-type answers, the next step is star chart (positive reinforcement), where the child receives a star on a chart every time he/she doesn't wet the bed, where after, e.g., 10 stars, he/she gets an extra dessert. - If star chart is not listed, enuresis alarm is the next best step (i.e., literally an alarm that detects moisture in a child's underwear, typically using a sensor, where if moisture is detected, indicating urination, the alarm sounds or vibrates child awake, thereby encouraging him or her to stop urinating and go to instead use the toilet). - Answers such as imipramine, desmopressin, and water deprivation are wrong on USMLE.
Encopresis	- Shitting your pants. Normal on occasion up through age 5.

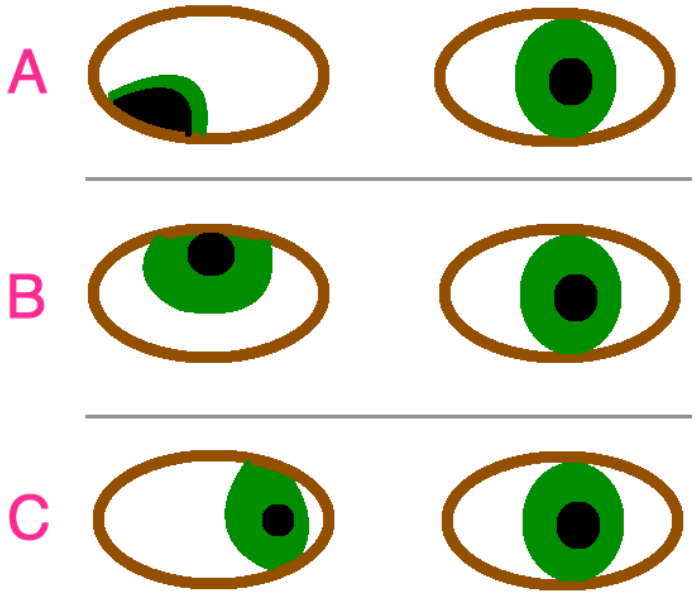
Medication-induced delirium	
Reye syndrome	- Acute encephalopathy and hepatic dysfunction caused by giving aspirin in the setting of a viral illness and/or fever. - Aspirin should be avoided under age 12, and some sources say avoid in teens for that matter. - Mechanism is obscure and thought to be related to impairment of β-oxidation.
Anti-cholinergic delirium	- Anti-cholinergic medications can cause confusion and a delirium-like presentation. Children have ↑ susceptibility. - Q can mention a kid was given an over-the-counter medication and now has low-grade fever and confusion. - Details such as flushed, dry, warm skin, or enlarged pupils, suggest anti-cholinergic delirium over Reye. - Can be confused with Reye syndrome if they just say "over-the-counter" med. - 1st-gen H1-blockers (i.e., diphenhydramine) are notoriously anti-cholinergic.
Dextromethorphan	- Can cause delirium and psychosis in children when taken in excessive amounts. - An opioid that is an anti-tussive (cough suppressant).

Drug abuse	
Glue	- Ataxia + cognitive decline in teenager. - Will sound a bit like alcohol abuse but the effects of alcohol don't occur so young.
Paint	- Cognitive decline. - Q can say teenager is seen with gold or silver coloration around the nose/mouth.
Butane (inhalant)	- Cognitive decline. - Classically inhaling computer cleaner (dusters) or whipped cream bottles. - Q will say high schooler found on floor in school bathroom + is brought into ED sluggish + pupils and vitals all normal → answer = butane.
Caffeine	- Most common drug addiction in the world. - Adenosine accumulates in the brain throughout the day and causes sense of fatigue. - Caffeine blocks adenosine receptors, promoting sense of wakefulness. - Intoxication can cause sense of over-stimulation, panic, and palpitations. - Withdrawal can cause headache, sense of depression, fatigue, anxiety (i.e., sense of worry/doom), and inability to concentrate.
Smoking/vaping	- Nicotine can promote sense of euphoria. - Withdrawal can cause many symptoms, including anxiety, depression, difficulty concentrating, and weight gain due to increased appetite.
Marijuana	- Injection (redness) of conjunctivae + dry mouth. - ↑ risk of developing psychosis and schizophrenia.
Cocaine	- Mydriasis, tachycardia. - High BP causing aortic dissection; can cause chest pain (coronary vasospasm). - Abruptio placentae if pregnant teens. - Give benzo if acutely intoxicated + observe in emergency.
Amphetamine	- Mydriasis, agitation, **insomnia** / staying up all night. - Can cause tactile hallucinations. - Give benzo if acutely intoxicated + observe in emergency.
PCP	- Bellicosity / pugnacity (if you're ESL, those mean wanting to fight + aggressive). - **Nystagmus** +/- mydriasis. - One 2CK NBME Q gives mutism + constricted pupils for PCP + nothing about pugnacity, so just be aware this presentation is rare but possible. - Give benzo if acutely intoxicated + observe in emergency.
MDMA (ecstasy)	- Euphoria, heightened sensory perception, low-grade fever, bruxism (teeth grinding). - An NBME Q floating around gives increased creatine kinase, so this is also possible. - Give benzo if acutely intoxicated + observe in emergency.
LSD (acid)	- Visual hallucinations. - Give benzo if acutely intoxicated + observe in emergency.
MPTP	- Synthetic heroin. - Causes Parkinsonism. - Shows up on a 2CK form, so if you think it's weird, take it up with NBME, not me.
Heroin/opioids	- E.g., oxycodone or dextromethorphan. - Respiratory depression + constricted pupils + constipation. - Naloxone (opioid receptor antagonist) for acute toxicity. - Methadone (opioid receptor agonist) to ↓ relapses.
Benzos	- E.g., diazepam. - Respiratory depression. - Flumazenil to treat acute toxicity (benzodiazepine receptor antagonist).
Barbiturates	- Respiratory depression. - Q will say naloxone and flumazenil had no effect, so you eliminate to get to barbiturates.
Acetaminophen	- Fulminant liver failure. - Give activated charcoal if ingested within 1-2 hours. - *N*-acetylcysteine must be given after to regenerate reduced glutathione to prevent liver damage from NAPQI (acetaminophen metabolite).
Aspirin	- Tinnitus + mixed metabolic acidosis-respiratory alkalosis. - Give sodium bicarb to treat (↑ excretion through urinary alkalinization).

TCAs	- E.g., amitriptyline. - CCCs (coma, convulsions, cardiotoxicity). - ECG changes seen frequently in vignettes. For example, you'll get a big vague paragraph about some drug overdose + they say in last line QT is prolonged → answer = the TCA. Then you remind student about cardiotoxicity and they're like Oh yeah. - Anti-cholinergic effects (i.e., delirium + hot, red, dry patient).

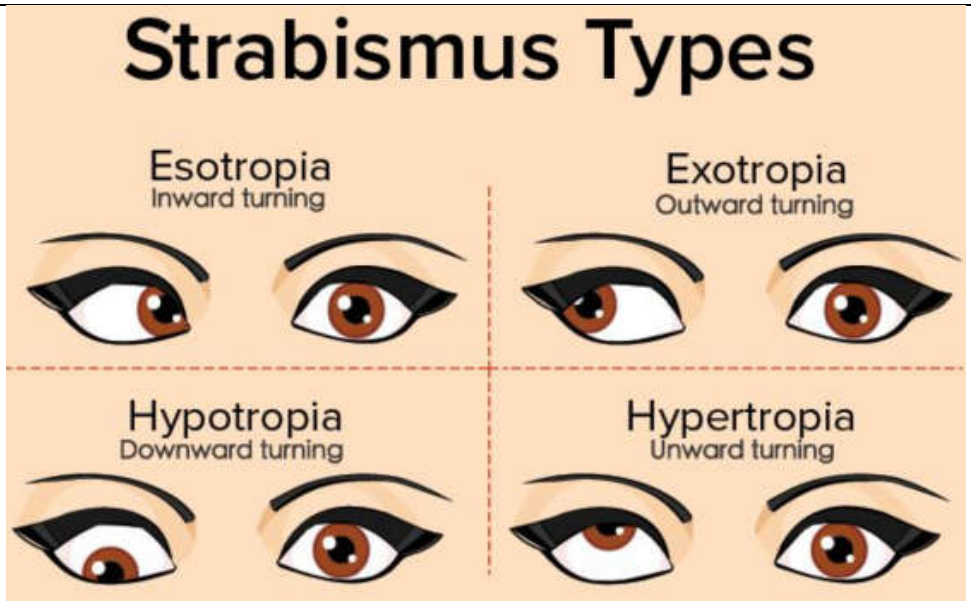
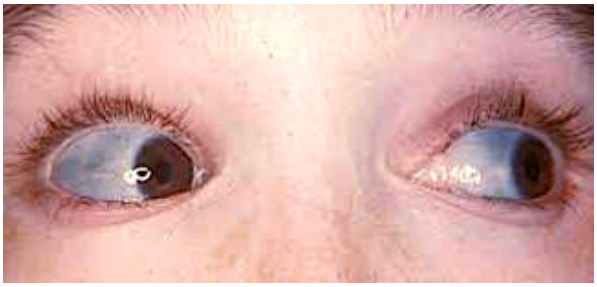
Depression meds	
SSRIs	- Selective-serotonin reuptake inhibitors. - Fluoxetine, escitalopram, sertraline, etc. - Cause sexual dysfunction (anorgasmia) and sleep disturbance. - The fact that they cause anorgasmia actually makes them the Tx for premature ejaculation. - Do not combine with drugs such as monoamine oxidase inhibitors or St John wort, as this can cause serotonin syndrome (discussed below). - There are unique side-effects of various SSRIs – e.g., sertraline is more likely to cause diarrhea; fluoxetine has a stimulating effect and is more likely to cause insomnia; citalopram can prolong QT interval at higher doses. But USMLE doesn't give a fuck. - Used for a variety of psych conditions external to depression, e.g., fibromyalgia, OCD, etc. - It can take 4-6 weeks for an SSRI to achieve desired effect. If after this time point the drug isn't working, the first step is increasing the dose. If this doesn't work, the next step is switching to a different SSRI, followed by switching to a different class agent.
SNRIs	- Serotonin and norepinephrine reuptake inhibitors. - Desvenlafaxine, duloxetine, etc. - Can increase blood pressure at higher doses.
TCAs	- Block reuptake of both serotonin and norepinephrine. - Amitriptyline, nortriptyline, clomipramine, doxepin, etc. - Can be used for neuropathic pain (e.g., from trauma). Gabapentin is also frequently used. - Have nasty anti-cholinergic side-effects (anti-DUMBBELSS; see HY Neuro PDF for discussion of anti-cholinergic vs pro-cholinergic effects if you're confused). - Coma, convulsions, cardiotoxicity. - Sometimes the Q can just mention prolonged QT interval in patient on an anti-depressant, and the answer is the TCA, since they're cardiotoxic. - TCAs, such as imipramine, for nocturnal enuresis are wrong on USMLE. - Doxepin is asked on 2CK form, where the Q rides on you knowing it's a TCA to get it right (i.e., the only drug listed that causes anti-cholinergic effects). I mention this because students get the Q wrong and then are like wtf is doxepin.
MAOIs	- Monoamine oxidase inhibitors. - Phenelzine, tranylcypromine. - Avoided 1st-line overwhelming majority of the time because of ↑ risk of serotonin syndrome. This can be in isolation, but also when commenced too soon following discontinuation of SSRI.
Bupropion	- Dopamine + norepinephrine reuptake inhibitor; has anti-depressant effects. - Also antagonizes nicotinic receptors, which disincentivizes smoking, since doing so won't produce rewarding effects. - Lowers seizure threshold. Don't give to patients with eating disorders (electrolyte abnormalities increase seizure risk). - Doesn't cause sexual side-effects, unlike SSRIs.
Mirtazapine	- α2-antagonist; stimulates appetite; used for patients with depression and anorexia.
Trazodone	- Serotonin antagonist and reuptake inhibitor (SARI). - Used for depression, but is frequently used for insomnia due to strong sedative effects. - Can cause serotonin syndrome.

Peds Ophthal

Extraocular nerve palsies
 - A = CN III (oculomotor) lesion → presents as down and out. - B = CN IV (trochlear) lesion → presents as upward gaze + patient will tilt head toward unaffected side. - C = CN VI (abducens) lesion → medially directed eye.

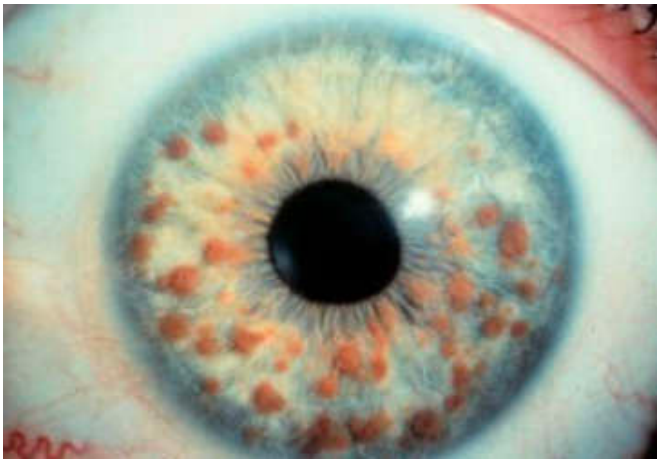
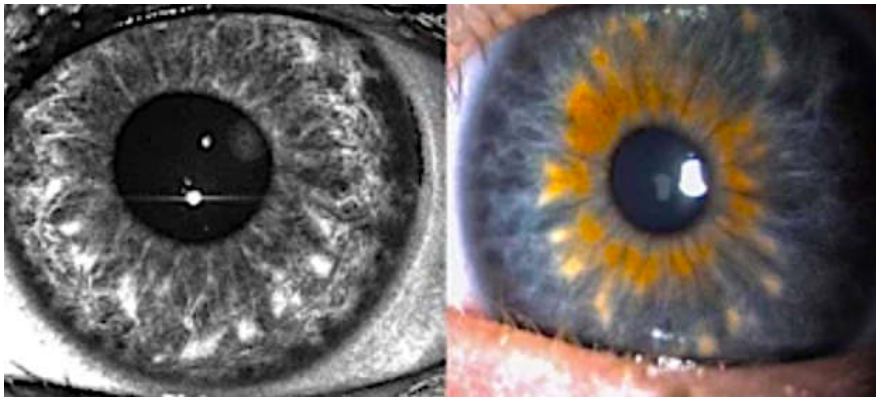
HY Peds Ophthal DDx	
Orbital cellulitis	 - Infection involving tissues posterior to orbital septum; *Staph aureus* most common cause; rare sequela of adjacent infection, e.g., from the paranasal sinus; medical emergency; requires IV antibiotics; preseptal cellulitis is a less severe version of orbital cellulitis.
Hordeolum (stye)	- Painful bump on the eyelid. - *Staph aureus* infection of sweat gland or oil duct; treat with warm compresses. - Can occur any age.

	
Chalazion	 - Painless bump on the eyelid. - Blocked oil duct; not an infection; treat with warm compresses.
Retinoblastoma	 - The image shows leukocoria, which is an abnormal white appearance of the eye on light reflex (should normally appear red). - USMLE wants you to know that *RB* gene mutations not only cause congenital retinoblastoma, but also ↑ risk of osteosarcoma. - In other words, classic Q is they show you image above + ask what child is at risk for later → answer = osteosarcoma.
Strabismus	- Means “lazy eye,” or misalignment of the eyes, due to weakened muscles in one eye.

	 - Can lead to amblyopia, which is reduced visual acuity in the weak eye due to the brain favoring the stronger eye during development. - Strabismus, however, is not reduced vision in the weak eye. It precedes amblyopia if not treated with eye exercises and/or penalization of (patching) the strong eye.
Peds cataracts	- Progressive clouding or opacification of the lens, leading to blurred or hazy vision, particularly in bright light, with increased glare sensitivity. - Can be seen in Peds in congenital rubella, congenital syphilis, and NF2. 
Osteogenesis imperfecta	- Collagen I mutation. - Can cause the very buzzy blue sclerae. 
Alport	- Mutation in type IV collagen; X-linked recessive. - The answer on USMLE for male with eye and/or ear problem + red urine.
Marfan	- Fibrillin mutation; AD; chromosome 15; can cause lens dislocations.
Homocystinuria	- Presents as school-age kid who's Marfanoid + thromboses +/- lens dislocation. - Marfan syndrome and Alport can also cause eye issues, but

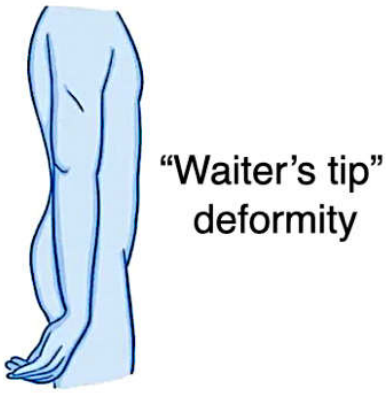
Subconjunctival hemorrhage
- Buzzy image / spot-diagnosis for USMLE. - Caused by bursting of superficial conjunctival vessels. Looks sinister but self-resolves uneventfully. - Can be caused by trauma or brief periods of increased pressure (i.e., severe vomiting or coughing). 

Iris abnormalities	
Aniridia	- Seen as part of **WAGR** complex with *WT1* mutations – i.e., **W**ilms tumor, **A**niridia, **G**enitourinary anomalies, **R**etardation. 
Coloboma	- Means hole in the eye. Seen in **CHARGE** syndrome, → **C**oloboma of the eye, **H**eart defects, **A**tresia of the choanae, **R**etardation of growth and development, **G**enitourinary anomalies, **E**ar defects. 
Albinism	- Can cause pale blue or pink irides (pleural for iris).

	
Lisch nodules	- Iris hamartomas seen in NF1.  - NF1 can present with axillary and groin freckling, café au lait spots (hyperpigmented macules), **optic glioma (tumor of CN II),** neurofibromas (nodules under the skin), pheochromocytoma, and weird brain cancers such as ependymoma and oligodendroglioma.
Brushfield spots	- Connective tissue deposits seen within the irides of some patients with Down syndrome. 

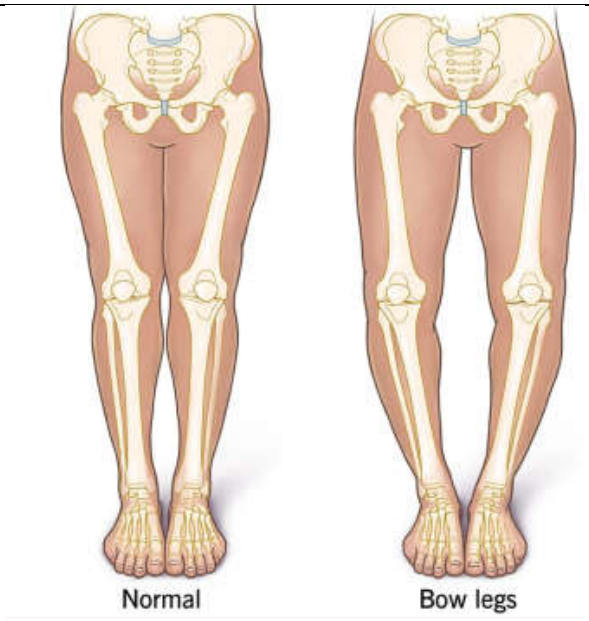
Ophthalmia neonatorum	
- Fancy way of saying neonatal conjunctivitis.	
Chlamydia	- Caused by *C. trachomatis* D-K (the STD). - Presents usually >1 week post-birth and can lead to **pneumonia** with a lymphocytic shift (since chlamydia is obligate intracellular, so requires lymphocytes for cell-mediated immunity to clear them, unlike extracellular bacteria which can be cleared humorally with neutrophils). - The chlamydia can drain through the nasolacrimal duct down to the lungs, causing the pneumonia. - Prophylaxis is treating the mom while pregnant if she has chlamydia; treatment in the neonate who already has the conjunctivitis is **oral** erythromycin. - NBME can give you vignette of wheezes, crackles, and fever in a 3-week-old who they say was treated for chlamydia conjunctivitis 2 weeks ago with topical erythromycin at birth. But this is not the correct treatment since topical will not cover any organism that has already entered the nasolacrimal duct.
Gonorrhea	- Gonococcal conjunctivitis will present in the first week of life with yellow discharge from the eye(s). - The USMLE vignette will be vague, where the student says, “But how are we supposed to know it’s not chlamydia, just because it’s first week of life, that’s it?” And my response is: if they give you chlamydia ophthalmia neonatorum, they’ll **always** tie it to subsequent pneumonia somehow, since that is such a HY point. - Prophylaxis for gonococcal is erythromycin ointment; treatment is IM 3rd-gen cephalosporin (usually cefotaxime in peds).
Chemical	- Chemical conjunctivitis is classically due to silver nitrate eyedrops that used to be given to neonates for gonococcal prophylaxis. However, they have been replaced by erythromycin ointment.
Viral	- Viral conjunctivitis will present as teary eye that is either unilateral or bilateral, and either itchy or non-itchy. I’ve seen variable presentation on USMLE. You just need to know this is the most likely diagnosis in both school-age kids and adults, and is caused by **adenovirus,** which is a DNA virus. Treatment is supportive with saline rinse.
Allergic	- USMLE wants you to know that “cobblestoning” of the tarsal conjunctiva is buzzy for allergic conjunctivitis. - Likewise, for allergic rhinitis, they can say cobblestoning of the nasal mucosa. - Some students have asked if allergic has to be bilateral, which it need not. And it doesn’t have to be itchy either.

Peds MSK

<table>
<tr><th colspan="2">Brachial plexus injuries</th></tr>
<tr><td colspan="2">- Both can be caused in neonates by breech / traumatic labor.
- Can also be caused by injuries such as grabbing onto tree branch while falling.</td></tr>
<tr><td>Erb-Duchenne</td><td>- Upper brachial plexus injury (C5-C6).
- Sometimes rather than Erb-Duchenne written as the diagnosis, the answer will just be “upper brachial plexus” when they ask for what’s injured.
 

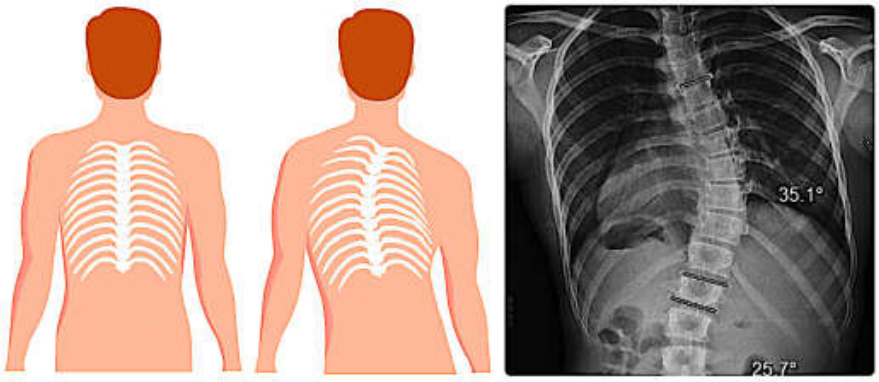
- Arm is adducted, pronated, and wrist flexed.</td></tr>
<tr><td>Klumpke</td><td>- Lower brachial plexus injury (C8-T1).
- Sometimes rather than Klumpke written as the diagnosis, the answer will just be “lower brachial plexus” when they ask for what’s injured.
 
</td></tr>
</table>

<table>
<tr><th colspan="2">Pediatric bone issues</th></tr>
<tr><td>Rickets</td><td>- Rickets = vitamin D deficiency in children. Osteomalacia = vitamin D deficiency in adults.
- Rickets = craniotabes (soft skull), rachitic rosary (bony knobs at costochondral junctions), genu varum (bowing of tibias).
- Activated vitamin D (1,25) is necessary to convert unmineralized osteoid into mineralized hydroxyapatite, therefore hardening bones.
- In both rickets and osteomalacia, patient will have “increased unmineralized osteoid,” or “deficient mineralization of osteoid.”
- Important cause of Vit D deficiency on USMLE is impaired intestinal malabsorption (i.e., CF, Crohn). For CF, answer can be written as “exocrine pancreas insufficiency.”
- “Pseudofracture” on x-ray is buzzy finding in osteomalacia/rickets.
- Patients have ↓ Ca^{2+}, ↓ $PO4^{3-}$, ↑ PTH (due to ↓ negative feedback at parathyroid glands).</td></tr>
</table>

	
Osteopetrosis	- Osteoclast dysfunction resulting in recurrent fractures in children due to bone density being **too high.** Sounds weird, but bone strength is based on balanced internal architecture of canals and networks, not just density alone. - HY DDx against osteogenesis imperfecta and child abuse. - Osteoclast dysfunction is due to deficiency of carbonic anhydrase II. This enzyme inside osteoclasts normally allows osteoclasts to form H^+ to resorb bone. - Granulocyte-macrophage colony-stimulating factor (GM-CSF) causes the differentiation of osteoclast precursors into osteoclasts. This is asked on one of the Step 1 NBME forms, where they give osteopetrosis and ask something along the lines of, "The cell lineage that's fucked up in this condition requires which of the following" → answer = GM-CSF.
Osteogenesis imperfecta	- Collagen I defect that results in recurrent fractures in a child; important DDx are child abuse and osteopetrosis. - Blue sclerae too buzzy and often not mentioned.  - Conductive hearing loss due to defective ossicles (middle ear bones). - Many different types of OI, some resulting in miscarriage. Harder vignette can mention child with multiple fractures, where the mom has Hx of recurrent miscarriage.

Connective tissue disorders	
Marfan	- Autosomal dominant; chromosome 15, *FBN1/2*. - Codes for fibrillin, which is a glycoprotein that forms a sheath around elastin. - Tall, lanky body habitus with flat feet, chest wall abnormalities (i.e., pectus excavatum or carinatum), flat feet (pes planus), scoliosis, mitral valve prolapse (mid-systolic click), increased risk for aortic dissection (can retrograde propagate toward aortic root, causing root dilatation and aortic regurgitation [decrescendo diastolic murmur]). - Is not associated with berry/saccular aneurysms (unlike Ehlers-Danlos and ADPKD).
Ehlers-Danlos	- Defect in collagen III synthesis; can be written as "Defect in synthesis of fibrillar collagen." - This can be confusing because fibrillin is completely unrelated and refers to Marfan syndrome. It is just coincidental that "fibrillar collagen" means type III collagen. - Wrong answer = "abnormal synthesis of extracellular glycoprotein" (refers to fibrillin for Marfan syndrome, which as mentioned above, is a glycoprotein that forms a sheath around elastin. - Presents as easy bruising + hyperextensible skin. 
Williams	- Autosomal dominant; chromosome 7. - Defect in elastin. - Elfin-like facies, hypercalcemia, well-developed verbal skills. - Can cause supravalvular aortic stenosis.

Scoliosis	
Scoliosis	- Sideways curvature of spine, creating an S- or C-shaped curve. - Usually idiopathic; affects 3% of population; girls 4:1. - Most common type is idiopathic adolescent thoracic scoliosis. - Can be associated with Marfan syndrome, Friedreich ataxia, NF1. - Adams forward bend test used to diagnose. - USMLE wants you to know most children do not need treatment, but that curvatures will remain throughout life. - Answer = bracing if curvature is >25 degrees and child is still growing. - I've never seen surgery as answer for scoliosis on NBME; literature says recommended only when curvature >40 degrees.

	 Normal Spine Scoliosis Rib Rotation Scoliosis X-Ray with Rib Rotation

HY Arthritides / joint pathologies for USMLE	
JIA/JRA	- Juvenile idiopathic arthritis (formerly juvenile rheumatoid arthritis; JRA). - 2CK Peds forms love JIA. It will sound like regular RA but just in a kid. - USMLE will structure these Qs where they want you to pick between JRA and septic arthritis (SA) as answer choices. This can be confusing since SA can occur in patients with JRA. They might say a kid has a low-grade fever and a warm, red, painful knee (sounds like SA), but then they say he's had similar **episodes** in the past (i.e., they want JRA over SA). This is because SA is usually a one-off event; for JRA, however, the vignette will say "intermittent" or "episodic." Low-grade fever can occur in autoimmune flares (not limited to JRA; HY for sarcoidosis as well). - "Salmon-pink" maculopapular rash only in ~50% of JRA Qs. Often described as a buzzy finding, but I'd say about half of JRA vignettes don't even mention rash. - 2CK forms are obsessed with anemia of chronic disease in JRA. HY point is that MCV can absolutely be low. Resources push normal MCV for AoCD. This is absolute nonsense. Plenty of 2CK NBME Qs give MCV as 70s in AoCD. - Can be associated with serous pericarditis. - If the Q asks what arthrocentesis will show in JRA, the answer is leukocytes. - For antibodies, anti-cyclic citrullinated peptide (anti-CCP) **is more specific** than rheumatoid factor. Both should be ordered for JRA (and RA in adults). - Tx for JRA has arms of management: 1) symptoms; 2) disease progression. For symptoms, give NSAIDs first, followed by steroids. These do **not** slow disease progression. NSAIDs and steroids are for symptoms only. For disease progression, we use disease-modifying anti-rheumatic drugs (DMARDs), which slow disease progression. Methotrexate is given first, followed by adding an anti-TNF-α agent (i.e., infliximab, adalimumab, or etanercept).
Psoriasis	- Can occur in children and teens. - 15% of patients with psoriasis will get arthritis before any skin findings. - Plaques are described as silvery and scaly and over extensor regions (elbows); plaques can also show up on the face (i.e., forehead and lip). - Auspitz sign is bleeding of the scales if removed. - HY point for USMLE is that psoriasis is part of the HLA-B27 constellation (**PAIR** → **P**soriasis, **A**nkylosing spondylitis, **I**BD, **R**eactive arthritis). - For example, if 17M has silvery plaque on elbow and forehead + bloody stool → the latter is most likely IBD due to HLA-B27 association. - Don't confuse psoriasis + IBD combo with dermatitis herpetiformis + Celiac disease combo. - Treatment for plaque psoriasis is topicals first → USMLE wants calcipotriene (vitamin D derivative), triamcinolone or hydrocortisone (both corticosteroids), and

	coal tar. Choose in that order if you are forced. Chronic application of topical steroids causes dermal collagen thinning, so they are not preferred prior to topical vitamin D. - Oral meds are given if patient fails topicals, OR if patient has systemic psoriasis (i.e., arthritis). **Oral methotrexate** is HY on new NBME material as the first-line oral agent used. - An old NBME has **oral acitretin** (a vitamin A derivative) as an answer, where methotrexate is not listed.
Sacroiliitis	- Sacroiliitis is broad term that refers to arthritis of lower back; ankylosing spondylitis (AS) is most severe form. - Vignette will almost always be male 20s-40s who has lower back pain worse in the morning that improves throughout the day. However, there is Q on 2CK Peds form where AS is diagnosis in an 8-year-old, so you need to know it's possible in peds. - Lower back pain in patient with IBD, psoriasis, or reactive arthritis points toward sacroiliitis (HLA-B27 PAIR). - High ESR and anemia of chronic disease high-yield. - Diagnose with x-rays of the lumbosacral spine and sacroiliac joints. - USMLE wants "slit-lamp examination" to look for anterior uveitis in ankylosing spondylitis. Any autoimmune disease can theoretically increase risk of this eye finding, but for whatever reason USMLE likes it for AS. - Treat same as JRA.
Lupus	- Can present in Peds. Shows up on the forms. - Arthritis is most common presenting feature of lupus (90%). USMLE vignette will pretty much always give arthritis in lupus Qs. - Anti-double-stranded-DNA (dsDNA) antibodies go up with acute flares and are most closely related to renal prognosis for lupus nephritis. - Anti-Smith (ribonucleoprotein) antibodies are **most specific** for SLE (more than dsDNA). - Anti-phospholipid syndrome due to lupus anticoagulant (antibodies against β2-microglobulin or cardiolipin in patient with SLE). - Anti-phospholipid syndrome can cause false-positive syphilis VDRL test (HY). - Malar rash is type III hypersensitivity. Harder Qs won't mention this finding because it's too buzzy. You need to know **thrombocytopenia** is frequently seen in lupus due to anti-hematologic cell line antibodies. Antibodies can also target WBC and RBC → looks like aplastic anemia, but it's not → answer = "increased peripheral destruction," not "decreased bone marrow production." - Can cause lupus cerebritis (confusion / delirium-like episodes) and transverse myelitis (presents as Brown-Sequard syndrome). Peds form gives lupus cerebritis. - Similar to RA, lupus can cause pericarditis. - Flares cause decreased serum complement protein C3. - Congenital complement protein C1q deficiency causes ↑ risk of developing SLE; sounds nitpicky but it's on new Step 1 NBME exam. - For lupus nephritis, **biopsy** as the first step in management; steroids first is wrong answer; biopsy first sounds wrong but it guides management. - Lupus nephritis = diffuse proliferative glomerulonephritis (DPGN) on USMLE. - Treat flares of lupus with steroids. USMLE doesn't care about other Txs.
Septic arthritis	- Shows up on Peds forms as young healthy patient with recent trauma (e.g., car accident) or high-intensity exercise causing microtrauma (e.g., kickboxing/soccer tournament, long hike), as well as in sickle cell disease. - USMLE will give hot, red, painful joint in one of the above patient groups, almost always with a fever. - First step in management is arthrocentesis. A high-yield point is that joint aspirate can be **negative** for organisms. Do not rule-out SA if NBME says no organisms. - *Staph aureus* is most common pathogen overall.

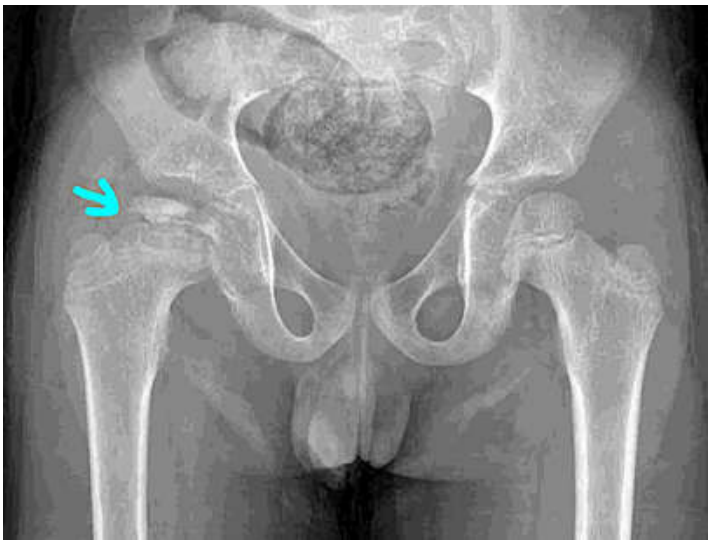
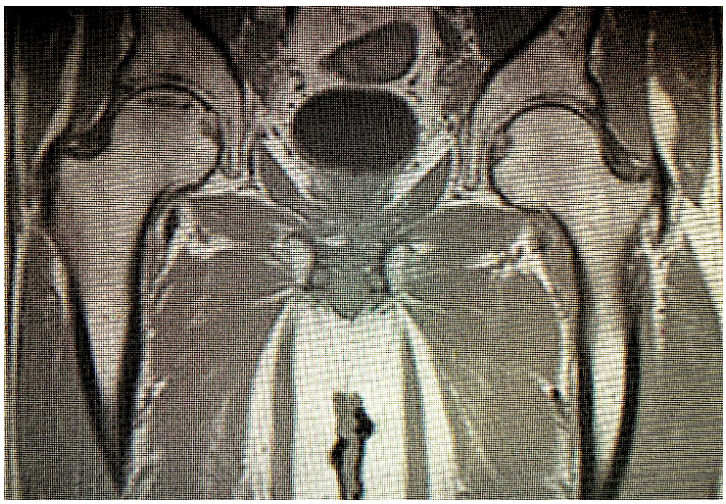
	- Choose *Salmonella* in sickle cell. But you have to use your head. *Salmonella* is a gram-negative rod, so if they say gram-positive cocci is cultured in sickle cell, the answer is still *staph aureus*. - *Gonococcus* is the answer for sexually active patients; presents two ways on USMLE: 1) monoarthritis of the knee in sexually active young patient, with no other information provided; or 2) as a triad: mono- or polyarthritis, cutaneous papules, and tenosynovitis (inflammation of tendon sheaths). - Treatment is surgical drainage of the joint + antibiotics. USMLE doesn't care about the exact antibiotics.
Toxic synovitis	- Aka transient synovitis. 2CK pediatrics forms are obsessed with this. - Presents as hip inflammation/pain in child after a viral infection. - Toxic synovitis is diagnosis of exclusion, meaning the vignette gives you various findings that make septic arthritis less likely. **Leukocytosis** and **inability to bear weight** make septic arthritis the likely answer over toxic synovitis. Hard questions will not mention warmth or redness as ways to differentiate. Fever can present in both. - Treatment is NSAIDs (ibuprofen) → asked on 2CK Peds.
Reactive arthritis	- Classically presents as triad of 1) urethritis or abdominal infection, 2) polyarthritis, and 3) "eye-itis" (i.e., conjunctivitis, episcleritis, or anterior uveitis). - *Chlamydia* is the classic organism that causes reactive arthritis. *Gonococcus* does **not** cause reactive arthritis on USMLE. - *Rubella*, Hep B+C, and *Yersinia* can also cause reactive arthritis. - Part of HLA-B27 constellation (PAIR), as mentioned above.

Other random joint/MSK disorders	
Lateral epicondylitis	- Tennis elbow. - Lateral elbow pain worsened when patient extends wrist against resistance. - NBME asks "extensor carpi radialis brevis" as answer for site of inflammation. - Tx = "forearm strap" on 2CK FM forms.
Medial epicondylitis	- Golfer elbow. - Medial elbow pain worsened when patient flexes wrist against resistance. - Inflammation at flexor carpi radialis and pronator teres. - Tx = forearm strap.
Radial head subluxation	- "Nursemaid elbow"; HY for 2CK Peds. - Child stops using arm + arm pronated and partially flexed. - Hx of child having arm pulled/yanked, or child holding hands and running with older sibling + the child falls, resulting in elbow pull. - Tx = hyper-pronation, OR supination when arm partially flexed. Either is correct. Both will not be listed at the same time as answers.
Patellar tendonitis ("Jumper's knee")	- Inflammation of patellar tendon. - The answer on USMLE if they describe anterior knee pain that initially occurs only after finishing sports (e.g., basketball game), then progresses to more chronic pain. - Tx on NBME = "quadricep strengthening exercises."
Patellofemoral instability	- Presentation is annoyingly similar to patellar tendonitis, but do not confuse. - Patellofemoral instability is misalignment of the patella at the trochlear groove of the femur. - Q can mention crepitus. - Shows up on 2CK Peds CMS form 6 as teenage girl who has knee pain worse after jumping or running + has crepitus → answer = "patellofemoral instability" (patellar tendonitis / "jumper's knee" not listed as answer).
Patellofemoral pain syndrome	- Aka chondromalacia patellae; name implies softening of cartilage in the knee. - The answer on NBME if they say pain that worsens when sitting for long periods of time, or when going up or down stairs. - Classic in obesity or those who squat heavy weight (knees think you're obese).

	- Tx = quadriceps strengthening exercises.
Anterior cruciate ligament injury	- ACL is answer if (+) anterior drawer test or Lachman test → excessive anterior displacement of tibia relative to femur. - Classically injured when knee is hyper-extended, or with a rotational force on a fixed, planted knee.
Posterior cruciate ligament injury	- PCL is answer if (+) posterior drawer test → excessive posterior displacement of tibia relative to femur. - Classically injured when knee hits the dashboard in car accident.
Lateral collateral ligament injury	- LCL is the answer if varus test induces excessive lateral motion of the knee compared to the unaffected side. - Varus test = hand placed on medial knee and pushing outward + other hand placed on lateral ankle and pushing inward.
Medial collateral ligament injury	- MCL is the answer if valgus test induces excessive medial motion of the knee compared to the unaffected side. - Valgus test = hand placed on lateral knee and pushing inward + other hand placed on medial ankle and pulling outward.
Lateral meniscal tear	- Lateral knee pain where patient experiences “locking” or “catching” of the knee in partial flexion. - Diagnosed with McMurray test → internal rotation of leg with concurrent knee extension causes lateral knee pain / “catching.”
Medial meniscal tear	- Medial knee pain where patient experiences “locking” or “catching” of the knee in partial flexion. - Diagnosed with McMurray test → external rotation of leg with concurrent knee extension causes medial knee pain / “catching.”
“Unhappy triad”	- Refers to a trio injury of the ACL, medial collateral ligament, and either the medial or lateral meniscus. - Students are sometimes fanatical about this triad as though it has yieldness. USMLE doesn’t give a fuck. I cannot recall a single NBME question that has ever assessed this. This Dx primarily resides within the domain of Qbank, not the NBME.
Iliotibial band syndrome	- The answer on USMLE if they say lateral knee pain, usually in a runner. - Iliotibial band runs from the hip to the knee. Pain may occur anywhere along the hip, lateral thigh, and lateral knee, but is worst in the latter. Tx = conservative with physiotherapy; NSAIDs for pain.
Osgood-Schlatter disease	- Buzzy vignette is knee pain in fast-growing teenage male who plays soccer. Don’t pigeon-hole things, but that’s classic vignette. - Inflammation of the patellar ligament at the tibial tuberosity. - Mechanism is repeated stress on the growth plate of the superior tibia.
Plantar fasciitis	- The answer on USMLE if they give severe heel pain that is worst when first getting out of bed in the morning.
Growing pains	- No, this is not a joke. This is the answer straight-up on a 2CK NBME form. - Vignette is healthy child age 3-12 who awakens from sleep with throbbing pain in the legs. - No treatment necessary. You just need to know this Dx is exists and isn’t a troll.
Tibia vara	- Aka Blount disease. - Bowing of the tibias after the age of 2 years in a patient whom rickets has been ruled out. - Can be unilateral or bilateral. - Bowing of one or both tibias is sometimes **normal** until age 2 years. - Treatment is surgery (osteotomy).

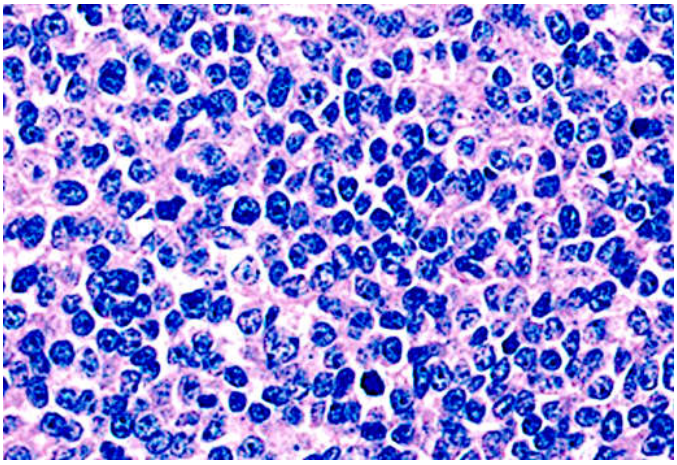
	
Talipes equinovarus	- Aka clubbed foot.  - USMLE just wants you to know that this is treated initially with **serial casting.** - Usually idiopathic; can be seen in Potter sequence. - Not the same as rocker-bottom foot (aka congenital vertical talus), seen in Edward syndrome.
Arthrogryposis	- You just need to know this is fancy name for a child born with multiple joint contractures. - If they give you a child with not just a clubbed foot, but also knee and/or elbow contractures, etc., the answer is arthrogryposis.

HY Pediatric hip disorders for USMLE	
Primary hip dysplasia	- Aka developmental dysplasia of the hip. - Mechanism is “poorly developed acetabulum.” - Initial diagnosis is with Ortolani and Barlow maneuvers, where a “clicking and clunking” is elicited on physical exam. - After the O&A maneuvers, next best step is ortho referral. Sounds wrong, but if it’s listed, it’s the answer before going to imaging. - Definitively diagnose with hip ultrasound if <6 months of age; hip x-ray if >6 months of age. - Treatment is “abduction harness,” aka Pavlik harness, which positions the child’s legs in a frog-leg-appearing manner.
Legg-Calve-Perthes	- Aka idiopathic avascular necrosis of the femoral head.

	- If the etiology for the avascular necrosis is known (i.e., Gaucher, sickle cell, corticosteroids), then the diagnosis is just “avascular necrosis,” not LCP. - Vignette will be child 5-8 years old with hip pain. - First step in diagnosis is hip x-ray, which will show a “contracted” or flattened femoral head. The word “contracted” is HY and synonymous with avascular necrosis. - If x-ray is negative, diagnose with bone scan or MRI (on 2CK Peds form). - Tx = hip replacement.  - Student says, “No idea what I’m looking at.” Relax. If we try to imagine on this x-ray, the right femoral head (left side of image) looks “contracted” or “flattened.” Once again, this is HY and buzzy for avascular necrosis.  - Exam can also show an MRI. In this case, the right femoral head (left side of image) appears hypointense (more black). The necrotic / lack of bone in the black superior portion of the femoral head means the remaining white part of the femoral head is “flattened” or “contracted.” The left femoral head (right side of image) shows a small area of necrosis as well (black medial portion).
Slipped capital femoral epiphysis (SCFE)	- Classic vignette is a 10-13-year-old (pre-adolescent) overweight boy with a painful limp.

	- NBME will write answer / mechanism as "displacement of the epiphysis of the femoral head." - Resources tend to emphasize obesity as the main risk factor, but maybe only ~1/2 of NBME Qs for SCFE give the child as overweight. This causes problems for students, where they rely on seeing high BMI to think SCFE. - This has led me to conclude that the **age** matters the most, since they will always give a kid who's about 10-13-ish. If they give you a kid who's younger, think LCP instead. - 2CK NBME Q gives 13M with painful gait + no mention of weight → answer is SCFE. - Another 2CK Q outright says BMI 20 in a 13-year-old who's 6 feet tall, and answer is SCFE. - X-ray shows "ice cream falling off the cone." - Offline 2CK gives Q where they say x-ray in 5-year-old shows "contracted capital femoral epiphysis" → answer is LCP, not SCFE. As I said above, "contracted" is HY for LCP. In this case, the younger age + the word "contracted" win over the words "femoral capital epiphysis." Tx = surgical pinning.  X-ray shows the "ice cream slipping off its cone."

Bone tumor points for Peds	
Osteosarcoma	- Most common primary bone cancer; usually in patients age 10-30. - As mentioned earlier, *RB* mutations (congenital retinoblastoma) are associated with osteosarcoma (HY) – i.e., 1-year-old boy has enucleation of eye for retinoblastoma; what is he at risk of developing later in life? → answer = osteosarcoma. - Buzzy findings are "Codman triangle" and "Sunburst appearance." - Codman triangle = periosteal reaction with lifting of periosteum off the bone. - Sunburst appearance = periosteal reaction described on NBME as "spiculated new born formation."

	 - The white arrow is the Codman triangle; the white star is the sunburst appearance.
Ewing sarcoma	- The answer on USMLE if they give bone tumor in a child that presents similarly to osteomyelitis (i.e., fever and bone pain in a kid). - If bone scan is performed, it is most likely to show uptake in the diaphysis; in contrast, osteomyelitis will show uptake in the metaphysis. - Histo will show “small blue cells of neuroendocrine origin” and/or “onion-skinning.” - I’ve seen an NBME answer choice say “closely packed, small, round, uniform neoplastic cells.”  - Associated with t(11;22) translocation; don’t confuse this with the 22q11 deletion in DiGeorge syndrome.
Unicameral bone cyst	- Underrated diagnosis for USMLE. Asked on 2CK exam. - Benign bone tumor that looks similar to osteoclastoma but age of onset usually birth to age 20, rather than 20-40.

	 - Unicameral means “one chamber”; it is fluid-filled.

HY bone fracture points for Peds	
Spiral fracture	- Pathognomonic for child abuse. - Caused by rotational/twisting force applied to a limb. - USMLE doesn’t expect you to diagnose based on imaging.
Greenstick fracture	- Bone bends and cracks instead of breaking completely into two pieces. - More common in Peds than adults. - Children’s bones are more flexible than adult bones adults’ bones because they have ↑ collagen content and ↓ mineral content, allowing them to bend more. - When a force that would cause a complete fracture in an adult bone is applied to a child's bone, it might only cause a greenstick fracture, which is an incomplete fracture with the bone bending and breaking only on one side, resembling a green branch of a tree that bends and splinters on one side without breaking completely.
Metatarsal stress fracture	- The answer on USMLE for pain in the metatarsal area of the foot in long-distance runners with low BMI. - Rather than asking diagnosis directly, USMLE will often give a vignette where a female long-distance runner with low BMI already has a metatarsal stress fracture, and then they’ll ask what she’s at greatest risk of developing → answer = osteoporosis.
Clavicular fracture	- Occurs with fall on outstretched hand (FOSH), or occasionally as a result of handlebar injury / impaction, with force transferred up to clavicle. - Most common site of break is the middle-third of the clavicle. - Tx = Figure-of-8 sling.
Scaphoid fracture	- Presents as pain over anatomic snuffbox in patient with FOSH. - X-ray will usually be negative acutely. Must do thumb-spica cast to prevent avascular necrosis of scaphoid, followed by repeat x-ray in 2-3 weeks.
Lunate fracture	- The answer on USMLE if FOSH with pain in central palm + **no** pain over anatomic snuffbox.

Hook of hamate fracture	- Cause of distal ulnar nerve injury / Guyon canal syndrome. - Often from handlebar injury / impaction.
Surgical neck of humerus fracture	- Causes axillary nerve injury → loss of deltoid function + sensation over deltoid.
Midshaft fracture of humerus	- Causes radial nerve injury → wrist-drop + pronated arm.
Supracondylar fracture of humerus	- Aka "distal shaft fracture." - Causes median nerve injury → motor/sensory dysfunction of forearm muscles, first three fingers and thenar region.
Pseudofracture	- Band of low-density bone that looks like fracture on x-ray but not actual fracture. - Synonymous with vitamin D deficiency (osteomalacia/rickets) on USMLE. - Can be seen in renal failure, since 1,25-D3 is low. Osteomalacia due to renal failure is called renal osteodystrophy.

HY myopathies / muscular dystrophies for USMLE	
Duchenne / Becker muscular dystrophy	- XR disorder caused by mutation in dystrophin (*DMD*) gene. - Mutation results in disruption of α-/β-dystroglycan, which is required for proper internal **cytoskeletal** anchoring of the muscle cell to the extracellular matrix. - Presents with pseudohypertrophy, where muscles appear large but are replaced with fibroadipose tissue (connective tissue stromal cells). - Duchenne presents in a young boy who implements Gower maneuver to stand up (uses arms to walk up off the floor because leg muscles are weak). - Becker presents in adolescence or young adulthood (less severe form of Duchenne). - Duchenne is classically frameshift mutation; Becker is classically not frameshift.
Polymyositis / Dermatomyositis	- Q on Peds form has dermatomyositis showing up in a kid. Weird, but just know it's possible. Usually, the patient is over 50. - Presents as proximal muscle pain and stiffness + **1) muscle weakness on physical exam, and/or 2) increased serum CK.** - If polymyositis presents with skin findings, it is called dermatomyositis – i.e., Gottron papules (violaceous papules on the knuckles), mechanics' hands (rough-surfaced hands), shawl rash (body rash), heliotrope rash (violaceous eyelids / periorbital rash; don't confuse with malar rash of SLE). - Patients often positive for anti-Jo1 antibodies. - USMLE wants "electromyography and nerve conduction studies" as first step in management for polymyositis/dermatomyositis. This is what they ask on 2CK NBME forms. I have not seen them ask anti-Jo1 antibodies vs EMG+NCS as two separate answer choices. Usually anti-Jo1 antibodies are mentioned in the vignette rather than as the test you need to order. - Muscle biopsy is confirmatory, showing CD8+ T cell infiltration. The histo can be described as "CD8 + T cells and macrophages surrounding muscle fibers." Tx = steroids.
Fibromyalgia	- This is a psych condition, not an actual muscle disorder. - Labs will be normal. ESR will not be elevated. Patient will not have fever. - Can occur in teenagers, although typically women 20s-50s with multiple, symmetric muscle tender points. - Treatment is SSRIs. USMLE can write this as "anti-depressant therapy." This confuses students ("But she doesn't have depression though.") → Right. But SSRIs are still anti-depressant medication.
Myotonic dystrophy	- Autosomal dominant, CTG trinucleotide repeat expansion disease. - Myotonia is inability to relax muscles.

	- The answer on USMLE if they say patient cannot relax grip on doorknob / handshake, or cannot let go of golf club. - Sometimes associated with early / frontal balding.
Mitochondrial myopathy	- Broad term that can refer to numerous mitochondrial diseases. - USMLE wants you to know the patient has a mitochondrial disorder when he/she presents with **hypotonia, ear/eye problems, and lactic acidosis.** You want to memorize this tetrad as synonymous with mitochondrial disorders. - "Ragged red fibers" can be a buzzy descriptor in mitochondrial myopathy Qs. - Mitochondrial disorders are maternally inherited only. - Heteroplasmy refers to offspring having varying disease severity based on variation in allocation of diseased mitochondrial genes (I talk more about this stuff in my HY biochemistry PDF).
Mixed connective tissue disease	- Can occur in kids. Answer on USMLE if they give you a patient who has anti-U1-ribonucleoprotein (U1-RNP) antibodies. - Presents as having combined features/symptoms from three different disorders → LPS → Lupus, Polymyositis, Scleroderma.

HY neck MSK masses for Peds	
Thyroglossal duct cyst	- The answer on USMLE if they say there's a painless, midline neck lump in a child that moves upward with swallowing or protrusion of the tongue. This buzzy description is seen for maybe only about half of Qs. - Can also be described as painless mass inferior to the hyoid bone that demonstrates uptake with a Technetium-99 scan. - USMLE wants "endoderm of foramen cecum" as the embryology.
Sternocleidomastoid injury	- The answer on USMLE if they say nodular mass in the lateral neck in an infant who had been born via forceps delivery (risk factor for damage to the muscle).
Branchial cleft cyst	- The answer on USMLE if they give idiopathic lateral neck mass in infant that may or may not have an opening to the skin.

Peds Biochem

- You do **not** need to memorize the specific enzyme deficiencies for the 2CK Peds shelf. What you do need to know, however, is how these conditions present + being able to diagnose. I write the enzyme names here nevertheless because, for some of you, it helps generate completeness/context.

Glycogen storage diseases		
- All autosomal recessive.		
Disease	**Enzyme deficiency**	**HY points**
Type I (Von Gierke)	Glucose-6-phosphatase	- **Lactic acidosis**, hypoglycemia, jaundice, hepatomegaly. - "Super sick kid with glycogen storage disease."
Type II (Pompe)	Debranching α1,4-glucosidase (lysosomal acid maltase)	- Cardiomyopathy, hepatomegaly. - The answer if they say anything about the heart in the context of glycogen storage disease.
Type III (Cori)	Debranching α1,6-glucosidase	- **No lactic acidosis** (in contrast to von Gierke). - "Not so sick kid with glycogen storage disease."
Type V (McArdle)	Myophosphorylase (glycogen muscle phosphorylase)	- Usually adolescent or adult. - Severe cramping/rhabdo after intense exercise, but normal serum lactate.

Lysosomal storage diseases			
- All AR except for Fabry and Hunter, which are XR.			
Disease	**Enzyme deficiency**	**Buildup product(s)**	**HY points**
I-cell	N-acetylglucosamine-1-phosphotransferase	Many lysosomal enzymes	- Cannot synthesize mannose-6-phosphate at the Golgi; M6P needed to target lysosomal enzymes to the lysosomes; instead, lysosomal enzymes are secreted into cytosol + serum. - Course facial features, joint contractures, hepatomegaly, recurrent ear infections.
Gaucher	Glucocerebrosidase	Glucocerebroside	- Avascular necrosis of the hip. - "Crumpled tissue paper" (i.e., lipid-laded) macrophages. - The answer if they give you lysosomal storage disease + a bone problem.
Tay-Sachs	Hexosaminidase A	GM_2 ganglioside	- Cherry red spot on macula (blindness). - **No hepatosplenomegaly.** - Neurodegeneration.
Niemann-Pick	Sphingomyelinase	Sphingomyelin	- Cherry red spot on macula (blindness). - **Yes, hepatosplenomegaly.** - Niemann-Pick is a longer name than Tay-Sachs, and hepatosplenomegaly is a long word, so Niemann-Pick is the one with HSM. - Neurodegeneration.
Metachromatic leukodystrophy	Arylsulfatase A	Cerebroside sulfate	- Progressive neurologic decline.
Krabbe	Galactocerebrosidase	Galactocerebroside, psychosine	- Aka Globoid cell leukodystrophy. - Globoid cells (giant, multinucleated cells). - Neurodegeneration.
Fabry **(XR)**	α-galactosidase A	Ceramide trihexoside	- Angiokeratomas (small, red/purple dots on the skin due to capillary dilation). - Heart and/or renal problems.

Hunter **(XR)**	Iduronate sulfatase	Glycosaminoglycans (GAGs) – i.e., Dermatan + heparan sulfates	- Gargoyle-like facies, stridor, aggressive behavior. **- No clouded corneas.** - "Hunters can see the X" → Therefore XR and no clouded corneas.
Hurler	α-L-iduronidase		- Gargoyle-like facies, stridor, aggressive behavior. **- Yes, clouded corneas.** - More severe form of Hunter.

Peds enzyme deficiencies	
Phenylketonuria (PKU)	- All autosomal recessive. - Deficiency of phenylalanine hydroxylase (or rarely THB, which is cofactor for the enzyme), causing inability to convert phenylalanine into tyrosine. - Screened for on heel-prick test at birth, since failure to detect PKU will result in mental retardation. - Phenylalanine-deficient diet must be given. - Presents as child with "mousy" or "musty" body odor + appears lighter skin than siblings (can cause partial albinism, since tyrosine is normally converted into melanin, and tyrosine can't be synthesized).
Maple syrup urine disease	- Deficiency of branched-chain ketoacid dehydrogenase. - Inability to breakdown leucine, isoleucine, and valine. - Causes a maple syrup-scent in the diapers from the urine.
Homocystinuria	- Marfanoid school-age kid + thromboses. - In other words, they will give a tall, lanky, 12-yr-old girl with lens dislocation (sounds like Marfan syndrome), but then they also mention thrombosis. - ↑ Homocystine can cause endothelial cell dysfunction and platelet activation. - Causes ↑ homocystine in the urine, due to either deficiency of cystathionine synthase or methionine synthase.

Sugar disorders for USMLE		
Condition	**Enzyme deficiency**	**HY point(s)**
Galactokinase deficiency	Galactokinase	- From birth (milk lactose → galactose + glucose); screened via heel-prick test . - Cataracts in neonate. - Aldose reductase converts galactose → galactitol, which has ↑↑ osmotic pull in lens. - Milder than classic galactosemia.
Classic galactosemia	galactose-1-phosphate uridyl transferase	- From birth (milk lactose → galactose + glucose); screened via heel-prick test. - Cataracts in neonate. - Hepatomegaly, jaundice, hypoglycemia. - Associated with *E. coli* sepsis. - Trapped G1P in liver sequesters phosphate. - Worse than GK deficiency.
Essential fructosuria	Fructokinase	- Benign condition. - Mere presence of "reducing sugars" in the urine. - Negative glucose oxidase urine test (indicating reducing sugars are not glucose).
Hereditary fructose intolerance	Aldolase B	- Begins with dietary introduction of juice, fruit, honey, and table sugar. - Hepatomegaly, jaundice, hypoglycemia. - Trapped F1P in liver sequesters phosphate.

Phakomatoses (neurocutaneous disorders)		
- I discussed NF1 and TSC a briefly earlier, but these conditions are exceedingly HY and need to be reinforced.		
Condition	**Genetics**	**HY points**
Neurofibromatosis type I (NF1)	*NF1*; AD; chromosome 17	- Neurofibromas, café au lait spots (hyperpigmented macules), axillary/groin freckling, Lisch nodules (iris hamartomas), pheochromocytoma, **optic nerve glioma,** oligodendroglioma, ependymoma.
Neurofibromatosis type II (NF2)	*NF2*; AD; chromosome 22	- In kids, sometimes causes bilateral cataracts. - Bilateral acoustic schwannomas + meningioma in adults.
Tuberous sclerosis (TSC)	*TSC1*; AD; chromosome 9; hamartin *TSC2* ; AD; chromosome 16; tuberin	- Periventricular nodules (tubers) on MRI of head, seizures, adenoma sebaceum (angiofibromas), subungual fibromas, renal angiomyolipoma, cardiac rhabdomyoma, hypopigmented macules ("ashleaf spots"), hyperpigmented velvety lesions ("shagreen patches").
Von Hippel-Lindau	*VHL*; AD; chromosome 3	- Cerebellar/retinal hemangioblastomas, bilateral renal cell carcinoma, pancreatic cysts. - Mutation causes constitutive activation of hypoxia-inducible factor.
Sturge-Weber	Not inherited; somatic mosaicism of *GNAQ* gene	- Port wine stain birth mark (nevus flammeus; may also present as violaceous papules in temporal distribution), leptomeningeal angioma (presenting as seizure); glaucoma.

Immunodeficiencies HY points	
- In contrast to the glycogen and lysosomal storage diseases, Peds shelf and 2CK *do* expect you to know the immunodeficiencies at a Step-1 level-degree of detail (i.e., know the mechanisms).	
Severe-combined immunodeficiency (SCID)	- X-linked recessive most common → usually due to common gamma chain mutation, or mutation in IL-2 receptor. - Autosomal recessive type less common → due to deficiency in adenosine deaminase (ADA). - Combined T and B cell deficiency. - T cell deficiency → absent thymic shadow. - B cell deficiency → scanty/absent lymph nodes and tonsils. - Q will give you sick kid who's had infections of all different types (i.e., viral, fungal, bacterial, protozoal) **since birth.** - This contrasts with Bruton XLA, which will be *just* bacterial infections (usually **since 6 months of age**). - Tx = bone marrow transplant.
Bruton X-linked agammaglobulinemia (XLA)	- X-linked recessive (as name implies) → caused by mutation in Bruton tyrosine kinase. - Isolated B cell deficiency. - Scanty/absent lymph nodes + tonsils. - Kid will have ↓ immunoglobulins of all classes (because B cells make Ab). - Q will almost always be a boy who's had only bacterial infections since 6 months of age (if they mention bacterial *and* viral, fungal, and/or protozoal, the Dx is SCID). - There is a Q on a Peds 2CK form where they say "bacterial infections since birth," but you're able to eliminate the other answers (literally 14/15 Qs will say from ~5-7 months-onward, so this is a highly annoying point to have to make, but I have to communicate it because it's on the NBME). - Tx on NBME is "monthly infusion of immunoglobulin."

	- Answer on USMLE will often be "deficiency of humoral immunity," rather than just "Bruton XLA."
IgA deficiency	- One of the highest yield conditions for the Step (hence I mention it repeatedly in this doc; purposeful redundancy). - Most common hereditary immunodeficiency. - 9/10 Qs will give you recurrent sinopulmonary infections in someone between high school and 30s; one 2CK Peds Q has it in a kid; the Q might say the patient has a sore cheek (sinusitis) + Hx of pneumonia. - Anaphylaxis with transfusion of blood products is ultra-HY detail but often omitted from real NBME Qs because it's too easy/buzzy. - **Atopy** is really HY (asthma, seasonal allergies, eczema). - Hx of Giardia infection. - Associated with other autoimmune diseases (i.e., vitiligo, Celiac). - Celiac Ab screen falsely negative in IgA deficiency. - IgA deficiency can be associated with false-positive pregnancy tests (yes, fucking weird, but due to heterophile Ab production apparently, with zero relation to EBV). - Answer on USMLE will often be "deficiency of mucosal immunoglobulin" or "deficiency of humoral immunity," rather than just "IgA deficiency."
DiGeorge syndrome	- T cell deficiency caused by 22q11 deletion. - Failure of development of 3rd and 4th pharyngeal **pouches** (not arches) → 3rd becomes thymus + two inferior parathyroids (these structures form a triangle → 3rd); 4th becomes two superior parathyroids. - Absent thymic shadow. - Hypocalcemia due to hypoparathyroidism (positive Chvostek and Trousseau signs of hypocalcemia). - Tetralogy of Fallot (VSD, overriding aorta, pulmonic stenosis, RVH) and truncus arteriosus are HY cardiac anomalies. - Cleft lip/palate may or may not be mentioned. - Patient will have *normal* antibody levels (even though T cells activate B cells, choose normal Ab levels).
Chronic mucocutaneous candidiasis	- T cell dysfunction. - Recurrent candidal skin infections since birth. - Choose "deficiency of cell-mediated immunity." - Associated with other autoimmune diseases (i.e., T1DM, thyroiditis). - 17F + recurrent candidal skin infections since birth + one year Hx of autoimmune thyroiditis + 2-yr Hx of T1DM; mechanism for infections? → answer = "deficiency of cell-mediated immunity."
Chronic granulomatous disease (CGD)	- Aka NADPH oxidase deficiency. - Usually X-linked recessive. - First enzyme of the respiratory burst; converts molecular O_2 → superoxide. - Recurrent infections with catalase (+) organisms: *Serratia, Pseudomonas, Aspergillus, Candida, E. coli, Staph, H. pylori.* - Condition results in ↓ ability of patient to synthesize enough H2O2 to overwhelm catalase (+) organisms. - Diagnose with dihydrorhodamine test (tetrazolium blue assay is atavistic and the **wrong** answer if both are listed, as per Step 1 NBME).
Myeloperoxidase (MPO) deficiency	- Autosomal recessive. - Last enzyme in respiratory burst; converts H_2O_2 → hydroxyl-halide radicals (bleach). - Q will literally give you a one-liner telling you someone can't produce hydroxyl-halide radicals, and the answer is myeloperoxidase deficiency. - Q can mention myeloperoxidase deficiency in vignette, and then the answer is "impaired conversion of hydrogen peroxide to hypochlorous acid." - Usually presents as mix of *Staph aureus* and *Candida* infections.

	- Q on Free 120 gives mix of *S. aureus* and *Candida* infections in 6-year-old girl with AR disorder; Q wants MPO deficiency as answer, not CGD. Even though CGD can present with same organisms, the literature not only says 70% of CGD cases are XR, but if CGD occurs in children, it's usually XR, since less-frequent AR CGD is often more mild and presents later.
Wiskott-Aldrich syndrome	- X-linked recessive (*WAS* gene). - T cell cytoskeletal dysfunction. - Triad of: immunodeficiency, eczematoid skin lesions, thrombocytopenia. - ↑ serum IgA/IgE; ↓ IgM. - 12M + *Pneumocystis* infection + Hx of nosebleeds + eczema; Q asks what cell is dysfunctional? → answer = T cell. - Answer can be "deficiency of cell-mediated immunity."
Leukocyte adhesion deficiency (LAD)	- Deficiency of CD18/LFA-1 integrin. - ↓ ability of leukocytes to leave blood vessels for sites of infection → ↓ pus. - Classically associated with delayed separation of umbilical cord. - Answer can be "defective leukocyte chemotaxis" instead of just "LAD."
Chediak-Higashi syndrome	- Phagolysosomal fusion disorder with disruption of microtubule function. - Melanosome function also impaired → partial albinism / fairer skin than siblings. - Giant granules in phagocytes in neutrophils and eosinophils.
Ataxia-telangiectasia	- Failure of double-stranded DNA break repair. - Cerebellar atrophy → ataxia. - Telangiectasias. - ↑ susceptibility to radiation → ↑ risk of leukemia and lymphoma.
Hyper-IgM syndrome	- Deficiency of CD40L on CD4+ T cells. - ↓ activation of CD40 on B cells → ↓ B cell activation → ↓ ability to differentiate into plasma B cells + isotype class-switch → Ab isotype "stuck" at IgM.
Hyper-IgE (Job) syndrome	- *STAT3* mutation. - **FATED** → abnormal **F**acies, staphylococcal cold **A**bscesses, retained primary **T**eeth, hyper Ig**E**, **D**ermatologic findings (e.g., eczematoid lesions).
IL-12 receptor deficiency	- Recurrent TB infections. - Important point for USMLE is that Th1 CD4+ T cells and macrophages activate each other in a cyclical loop via IL-12 and IFN-γ. Th1 cell lacking IL-12 receptor cannot get activated and therefore does not secrete IFN-γ → macrophages not activated → ↓ granuloma formation.
Common variable immunodeficiency (CVID)	- Normal B cell number but defective maturation into plasma cells. - ↓ immunoglobulin production. - Will be the answer if the Q gives you an *adult* who has had recurrent sinopulmonary infections (similar to IgA deficiency), except ↓ Ig not limited to IgA; in addition, if vignette gives you atopy or Giardia, answer is IgA deficiency not CVID.
Terminal complement deficiency	- Deficiency of C5-C9 complement proteins. - Cannot form membrane attack complex (MAC). - Recurrent *Neisseria* infections (both gonococcal and meningococcal). - Q will often have deficiency of "C7" or "C8" as the answer (i.e., they choose a random terminal complement protein), or "terminal complement deficiency," or "complement-mediated immunodeficiency."

Peds Endocrine

SIADH vs DI vs PP	
SIADH	- Syndrome of Inappropriate Anti-Diuretic Hormone secretion → means too much ADH (vasopressin secretion). - Central SIADH → follows head trauma, meningitis, brain cancer, and pain (Peds or adults). - Ectopic SIADH → small cell lung cancer secreting ADH (adults). - Drug-induced ADH → ultra-rare on USMLE, but carbamazepine can do it (Peds or adults). - Patient will have dilute serum and concentrated urine: - ↓ Serum sodium, ↓ serum osmolality, ↓ serum specific gravity. - ↑ Urinary osmolality, ↑ urinary specific gravity. - You must know serum sodium is normally 135-145 mEq/L. So in SIADH, it's <135. - Serum vs urinary osmolalities will be all over the place and you do *not* need to memorize values. They might say serum osmolality is 250 and urinary is 750, and then you say, "Well I can tell serum is dilute compared to urine, which sounds like SIADH." - Specific gravity will be 1.000-1.030 on USMLE. Sounds obscure, but it shows up quite a bit, particularly on 2CK Qs. It's to my observation that values 1.000-1.006ish are "dilute"; 1.024-1.030 are "concentrated." For 9/10 Qs, values will be what you expect. - "Fluid restriction" is first answer in diagnosis on 2CK. We want to see how serum/urinary values change in response. - Demeclocycline is answer on 2CK offline NBME 8 for treatment of SIADH. Demeclocycine is technically a tetracycline antibiotic but isn't used because it can cause insensitivity to ADH at the kidney (i.e., nephrogenic diabetes insipidus). So we essentially cause a 2nd problem that cancels out the 1st problem.
DI	- Central diabetes insipidus → means not enough ADH secretion by hypothalamus, or the posterior pituitary is unable to release it properly. - Nephrogenic DI → insensitivity to ADH at the kidney (serum ADH is ↑). - Similar to central SIADH, central DI can be caused by head trauma, meningitis, and cancer. - Nephrogenic DI is caused by **lithium, demeclocycline, hypercalcemia,** and **NSAIDs.** - There is an NBME Q that asks about a patient's response to ADH who is on chronic NSAIDs, and the answer is "↔ Response to ADH" and "↔ urinary osmolality" (meaning no change). - 2CK Q gives patient with primary hyperparathyroidism and ↑↑ serum calcium + nephrogenic DI; Q asks cause of the DI → answer = hypercalcemia. High calcium can cause renal insensitivity to vasopressin. - Serum vs urinary values are the opposite of SIADH: - ↑ Serum sodium (>145 mEq/L), ↑ serum osmolality, ↑ serum specific gravity. - ↓ Urinary osmolality, ↓ urinary specific gravity. - When we are trying to first diagnose DI, the first thing we do is fluid restriction, same as with SIADH. We want to see how serum/urinary values change first. - After we determine that the urine is staying dilute + the serum is staying concentrated, the next best step is desmopressin (analogue of vasopressin). If the urine gets more concentrated, (i.e., if the drug works), we know central DI is the diagnosis and we're merely deficient in ADH. - If desmopressin doesn't work, we know we have nephrogenic DI. I should point out that even in nephrogenic DI, desmopressin might work *but only very little,* whereas with central DI, administration will ↑↑ urinary osmolality robustly. It will be obvious on USMLE. But my point is, don't say, "Oh well desmopressin worked like 5% so we can't have nephrogenic DI here." - Treatment for central DI is therefore desmopressin. - Treatment for nephrogenic DI is NSAID + a thiazide. Sounds weird, but ↓ Na^+ reabsorption induced by thiazides in the early-DCT promote compensatory ↑ Na^+ reabsorption in the PCT, where water follows Na^+ and our net loss of fluid is less than without the thiazide. In healthy

	individuals, however, they *will* lose more net fluid with the thiazide. The NSAID presumably ↓ renal blood flow, which will ↓ GFR and ↓ net fluid loss. - There is difficult 2CK NBME Q where they say patient is on lithium + has ↑ urinary output, and fluid restriction is wrong answer to this question (I say hard because 9/10 times, fluid restriction is correct when it's listed); answer = NSAID + thiazide diuretic. The implication is, if it's obvious what the patient's diagnosis is already nephrogenic DI, going straight to Tx is acceptable.
PP	- Psychogenic polydipsia means the patient is simply drinking too much. - Both the urine and serum will be dilute. - Serum vs urinary values are the opposite of SIADH: - ↓ Serum sodium (<135 mEq/L), ↓ serum osmolality, ↓ serum specific gravity. - ↓ Urinary osmolality, ↓ urinary specific gravity. - I'd say 3/4 Qs on USMLE are obvious and will say some psych patient is drinking lots to "clear himself from evil spirits," etc. - Probably 1/4 Qs won't be an obvious psych vignette, but will just show you the lab values where you have to say, "The urine and serum are *both* dilute, so this is psychogenic polydipsia." - First step in diagnosis is fluid restriction in order to see how urinary/serum values change.

HY Thyroid diagnoses	
Hashimoto thyroiditis	- Autoimmune destruction of thyroid gland. - Shows up in teenager on a Peds form. - Lymphocytic infiltrate seen on biopsy (asked on 2CK NBME, even though sounds like Step 1 detail). - Anti-microsomal (aka anti-thyroperoxidase) + anti-thyroglobulin antibodies. - ↓ T3, ↓ T4, ↑ TSH. - ^{131}I uptake scan is ↓ or patchy (i.e., the gland is not producing hormone). - Buzzy findings are weight gain, brittle hair, dry/doughy skin, and cold intolerance. The USMLE often will omit these findings because they're too easy. - HY additional findings I've seen show up in NBME Qs are: bradycardia (55-60), menstrual irregularity, dysthymia/depression/apathy, carpal tunnel syndrome (due to glycosaminoglycan [GAG] deposition), ↑ cholesterol, proximal muscle weakness with ↑ creatine kinase (hypothyroid myopathy); transaminitis (sounds weird, but LFTs can be ↑). - Associated with other autoimmune diseases (e.g., type I diabetes, pernicious anemia, vitiligo) and immunodeficiencies (e.g., IgA deficiency). For USMLE, a HY big-picture concept is that "Autoimmune diseases go together," where if you have one autoimmune disease, there's ↑ risk of other autoimmune diseases. The strict HLA associations aren't important (outside of HLA-B27). In addition, "Autoimmune diseases and immunodeficiencies to together," where having autoimmune disease in general ↑ risk of immunodeficiencies, and vice versa. - ↑ risk of non-Hodgkin lymphoma (e.g., primary CNS lymphoma). - Tx = levothyroxine or triiodothyronine.
Graves disease	- Activating autoantibody against TSH receptor. This antibody is called thyroid-stimulating immunoglobulin (TSI). - ↑ T3, ↑ T4, ↓ TSH. - Lymphocytic infiltrate + hyperplastic thyroid follicles seen on biopsy. - ^{131}I uptake scan shows diffusely ↑ uptake. - Exophthalmos (aka proptosis; protrusion of eyes) and pretibial myxedema (non-pitting edema of shins due to GAG deposition) are specific findings in Graves. The vignette need not mention either of these findings, but if the Q specifically tells you "there's no exophthalmos," they're saying it's not Graves. - Buzzy findings are tachycardia, heat intolerance, sweating, and palpitations.

	- Myopathy can also occur with ↑ CK, but USMLE tends to give this finding in hypothyroidism. - Can cause atrial fibrillation in younger patients (normally AF is elderly). - "Thyroid storm" is an acute exacerbation of Graves, often precipitated by a stress factor such as surgery, trauma, or infection, where thyroid hormone production goes ↑↑↑, causing dangerous physiologic responses, such as hyperthermia, tachycardia, and altered mental status. - Thyroid storm can also cause adrenal crisis (i.e., acute ↓ in blood pressure due to rapid consumption of cortisol), especially in those with concurrent adrenal insufficiency taking exogenous steroids. Tx for the low BP is IV glucocorticoid (i.e., hydrocortisone or methylprednisolone). - Tx for general Graves is thionamides (propylthiouracil [PTU] and methimazole). - Both PTU and methimazole can cause neutropenia/agranulocytosis. - Propranolol is used for tachycardia in hyperthyroidism (beta-blockade inhibits 5-deiodonase). - Oral prednisone is specifically used for Tx of exophthalmos, not thionamides. - Tx of thyroid storm on USMLE is tetrad of: 1) PTU, 2) propranolol, 3) steroids, and 4) potassium iodide.
deQuervain thyroiditis	- Aka subacute granulomatous thyroiditis, or just simply subacute thyroiditis. - Mechanism is viral infection followed by a **painful/tender** thyroid. There is inflammation of the thyroid gland, which causes the spacing between the cells to increase slightly, allowing for the release of pre-formed thyroid hormone into the blood. Therefore we have ↓ TSH, ↑ T3, ↑ T4, ↓ ^{131}I uptake. - The gland is not over-producing thyroid hormone. This is why uptake is not increased. - Subacute granulomatous thyroiditis can be either hypo- or hyperthyroid. The key detail you need to know is that **uptake is always decreased *even if the patient is hyper-*.** - DeQuervain vignettes will almost always be given as hyperthyroidism because the USMLE wants to specifically assess that you know uptake is decreased. If they give you hypo-, of course you'll select decreased for uptake. - Decreased uptake applies to all thyroiditis conditions (i.e, deQuervain, drug-induced, and postpartum). - Viral infections are often asymptomatic, so most deQuervain vignettes will not mention the viral infection.
Factitious thyrotoxicosis	- Aka surreptitious thyrotoxicosis. - USMLE need not mention the patient is a pharmacist (implying access to the hormone). - Q will give hyperthyroid patient with small, non-palpable thyroid gland (atrophic due to suppressed TSH). - As mentioned earlier, T4 is converted to T3, but T3 isn't converted to T4. - So if thyroxine is administered, we will have ↑ T3, ↑ T4, ↓ TSH. - If triiodothyronine is administered, we will have ↑ T3, **↓ T4,** ↓ TSH.

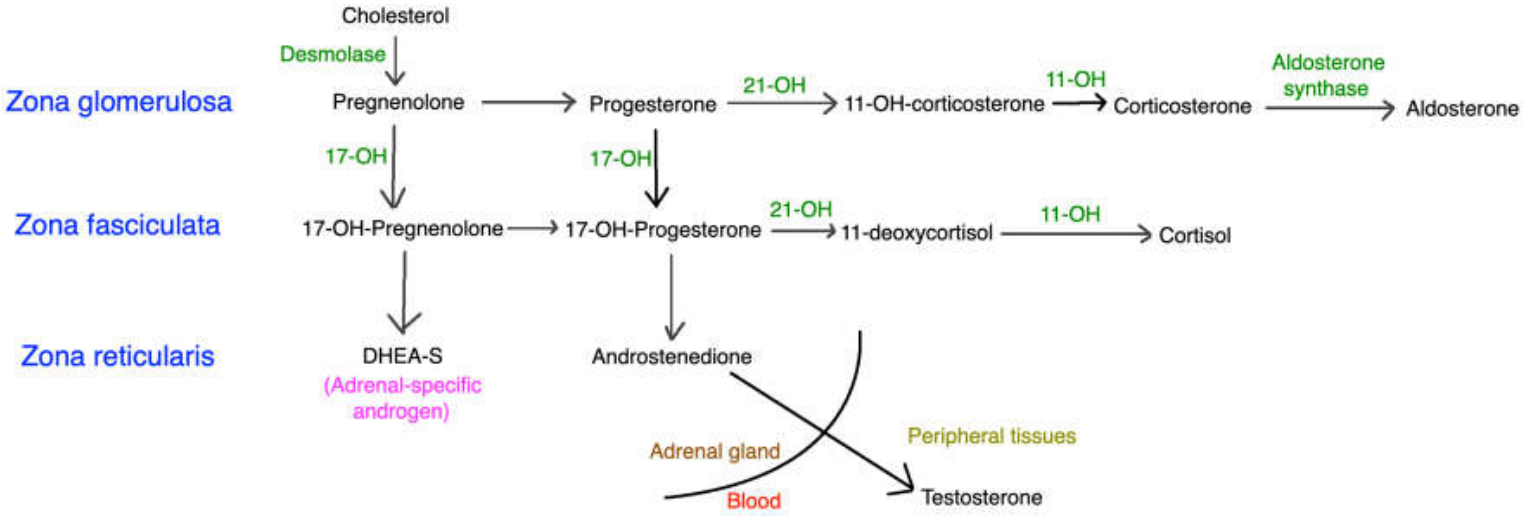
Thyroid in pediatrics	
Thyroglossal duct cyst	- As mentioned earlier, benign, painless, midline neck mass in a school-age kid that arises from a persistent thyroglossal duct (i.e., the embryologic remnant of the thyroid gland's descent from the base of the tongue to its final position in the neck). - 2/3 of Qs will give a painless midline neck lump that moves upwards with swallowing or protrusion of the tongue (due to its attachment to the hyoid bone and/or the base of the tongue).

	- 1/3 of Qs won't mention the buzzy upward movement with swallowing or protrusion of the tongue; instead, it will just say a kid has a painless midline neck mass just inferior to the hyoid bone that demonstrates ^{99}Tc uptake.
Lingual thyroid	- Obscure cause of hypothyroidism that shows up on an offline NBME. - Refers to presence of ectopic thyroid tissue located at the base of the tongue due to failure of the thyroid gland to descend from the foramen cecum. - This ectopic tissue may be the only functioning thyroid tissue in the body, resulting in hypothyroidism. - The USMLE Q will say a kid has hypothyroidism + a midline neck lump located high in the neck. They say nothing about protrusion of the tongue or uptake into the mass. - Can sometimes cause dysphagia (trouble swallowing), dysphonia (voice changes), or dyspnea (difficulty breathing).
Cretinism	- Aka congenital hypothyroidism. - Most common causes are iodine deficiency in the mother during pregnancy (worldwide) and fetal thyroid dysgenesis (western countries). - Leads to mental retardation (poor myelin sheath development), impaired bone growth, hypotonia, macroglossia, and protuberant abdomen (due to umbilical hernia; can be confused with kwashiorkor, which is protein-calorie malnutrition causing ascites). - Screened for at birth using the heel-prick test to prevent exacerbation.
TBG deficiency	- The USMLE does not expect you to know about some obscure condition called thyroid-binding globulin deficiency. The reason they ask about this is because the role of TBG on thyroid hormones in pregnancy is exceedingly high-yield, so if you know the reasoning/mechanism behind the latter, you can easily infer what would be seen in TBG deficiency (i.e., the inverse). - ↔ TSH, ↔ T3, ↔ free T4, ↓ total T4. - In pregnancy, since TBG is high, this ultimately results in high total T4 despite a normal free T4. So the student can easily infer, "Well, if our TBG merely is low, rather than high, then total T4 must be low while free is same. Sort of like pregnancy but just the opposite direction."

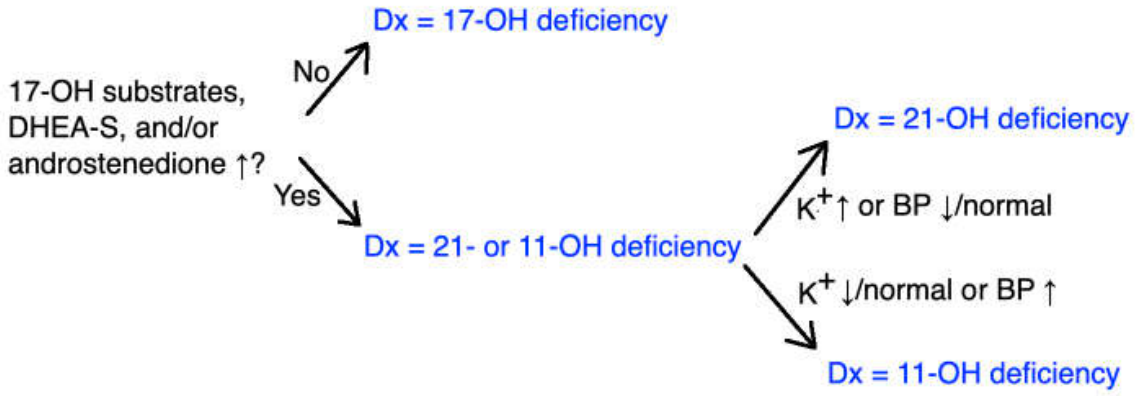
Primary adrenal DDx affecting aldosterone	
Conn syndrome	- Aldosterone-secreting tumor. - Causes ↑ Na^+, ↓ K^+, ↑ HCO_3^-, ↑ pH. - Renin is ↓ due to negative-feedback at the kidney. In other words, the ↑ aldosterone from the tumor ↑ fluid retention, which causes ↑ renal blood flow, which shuts off renin production. - In ~50% of aldosterone-derangement Qs, Na^+ will be normal. This confuses students, but it's because the body has strong ability to maintain Na^+ (i.e., also via ADH). - Tx = spironolactone (aldosterone receptor antagonist) prior to surgery.
Addison disease	- Autoimmune destruction of the adrenal cortex, leading to ↓ aldosterone *and* ↓ cortisol. - I haven't seen the USMLE ever care about "decreased androgens," so the effect is predominantly on the zona glomerulosa and fasciculata. - Called "primary adrenal insufficiency" because the adrenal gland itself is under-producing hormones. - Addison is idiopathic / can be seen in patients with Hx of other autoimmune diseases, either in themselves or family members (i.e., the vignette throws in the side-detail that the brother has RA, or the sister has IBD, or the patient has Hx of thyroiditis). - ↓ Na^+, ↑ K^+, ↓ pH, ↓ bicarb. - Blood pressure will either be low-normal or low, due to both ↓ aldosterone as well as ↓ cortisol. I explicate this because students tend to just mention the ↓ aldosterone as the reason BP is ↓, but the effect of ↓ cortisol is HY for USMLE.

	- ACTH and α-MSH are both ↑ due to ↓ cortisol → hyperpigmentation. Although I'd say they only mention this in ~50% of vignettes. - Patients can present with eosinophilia. This is because cortisol normally plays a role in helping to sequester eosinophils within the spleen + facilitates eosinophil apoptosis, so ↓ cortisol means ↑ eosinophils. Don't go chasing stool ova and parasites. Qs can give you eosinophils of 15-25%, where you're like wtf? They should normally be <5%. - Classic Addison vignette is patient (young or old, I've seen both) who has ongoing fatigue + hyperpigmentation, and then the answer is just "ACTH stimulation test." - Exogenous ACTH should normally cause a robust ↑↑ in cortisol from the adrenal fasciculata. In the setting of primary adrenal insufficiency, since the adrenal gland is fucked up, giving exogenous ACTH won't do jack shit. - ↓ cortisol causes chronic fatigue syndrome. This is one of the most over-looked details in NBME Qs and in real life. Any time a patient has ongoing chronic fatigue that is unexplained, cortisol levels should be checked. If they are ↓, hydrocortisone will immediately boost the patient. - Tx for Addison = fludrocortisone +/- hydrocortisone. Fludrocortisone is a mineralocorticoid similar to aldosterone. Hydrocortisone is a glucocorticoid similar to cortisol.
Waterhouse-Friderichsen	- Meningococcal septicemia causing bilateral hemorrhagic necrosis of the adrenal glands, resulting in ↓ aldosterone and ↓ cortisol. - Vignette will be ↓ BP + ↓ Na^+ + ↑ K^+ + ↓ bicarb in patient with meningococcal infection (e.g., recent meningitis + non-blanching rash), where answer = ACTH stimulation test.

Adrenal crisis
- Patients who are on chronic glucocorticoids (i.e., prednisone) for autoimmune conditions such as IBD, JIA, SLE, etc., will have a chronically suppressed adrenal gland that is atrophied, since prednisone is a cortisol analogue that induces negative-feedback and shuts off ACTH production. - Can occur in Peds in setting of infection or surgery. - This means that if these patients encounter a stressor such as surgery, trauma, infection, or thyroid storm, where glucocorticoids normally experience ↑ rate of consumption, they are unable to "mount a stress response," (i.e., unable to produce an acute ↑ production of cortisol), so they are prone to acute glucocorticoid deficiency and a precipitous ↓↓ in BP. - Q will tell you kid has SLE treated with prednisone who goes into surgery and gets a ↓↓ drop in BP to 80/40 + fluids and pressors don't help; Q asks next best step → answer = IV hydrocortisone or methylprednisolone. What we are doing is replenishing the glucocorticoid. "Pressors" can refer to NE, E, or even desmopressin. - Remember that without cortisol (or cortisol equivalent, i.e., exogenous glucocorticoid), we can't get adequate α1-recepor expression on arterioles, so even if NE and E are floating around in normal amounts (or given exogenously), they can't bind to α1-receptors to maintain BP. - This concept is referred to as: "glucocorticoids are permissive of the effects of catecholamines," where they literally "permit" NE and E to do their job. - The Q can also tell you patient being treated with prednisone for SLE or RA experiences an acute thyroiditis (remember autoimmune diseases go together) and gets ↓↓ in BP → answer = hydrocortisone (after fluids). Or patient with autoimmune disease gets an infection + ↓↓ in BP → answer = hydrocortisone (after fluids). - I reiterate this concept because it is exceedingly HY on 2CK content in particular and is missed by students all of the time. If you're studying for Step 1, you still need to know the physio / pharm behind this concept.



Congenital adrenal hyperplasia (CAH)	
- Autosomal recessive. -21-, 11-, and 17-hydroxylase deficiencies. 21-hydroxylase deficiency is most common. - ACTH is ↑ in all three types. This is the mechanism USMLE wants for the hyperplasia. That is, cortisol production is impaired in all three, so ACTH goes ↑ to compensate → stimulates hyperplasia of adrenals. - The USMLE will give a vignette of CAH, followed by "which of the following is the most likely explanation for these findings?" → answer = "Increased ACTH secretion." Sounds weird, but it's on NBME. - The relevance of these conditions is that they are a cause of ambiguous genitalia and/or virilization. - Presentations are variable, with Qs giving vignettes in neonates, children, and adults. So the way to differentiate them comes down to electrolytes and BP. - For both 21- **AND** 11-hydroxylase deficiencies: ○ **↑ 17-OH substrates (i.e., 17-OH-progesterone and 17-OH-pregnenolone).** ○ **↑ DHEA-S, ↑ androstenedione.** ○ If you see any of these up in the Q, you know right away answer is either 21- or 11-deficiency. This is because it's impossible to get 17-OH substrates in 17-hydroxylase deficiency.	
21-hydroxylase deficiency	- **↑ Serum K^+, ↓ Na^+, ↓ pH, ↓ $HCO3^-$, ↓ glucose.** - **↓/↔ BP.** - Cortisol and aldosterone, as well as their precursors, are all low.
11-hydroxylase deficiency	- **↓/↔ Serum K^+, ↑/↔ Na^+, ↑/↔ pH, ↑/↔ $HCO3^-$, ↑/↔ glucose.** - **↑ BP.**



	- Cortisol and aldosterone are low, but their precursors, which have some effect, are high.
17-hydroxylase deficiency	- ↓ **17-OH substrates (i.e., 17-OH-progesterone and 17-OH-pregnenolone)** - ↓ **DHEA-S,** ↓ **androstenedione** - ↓ **Serum K^+,** ↑ **Na^+,** ↑ **pH,** ↑ **$HCO3^-$** (high aldosterone) - ↓ **glucose** (cortisol and its precursor are low).

Neuroblastoma
- Essentially the "pediatric pheo"; *N-myc* gene. - Secretes catecholamines; can calcify (same as pheochromocytoma in adults). - Grows from the median (midline) sympathetic chain. Usually intra-abdominal but can sometimes be in the posterior mediastinum. Sounds odd, but the latter shows up on a Peds CMS form. - Causes ↑ BP and findings such as opsoclonus-myoclonus syndrome (dancing eyes) and violaceous eyelids (sounds like heliotrope rash of dermatomyositis but unrelated). - Diagnose with urinary homovanillic acid (HVA) and vanillylmandelic acid (VMA), which are metabolites of catecholamines. - Metaiodobenzylguanidine scan can be done after HVA and VMA detected.

Diabetes mellitus (DM)	
- The word diabetes means polydipsia + polyuria (i.e., drinking a lot and peeing a lot). - I see students frequently misconstrue diabetes mellitus with diabetes insipidus, since both vignettes will give patients who have increased thirst and urination. - A key detail that distinguishes acute diabetes mellitus from diabetes insipidus is that DM vignettes can often mention weight loss due to ↑ catabolism. Weight loss is also seen in acute type II, even though the patient is usually overweight. Vomiting is another HY finding in acute DM. - So if vignette says ↑ thirst + ↑ urination + weight loss +/- vomiting, this is DM, not DI. - For extensive info on the effects of diabetes on different organ systems (i.e., cardio, renal, ophthal, GIT, pregnancy, etc.), use those corresponding HY PDFs I've made. I will keep the focus here on endocrine and electrolytes / acid-base. - Diagnosis of DM is made by measuring two fasting glucoses >126 mg/dL, any one random fasting glucose >200 mg/dL, or an HbA1c of >6.5%.	
Type I	- Absence of insulin; autoimmune-mediated; no one specific gene, although HLA-DR3/4 are associated. - Often described as "genetic susceptibility with environmental trigger" (i.e., viral infection, usually Coxsackie B). - Occurs almost always in children or teenagers of normal BMI. - Pathogenesis is usually combo of an antibody- (anti-GAD65 or -ZnT8) and T-cell-mediated process that leads to destruction of β-islet cells in the tail of the pancreas → ↓ insulin secretion → inability to recruit GLUT4 to cell surface of skeletal muscle and adipose tissue → inability to take up glucose into cells → cells are starved of energy → ↑ gluconeogenesis + ketogenesis. - Hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue. - Biopsy of pancreas shows inflammatory infiltrate in acute type I DM; in late type I (i.e., years later), it shows atrophy and fibrosis. - It is the absence of insulin in type I DM that enables ketogenesis. In type II DM, ketones are low, not high, because insulin is present. - Ketone bodies (USMLE likes β-hydroxybutyrate) are highly acidic, so will drop the blood pH (high anion-gap metabolic acidosis). - Vignette will give kid or teenager with polyuria, polydipsia, and sometimes vomiting, who has severe metabolic acidosis (bicarb ↓), which is due to diabetic ketoacidosis. Breath can be fruity in odor (smell of ketones). DKA will have:

	- **↓ pH, ↓ HCO_3^-, ↓ CO_2.** - Ketone bodies are acidic → causes metabolic acidosis (decreased bicarbonate). - CO_2 is acidic, so we blow that off to compensate. Kussmaul breathing is deep, labored breathing seen in DKA. - **↑ β-hydroxybutyrate, ↑ serum osmolality, ↑ anion gap.** - β-hydroxybutyrate is a ketone body. Since we have ketosis (ketone production), it will be increased. I have seen this specific ketone body as an arrow on an offline Step 1 NBME. - Serum osmolality is increased because serum glucose is markedly increased (insulin isn't present to drive glucose into cells). Even though serum sodium is low (so you might think osmolality is low similar to SIADH), the ↑ glucose makes serum osmolality high. - DKA is the D in MUDPILES, so anion gap is up. - **↓ Serum Na^+, ↑ serum K^+, ↓ total body K^+, ↑ GFR.** - Serum sodium is decreased mostly due to dilutional hyponatremia. As mentioned earlier, serum osmolality is high, which causes fluid retention intravascularly → dilutes out serum sodium. In addition, low insulin means less glucose enters cells → less ATP production → less cellular Na^+/K^+-ATPase activation → sodium not pumped out of cells. - We describe the state of K^+ as: hyperkalemia (high serum K^+) despite a low total body potassium. - Serum potassium is high for two main reasons: 1) **Potassium-proton shift:** Acidosis (due to ketosis) causes protons in the blood to exchange with potassium in the cell via the H^+/K^+-antiporter (i.e., H^+ goes into cell; K^+ moves out). 2) **Insulin drives potassium into cells (and we don't have insulin in DKA):** Insulin normally leads to the upregulation of a Na^+/K^+-ATPase antiporter that moves $3Na^+$ out for every $2K^+$ in. If insulin is low, then less K^+ will be moved into the cell → hyperkalemia (high potassium in blood). - Total body potassium is low because of increased losses at the kidney. Since serum potassium is high, more potassium is filtered through the glomerulus, so more is excreted (i.e., the kidney thinks there's too much potassium in the body since more is being filtered, so it tries to get rid of more of it). However, this causes bodily depletion of K^+ because the processes that cause hyperkalemia are unmitigated, so the kidney continues to excrete potassium. - GFR is increased because of hyperfiltration secondary to hyperglycemia. That is, since serum glucose is high, more glucose is filtered through the glomerulus, which pulls water with it → hyperfiltration. - Tx for DKA is normal saline first (i.e., 0.9% NaCl), prior to giving insulin. - Serum potassium will drop as insulin is given. If the Q tells you patient has DKA + shows you normal potassium (even though it's usually high) + insulin is given + patient now has arrhythmia, the answer is potassium for which electrolyte is disturbed (i.e., potassium was normal prior to giving insulin, but now it's been driven into cells and the patient's developed hypokalemia). - Bicarb is always a wrong answer for low pH in DKA. - Don't confuse DKA with a hypoglycemic episode. The latter will be a type I diabetic who went out to do exercise or who was cleaning the roof of his house, who is now found unconscious → if insulin dose is not decreased prior to exercise, patient can get hypoglycemia. The Tx is intramuscular glucagon (2CK likes this) or dextrose in water. - Sometimes the USMLE can give a bit of a weird Q where they say a patient is given insulin for diabetes but has prolonged hypoglycemia + they ask why → answer = "deficiency of counter-regulatory glucagon." You need to know that in advanced diabetes, the glucagon-producing α-islet cells of the pancreas can also get fucked up. I've also seen this Q asked for chronic alcoholism and pancreatectomy; in both cases, if insulin is given, prolonged hypoglycemia can result from "deficiency of counter-regulatory glucagon."
Type II	- Insulin resistance; seen usually in high-BMI adults, but can rarely show up in Peds (2CK Peds form gives type II in a 17-yr-old).

	- High BMI causes insulin resistance because adipose tissue secretes adipokines that impair insulin signaling. - Insulin is *high* initially (hyperinsulinemia) due to β-cell hypertrophy in order to compensate for peripheral insulin resistance. - Ketones are absent/low because the presence of insulin inhibits ketogenesis. - The two above points are the highest yield pieces of info for USMLE. - Biopsy of the pancreas initially shows β-cell hypertrophy, but years later will show amyloid deposition and cell shrinkage. - Same as with type I, hyperglycemia occurs as a result of continued glucose production by the liver (gluconeogenesis) in combination with ↓ uptake by skeletal muscle and adipose tissue. - Hyperosmolar hyperglycemic state (HHS) is seen in type II; DKA is strictly type I on USMLE. - In HHS, glucose is usually >600 mg/dL (in DKA, it need not be this high) and bicarb is usually not below 20 mEq/L (NR 22-28); in DKA, bicarb can be as low as it wants (I've seen 5 mEq/L on NBME). - The USMLE vignette will give massive paragraph Q where they mention glucose is 700 and then ask for the cause of mental status change → answer = "hyperosmolality," or simply "hyperglycemia," or "osmotic diuresis" (causes severe dehydration). I point this out because this is particularly prevalent on 2CK NBMEs, where they'll give a 15-line bullshit paragraph with tons of lab values that the student spends 9 minutes reading, but meanwhile you can see glucose is 650 and then the answer is just "osmotic diuresis" or "hyperosmolality"; should take two seconds to answer. - Mental status change is technically due to hyperosmolality in the ECF of the brain pulling water out of the ICF (i.e., the cells). - Treat HHS with normal saline first prior to insulin.

Bullet points

- 5F + 1-year Hx of chronic nonproductive cough + all asthma Txs not effective + "CXR shows no abnormalities except for a linear consolidation in the right middle lobe"; Dx? → Peds NBME = bronchiectasis → students says wtf? (because normally bronchiectasis is "cups and cups" of foul-smelling sputum in CF, TB, or COPD) → Dx is called "right middle lobe syndrome" (Google it) and can present as bronchiectasis in peds characterized by nonproductive cough and linear consolidation on CXR. You don't have to agree that it's HY, but that doesn't change the fact that it's on the Peds NBME.
- 7F + seizure-like episode + during episode, ECG shows P waves at 80bpm but not QRS complexes + normal sinus rhythm resumes after 20 seconds + patient fully alert after one minute; Dx? → answer on Peds NBME = Adam-Stokes attack → not true seizure disorder as per EEG; arrythmia leads to hypoxia of brainstem → seizure-like fits ensue.
- 8M + low-pitched vibratory murmur heard throughout cardiac cycle + loudest at left upper sternal border when sitting + murmur disappears when supine and neck rotated; Dx? → answer on Peds NBME = venous hum → benign/innocent pediatric murmur that will disappear as child grows; caused by turbulent venous return in SVC and jugular veins.
- 2M + fever 101F + diffuse crackles across both lungs + CXR shows diffuse interstitial infiltrates + ongoing Hx of thrush, failure to thrive, and frequent diarrhea + mother used drugs during pregnancy; what's the next best step in diagnosis? → answer = "silver stain of bronchoalveolar fluid"; this is pneumocystis jirovecii pneumonia (PJP) in child with AIDS secondary to maternal drug use (presumably IV); NBME will ask any number of Qs on this, but important take-home point is bronchoalveolar lavage is done after the CXR; PJP is also notably bilateral diffuse infiltrates with a ground-glass appearance; if NBME gives you lobar presentation, it's not PJP (usually S. pneumo).
- 6-month-old boy + runny nose for 2 days + respiratory stridor + temp 101F; what is the most likely location of the pathology? → answer = larynx; Dx is laryngotracheobronchitis (croup) caused by parainfluenza virus; classically "seal-like, barking" cough in school-age child; Tx = supportive, or nebulized racemic epinephrine if severe.
- 10-month-old + fever 101F + bilateral wheezes; Dx? → answer = "community-acquired viral infection" → RSV bronchiolitis; Tx = supportive (ribavirin and palivizumab are almost always wrong).

- 8M + from immigrant family + sitting forward in tripod position + drooling; next best step? → answer = intubation; Dx = epiglottitis → Haemophilus influenzae type B (patient not vaccinated); if Q asks for antibiotic, give ceftriaxone; for close contacts, give rifampin.
- 8M + recently convalesced from URTI + now has stridor + fever 101F; Dx? → answer = bacterial tracheitis → usually S. aureus; Dx w/ bronchoscopy; Tx with third-gen cephalosporin + aminopenicillin (e.g., IV flucloxacillin).
- 8-month-old girl + stridor that improves with neck extension; Dx? → answer = vascular ring → weird but HY diagnosis for peds → aberrant embryologic development where the aorta and/or surrounding vessels form a ring around the esophagus and/or trachea.
- 8-month-old girl + stridor that improves when prone or upright; Dx? → laryngomalacia → most common cause of stridor in peds → soft cartilage of upper larynx collapses during inhalation.
- 3F + 2-wk Hx of cough and nasal congestion + snoring loudly past 6 months + P/E shows she breathes predominantly through her mouth + 1/6 holosystolic murmur and loud S2 + CXR shows cardiomegaly + increased pulmonary vascular markings + echo shows RV hypertrophy and mild tricuspid regurg; what is the most appropriate long-term Mx for this patient? → answer = adenoidectomy and tonsillectomy → sounds weird, but HY for Peds shelf → can cause obstructive lung disease with cor pulmonale → loud S2 and increased pulmonary vascular markings suggest pulmonary HTN; tricuspid regurg can be seen sometimes in pulmonary HTN.
- 2F + stridor + laryngoscopy shows small growths of larynx; Dx? → answer = HPV 6/11 → laryngeal papillomatosis.
- 9M + grey pseudomembrane seen in oropharynx; Tx? → answer = diphtheria antitoxin.
- 12M + steatorrhea + recurrent URTIs + new-onset stridor; why the new-onset stridor? → answer = nasal polyps (common in cystic fibrosis); steatorrhea due to exocrine pancreas insufficiency.
- 7F + painless midline neck lump that moves upward when swallowing; Q asks what would be seen on biopsy of the lesions → answer = thyroid tissue; Dx = thyroglossal duct cyst.
- 7F + painless lateral neck mass over sternocleidomastoid; Dx? → answer = brachial cleft cyst.
- Neonate born by forceps delivery + crooked neck + 1.5cm mass palpated in left side of neck; Dx? → answer = “sternomastoid injury.”

- 8F + hypothyroidism + dysphagia; Dx? → answer = lingual thyroid (ectopic thyroid location); 4-7x more common in girls; associated with hypothyroidism.
- 1F + Hct 26% + first six months of life on commercial formula + past 6 months of 2% cows milk and table food + smear shows pale RBCs; Dx? → answer = iron deficiency anemia → infants (especially age < 1 year) fed more than 24 ounces of cows milk daily at risk for iron deficiency anemia.
- 5F + pale conjunctivae + low Hb + low MCV + WBCs and platelets normal + low red cell distribution width (RDW); Dx? → answer = "decreased synthesis of globin chains"; thalassemia has low RDW, whereas iron deficiency anemia has high RDW (thalassemia → defective Hb leads to RBCs that are uniformly small; IDA creates non-uniformity of RBC size where some can escape marrow at normal size → high RDW); target cells are HY finding on smear (giveaway); next best step = hemoglobin electrophoresis.
- 4M + itchy head + small oval areas of alopecia on scalp; most likely causal organism? → answer = Trichophyton tonsurans → Dx is tinea capitis (cradle cap) → Tx with oral griseofulvin for patient only; reduce risk by not sharing hats (three Qs on the peds forms right there).
- 17M + lives in Florida + hypopigmented areas on shoulders and torso; Tx? → answer = topical selenium; Dx is tinea versicolor (Malassezia furfur).
- 7M + one-week Hx of low-grade fever and fatigue + 3-day Hx of rash and swelling starting at ankles and spreading upward + ankles tender + exam shows "several palpable petechiae and confluent purpuric areas over lower extremities"; Dx? → answer = Henoch-Schonlein purpura → can present with tetrad of 1) palpable purpura (and apparently petechiae), 2) IgA nephropathy (red urine), 3) abdominal pain (mesenteric adenitis), and 4) arthralgias; classically following viral infection, however 2CK vignettes will often omit mentioning the infection altogether; self-limiting.
- 4-month-old girl + asplenia + dextrocardia; which antibiotic should be used for prophylaxis? → answer = penicillin (most important for Strep pneumo due to asplenia); Dx is Ivemark syndrome (dextrocardia/situs inversus + hypoplasia of various organ systems).
- 3-month-old female + sepsis + Q asks for which antibiotic to give; answer = cefotaxime → give third-generation cephalosporin for sepsis; for young children (generally <6 years), use cefotaxime over ceftriaxone. Q on 2CK NBME 8 has 6M with sepsis and answer is ceftriaxone (cefotaxime isn't listed).

- 12-month-old female + sickle cell + missed dose of penicillin prophylaxis + now has sepsis; which Abx to give? → answer = cefotaxime; wrong answers are penicillin and ceftriaxone.
- Neonate with APGARs good at birth + 3-10 days later has BP 60/35 in upper extremities and unobtainable in lower extremities + O2 sats 98 mmHg but nails appear dusky + 3/6 holosystolic murmur at left sternal border; Q asks what's responsible for current presentation? → answer = "closure of ductus arteriosus" → Dx = preductal coarctation (the type seen in neonates); CoA can be associated with tricuspid regurgitation and other cardiac abnormalities. Dusky nailbeds don't equate to cyanosis if O2 sats are normal.
- 7-day-old neonate + not cyanotic + 4/6 holosystolic murmur at left sternal border + murmur was not present (or soft) at birth; what best explains the current presentation? → answer = "decreased pulmonary vascular resistance"; this is a VSD with L → R shunt; pulmonary vessels have gradually opened up over first week of life as ductus arteriosus closes; the decrease in pulmonary vascular resistance causes ↓ right-heart pressure → enables a favorable L → R pressure gradient for the VSD murmur to become more salient; lack of cyanosis means there is no R → L shunt + Eisenmenger wouldn't occur till later in life anyway.
- 5-day-old neonate + not cyanotic + 3/6 holosystolic murmur at left sternal border + murmur was not present (or soft) at birth; what best explains the child's presentation at birth? → answer = "increased pulmonary vascular resistance"; lungs still hadn't opened up so VSD L → R pressure gradient was not yet conducive to salient murmur; Peds shelf asks both versions of this question (i.e., increased vs decreased pulmonary vascular resistance as answers).
- 14F + Down syndrome + polycythemia + moderate cyanosis and digital clubbing + no murmur on cardio exam but loud S2 + echo shows large VSD and dilated main pulmonary artery; mechanism for polycythemia? → answer on Peds NBME = pulmonary artery hypertension"; Dx is Eisenmenger (large VSD with reversal R → L); large VSDs may present without murmur; pulmonary vessels constrict to compensate for high preload from previous L → R VSD shunt; constriction leads to hypoxia and secondary polycythemia with high EPO; should also be noted that Down syndrome is associated with endocardial cushion defects (AVSD > VSD > ASD; Step 1).
- Neonate + pulmonary valvular stenosis; Dx? → answer = Noonan syndrome; second most common syndrome associated with congenital heart disease after Down.

- Neonate <34 weeks gestation + dyspnea; Dx? → neonatal respiratory distress syndrome (NRDS; hyaline membrane disease); healthy lecithin (dipalmitoyl phosphatidylcholine) to sphingomyelin ratio is >2; ratio is reduced in NRDS.
- Neonate born at 39 weeks gestation via C-section + RR of 70 (normal 40-60) + CXR shows bilateral mild hyperinflation and prominent perihilar interstitial markings ; Dx? → answer = transient tachypnea of the newborn → TTN is the answer when the vignette "sounds like NRDS but the kid is term"; **seen in C-sections and fast vaginal deliveries in term neonates;** mechanism is delayed absorption of the fetal lung fluid by pulmonary lymphatics; Tx is supportive.
- Neonate born vaginally at 43 weeks gestation + cyanotic + pO2 is 26mmHg on 100% oxygen + CXR normal + echo shows normal cardiac anatomy with a right-to-left shunt across the foramen ovale; what is the mechanism for the child's condition? → answer = "failure of pulmonary vasodilation" → Dx = persistent fetal circulation (aka persistent pulmonary hypertension of the newborn).
- Neonate born at 26 weeks + required oxygen in ICU for several weeks + is now on home oxygen; child is at increased risk for what? → answers = bronchopulmonary dysplasia, retinopathy of prematurity, germinal matrix bleed → all can be seen in neonates on oxygen therapy.
- 6-hour-old newborn born at term + cyanosis of upper and lower extremities but no circumoral cyanosis + O2 sats normal + not in acute distress; next best step? → answer = "placement of the newborn under warming lights" → Dx = acrocyanosis → benign peripheral cyanosis that improves with warming; lack of circumoral cyanosis means no central cyanosis (i.e., heart/lungs OK).
- Newborn with slightly blue upper and lower extremities + placed under warming lights; next best step? → answer = "tactile stimulation and oxygen therapy." So for acrocyanosis: place under warming lights *then* do tactile stimulation + oxygen therapy.
- Three ways VSD sounds on Peds shelf?
    - Holosystolic murmur at left sternal border PLUS either parasternal heave or palpable thrill.
    - Holosystolic murmur at left sternal border PLUS left atrial enlargement (L → R shunt leads to increased preload back to LA in addition to the systemic preload).
    - Holosystolic murmur at left sternal border PLUS diastolic rumble (volume overload of LA).
- 2-month-old boy + not cyanotic + to-and-fro murmur; Dx? → answer = "extracardiac left-to-right shunt"; Dx = PDA.

- What are the four ways PDA sounds on Peds shelf?
    - Continuous, machinery-like murmur (easy; the one everyone knows).
    - Pan-systolic-pan-diastolic (if it's continuous, then it's present throughout cardiac cycle).
    - To-and-fro murmur (student says wtf? → HY; also on NBME 6 for 2CK, where the Q rides on you specifically knowing the description).
    - Bounding pulses (**normally refers to aortic regurg, but also a rare descriptor for PDA** → blood leaves aorta quickly L → R); in this scenario, the Q will say there's a continuous murmur with bounding pulses; in contrast, AR is a decrescendo holodiastolic murmur.
- 2-month-old boy + fixed splitting of S2; Dx? → answer = patent foramen ovale (ASD).
- Heart problem in neonate of mom with SLE? → congenital third-degree heart-block.
- 3M + elfin-like facies + hypercalcemia + well-developed verbal skills; Dx? → answer = William syndrome.
- Heart problem in William syndrome? → supravalvular aortic stenosis.
- Bicuspid aortic valve; most likely to cause which murmur? → aortic stenosis.
- Mid-systolic (crescendo-decrescendo) murmur + gets worse with Valsalva? → HOCM.
- Mid-systolic (crescendo-decrescendo) murmur + no change or softens with Valsalva? → aortic stenosis.
- Myxomatous degeneration of mitral valve? → mitral valve prolapse (MVP).
- Mid-systolic click? → MVP.
- Marfan or Ehlers-Danlos syndrome; heart murmur(s)? → MVP or aortic regurg.
- Late-peaking systolic murmur with ejection click? → another way they describe aortic stenosis.
- 2-week-old girl + HR of 240 + QRS duration of 50ms + no P waves + awake and alert + ice pack to the face does not change the HR; next best step? → answer = adenosine; Dx is paroxysmal supraventricular tachycardia (PSVT); complexes are needle-shaped (short QRS); in contrast, ventricular tachycardia has wide complexes (>120ms).
- 10-month-old girl + sore throat for 2 days + fever 101F + S3 gallop heard on auscultation; what best explains these findings? → answer = myocarditis (rheumatic fever); should be noted that PSGN will take 1-2 weeks to occur after Group A Strep infection, but NBME has RF occurring as early as 2 days

after the Strep pharyngitis; type II hypersensitivity via molecular mimicry with Group A Strep M protein (Step 1); Tx with penicillin.

- 8M + sore throat + new-onset tic; Dx? → answer = PANDAS (Pediatric Autoimmune Neuropsychiatric Disorder Associated with Streptococcus) → Group A Strep infection can precipitate OCD, Tourette, ADHD; answer = "check Streptolysin O titers"; student says wtf? → sounds weird when you first hear about it, but this is actually HY for Psych shelf, Peds shelf, and 2CK.
- 7F + facial grimaces past 5 months + no other motor findings or abnormal sounds + mental status normal; next best step in Mx? → answer = "schedule a follow-up examination in 3 months" → Dx = provisional tic disorder → 1/5 children experience some form of tic disorder; most common ages 7-12; usually lasts less than a year; "watch and wait" approach recommended. Provisional tic disorder used to be called transient tic disorder; the name was changed because a small % go on to develop chronic tics.
- 11-month-old boy + red tongue + maculopapular body rash + fever 100.5F; Dx? → answer = Scarlet fever; Group A Strep; give penicillin to prevent RF; tongue colloquially referred to as Strawberry.
- 4F + maculopapular body rash + fever + white spots visualized on buccal mucosa; Dx? → answer = rubeola (measles).
- 4F + maculopapular body rash + fever + suboccipital and/or post-auricular lymphadenopathy; Dx? → answer = German measles (rubella).
- 4M + swollen lower face bilaterally + irritable + stiff neck; Dx? → answer = mumps; POM → Parotitis, Orchitis, Meningitis; rash not classically associated.
- 4F + spiking fever followed by a rash + no other findings; Dx? → roseola (HHV-6); all you need to remember is the "spiking fever followed by a rash."
- 17F + itchy maculopapular rash on back + pink 2-cm ellipse on lower back; next best step in Mx? → answer = reassurance and observation → Dx = pityriasis rosea → caused by HHV-6 and HHV-7; self-limiting and lasts 5 weeks on average.
- 6M + Q shows you two pics: one of a circular rash (not a target) on the forearm; the other pic is a Bell palsy (asymmetric smile); Q asks you to pick the antibiotic → answer = amoxicillin; Dx = Lyme disease; rash need not be target (erythema chronicum migrans); **don't give doxycycline to kids age <8.**

- 3M + rapid breathing for 1 hour + hyper-resonance on right side of chest + CXR shows overexpansion of right lung; next best step in Mx? → answer = "endoscopic examination of the patient's airway" (bronchoscopy for foreign body aspiration); this Q is exceedingly HY for Peds shelf; easier Qs will mention a 14-month-old crawling around on the floor beforehand.
- 2M + writing movements observed in sleep by parents + MRI of head shows periventricular nodules; what else is most likely to be seen in this patient in the future? → answer = cardiac rhabdomyoma; Dx = tuberous sclerosis (TSC) → periventricular nodules are the "tubers"; can also see renal angiomyolipoma, adenoma sebaceum (angiofibromas in a butterfly distribution, along the nasal bridge and nasolabial folds), subungual fibromas (nailbed fibromas); writhing movements in sleep = seizures = classic first presenting feature.
- 17M + ataxia + cerebellar vascular lesion excised + red urine; Dx? → answer = von Hippel-Lindau → causes cerebellar and retinal hemangioblastomas; can eventually lead to renal cell carcinoma (which may become bilateral).
- 6F + axillary and groin freckling + small growth seen in the eye by mom? Dx? → neurofibromatosis type I (NF1) → Lisch nodules (iris hamartomas); optic glioma; axillary + groin freckling; neurofibromas (cutaneous lumps); café au lait spots (hyperpigmented macules); pheochromocytoma.
- 2F + congenital cataracts + mom has Hx of meningioma removed; Dx? → NF2 → congenital cataracts can be seen; meningioma; classically bilateral acoustic schwannomas.
- 16M + violaceous papules in a temporal distribution + Hx of seizure; Dx? → Sturge-Weber; Port wine stain birthmark classic, but violaceous papules is a newer presentation on USMLE.
- What are phakomatoses? → neurocutaneous disorders = TSC, VHL, NF1/2, Sturge-Weber; all autosomal dominant, but SW is *de novo*.
- Neonate + large facial hematoma + echo shows aortic coarctation; Dx? → answer = PHACES syndrome → Posterior fossa anomalies, Hemangiomas (classically large facial hemangioma), Arterial anomalies, Cardiac anomalies, Eye anomalies, Sternal defects.
- Neonate + becomes blue during breastfeeds + cries and becomes pink when detaching from breast + examination shows a pupil that appears to "bleed" into the periphery of the iris; Dx? → answer = CHARGE syndrome → **Coloboma** (hole) of the eye, Heart defects, **Atresia of the choanae,** Renal defects, Genitourinary defects, Ear abnormalities.

- 3F + brought to ED 30 mins after syncopal episode + mouth turned blue during temper tantrum + fell to floor, shook, and became limp + after a few seconds she regained normal activity + exam shows alert, active, well-hydrated patient; Dx? → answer = breath-holding spell.
- 4M + pale conjunctivae + 2/6 systolic murmur + Hb 6.4 g/dL + WBCs 50,000 + platelets 10,000; Dx? → answer = acute lymphoblastic leukemia (ALL) till proven otherwise; murmur is flow murmur (from low Hb → low O2 delivery → HR increases to compensate; **this is extremely HY for peds**).
- Leukemia in a kid; Dx until proven otherwise? → answer = ALL.
- 6M + just diagnosed with ALL + parents don't want him to know + you're now alone with the kid and he asked you if he's going to die; what do you say? → answer = "Have you talked to your parents about this?"
- 6M + ALL + facial flushing and/or positive Pemberton sign; Dx? → answer = TALL (T cell ALL) → thymic lesion can lead to SVC-like syndrome.
- 3F + 10-day Hx of coughing spells followed by vomiting; Dx? → answer = Pertussis; post-tussive vomiting and hypoglycemia sometimes seen; coughing may also be described as "paroxysms."
- 14M + WBCs 32,000 (85% lymphocytes) + two-week Hx of paroxysmal cough followed by vomiting; Dx? → answer = pertussis. Student says wtf? → pertussis can cause absurdly elevated WBC count that is almost all lymphocytes and resembles ALL.
- 17F + 8-month Hx of hirsutism + 1-yr Hx of menses occurring at 45-day intervals (prior was 28-day) + increasingly moody and depressed according to mom + striae over abdomen; next best step? → answer on Peds NBME = "measurement of morning fasting serum cortisol concentration"; students frequently say wtf at this answer but it's what the Peds NBME wants; wrong answers are measurements of prolactin and testosterone.
- 12F + recent viral infection + lethargic + polyuria + polydipsia + high serum and urinary glucose; what is the biochemical / acid-base disturbance most expected? → answer = high serum K (hyperkalemia) but low total body K; low Na; low bicarb (metabolic acidosis); low pH; low CO2 (compensation).
- 12F + DKA + HR 56 + BP 146/88 + headache and confusion; Q asks cause of the mental status change; answer = cerebral edema; hyperglycemia leads to osmotic changes with water moving out of cells → cerebral edema (serious complication of DKA) → give IV normal saline first before adding insulin.

- 12-month-old female + fever + 8-10 watery stools daily + low BP + high HR + low bicarb; why the low bicarb? → answer = lactic acidosis → any scenario with decreased perfusion (oxygen delivery) to muscle leads to increased anaerobic respiration; this is HY for any type of shock – septic, cardiogenic, hypovolemic (can also be seen with severe diabetes insipidus and pulmonary embolism).
- 3F + fell of tricycle two weeks ago + polyuria + polydipsia + high serum sodium + high urinary output; Dx? → answer = diabetes insipidus → head trauma can cause both central DI as well as SIADH; DI = high serum sodium and low urinary osmolality; SIADH = low serum sodium + high urinary osmolality.
- 3F + two-week Hx of fever, pallor, and decreased appetite + mouth ulcers + RBCs, WBCs, and platelets all low; Dx? → answer = aplastic anemia (presumably from Parvo, but can also be other viruses); next best step = immediate IV antibiotics for neutropenic fever (any time you have low neutrophils in the setting of fever; if WBCs are low, then neutrophils also low; in addition, mouth ulcers on USMLE = hugely HY presentation of neutropenia).
- 4F + treated for three weeks with TMP/SMX + now has isolated neutropenia; cause of neutropenia? → answer = TMP/SMX → just be aware this can cause neutropenia.
- 3F + fever + RBCs, WBCs, platelets all low; IV antibiotics are administered; next best step in Dx? → answer = bone marrow aspiration (sounds overly invasive, but it's what they want to Dx the aplastic anemia); there's a UWorld Q where "Parvo IgM/IgG titers" is an answer, but it's in an adult daycare worker with a lacy body rash and no drop in her hematologic cell lines.
- 4M + hypoplastic thumbs + RBCs, WBCs, platelets all decreased; Dx? → answer = Fanconi anemia; AR aplastic anemia characterized by absent or hypoplastic thumbs or radii.
- 4M + triphalyngeal thumbs + low RBCs; Dx? → answer = Diamond-Blackfan anemia; pure red cell aplasia characterized by triphalyngeal thumbs.
- 4F + immunodeficiency + fairer skin than siblings + giant granules seen in phagocytes; Dx? → answer = Chediak-Higashi syndrome; phagolysosomal fusion disorder with disruption of microtubule function; melanosome function also impaired.
- 2M + immunodeficiency + delayed separation of umbilical cord; Dx? → answer = leukocyte adhesion deficiency → deficiency of LFA-1/CD18 integrin.
- 2M + viral, fungal, bacterial, protozoal infections since birth + absent thymic shadow + scanty/absent lymph nodes/tonsils; Dx? → answer = severe-combined immunodeficiency (SCID) → XR type is most

common and due to common gamma-chain mutation or IL-2 receptor deficiency; AR is due to adenosine deaminase (ADA) deficiency; Tx is bone marrow transplant; absent thymic shadow means T cell deficiency; scanty/absent lymphoid tissue means B cell deficiency.

- 2M + bacterial infections from ~6 months of life + scanty/absent lymph nodes/tonsils; Dx? → answer = "deficiency of humoral immunity" (Bruton X-linked agammaglobulinemia); it must be pointed out that a Q on one of the newer Peds forms has Bruton as the answer for a kid who has bacterial infections *since birth* (which throws a dagger in the classic contrast with SCID); therefore the stronger emphasis must be that Bruton is bacterial only, whereas SCID is all types of infections (viral, fungal, bacterial, protozoal); Tx = "monthly infusion of immune globulin."
- 2M + mental retardation + fairer skin than siblings + deficiency of hydroxylase enzyme; how could this patient's presentation have been prevented? → answer = "routine newborn screening" via heel-prick test at birth for phenylketonuria (PKU; autosomal recessive; deficiency of phenylalanine hydroxylase) → tyrosine becomes essential; phenylalanine must be avoided in diet; malignant PKU = tetrahydrobiopterin (THB) deficiency (cofactor for phenylalanine hydroxylase).
- 8-month-old girl + oropharyngeal candidiasis + Hx of parainfluenza infection + 3/6 harsh systolic murmur at left sternal border; what does she have a deficiency of? → answer = T lymphocytes → possibly DiGeorge syndrome in this case (tetralogy of Fallot with murmur being a mix of VSD and pulmonary stenosis); T lymphocyte deficiency will cause viral, fungal and protozoal infections; if the USMLE wants B cell deficiency as part of answer, they'll mention bacterial infections.
- 8-day-old newborn + truncus arteriosus + serum calcium of 7mg/dL + no thymic shadow; Dx? → answer = hypoparathyroidism (DiGeorge; 3rd and 4th **pouch** agenesis → 3rd = thymus and inferior parathyroids; 4th = superior parathyroids).
- 16F + 3-day Hx of pain and pressure over left cheek + Hx of strep pneumonias at age 6 and 10 + seasonal allergies; Dx? → answer = "impaired humoral immunity" or "deficiency of mucosal immunoglobulin"; Dx is IgA deficiency (one of the highest yield presentations for not just the Peds shelf but also USMLE); sore cheek = sinusitis; presents as recurrent sinopulmonary infections; also associated with **Hx of Giardia infection, autoimmune diseases (e.g., vitiligo), and atopy** (dry cough in winter [cough-variant asthma], hay fever in spring, eczema in summer); anaphylaxis with blood transfusion is "too easy" for most 2CK IgA deficiency Qs but will rarely show up, yes.

- 3M + recurrent OM + recently underwent placement of tympanostomy tube + mom HIV negative; Dx? → answer = "antibody deficiency" → IgA deficiency.
- 6M + recurrent Staph infections + diarrhea; Dx? → answer = "neutrophil oxidation burst" → chronic granulomatous disease (NADPH oxidase deficiency) → infections with catalase (+) organisms (Staph infections are signature complication); SPACES → Staph, Psuedomonas, Aspergillus, **Candida, E. coli, Serratia;** Aspergillus most common fungal infection seen in CGD; 2CK has Q where child gets Serratia sepsis as part of CGD.
- 8F + fever + purpuric lesions over trunk and extremities + brother died of fulminant meningococcemia four years ago; Dx? → answer "complement system immunodeficiency" → terminal complement deficiency (C5-9) is associated with recurrent Neisseria infections (gonococcal and meningococcal).
- 14M + fever + stiff neck + non-blanching purpura on abdomen + BP of 60/35 + IV fluids and norepinephrine have limited effect; Tx? → answer = hydrocortisone; Dx = Waterhouse-Friderichsen syndrome; hemorrhagic necrosis of adrenal cortices secondary to meningococcal septicemia; non-blanching rash in the setting of meningitis = meningococcus; hydrocortisone is the answer because cortisol is deficient in this setting → cortisol normally needed to upregulate alpha-1 receptors on arterioles, thereby permitting NE and E to do their job; that's why NE has limited effect.
- 17F + Hx of cutaneous candida infections since childhood + 2-year Hx of type I diabetes mellitus + Hx of autoimmune thyroiditis; Dx? → answer = chronic mucocutaneous candidiasis → answer = "T cell dysfunction" = "impaired cell-mediated immunity" (both answers); although candida infections can be seen in diabetes, the durations don't match up here.
- 1-month-old male + diffuse white plaques in oral cavity that bleed when scraped + popular erythematous rash with satellite lesions over the diaper area; mechanism? → answer = "decreased T lymphocyte activity" → chronic mucocutaneous candidiasis.
- 8M + super high IgM on lab report; what's the mechanism? → answer = deficiency of CD40 ligand on T cells (can't activate CD40 on B cells to induce isotype class switching) → Dx = Hyper IgM syndrome.
- 6F + recurrent Staph abscesses + eczematoid skin lesions + abnormal dentition; Dx? → answer = hyper IgE syndrome (Job syndrome) → FATED → abnormal Facies, staphylococcal cold Abscesses, retained primary Teeth, hyper IgE, Dermatologic findings (e.g., eczematoid lesions).

- 12M + eczematoid skin lesion on forehead + nosebleeds + Hx of infections; Q asks which cell is dysfunctional → **answer = T cell (on student's 2CK, not Step 1);** Dx = Wiskott-Aldrich syndrome (XR) → classic triad of eczematoid skin lesions + thrombocytopenia + immunodeficiency.
- 3F + dilated superficial blood vessels on face + wobbly gait; mechanism? → answer = failure of double-stranded DNA break repair" → Dx = ataxia-telangiectasia.
- 8M + Q shows you a pic of a red dot on the tongue or nailbed; Dx? → answer = hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu); autosomal dominant; HY tangent outside peds: should be noted that this can lead to pulmonary arteriovenous fistulae in adults with high-output cardiac failure (NBME for Step 1 has 45M + epistaxis since childhood + shows pic of red dot on tongue + high ejection fraction; answer = "pulmonary arteriovenous fistula" [due to HHT]).
- 17M + 3-wk Hx of lymphadenopathy, fever, and loose stools + hepatosplenomegaly + low RBCs and WBCs + high IgM and IgG; Dx? → answer = HIV infection; wrong answers are CVID, IgA deficiency, SCID, DiGeorge, Bruton (so you can eliminate to get there); CVID is can rarely occur as young as adolescence but Dx is with IgG and IgA >2SD **less than** the mean (IgG high in this stem).
- Mother is HIV (+); what should be given to neonate following C-section? → answer = "A 6-week course of zidovudine (AZT) within 12 hours of delivery"; some students argue this is outdated, but it's the answer on Peds NBME form 1; no newer Q exists yet for this.
- 6F + pruritic rash in intertriginous areas of elbows and knees + often appears with episodes of wheezing and respiratory distress; Tx? → answer on NBME = topical triamcinolone (steroid); Dx is atopic dermatitis (eczema) as part of atopy; should be noted "emollients" (lotion; should be oil-based) are tried before topical steroids if both are listed.
- 17M + 12-month Hx of intermittent swelling of face/lips, chest, and arms + during episodes has abdominal pain and diarrhea; next best step in Dx? → answer = "measurement of C1 esterase inhibitor concentration"; **not C1 esterase; C1 esterase *inhibitor;*** Dx = hereditary angioedema; acute Tx = C1 esterase inhibitor administration; danazol also used (induces liver to produce more C1EI).
- 3F + brown blood; Dx? → answer = congenital methemoglobinemia due to deficiency of cytochrome reductase B5.
- 10M + ate new batch of sausage from his father, who is a butcher + high methemoglobin level; Tx? → answer = "administration of methylene blue"; can also add vitamin C to regimen.

- 9-month-old boy + “can only sit with support” + “bangs two blocks together but does not scribble” + “says ‘mama’ and ‘dada’ but not specifically to his parent”; Q wants “normal” or “delayed” for Gross motor, Fine motor, and Language development; answer = Delayed Gross motor + Normal fine motor and language development.
- 10-month-old boy + grade IV GERD; Q asks for what surgical Mx to alleviate Sx; answer = fundoplication.
- 12-hour-old newborn + excessive oral secretions and coughing after first feed + Sx resolved after suctioning + pregnancy was characterized by polyhydramnios; next best step in Mx? → answer = insertion of a nasogastric tube → Dx = tracheoesophageal fistula.
- 2-month-old boy + non-bloody non-bilious vomiting with feeds + increasing in amount and force; Dx? → pyloric stenosis → associated with first-born males + oral erythromycin use (motilin receptor agonist; Tx for chlamydia ophthalmia neonatorum); another Peds NBME Q wants you to know pyloric stenosis is “single developmental defect,” implying it’s almost always not part of a broader syndromic constellation; Dx with abdominal ultrasound; Tx = myomectomy; biochemical / acid-base disturbance = low K, low Cl, high bicarb; detail such as “olive-shaped” abdominal mass too easy for 2CK but not unheard of.
- 2-month-old boy + bilious vomiting + Down syndrome; Dx? → answer = duodenal atresia; do AXR which shows double bubble sign → answer = “failure of canalization of proximal bowel”; bear in mind that annular pancreas can be alternative cause of bilious vomiting + double bubble sign (not Down syndrome).
- 2-month-old boy + bilious vomiting + chronic constipation; Dx? → answer = Hirschsprung; also associated with Down syndrome; failure of neural crest cell migration distally; answer sometimes = “rectal manometry” as next best step in Dx; notable that this can cause bilious vomiting.
- Acid-base / biochemical disturbance in vomiting? → hypochloremic, hypokalemic metabolic alkalosis.
- 18-month-old girl + intermittent abdominal pain for 24 hours + mass palpated in RLQ + fecal occult blood test is positive; next best step in diagnosis? → answer = **air-contrast enema** (or contrast, or air, etc.); imaging modalities such as USS (target sign) may be performed, but on the NBME, the answer for how to Dx intussusception is always enema (Dx and therapeutic); presentation for intussusception will almost always be under the age of 2 and be characterized by an “intermittent” nature to the

child's presentation (i.e., intermittent crying; intermittently bringing the legs to the chest; intermittently squatting; intermittently vomiting).

- 7-day-old boy + bilious vomiting + blood in stool + abdo slightly distended + bowel sounds decreased + digital rectal exam shows gross blood + lateral x-ray of upper GI tract with barium contrast shows corkscrew appearance; Dx? → answer = "failure of normal bowel rotation" → congenital midgut volvulus → diagnosed with **upper-GI series** (x-ray with contrast swallow).
- 1-month-old boy + bilious vomiting + normal rectal tone + fecal occult blood positive; Dx? → answer = midgut volvulus (Peds NBME form 2); intussusception and Hirschsprung are wrong answers.
- 18-month-old girl + intermittent abdominal pain for 24 hours + fecal occult blood test is negative + AXR shows dilated loops of small bowel and air-fluid levels; Dx? → answer = "volvulus" → congenital midgut volvulus → originally I conjured that the negative fecal occult blood is the reason this is volvulus (i.e., "sounds like intussusception but FOB is negative"), but the above 18-month-old from Peds form 2 has positive FOB for midgut volvulus, so the point of distinction must be these specific AXR findings, which would not be seen for intussusception → **in other words, if vignette sounds like intussusception but AXR shows dilated loops of small bowel and air-fluid levels, answer = midgut volvulus, not intussusception, and FOB isn't specific for either.**
- 9M + blood in stool + Tc99 uptake scan confirms Dx; what's the mechanism for patient's presentation? → answer = "heterotopic gastric tissue"; NBME has Qs with Meckel presenting even in adults (do not fixate on age 2).
- 3F + failure to thrive + 3-month Hx of steatorrhea + did not pass meconium till 3 days of life; mechanism for failure to thrive? → answer = "exocrine pancreatic insufficiency"; this is frequently how the Peds shelf will refer to cystic fibrosis (can also refer to chronic pancreatitis in adults).
- 4-month-old girl with 5-cm melanoma-appearing lesion on thigh; Dx? → answer = congenital melanocytic nevus (most common type of birthmark; benign).
- 4-month-old girl with 3-cm "bruise" on her back + vignette doesn't sound like abuse; Dx? → answer = blue nevus (Mongolian spot); more common in Asian descent; melanocytes don't migrate to basal layer + get trapped in dermis.
- Neonate + spongy 1-cm red lesion on the chest; Dx? → strawberry hemangioma.

- Strawberry hemangioma Tx? → don't treat; will grow slightly then regress spontaneously over a few years.
- Neonate + large vascular lesion on the leg + thrombocytopenia; Dx? → Kasabach-Merritt syndrome (aka hemangioma with thrombocytopenia) → this is on the pediatric 2CK forms **three times** asked in different ways; students always say wtf and I have to explain that, yes, it's weird, but it's HY for some magical reason; this is not a strawberry hemangioma and requires surgical Tx.
- 1-month-old male + 12x8cm vascular lesion on buttocks since birth + examination shows scattered petechiae and bruises; what best explains these findings? → answer = "platelet sequestration." I've memorized this from the NBMEs → similar to splenomegaly, which can cause thrombocytopenia from sequestration within the red pulp, the implication that the large vascular lesion of KMS is that platelets simply get caught within it.
- What is Potter sequence? → renal agenesis leading to oligohydramnios → causes pulmonary hypoplasia in fetus (amniotic fluid needed for lung development).
- 12-hour-old male + abdomen is distended with a midline mass; Dx? → answer = posterior urethral valves → most common genitourinary (GU) abnormality in neonatal males → thin membrane(s) in the urethra preventing the outflow of urine.
- 2-day-old male + urinary dribbling + decreased urinary output + 6-cm suprapubic mass; Dx? → answer = posterior urethral valves; two different Qs on different Peds forms.
- 3-month-old boy + fever + pyuria + few bacteriuria; next best step in Dx? → answer = renal ultrasound; Dx most likely pyelonephritis (few bacteria typical on U/A despite pyuria); must do renal and bladder ultrasound for all kids age 2-24 months who have febrile UTI → screens for congenital genitourinary tract abnormalities.
- 6-month-old boy + one-day Hx of fever + pyuria + E.coli grown from urine culture + Abx initiated + renal ultrasound shows no abnormalities; next best step? → answer = voiding cystourethrogram (check for posterior urethral valves).
- What is phimosis vs paraphimosis? → phimosis = inability to retract the foreskin; paraphimosis = inability to reduce (put back) the foreskin.
- Neonatal male with PUV; next best step in Dx? → voiding cystourethrogram. Tx must do surgery.

- 34M falls on balance beam + saddle injury + blood at urethral meatus; Dx + next best step in Mx? → urethral injury; do retrograde urethrogram.
- 16M + skiing accident + bruising and/or pain over a flank + red urine; Dx + next best step in Mx? → kidney injury (renal contusion); must do CT with contrast, **not** ultrasound. This is exceedingly HY on surgery forms and sounds wrong (i.e., "why would you do CT with contrast in someone with suspected renal injury if contrast can cause nephrotoxicity?") → great question, but CMS/NBME/USMLE still want CT with contrast, so don't pick ultrasound.
- 16M + skiing accident + bruising and/or pain over a flank + no gross blood of urine (i.e, urine not red); next best step in Mx? → urinalysis to look for blood (super HY) → if urinalysis shows blood, do CT scan to check for renal injury. If U/A negative, renal injury unlikely, so choose "no further investigations indicated," **not** CT.
- 16M + skiing accident + bruising and/or pain over a flank + urinalysis is normal; next best step? → no further investigations indicated.
- 3M + ran into corner of table + 1-day Hx of increasingly severe abdominal pain + vitals normal + examination shows 2x2cm mass in the midline + serum lipase and amylase both elevated; next best step? → answer = CT of abdomen → Dx = traumatic pancreatitis.
- Kid + bloody diarrhea + petechiae + red urine; Dx? → hemolytic uremic syndrome (HUS) caused by EHEC O157:H7 or Shigella → HUS = triad of thrombocytopenia + schistocytosis (microangiopathic hemolytic anemia) + renal insufficiency; toxin will inhibit ADAMTS13 in afferent arterioles + cause endothelial damage → platelet consumption + clumping (thrombocytopenia) → RBCs shear as they pass by the platelet clumps (schistocytosis).
- How does HUS contrast with TTP? → TTP is caused by a mutation that results in defective ADAMTS13, or antibodies against ADAMTS13, resulting in the inability to cleave vWF multimers → platelet clumping → similar progression as HUS. One of the points of contrast is that TTP is not toxin-induced, and TTP also tends to be a pentad of the HUS findings + fever + neurologic signs.
- 30-month-old boy + fever 101F + loose, bloody stools + tonic-clonic seizure; Dx? → answer = shigellosis on Peds NBME; wrong answers are HUS and "idiopathic seizure disorder."
- 14F + lower abdominal cramps and intermittent knee pain for 6 weeks + 6-10 bloody stools per day + WBCs and platelets elevated + arthrocentesis is normal; Dx? → answer = ulcerative colitis (UC) → IBD

frequently associated with arthritis; high WBCs characteristic of autoimmune flares; thrombocytosis can be seen in infections and/or autoimmune flares; next best step in Dx = colonoscopy; Tx is initially with 5-ASA compounds (mesalamine) and steroids; colectomy can be curative in severe cases.

- 16M + mouth ulcer + bloody diarrhea; Dx? → answer = Crohn disease → mouth to anus; mouth ulcers are HY.
- HY differences between Crohn and UC?
    - Crohn → transmural (can lead to perianal fistulae); granulomatous (non-caseating); skip lesions (string sign on enema); cobblestoning (appearance on colonoscopy); "mouth to anus," but classically terminal ileum; increased risk of colorectal cancer (CRC) if colon affected (lower risk compared to UC overall); erythema nodosum and anterior uveitis common.
    - UC → not transmural; not granulomatous; loss of haustra (lead-pipe appearance on enema); pseudopolyps (appearance on colonoscopy); spares the anus; distribution is rectum-ascending; greater risk of CRC compared to Crohn; associated with pyoderma gangrenosum (crater-appearing lesion on [usually] limb with necrotic debris); associated with primary sclerosing cholangitis more than Crohn.
- 7F + vignette shows pic of hyperpigmentation around the lips; Dx? → answer = "hamartomatous polyps" → Dx = Peutz-Jeghers syndrome → perioral melanosis + hamartomatous colonic polyps → peds form has "colonoscopy" as the answer in a 7F with PJ syndrome for next best step in Dx.
- 8F + recently finished course of antibiotics + bloody diarrhea; Dx? → answer = pseudomembranous colitis (C. difficile) → presentations across NBMEs are both bloody and watery.
- 12M + epistaxis + bleeding time 9 minutes + platelet count 50,000; Dx? → immune thrombocytopenic purpura (ITP); mechanism = antibodies against GpIIb/IIIa following viral infection (2CK Qs will often **not** mention preceding viral infection for such diagnoses).
- 12M + epistaxis + bleeding time 9 minutes + platelet count 50,000; next best step? → answer = corticosteroids, then IVIG, then splenectomy; Qbank has a Q floating around where answer is observation only (apparently for mild cutaneous findings only in peds), but on NBME the answer is always steroids (you don't just watch the kid; you give him/her fucking steroids; not harmful).

- 17F + heavy periods + occasional epistaxis + finger takes a while to stop bleeding after being cut + normal PT + normal aPTT + normal platelet aggregation studies; Dx? → answer = von Willebrand disease (vWD); **autosomal dominant;** student says, "Wait, I thought aPTT and bleeding time are both elevated in vWD. Why would aPTT be normal here?" → BT is always increased, as the main function of vWF is to bridge GpIb on platelets to underlying collagen (so platelet dysfunction is guaranteed), but the stabilizing effects on factor VIII are ancillary, not primary, which is why aPTT is normal in about half of questions; it's to my observation that most vWD Qs will have aPTT around the upper limit of normal (NR 25-40s; where the Q might say aPTT is 38, or 43, seconds; only one Q I have seen where BT was 60s for vWD); vWD will always have a mix of a platelet problem (cutaneous; generally mild – i.e., epistaxis, petechiae) and a clotting factor problem (more severe – i.e., excessive bleeding after tooth extraction; menorrhagia); platelet aggregation studies are normal because vWD doesn't have anything to do with GpIIb/IIIa, which mediate *aggregation* between platelets, not *adherence* to underlying collagen; Tx is with desmopressin (DDAVP; stimulates production and release of vWF by endothelial cells).
- 5M + hemarthrosis; Dx till proven otherwise? → answer = hemophilia A or B (A way more common, so choose this if both listed and no way to differentiate); **both X-linked recessive;** isolated increase in aPTT → hemophilia A is VIII deficiency; hemophilia B (Christmas disease) is factor IX deficiency; hemarthrosis classic for hemophilia but rare in wVD (factor deficiency creates greater impediment of clotting than the auxiliary deficit seen in vWD); Tx is with DDAVP and factor VIII replacement in hemophilia A; DDAVP cannot be used for hemophilia B (factor IX replacement only).
- 5M + hemophilia A + Hx of several Txs of factor VIII replacement + Txs having increasingly reduced ability to help + aPTT is now >120s; Dx? → answer = antibodies against factor VIII (occurs with repeated Txs).
- 5F + hemophilia A; Q asks you directly how this is possible; answer = lyonization (skewed X-inactivation); you will never see an XR disorder in a female unless the explicit purpose of the question is lyonization.
- 12M + epistaxis + bleeding time 9 minutes + platelet count 50,000 + most effective way to decrease recurrence? → answer = splenectomy. Student says, "Wait, I thought we give steroids first." Yeah, you're right, but splenectomy is still *most effective* at decreasing recurrence.

- 10F + tonic-clonic seizure + 2-wk Hx of fever and joint pain + Hb, WBCs, and platelets all down + Coombs test positive; Dx? → answer = lupus; **cell lines are down due to anti-hematologic cell line antibodies** (i.e, this is not aplastic anemia; answer = "increased peripheral destruction"); Tx = steroids.
- 15F + arthritis + malar rash + urine positive for blood and protein; which of the following would help determine the course of management or this patient? → answer = renal biopsy → guides the management of lupus.
- 3F + viral infection + fever + tonic-clonic seizure; Dx? → answer = febrile seizure; fever can precipitate idiopathic seizure in 2-4% of children ages 6 months – 5 years; there is about a two-fold risk progression to epilepsy compared to general population; Tx is with benzodiazepine.
- 10-month old boy + jerking movements of the limbs + EEG shows chaotic high-amplitude spikes with no recognizable pattern; Dx? → answer = West syndrome (infantile spasms) → triad of spasms + hypsarrhythmia on EEG (no recognizable pattern with high amplitude spikes) + developmental regression; international definition of the diagnosis requires two out of three; starts age 3-12 months; seen in 1-5% of Down syndrome kids; Tx = **ACTH,** prednisolone, or vigabatrin; ACTH is thought to act by two mechanisms: 1) stimulates steroid (cortisol) release; 2) direct (cortisol-independent) effect on melanocortin receptors.
- 13M + tonic-clonic seizure + 4-month Hx of hypnagogic/hypnopompic jerking of left arm + uncle has epilepsy; Dx? → answer = juvenile myoclonic epilepsy; genetic with unclear inheritance pattern; characterized by myoclonic jerks (usually hypnagogic and/or hypnopompic) that progress to tonic-clonic seizures after several months; age of onset is usually 10-16, but can also start in adulthood; Tx is valproic acid.
- 4F + few-month Hx of near-daily seizures + seizures typically occur while she's sleeping + has started putting objects in her mouth and making less eye contact + seizures not responding to anti-epileptic meds; Dx? → answer = Lennox-Gastaut syndrome → severe childhood-onset epilepsy characterized by near-daily seizures and cognitive decline (hyperoralism is a sign of cognitive regression [babies put things in their mouths]); poor prognosis, with 5% mortality rate in childhood; 80-90% persistence of seizures into adulthood.
- Child <5 years + watery diarrhea → rotavirus; classically seen < age 5.

- Child <5 years + watery diarrhea + 12-year-old brother and parents have similar Sx; Dx? → answer = Norwalk virus, not rotavirus, because rest of family wouldn't get rotavirus.
- 4M + watery diarrhea two weeks ago + now has heavier diarrhea and bloating and flatulence with meals; Dx? → secondary lactose intolerance caused by sloughing of brush border following gastroenteritis (rotavirus).
- 8M + bloating and flatulence with dairy products; next best step in diagnosis? → answer = hydrogen breath test, or testing for decreased stool pH; Dx = lactose intolerance.
- 2F + iron deficiency anemia + bloating + diarrhea + eating table food; Dx? → answer = Celiac disease; can cause iron deficiency anemia; Dx with Ab screen (anti-endomysial/-gliadin; anti-tissue transglutaminase IgA).
- 5F + positive screen for Celiac Ab; next best step? (answers are either duodenal biopsy, or no further studies indicated) → answer = duodenal biopsy (shows flattened villi); sounds wrong, but it's what NBME wants.
- "Can you explain that milk protein allergy stuff? Like the cow milk, soymilk, hydrolyzed formula stuff."
    - Peds NBME / 2CK wants you to know that best way to decrease chance of developing allergy to food protein → answer = "exclusively breastfeed for the first 6 months." Milk protein allergy is one type of food protein allergy.
    - Milk protein allergy affects 2-3% of infants and young children; usually self-resolves, but 0.4% of adults retain the allergy.
    - **10-15% of infants with cow milk protein allergy demonstrate allergic crossover with soymilk protein.**
    - Dx can be made with skin prick and patch tests.
    - Treatment for baby/infant with milk or soy protein allergy on USMLE is "switch to hydrolyzed formula" → proteins processed into fragments → FDA labels hydrolyzed formulas as hypoallergenic.
    - Different types of milk protein allergy:
        - Food protein-induced enterocolitis syndrome (FPIES)

        - Very severe; incessant vomiting and diarrhea within hours of consumption; some may experience shock due to dehydration.
    - **Food protein-induced allergic proctocolitis (FPIAP)**
        - **Milder; characterized by intermittent bloody stools.**
        - **This is how cow milk allergy frequently presents on Peds NBME.**
    - Food protein-induced enteropathy (FPE)
        - Chronic diarrhea that resolves when milk protein removed from diet.

- 2-month-old boy + started on cow milk-based formula last week + 1-wk Hx of vomiting and diarrhea hours after feeds; next best step in Mx? → answer = "switch to hydrolyzed formula"; Dx = FPIES.
- 6-week-old girl + on cow milk-based formula for first 4 weeks of life + switched to soy milk-based formula 2 weeks ago + 2-wk Hx of stool streaked with blood and mucous + no vomiting; next best step in Mx? → answer = "switch to formula with hydrolyzed casein"; Dx = FPIAP.
- Bloody diarrhea + appendicitis-like pain (pseudoappendicitis) in a child → Yersinia enterocolitica.
- 3F + finished 10-day course of Abx + fever 101F + LLQ tenderness + fecal occult blood is positive; Dx? → C. difficile; Tx = vancomycin. Two important points: 1) C. diff causes bloody stool frequently on 2CK forms; 2) pseudoappendicitis seen with Y. enterocolitica is indeed RLQ (terminal ileitis / mesenteric adenitis), not LLQ.
- 16-month-old male + Giardia infection + failure to thrive; Q asks mechanism for failure to thrive; answer = "malabsorption of fats and carbohydrates"; Tx Giardia infection with metronidazole.
- 16M + painful testes + fever + positive cremasteric reflex; Dx? → answer = epididymitis
- Most common organism causing epididymitis? → Chlamydia in sexually active younger males; E. coli in elderly males. This is also the same for prostatitis. If the vignette tells you no organisms grow on culture, choose Chlamydia.
- 16M + acutely painful testes + negative cremasteric reflex; Dx? → answer = testicular torsion; do Doppler ultrasound followed by surgical detorsion.
- 6M + painful testis + superior pole shows blue/black discoloration + bowel sounds are decreased + abdomen is rigid; Dx? → answer = **strangulated hernia, not testicular torsion** → answer = "operative management."

- 6M + painful testis + superior pole shows blue dot + cremasteric reflex is intact; Dx? → answer = torsion of **appendix testis** → this is on the new peds form but is fair game for FM → torsion of appendix testis is different from testicular torsion; the latter presents with negative cremasteric reflex; but torsion of appendix testis causes "blue dot sign."
- 8-month-old boy + unilateral testicular enlargement + light shone to it causes transillumination; answer = "persistent processus vaginalis" (hydrocele).
- Tx for hydrocele? → **observe** until the age of one as most spontaneously resolve; this is almost always the answer; after the age of one, surgical management can be considered.
- Cryptorchidism; Tx? → observe; don't do orchiopexy until at least age 1; most spontaneously descend in first year of life.
- 3M + hard nodule on testis; Dx? → yolk sac tumor (endodermal sinus tumor) → serum AFP may be elevated.
- 3M + hard nodule on testis + serum AFP + beta-hCG are elevated; Dx? → answer = mixed germ cell tumor (embryonal cancer), **not** yolk sac tumor (yolk sac tumor is *only* high AFP; in mixed germ cell, beta-hCG is also elevated).
- 14M + heaviness and/or bogginess of testes; Dx? → varicocele → one FM shelf Q literally says "bag of worms" (normally this is so buzzy that we'd say this wouldn't show up on an actual form, but it does, so it must be mentioned here) → Dx with Doppler ultrasound → elective surgical intervention may be performed to reduce risk of sub-fertility due to increased testicular temperature.
- 14M + dragging sensation of left testis + soft tissue mass at top of left testis that disappears when supine; patient is at increased risk of what? → answer = infertility; Dx = varicocele.
- 16M + exam for school sports + both testes descended + painless mass in scrotum that can be "milked" into the abdomen through the inguinal ring; Dx? → answer = indirect inguinal hernia.
- 8-month-old boy has undescended testis; Tx? → answer = observation until at least the age of 1; do not do orchiopexy before age 1 for USMLE purposes.
- Tx for scabies? → answer = topical permethrin.
- Tx for pediculosis (lice)? → answer = topical permethrin.
- Tx for acne:

  - Topical retinoids first (i.e., topical tretinoin; NOT oral isotretinoin); **cause photosensitivity (rash);** also used for photoaging; mechanism is decreasing sebum production; topical tretinoin (not oral isotretinoin) is not a teratogen and does ***not*** have any effect on pregnancy or male sperm.
  - Benzoyl peroxide used second; often coadministered with topic retinoids; mechanism is the killing of bacteria.
  - Topical clindamycin.
  - Oral tetracycline; **causes photosensitivity (blistering).**
  - Oral isotretinoin; must do beta-hCG in women; recommend barrier contraception even if on OCP; can cause elevations in LFTs; can cause dyslipidemia; main complaint is dry skin + peeling; takes several weeks to really start working but ultra-effective according to most patients; can be commenced earlier in patients with severe nodulocystic acne; works by diffusely shutting of sebum production.

- 3M + large calves + uses arms in order to crawl up off the floor; mechanism for disorder? → answer = defective cystoskeletal connection to extracellular matrix → Dx = Duchenne muscular dystrophy (XR); caused by frameshift mutation; Becker is less severe form presenting in adolescence or early-adulthood; Becker not caused by frameshift (i.e., missense, etc.).
- 4F + intermittent cramping past several months + difficulty relaxing grip when squeezing physician's fingers; Dx? → answer = myotonic dystrophy → CTG trinucleotide repeat (TNR) expansion.
- 13M + 3-month Hx of decreasing muscle coordination + impaired speech + high-arched feet; Dx? → answer = Friedreich ataxia → GAA TNR expansion; development of cardiomyopathy in >90%; pes cavus (high-arched feet) a classic finding; in contrast, pes planus (flat feet) is Marfan syndrome.
- 14F + tripping and incoordination past 10 months + high-arched feet + hammer toes + weakness and wasting of hand and foot muscles + deep tendon reflexes absent at ankles + vibration and proprioception decreased on lower extremities + motor nerve conduction studies decreased in lower extremities with prolonged distal motor latencies; Dx? → answer = Charcot-Marie-Tooth disease (aka hereditary motor and sensory neuropathy); mechanism is defective protein production for myelin sheaths and nerve fibers; hammer toes classic; pes cavus also seen.

- 2M + "scissoring" of the legs + increased tone in all extremities; Dx? → cerebral palsy; posited to have peripartum fetal hypoxia as precipitating cause; spastic cerebral palsy accounts for 708-80% of cases.
- 10F + violaceous erythema of upper eyelids + scaly knuckles and finger joints; Dx? → answer = dermatomyositis → asked on Peds NBME; heliotrope rash + mechanic's hands; Gottron papules and shawl rash can also be seen; Dx with electromyography and anti-Jo1/-Mi2 (same as polymyositis); definitive Dx is muscle biopsy; Tx acute flares with steroids.
- 15M + 5'11" + plays soccer + knee pain; Dx? → Osgood-Schlatter → inflammation of patellar ligament at the tibial tuberosity; occurs in fast-growing, active teenagers; USMLE wants "repeated avulsion microfractures" as an answer.
- Kid + recurrent knee redness, warmth, pain + fever → Juvenile rheumatoid arthritis (JRA; called Still disease if there are extra-articular manifestations like fever and rash).
- Kid + recurrent joint pain +/- high ESR +/- rash → JRA.
- Kid + sore throat two days ago + high ESR + Hx of intermittent knee pain + presents today with knee pain + afebrile; Dx? → JRA → infection can be a precipitating factor for a flare.
- Tx for JRA? → NSAIDs + corticosteroids.
- 7F + 1-yr Hx of occasional fever and knee pain + low Hb + MCV 75; Dx? → anemia of chronic disease (AoCD) secondary to JRA → low MCV seen AoCD in various Qs on 2CK NBMEs (resources classically say normocytic [80-100]).
- 6M + viral infection + now has hip pain +/- fever; Dx? → answer = **toxic synovitis (aka transient synovitis), not septic arthritis** → inflammation of the synovial lining of hip joint; Tx is supportive.
- 6M + suspected JRA + red, hot, painful knee → **must do arthrocentesis** to rule out septic arthritis. If the vignette sounds like classic transient synovitis (affects *hip, not knee*), you do not need to do an arthrocentesis.
- 3F + recurrent joint pain + fever + rash; Q says "in addition to naproxen, which of the following is the next best step in Mx?" → answer = slit-lamp exam → indicated annually (high risk of anterior uveitis).
- 5F + 2-day Hx of limp and left hip pain + a week ago had watery stools and a temp of 100F + pain with weight-bearing and movement + no swelling or erythema; Tx? → answer = ibuprofen (toxic synovitis).
- 17F had kickboxing tournament last weekend + knee is red, warm, tender → arthrocentesis → septic arthritis.

- 15F + severe pain of sternoclavicular joint + fever + arthrocentesis yields thick, yellow fluid + gram stain shows gram-negative diplococci; next best step? → answer = "culture of the aspirate fluid"; student says "wtf why? You've already determined the organism" → determine sensitivities; but empirically she would receive IM ceftriaxone and oral azithromycin or doxycycline; if septic, do IV therapy.
- 3M + fever + bone pain of the tibia + Tc99 bone scan shows uptake in diaphysis; Dx? → answer = Ewing sarcoma → presents like osteomyelitis with bone pain and fever; t(11;22); onion-skinning on histo; blue cells.
- 16M + soccer tournament yesterday + fever + high WBCs + bone pain + Tc99 bone scan shows uptake in the metaphysis; Dx? → answer = osteomyelitis; uptake in metaphysis, not diaphysis (Ewing).
- 1M + white-eye reflex; Q asks what child is at increased risk for later in life; answer = osteosarcoma; this is a HY Q → children with hereditary retinoblastoma are also prone to osteosarcoma (generally age 15-25 years); Q might also say a 15-year-old with Hx of enucleation from childhood cancer has a bone mass → osteosarcoma.
- 14M + on knees helping dad with plumbing under kitchen sink for several hours + knee pain + joint shows no effusion + no fever; Dx? → answer = prepatellar bursitis.
- 4M + increased bilirubin following viral infection or drug; blood smear shows degmacytes; mechanism? → answer = decreased NADPH production; Dx = G6PD deficiency (XR); degamacytes = bite cells; Heinz bodies (denatured/oxidized Hb) also seen in RBCs.
- 4M + hemolytic anemia due to enzyme deficiency + G6PD not listed as answer; Dx? → answer = pyruvate kinase deficiency (second most common cause of hemolytic anemia secondary to enzyme deficiency, after G6PD).
- 7M + viral infection or took drug + spherocytes seen on smear + Coombs test positive; Dx? → answer = hemolytic anemia (caused by drug or infection); in contrast, if Coombs negative, answer = hereditary spherocytosis.
- 3M + Hx of pathologic jaundice treated with phototherapy for 5 days + father had splenectomy when younger + Coombs test negative + high reticulocyte count (10%); what would a blood smear show? → answer = "normocytic and normochromic"; Dx = hereditary spherocytosis; autosomal dominant; Tx ongoing hemolysis with splenectomy.

- 8F + viral infection + appears pale + spleen enlarged + low Hct + spherocytes on smear; patient is at increased risk for what? → answer = **cholelithiasis** due to pigment stones.
- Kid with suspected JRA has sore knee → must do arthrocentesis to rule out septic arthritis.
- 3F + 3-month Hx of leg pain predominantly in calves + occurs at night and wakes her from sleep + exacerbated by daily activity + relieved by acetaminophen + vitals normal + P/E normal; Dx? → answer on Ped NBME = growing pains (weird Dx you need to know).
- Patient has "knock-knees" (i.e., knees touch); Dx? → genu valgum.
- 9F + both legs bowed + small child and/or malabsorptive disease (i.e., Crohn, CF); Dx? →answer = rickets (vitamin D deficiency) → will also cause low calcium, low phosphate, high PTH, high ALP (because PTH high).
- 9F + both legs bowed + parents noticed bowing since she started to walk + recently bowing worse in right leg + x-ray while standing shows collapse of the medial aspect of the metaphysis of proximal tibia + rest of vignette describes healthy, thriving patient; Dx? → answer = tibia vara (Blount disease); wrong answer is rickets; should be noted that bowing is physiologic age < 2 years; tibia vara.
- 11F + spina bifida + paraplegic and wheelchair-bound + swelling and pain in thigh for two days + afebrile; next best step? → answer = "x-ray of lower extremity"; fracture may indicate child abuse.
- 4-month-old + "clicking/clunking" on physical exam → (+) Ortolani and Barlow maneuvers → primary hip dysplasia (congenital hip dysplasia) → once these are positive, **the next best step is ORTHO REFERRAL if it is listed** → referral always sounds wrong, but this is the correct answer if it's listed; if it's not listed, do ultrasound if under 6 months, or x-ray if over 6 months. Tx is with abduction harness (Pavlik harness; looks frog-leg-like).
- Newborn girl + palpable clunk when the hip is abducted, flexed, and lifted forward; what is the most likely mechanism of the disease? → answer = "shallow, poorly developed acetabulum" (congenital hip dysplasia).
- 5-8-year-old boy with painful limp; no other risk factors; x-ray shows contracted capital epiphysis; Dx? → Legg-Calve-Perthes (idiopathic avascular necrosis); the word "contracted" **wins** over "capital epiphysis" → this is a Q on one of the NBME forms where everyone selects slipped capital femoral epiphysis (SCFE), but it's Legg-Calve-Perthes; Tx = hip replacement.

- 5-8-year-old boy with painful limp + sickle cell disease; Dx? → avascular necrosis (but **not** Legg-Calve-Perthes, because LCP is idiopathic).
- 5-8-year-old boy + painful limp + x-ray is negative + bone scan confirms diagnosis; answer? → USMLE wants you to know that x-ray can be negative initially in avascular necrosis, **but bone scan or MRI can also pick it up.**
- 11-13-year-old overweight boy with a painful limp; Dx? → SCFE; Tx = surgical pinning.
- 13M + painful limp + walks with antalgic gait + vignette mentions zero about his weight; mechanism? → answer = "displacement of the epiphysis of the femoral head" → SCFE.
- 2-year-old boy running + playing with 8-year-old sister + they were holding hands and he fell + now he holds arm pronated by his side; Dx? → **nursemaid's elbow** → radial head subluxation.
- Tx for nursemaid's elbow → hyperpronation OR gentle supination (both are correct answers; only one will be listed).
- Kid falls on outstretched arm + pain over anatomical snuffbox; Dx + next best step in Mx? → scaphoid fracture → do x-ray.
- Kid falls on outstretched arm + pain over anatomical snuffbox + x-ray is negative; next best step in Mx? → **thumb-spica cast** → x-ray is often negative in scaphoid fracture; must cast to prevent scaphoid avascular necrosis → re-x-ray in 2-3 weeks.
- 14F + low T3/T4 + high TSH; Q asks what you'd see on biopsy of thyroid; answer on Peds NBME = lymphocytic infiltration (Hashimoto, aka chronic lymphocytic thyroiditis); no idea why this is asked specifically on the Peds form but it is.
- 17F + BMI 33 + irregular periods; Dx? → answer = anovulation; insulin resistance → abnormal GnRH pulsation → high LH/FSH → insufficient FSH means follicle isn't adequately primed by the time the LH spike triggers ovulation → follicle retained as cyst; add hirsutism → Dx = polycystic ovarian syndrome (PCOS; must have 11 cysts bilaterally as per Amsterdam criteria); **should be noted that on Peds form 5 the answer for the location of the pathology in PCOS is ovary, not hypothalamus.**
- 17F + polyuria + polydipsia + high BMI; Dx? → answer = type II diabetes (yes, in a 17-year-old).
- 10F + polyuria + polydipsia + sore throat two weeks earlier; most likely infective Dx? → answer = coxsackie B virus → can precipitate type I diabetes mellitus via molecular mimicry → antibodies

against six-oligopeptide viral sequence cross-react with beta-islet cell glutamic acid decarboxylase 65 (anti-GAD65 Abs).

- 10F + vesicles in posterior oropharynx; Dx? → answer = herpangina (coxsackie A) → can also cause hand-foot-mouth disease (vesicles at these locations).
- Young child with normal free T4 and low total T4 → thyroid-binding globulin deficiency (opposite of pregnancy).
- Young child + large belly + large tongue + hypotonia → cretinism (congenital hypothyroidism).
- Young child + large belly + family feeding him rice gruel; Dx? → answer = "protein-calorie malnutrition" → kwashiorkor.
- Young child in third-world country + severe wasting; Dx? → marasmus (total calorie malnutrition).
- Nocturnal enuresis; when is it pathologic? → after age 5.
- 6M + nocturnal enuresis; next best step? USMLE / NBME / shelf wants the following order:
    - Behavioral answer first; e.g., spend more time with child; decrease overt stressors as much as possible.
    - If the above not an answer, do **star chart** (positive reinforcement therapy; i.e., don't wet the bed and get a star; get 5 stars for extra dessert; 100 and we go to Disneyland).
    - If star chart not listed or already attempted, next answer is **enuresis alarm.**
    - Medications like imipramine and desmopressin are always wrong; water deprivation after 5pm is also always wrong.
    - Students mess these Qs up because they'll see enuresis alarm as correct on one form, but on a different form it's star chart, so know the above order.
- How to differentiate viral from bacterial upper respiratory tract infection (URTI)? → CENTOR criteria
    - If 0 or 1 point, the URTI is unlikely to be bacterial (i.e., it's likely to be viral). If 2-4 points, chance is much greater that URTI is bacterial.
    - **1) <u>Absence of</u>** cough (i.e., no cough = 1 point; if patient has cough = 0 points).
    - **2)** Fever.
    - **3)** Tonsillar exudates.
    - **4)** Lymphadenopathy (cervical, submandibular, etc.).

- There is a version of the criteria that includes age, but on the USMLE it can cause you to get questions wrong. So just use the simplified above four points.
    - If 0-1 point, answer = “supportive care”; or “no treatment necessary”; or “warm saline gargle” (same as supportive care); or “acetaminophen.” Latter is answer for 3M with viral URTI + fever on Peds NBME form 2.
    - If 0-2 points, next best step = “rapid Strep test.” If rapid Strep test is negative, answer = throat culture, **NOT** sputum culture.
    - While waiting on the throat culture results, we send the patient home with amoxicillin or penicillin for presumptive Strep pharyngitis.
    - If child is, e.g., 12 years old, and develops a rash with the beta-lactam, answer = beta-lactam allergy.
    - If the vignette is of a 16-17 year-old who has been going on dates recently (there will be no confusion; the USMLE will make it clear), the answer = EBV mononucleosis; therefore do a heterophile antibody test (Monospot test).
    - **EBV is the odd virus out that usually presents with all four (+) CENTOR criteria.**
    - This is why it’s frequently misdiagnosed as Strep pharyngitis. It is HY to know that beta-lactams given to patients with EBV may cause rash via a hypersensitivity response to the Abx in the setting of antibody production to the virus. EBV, in a patient who does not receive Abx, can cause a mild maculopapular rash. But the rash with beta-lactam + EBV causes a more intense pruritic response generally 7-10 days following Abx administration on the extensor surfaces + pressure points.
- 9-month-old boy + fever for five days + edema of dorsa of hands + cervical lymphadenopathy + cracked, fissured lips + redness of oral mucosa and conjunctivae; Tx? → answer = high-dose aspirin + IVIG; Dx = Kawasaki disease; never give aspirin to kid (Reye syndrome) unless Dx is Kawasaki; other features are palms and soles desquamation + coronary artery aneurysms.
- “Waiter tip” position in kid → upper brachial plexus injury (C5/6) → Erb-Duchenne palsy.
- Claw hand → lower brachial plexus → C8/T1 → Klumpke palsy.
- Asthma (outpatient) → albuterol (short-acting beta-2 agonist; SABA) inhaler for immediate Mx → if insufficient, start low-dose ICS (inhaled corticosteroid) preventer → if insufficient, maximize dose of

ICS preventer → if insufficient, add salmeterol inhaler (long-acting beta-2 agonist; LABA); in other words:

- **1) SABA; then**
- **2) low-dose ICS; then**
- **3) maximize dose ICS; then**
- **4) LABA.**
- That initial order is universal. Then you need to know last resort is **oral corticosteroids, however they are most effective.** In other words:
- 12M has ongoing wheezing episodes + is on albuterol inhaler; next best step? → add low-dose ICS.
- 12M has ongoing wheezing episodes + is on albuterol inhaler; what's most likely to decrease recurrence → oral corticosteroids (student says "wtf? I thought you said ICS was what we do next and oral steroids are last resort" Yeah, you're right, but they're still **most effective** at decreasing recurrence. This isn't something I'm romanticizing; this is also assessed on the FM NBME forms.
- After the LABA and before the oral steroids, any number of agents can be given in any order – i.e., nedocromil or cromolyn sodium, zileuton, montelukast, zafirlukast.
- MOA of nedocromil and cromolyn sodium? → mast cell stabilizers.
- MOA of zileuton? → lipoxygenase inhibitor (enzyme that makes leukotrienes from arachidonic acid).
- MOA of the -lukasts ?→ leukotriene LTC, D, and E4 inhibitors. LTB4 receptor agonism is unrelated and induces neutrophilic chemotaxis (LTB4, IL-8, kallikrein, platelet-activating factor, C5a, bacterial proteins [Step 1]).
- 16M goes snowboarding all day + takes pain reliever for sore muscles afterward + next day develops wheezing out on the slopes again; what's going on? → took **aspirin + this is Samter triad** (now cumbersomely known as aspirin-exacerbated respiratory disease [AERD]) → triad of aspirin-induced asthma + aspirin hypersensitivity + **nasal polyps**). Just to be clear, other NSAIDs can precipitate Samter triad, but the literature + USMLE will make it explicitly about aspirin.
- 16M takes aspirin + gets wheezing; what are we likely to see on physical exam? → answer on USMLE = nasal polyps.
- 8M + non-productive cough worse at night + SoB during the day + CXR shows mild hyperinflation; next best step in Dx? → answer on NBME = spirometry → Dx = asthma.

- 8M + Hx of asthma + recent dysphagia to solids + upper endoscopy shows many concentric rings; Dx? → answer = eosinophilic esophagitis → dysphagia and/or weight loss in patient with asthma; endoscopy shows "trachealization" of the esophagus; biopsy shows dense eosinophil infiltrates.
- 6F + attended summer camp for one month + comes home with peach-colored papules with central umbilication; Dx? → answer = molluscum contagiosum.
- 16M + large red, non-vesicular rash on the chin; Q asks for Dx based on image; answer = impetigo (don't go chasing HSV or tinea faciei).
- 11M + 7-day-Hx of yellow crusties around his lips + red urine; what's the mechanism? → answer = "antigen-antibody immune complex" (PSGN) caused by Strep impetigo (be aware that S. aureus exceeds Group A Strep for both bullous and non-bullous impetigo).
- 11M + yellow crusties on face; Tx? → answer = topical mupirocin (USMLE will not force you to choose between topical mupirocin and oral dicloxacillin / cephalexin for bullous impetigo; I point this out because more extensive impetigo can be managed with oral Abx, but NBME wants mupirocin; if the latter is not listed, choose oral dicloxacillin or oral cephalexin; these orals have MSSA coverage; amoxicillin and penicillin do not).
- Beefy red, well-demarcated skin plaque; Dx? → erysipelas → Group A Strep (S. pyogenes) >>> S. aureus.
- More diffuse pink skin lesion + tenderness + fever → cellulitis → S. aureus exceeds S. pyogenes.
- Tx of erysipelas + cellulitis → oral dicloxacillin or oral cephalexin.
- Most common cause of otitis media? → Strep pneumoniae.
- Tx of otitis media? → oral amoxicillin only.
- Tx of recurrent OM → amoxicillin/clavulanate.
- When to do tympanostomy tube? → three or more OM in 6 months, or 4 or more in a year.
- Most common cause of otitis externa? → Pseudomonas.
- Tx of otitis externa? → topical ciprofloxacin + hydrocortisone drops.
- 17M + on the crew team + repeated water exposure; how best to prevent otitis externa? → prophylactic alcohol-acetic acid drops; in contrast, carbamide peroxide drops are for cerumen (wax) buildup.

- 8M + Hx of ear infection one month ago + slightly reduced hearing in one ear + otoscopy shows fluid behind the tympanic membrane + afebrile and well-appearing; Dx + Tx? → answer = otitis media with effusion (serous otitis media); fluid accumulation seen occasionally in middle ear after resolution of OM (need not be recurrent OM); Tx is observation, as will usually self-resolve in 4-8 weeks.
- 10-month-old female + fussy + fever + no movement of left tympanic membrane on pneumatic otoscopy; Dx? → answer = otitis media; the wrong answer is otitis media with effusion; **immobility** of tympanic membrane = most sensitive finding for otitis media – i.e., if a Q tells you the tympanic membrane is mobile, the Dx is not OM.
- 7M + otoscopy reveals squamous proliferation seen behind tympanic membrane; Dx + Tx? → cholesteatoma; will gradually grow and invade inner ear, causing irreversible hearing loss; Tx is surgical excision.
- 2F + fever 103 + tugging on pinna + pinna is displaced upward and outward + tenderness of mastoid process; next best step? → answer = "CT of the temporal bone"; sounds incredibly wrong to do a CT on a kid in this scenario, but this is the answer on the Peds NBME; mastoiditis (malignant otitis externa) is sometimes associated with a temporal bone fluid collection that must be drained to prevent brain abscess; MRI or CT must be done; x-ray is the wrong answer; fluid collection must be drained as best answer; IV broad-spectrum antibiotics are then administered.
- 30-month-old male + knows 100 words but does not use them in phrases + parents concerned he's not developing verbally as well as older siblings did + other milestones met appropriately; next best step? → answer = audiometry → hearing deficits are common cause of language delay.
- 6M + bilateral pneumonia + no other risk factors; Dx? → answer = Mycoplasma.
- 6M + lobar pneumonia + no other risk factors; Dx? → answer = Strep pneumoniae.
- 6M + lobar pneumonia + CXR shows interstitial markings; Dx? → answer = Mycoplasma (the word "interstitial" wins over lobar distribution).
- 2F + lobar pneumonia + no other risk factors + Strep pneumo not listed as an answer choice; Dx? → answer = Staph aureus (Peds NBME).
- 5F + recent convalescence from influenza infection + now has bacterial pneumonia; organism? → answer = Staph aureus (answer following influenza infection).
- 17F + painless lateral neck mass + mediastinal mass; Dx? → Hodgkin lymphoma.

- 8M + puffiness around eyes and ankles; vignette gives you no other information; Dx? → answer = minimal change disease (lipoid nephrosis) → 2CK-level Qs will often **not** mention anything about preceding viral infections → classically periorbital and pedal edema +/- ascites; urine does not contain blood as this is nephrotic syndrome; LM shows nothing; EM shows effacement of podocytic processes.
- 8M + nephrotic syndrome + fever 103F + abdominal fluid wave + diffuse abdominal pain; next best step? → answer = abdominal paracentesis → Dx = spontaneous bacterial peritonitis secondary to ascites caused by [probably] minimal change disease; do gram stain of peritoneal fluid + check for >250 WBCs/hpf; Tx with third-gen cephalosporin (no hard rule, but proclivity for cefotaxime age <6; ceftriaxone >6).
- 44M + Hodgkin lymphoma + nephrotic syndrome; Dx? → answer = minimal change disease; I'm including this in this peds discussion because this is HY and everyone gets it wrong; "Wait wtf? MCD is peds; how does that make sense for 44M? Erratum?" → MCD is also answer for nephrotic syndrome in Hodgkin.
- 8M + sickle cell + nephrotic syndrome; Dx? → answer = focal segmental glomerulosclerosis (FSGS).
- 8M + sickle cell + dark urine; Dx? → answer = renal papillary necrosis.
- 12F + high BUN and Cr + epistaxis + low Hb + normal platelet count; Dx? → answer = "acquired platelet dysfunction" (uremic platelet dysfunction) → qualitative, *not quantitative,* platelet disorder caused by high BUN decreasing platelet efficacy; low EPO is wrong answer; low Hb due to epistaxis, not low EPO. If low EPO is the answer (i.e., anemia of chronic disease secondary to chronic renal failure), then the patient wouldn't have epistaxis.
- 12M + red urine 1-3 days after upper respiratory tract infection (URTI) → IgA nephropathy, not PSGN; can also get IgA nephropathy from GI infections.
- 12M + red urine 1-2 weeks after URTI or skin infection → PSGN → can get it from Group A Strep skin infections
- 10F + sore throat two weeks ago + red urine + peripheral edema + HTN; what's the mechanism for the HTN? → answer = "volume overload" → microvascular disturbance from the glomerulonephritis → RAAS stimulation.

- 6F + red urine + abdo pain + arthralgias + violaceous lesions on buttocks + thighs; Dx? → Henoch-Schonlein purpura; red urine = IgA nephropathy → HSP is tetrad of 1) IgA nephropathy, 2) palpable purpura, 3) arthralgias, 4) abdo pain.
- 16F + motor vehicle accident (MVA) + bruising/pain over flank + red urine; next best step? → answer = CT of abdomen with contrast to evaluate for renal injury; sound wrong, as you'd say "Really? A CT with contrast for suspected renal injury? Yeah. Ultrasound is the wrong answer.
- 16F + MVA + bruising/pain over flank + urine shows no gross blood + urinalysis performed for what? → answer = urine blood → presence of blood suggests renal injury in this setting.
- 16F + MVA + bruising/pain over flank + urine shows no gross blood + urinalysis shows no blood; next best step? → answer = "no further diagnostic studies indicated" → if accident + bruising/pain over flank, first look for gross blood; if positive, go straight to CT; if negative, do urinalysis looking for blood; if positive, do CT; if negative, no further studies indicated. This concept is HY on surgery NBMEs but also shows up on Peds form.
- 17F + HTN + low K + high bicarb; next best step? → answer = MR angiogram of renal arteries; Dx = fibromuscular dysplasia; another answer = "increased renin production."
- 13F has never had a period + has suprapubic mass + nausea + vomiting; next best step in Mx? → answer = **do beta-hCG** → she's pregnant; this is HY. Correct, girls can get pregnant without ever having had a period → must rule out.
- 14F has massive unilateral breast mass + mom is freaking out bc her sister died of breast cancer → answer = follow-up in six months → **virginal breast hypertrophy** is normal during puberty.
- 15M has unilateral mass behind his nipple +/- tenderness of it → answer = reassurance → physiologic gynecomastia of puberty (higher androgens are aromatized to estrogens).
- 14F + Tanner stage 3; which of the following is true? → **answer = menarche is imminent** → USMLE asks this Q straight up and it's exceedingly HY and frequent.
- 17F + often really pad period pain + needs to miss class sometimes because it's so bad + physical exam is normal → answer = primary dysmenorrhea = prostaglandin hypersecretion (PGF2alpha) → give NSAIDs.
- 17F + 2-day Hx of right-sided pelvic pain + vitals WNL + beta-hCG negative + USS shows 3.5cm simple cyst; next best step in Mx? → answer on Peds NBME = observation (should be noted OCPs is not an

answer here; on one of the Obgyn NBME forms, observation is wrong; correct answer is "oral contraceptive pills and re-ultrasound in 6 weeks) → therefore, choose the latter over observation if both listed (and no contraindications to OCPs); if only observation is listed, go with that.

- 14F + never had menstrual period + one-wk Hx of constant, severe pelvic pain + 6-month Hx of intermittent pelvic pain + BP of 90/50 + bluish bulge in upper vagina; Dx? → hematometra → imperforate hymen with blood collection in the uterus → vagal response causes low BP → Tx = cruciate incision of the hymen.
- 14F + never had menstrual period + 6-month Hx of intermittent pelvic pain + BP normal + bluish bulge in upper vagina; Dx? → hematocolpos → blood collection in the vaginal canal, but not backed up to the uterus like hematometra → Tx = cruciate incision of the hymen.
- 16F + vagina ends in blind pouch + no cervix or uterus + absent/scanty pubic and axillary hair + Tanner stage 4; Dx? → answer = androgen insensitivity syndrome → next best step in Mx = karyotyping (46XY).
- 16F + vagina ends in blind pouch + no cervix or uterus + coarse pubic and axillary hair + Tanner stage 4; Dx? → answer = Mullerian agenesis → next best step in Mx = karyotyping (46XX).
- 15M + 4'10" + normal hormone levels + bone parents normal height; next best step? → answer = get bone age → if bone age < chronologic age, Dx = constitutional growth delay (growth curve shifted to the right; will eventually become average height); if bone age = chronologic age, patient has genuine short stature. Variant of constitutional growth delay Q is: rather than saying patient's bone age is low, Q will say he is Tanner stage 2 (implying he hasn't yet gone through puberty so will catch up).
- 16F + 4'11" + Tanner stage 2 + scattered nevi + webbed neck + hasn't yet had menstruation + bone age = chronologic age; Dx? → answer = Turner syndrome (45XO); genuine short stature in Turner; LH and FSH are both high; can achieve eventual pregnancy via IVF (surrogate); webbed neck = cystic hygroma (lymphatic insufficiency).
- 15-month-old boy + 3 months ago was 25th %tile for length and weight + today is still 25th %tile for length but 10th %tile for weight; next best step in evaluation? → answer = "recommend a diet diary with calorie counts."
- 13F + went running outside in hot weather + very exhausted + body temp 102.2F + pulse 100; Dx? → answer = heat exhaustion (high body temp + no end-organ damage/failure); in contrast, heat stroke =

high body temp + end-organ damage; acanthocytes in blood smear = liver failure (HY for heat stroke); acanthocytes not limited to abetalipoproteinemia.

- Renal cancer in children? → Wilms tumor.
- 3M + painless flank mass + no other info; Dx? → Wilms tumor → answer on peds form = "ultrasonography of the abdomen."
- 6M + painless flank mass + seizure + MRI of brain shows periventricular nodules; Dx? → Tuberous sclerosis with renal angiomyolipoma.
- Anything else about Wilms tumor? → increased incidence in horseshoe kidney in Turner syndrome.
- Horseshoe kidney key point? → not only increased risk of Wilms tumor, but the kidney gets caught under the IMA. USMLE Step 1 likes this factoid for some dumb reason.
- What is WAGR complex? → Wilms tumor, Aniridia, Genitourinary malformation, Retardation; caused by WT1 gene mutation.
- What is Denys-Drash syndrome? → Wilms tumor + male pseudohermaphroditism (secondary sex characteristics are female; primary sex characteristics are male); caused by WT1 gene mutation.
- Neonate born at 4800g + macroglossia + omphalocele + visceromegaly; child is most likely to develop what? → answer = hypoglycemia → Beckwith-Wiedemann syndrome? → Wilms tumor + neonatal hypoglycemia + macrosomia / macroglossia + hemihypertrophy +/- omphalocele; caused by WT2 gene mutation.
- 1M + 2-month Hx of increasing weakness and difficulty feeding + examination shows profound hypotonia and tongue fasciculations + absent reflexes; Dx? → answer = "progressive infantile spinal muscular atrophy" → characterized by decreased reflexes, muscle weakness and decreased tone, and tongue fasciculations.
- Comment on AR vs AD polycystic kidney disease. → ARPKD is chromosome 6 and a pediatric condition; it's associated with hepatic fibrosis. ADPKD is chromosome 16 and presents in 30s or 40s; associated with HTN and circle of Willis saccular (berry) aneurysms.
- 4F + third case of pyelonephritis + high creatinine; most likely cause of high Cr? → "congenital urinary tract obstruction" or "vesicoureteral reflux"; Dx is chronic pyelonephritis → caused by recurrent acute pyelonephritis → chronic pyelo is characterized by "thyroidization of kidney" + blunting and scarring of renal calyses.

- 9F + Hx of recurrent pyelonephritis and simple UTIs + BP of 160/100; next best step in Mx? → answer = "measurement of serum urea nitrogen and creatinine concentrations" → likely chronic pyelonephritis leading to renal scarring + disruption of renal microvasculature + upregulation of RAAS.
- 6M + dancing eyes + HTN + lesion visualized in posterior mediastinum on CXR; Dx? → neuroblastoma; students says wtf? → can occur anywhere in the median sympathetic chain, although classically intra-abdominal. Dancing eyes = opsoclonus-myoclonus syndrome.
- Neuroblastoma; how to Dx? → answer on NBME is "urinary homovanillic acid (HVA) and vanillylmandelic acid (VMA); mIBG scan may also be used; n-myc gene.
- 5M + 4-day Hx of increased shortness of breath + CXR shows mass in superior mediastinum + thoracentesis yields 400mL serosanguinous fluid; most likely cause of pleural effusion? → answer = "malignant pleural effusion" → T cell ALL (mediastinal mass leading to SVC-like syndrome).
- Pathologic jaundice in peds = must have at least one of the following:
    - Any jaundice on the first day of life (first 24 hours of life), period = pathologic.
    - Jaundice present **after** one week if term, or after two weeks if preterm = pathologic.
    - Total bilirubin >15 mg/dL.
    - Direct bilirubin >10% of total bilirubin, *even if total bilirubin is <15 mg/dL.*
    - Rate of change of increase in bilirubin >0.5 mg/dL/hour.
- Tx for pathologic jaundice? → choose "phototherapy" first, followed by "exchange transfusion"; some literature makes a case for IVIG after phototherapy, but USMLE has exchange transfusion as correct, without listing IVIG.
- Neonate + total bilirubin of 14 mg/dL + direct bilirubin of 13 mg/dL; Dx + next best step in Mx + Tx? → answer = biliary atresia; next best step = liver biopsy; Tx = liver transplant. Notice that total bilirubin is OK in this particular scenario but direct is severely pathologic because >10% of total (i.e., >1.4 mg/dL).
- Neonate + total bilirubin of 20 mg/dL, with all of it unconjugated + continually worsens despite Mx; Dx? → answer = Crigler-Najjar → absence of UDP-glucuronosyltransferase (conjugating enzyme); Tx = liver transplant; type II of the disease is deficiency, rather than absence of enzyme, and can be Mx with phenobarbital, which leads to a reduction in bilirubin.
- Difference between breastmilk jaundice vs breastfeeding jaundice?

- **Breastmilk jaundice =** due to beta-glucuronidase in breast milk, which leads to de-conjugation of intestinal bilirubin + increased enterohepatic circulation –> jaundice that starts on day 3-5 and peaks at 2-3 weeks –> Tx = stop breastfeeding for ~48 hours (and do bottle feeding), which leads to a rapid decrease in bilirubin; once breastfeeding is resumed, bilirubin might rise, but not back to pathologic levels.
- **Breastfeeding jaundice =** insufficient feeding (e.g., failure of suckling, etc.) + decreased milk intake leads to reduced intestinal clearance of bilirubin –> increased enterohepatic circulation –> jaundice that peaks at 3-5 days –> Tx = formula feeding (fluid + caloric supplementation).

- Neonate born with jaundice (pathologic) + mom is O+ blood type + neonate is A or B blood type; Dx? → hemolytic disease of the newborn (ABO type) → mothers with O blood type will have fractional IgG (instead of all IgM) against A and B antigens → crosses placenta → fetal hemolysis → severity highly variable.
- Neonate born with jaundice (pathologic) + born from 2nd pregnancy of Rh negative mom; Dx? → answer = hemolytic disease of the newborn (Rh type) → presumably mother made antibodies against fetal Rh antigen from prior pregnancy following mixing of circulations.
- Neonate + ambiguous genitalia + hypoglycemia + high potassium + low blood pressure; Dx? → answer = congenital adrenal hyperplasia (CAH) due to 21 hydroxylase deficiency → low glucose because cannot make cortisol or 11-deoxycortisol; high potassium because cannot make aldosterone or corticosterone; ambiguous genitalia because aldosterone and cortisol precursors (pregnenolone and progesterone, plus 17-OH types) are shunted to DHEA-S and androstenedione production; androstenedione can be converted peripherally into testosterone; DHEA-S is adrenal-specific androgen. 21 hydroxylase deficiency is most common cause of CAH.
- Neonate + ambiguous genitalia + potassium not high + BP normal + DHEA-S elevated; Dx? → CAH due to 11 hydroxylase deficiency → patient can still make corticosterone (precursor to aldosterone) and 11-deoxycortisol (precursor to cortisol), so still has mineralo- and glucocorticoid synthesis, despite not full-blown aldosterone and cortisol.
- Neonate + potassium low + glucose low + DHEA-S low; Dx? → CAH due to 17 hydroxylase deficiency → potassium wasting from high aldosterone; cannot effectively make zona fasciculata (cortisol layer)

and reticularis (androgen layer) precursors and products; pregnenolone shunted purely to aldosterone in zona glomerulosa.

- 7F + progressive fatigue past 6 months + low BP + darkening of skin; Q asks "most effective test to determine diagnosis and management?" → answer on Peds NBME = "plasma cortisol"; Dx = Addison. On IM NBME the answer is "ACTH stimulation test" for next best step in adult with Addison. Literature says random plasma cortisol >25mcg/dL excludes Addison. It may be presumed that random cortisol may be performed prior to the ACTH stimulation test, since the latter will entail serial measurements (i.e., over time) of serum cortisol anyway.
- **Normal vaccine schedule by timeline:**
    - **At birth:**
        - Hepatitis B
    - **At 2, 4, 6 months (3 doses total):**
        - Hepatitis B (new 2020 guidelines: skip at 4 months; so give at birth, 2 months, and 6 months)
        - Strep pneumo PCV13
        - Rotavirus (oral; live-attenuated)
        - Polio (Salk; IM killed)
        - TDaP (tetanus, diphtheria, pertussis)
        - Haemophilus influenzae type B
    - **At 1 year:**
        - Strep pneumo (4th dose)
        - Haemophilus influenzae type B (4th dose)
        - MMRV (mumps, measles, rubella, varicella; 1st dose)
        - Hepatitis A (1st dose)
    - **At 18 months:**
        - Hepatitis A (2nd dose)
        - TDaP (4th dose)
    - **At 4-6 years:**
        - TDaP (5th dose)Polio (4th dose)

        - MMRV (2nd dose)
    - **Age 9-45:**
        - HPV (2 doses if first given age 9-14; 3 doses if first given age 15+)
    - **School age 11-13:**
        - TDaP (6th dose)
        - Meningococcal (1st dose)
    - **School age 16:**
        - Meningococcal (2nd dose)

- **Vaccines by organism:**
    - **HepB:** birth, 2 + 6 months (3 doses)
    - **Rotavirus (oral; live-attenuated):** 2, 4, 6 months (3 doses)
    - **Polio (Salk; IM killed):** 2, 4, 6 months; 4-6 years (4 doses)
    - **Strep pneumoniae PCV13:** 2, 4, 6 months; 1 year (4 doses)
    - **Haemophilus influenzae type B:** 2, 4, 6 months; 1 year (4 doses)
    - **Tetanus, diphtheria, pertussis (TDaP):** 2, 4, 6 months; 18 months; 4-6 years; 11-13 years (6 doses)
    - **Hepatitis A:** 1 year; 18 months (2 doses)
    - **Mumps, measles, rubella, varicella (MMRV):** 1 year; 4-6 years (2 doses)
    - **HPV:** 9-45 years (2 doses if first given age 9-14; 3 doses if first given age 15+)
    - **Meningococcal:** 11-13 years; 16 years (2 doses)
- **Special notes on vaccines:**
    - **Influenza**
        - Two options: either killed (IM) or live-attenuated (intranasal)
        - Only give in fall or winter **(if they say, e.g., April, the answer is don't give)**
        - Give IM killed starting at age 6 months, then every year (annually)
        - Live-attenuated may be given annually in the fall or winter **only to non-pregnant, non-immunocompromised persons age 2-49**
    - **Strep pneumoniae (additional administrations)**
        - Give one dose of PCV13 to all persons age 65, then PPSV23 6-12 months later

    - To patients with asplenia, sickle cell, and cochlear implants, give PPSV23 6-12 months after last dose of PCV13.

- Neonate + diffuse pink body rash + desquamation of palms and soles; mechanism? → answer = exotoxin → Dx = Staphylococcal scalded skin syndrome due to exfoliative toxin (produced by ~5% of community staph).
- Neonate born at home + umbilical cord cut with kitchen knife and tied with twine + trismus; Dx? → answer = tetanus; Tx = antitoxin; trismus = lockjaw (HY "vocab word" for USMLE); don't confuse this with tenesmus, which is the intractable feeling of needing to defecate; C. tetani classically enters through umbilical stump.
- Neonate born at home + bleeding from umbilical stump; BT, aPTT, PT? → answer = vitamin K deficiency → BT normal; PT and aPTT both elevated; vitamin K deficiency due to sterile bowel in the absence of injection at birth.
- 6M + crops of vesicles on the trunk at different stages of healing; Dx? → answer = varicella (VZV) → presumably not previously vaccinated; know this description.
- Neonate dies *in utero* + mother owns a deli; Dx? → granulomatosis infantiseptica (severest form of Listeriosis); deli meats and soft cheeses are sources during pregnancy.
- Neonate born at term + fever + high WBCs (65% neutrophils) + CSF shows trace cells (90% lymphocytes); Dx? → answer = Group B Strep (GBS; Strep agalactiae) sepsis; students can get confused over the CSF findings, but this apparently is a normal finding; 65% neutrophils is slight left-shift for bacterial infection.
- Neonate + GBS sepsis; Tx? → answer = ampicillin + gentamicin; wrong answer is ceftriaxone + vancomycin.
- 6M + diagnosed with ALL + on chemotherapy + vesicles beneath the scapula; Dx? → herpes zoster (immunosuppressed; you need to know pediatric shingles "is a thing" – i.e., it exists).
- Neonate + microphthalmia + limb deformities + zig-zag skin lesions; Dx? → congenital varicella syndrome.
- Neonate + intracranial/periventricular calcifications + hydrocephalus + chorioretinitis; Dx? → answer = congenital toxoplasmosis (know this triad).

- Neonate + patent ductus arteriosus +/- deafness +/- cataracts; Dx? → answer = congenital rubella syndrome.
- Neonate + saddle nose + saber shins + Hutchinson teeth +/- deafness +/- cataracts; Dx? → answer = congenital syphilis.
- Neonate + intracranial calcifications + deafness +/- hepatosplenomegaly +/- blueberry muffin rash; Dx? → answer = congenital CMV; calcifications are non-specific, but the triad is not for toxo, so you can rule that out; deafness very HY for CMV, although also not specific; CMV diagnoses from vignette after you've eliminated the others, as per the descriptions for toxo, syphilis, rubella, and varicella.
- 16F + sexually active; what do we do – Pap smears and/or STI checks? → Don't do Pap smears. Start them at age 21 regardless of age of onset of sexual activity. However, yes, check for chalmydia and gonorrhea starting from age of sexual onset. Also make sure she has HPV vaccine if not received already.
- 16F + mucopurulent discharge + adnexal pain; Tx? → answer = IM ceftriaxone, PLUS either oral azithromycin (one-off stat dose) or oral doxycycline (BID for a week) → always cotreat for chlamydia and gonorrhea; if patient is septic, must give IV Abs (they ask this contrast on the NBME); Dx here is pelvic inflammatory disease (PID).
- 17F + completely asymptomatic + male partner recently tested positive for gonorrhea; tests are ordered for chlamydia and gonorrhea; do we treat her or not treat her? → answer = "intramuscular ceftriaxone and oral doxycycline"; the wrong answer is "no treatment indicated because the patient is asymptomatic."
- 16M + redness of both eyes + rash over extremities + mucopurulent urethral discharge; Dx? → answer = reactive arthritis → due to chlamydia; gonococcus does **not** cause reactive arthritis.
- 16M + sexually active + left knee pain + fever + S. aureus not listed as answer; Dx? → answer = gonococcal arthritis.
- 16M + polyarthritis + positive Finkelstein test + cutaneous papules over the wrist; Dx? → answer = gonococcal arthritis → will present either as monoarthritis of large joint, such as the knee, or as a triad of polyarthritis, tenosynovitis, and cutaneous papules.

- 16F + thin, grey, watery discharge + Dx is made by wet mount; Dx? → answer = bacterial vaginosis (Gardnerella); clue cells seen on wet mount (squamous cells covered in bacteria); KOH whiff test with fishy odor also classic; Tx with topical metronidazole gel.
- 16F + yellow-green discharge + Dx made by wet mount; Dx? → answer = trichomonas vaginalis → wet mount shows flagellated protozoa; strawberry cervix (punctate hemorrhages) or erythematous vaginal canal might be seen; Tx with topical metronidazole for patient *and partner* (high risk of reinfection).
- 16F + vesicular lesions on vulva + stinging/burning pain; Q asks "What is the most likely trajectory of these lesions?" → answer = "gradually disappear over one week" → can agree or disagree with this exact timeline, but it's on the NBME; wrong answer is "gradually worsen then disappear over three weeks"; Tx is clearly with oral acyclovir (or oral valacyclovir, but NBME will often have "oral acyclovir" as answer); topical acyclovir is the wrong answer (even if co-prescribed in real life, USMLE wants oral over topical if both listed).
- 17M + painless papule that ruptures into painless crater on penile base; next best step? → answer = darkfield microscopy → Dx = primary syphilis; VDRL/RPR can be performed but sometimes negative in primary syphilis; answer on one NBME form = "spirochete."
- 17M + sexually active + maculopapular rash on upper back + palms and soles rash; next best step? → answer = VDRL/RPR, followed by fluorescent treponemal antibody; Dx = secondary syphilis.
- 18M + positive Romberg sign + sexually active; Dx? → answer on Neuro NBME = neurosyphilis → sounds wrong at first because you say, "Wait, really? Neurosyphilis in someone as young as 18?" But neurosyphilis can occur at any stage of syphilis, and often appears during secondary. Positive Romberg refers to tabes dorsalis. Can also present with Argyll-Robertson pupil (accommodates but doesn't react).
- 17M + went to Uganda over summer vacation + has painful crater on base of penile shaft; Dx? → answer = haemophilus ducreyi → Tx = azithromycin.
- 17M + skin-colored papules on penile shaft; Tx? → answer = podophyllum resin → Tx for HPV (this is also the answer on the Obgyn NBME for vulvar condylomata acuminata).
- 3F + purulent bloody discharge on underwear; Dx? → answer = vaginal foreign body.

- 17F + BMI 28 + erythematous and itchy vulva + no mention of discharge but Q asks for organism; Dx? → answer = candidiasis; classically curd-like, white discharge, but one 2CK-level Q doesn't mention this; Tx = topical nystatin first or oral fluconazole (topical nystatin usually tried first).
- 7M + 6-wk Hx of headaches and difficulty walking + bilateral papilledema; Dx? → answer = medulloblastoma → need to know this is **cerebellar,** not medullar; medulloblastoma of cerebellar vermis → truncal ataxia; morning vomiting classic, but papilledema = increased intracranial pressure.
- 4M + morning vomiting + MRI shows infratentorial lesion that's mixed hyper- and hypointense; Dx? → pilocytic astrocytoma → mixed solid and cystic grossly → most common peds primary brain tumor.
- 4M + upward-gaze palsy; Dx? → Paranaud syndrome due to pinealoma.
- 4M + bitemporal hemianopsia + MRI shows calcification; Dx? → answer = craniopharyngioma → most common pituitary tumor in peds (prolactinoma in adults); some sources argue it's not a true pituitary tumor since it's derived from Rathke pouch (roof of primitive pharynx); can calcify and have motor-oil appearance; frequently recurs following surgery.
- 6M + Q shows you image where kid looks like he got hit by a softball on his eye; Dx? → answer = orbital cellulitis → must give IV antibiotics; in contrast, pre-septal cellulitis is less severe and can be treated with orals.
- 6M + painful bump on eyelid; Dx? → hordeolum (aka stye); S. aureus infection; Tx with warm compresses.
- 6M + painless bump on eyelid; Dx? → chalazion (blocked oil duct); Tx with warm compresses.
- 6M + painless bump on eyelid + warm compresses not working; Dx? → dermoid cyst of eyelid; Tx is surgical.
- Comment on neonatal conjunctivitis (ophthalmia neonatorum):
    - Chemical (first 24 hours of life)
        - Frequently caused by silver nitrate eyedrops (gonococcal prophylaxis), although this is more historical and has been replaced by topical erythromycin drops (most states require by law).
        - Erythromycin (and silver nitrate) drops are only prophylaxis against gonococcus; they are *not* prophylaxis against chlamydia.

        - Best way to prevent chlamydial and gonococcal ophthalmia neonatorum is effective Tx in the mother prior to delivery.
    - Gonococcal (2-5 days of life)
        - Purulent discharge; Tx with IM cefotaxime (choose over ceftriaxone for neonates).
    - Chlamydial (5-14 days of life)
        - Watery or purulent discharge.
        - **Can classically lead to chlamydia trachomatis D-K pneumonia** (i.e., not chlamydia pneumoniae); can drain through nasolacrimal duct down into the nasopharynx and lungs; diffuse crackles auscultated 1-2 weeks after the eye infection.
        - Tx with **oral** erythromycin (topical not only less effective, but also does not account for any organism having entered the nasopharynx and/or lungs).

- 4M + goes to daycare + red, itchy eyes; Dx? → answer = adenovirus (most common cause of viral conjunctivitis).
- 4M + goes to daycare + red, itchy eyes; the organism most likely responsible can also cause what? → answer = hemorrhagic cystitis (red urine) → adenovirus.
- 6M + playing with firecracker + pain and tearing of eye + ocular movements full + visual acuity 20/20; next best step? → answer = "fluorescein instillation of eye"; Dx = corneal abrasion; other vignette will say kid playing in sandbox or near dad in metal shop; corneal abrasion shows up green on fluorescein staining (normal areas are blue); cornea heals super-fast + patient will normally have incessant tearing and scratchy sensation for a day and then wake up next morning just fine.
- 6M + itchy eye + vesicles present infraorbitally + fluorescein staining of eye shows dendritic pattern; Dx? → answer = herpes keratitis; similar Dx is herpes zoster (VZV) ophthalmicus (immunocompromised if child).
- 6M + being treated with chemo for ALL + vesicles around ear + Bell palsy; Dx? → herpes zoster oticus, aka Ramsey-Hunt syndrome type II (VZV).
- Newborn + cystic dilation of 4th ventricle + absent cerebellar vermis; Dx? → Dandy-Walker syndrome.
- Newborn + myelomeningocele + downward herniation of cerebellar vermis; Dx? → Arnold-Chiari malformation (term only applies to Chiari type II).

- 18M + bilateral loss of pain and temperature sensation; Dx? → syringomyelia (can be seen secondary to syrinx from Chiari type I [less severe and presents later than type II]).
- Newborn + prominent occiput + micrognathia + overlapping fingers; Dx? → answer = Edward syndrome (Trisomy 18).
- 8-month-old boy + 3rd-centile for weight + slanted palpebral fissures + epicanthal folds + single palmar crease + thin upper lip with a "fish mouth" appearance + indistinct nasal philtrum; Dx? → answer on Psych NBME = **fetal alcohol syndrome (FAS), not Down syndrome;** everyone says wtf about this question, so what I tell my students is: if Q sounds like Down syndrome but they mention anything about the philtrum (i.e., long, smooth, indistinct, etc.), the answer is FAS, not Down.
- 6M + IQ of 60 + small for age + born to female age 41 + no other information given; Q asks: most likely cause of mental retardation? → answer = Down syndrome, not FAS; although FAS is most common cause of MR overall, two points: 1) if they want FAS, they'll mention the philtrum as per above, and 2) most common cause of MR *over the age of 40* is Down, not FAS.
- 8M + prominent jaw + protruding ears + IQ of 65; most likely explanation? → "Fragile site on the X chromosome" → Fragile X (CGG TNR disorder).
- 7M + prominent ears + flattened nasal bridge + long philtrum + low IQ; Dx? → FAS, not Fragile X; the Q mentions the philtrum.
- 4F + wringing of the hands + putting objects in her mouth + less eye contact; Dx? → answer = Rett syndrome; only seen in girls; hyperoralism may reflect cognitive regression (babies put everything in their mouths).
- 5M + talks to imaginary friends; Dx? → age-appropriate behavior.
- Newborn + multiple joint contractures; Dx? → answer = arthrogryposis (just be aware of this).
- Newborn + clubbed feet; Tx? → answer = **serial casting** in order to reposition feet (followed by minor surgical Tx).
- 16M + belligerent + nystagmus; Tx? → answer = lorazepam (benzo for PCP toxicity).
- 16M + mutism + constricted pupils; Dx? → answer = PCP intoxication; student says "Wtf? I thought there was belligerence / bellicosity?" This is one of the presentations on the NBME; if you don't believe me, Google "Mutism constricted pupils PCP."

- 16M + found on floor in school bathroom + sluggish + vitals and pupils normal; Dx? → answer = butane toxicity (inhalant toxicity); **student had this on actual USMLE.**
- 14M + ataxia + cognitive decline; Dx? → glue toxicity, not EtOH.
- 14M + dilated pupils + tachycardia; Tx? → answer = benzo (cocaine).
- 3M + microcytic anemia + one-month Hx of poor coordination, anorexia, irritability, and apathy; what could have prevented this? → answer = "elimination of lead from the child's home"; Tx = succimer for kids (if lead levels >44 ug/dL); lead can cause microcytic anemia.
- 14M + dilated pupils + visual hallucinations + staying up all night; Tx? → answer = benzo (amphetamine).
- 2M + blisters on tongue/mouth + next best step? → answer = "fiberoptic endoscopy" → Dx? = exposure to drain cleaner.
- 16F + found by mom 20 minutes ago + consumed bottle of aspirin in suicide attempt + lethargic; what is the acid-base disturbance she has? → answer = mixed metabolic acidosis-respiratory alkalosis → the presumption is we don't know how long ago she ingested + the lethargy might imply the acidosis has taken effect (acutely, salicylate toxicity causes isolated respiratory alkalosis); another Q wants you to select the literal values for mixed: normal Na, Normal K, normal Cl, low bicarb, low CO2, normal or low pH (the metabolic acidosis will eventually "win" and the patient will become severely acidotic, but rarely a Q might give you a pH in the lower end of normal range if patient is in transition).
- Tx for aspirin toxicity? → answer = bicarb → increases excretion via urinary alkalinization → cannot reabsorb the oxalate (deprotonated, negative charge) form through the tubular walls.
- 16F + long Hx of depression + taking new medication + large pupils + dry skin + ECG shows QT changes; Tx? → answer = sodium bicarb → Tx for TCA toxicity → causes dissociation of TCA from myocardial sodium channels → TCA is basic, not acidic (in contrast to salicylates), so bicarb actually *decreases* urinary excretion, but the mechanism as an antidote is different; TCAs cause CCC → coma, convulsions, cardiotoxicity + triad of anticholinergic, anti-alpha-1-adrenergic, and anti-H1-histaminergic side-effects.
- Difference between caput succedaneum and cephalohematoma?
    - Caput succedaneum is poorly defined soft tissue edema on the scalp; caused by pressure of fetal scalp against cervix during parturition, leading to transient decreased blood flow and

reactive edema; crosses suture lines; can be purplish in color similar to cephalohematoma (i.e., don't use color to distinguish); complications rare; disappears in hours to few days.

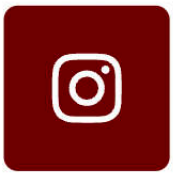
- Cephalohematoma is well-defined, localized, fluctuant swelling; caused by subperiosteal hemorrhage (answer on Peds NBME = "blood under periosteum of parietal bone"); does not cross suture lines; may be associated with underlying skull fracture, clotting disorders, jaundice; disappears in weeks to months.

- Classic five findings that are seen in neonates of diabetic moms?
  - Hypoglycemia → high neonatal insulin secondary to high maternal glucose crossing placenta; insulin does not cross the placenta; if they tell you the neonate has hypoglycemia and then ask for the mechanism, the answer is "increased insulin secretion"; hypoglycemia is also the answer if the Q describes the neonate as jittery.
  - Hypocalcemia → answer if they mention QT interval changes.
  - Hypomagnesemia → just be aware this is possible; hypomagnesemia can be a cause of hypocalcemia and/or hypokalemia refractory to supplementation (i.e., non-responsive to supplementation); this is more established in adult alcoholics who are dietary Mg deficient but can theoretically be applied to pediatrics.
  - Hyperbilirubinemia → increased risk for jaundice and kernicterus.
  - Polycythemia → increased RBC turnover is normal in neonates as fetal Hb declines and adult Hb rises; physiologic jaundice Q will often give Hct in the low-50s% **as normal;** polycythemia in neonates is defined as venous Hct >65%.
  - Should be noted that babies born to diabetic moms are often large (>4000g; macrosomia).



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